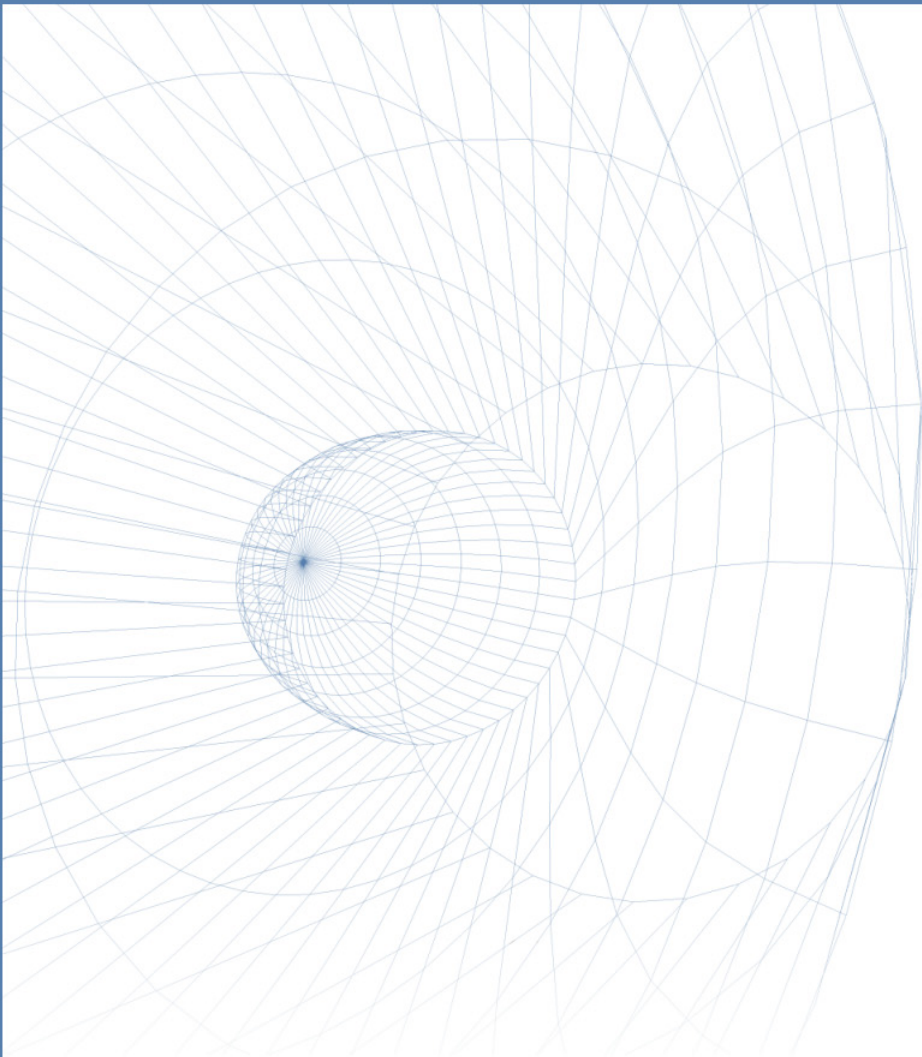


A large, light blue wireframe sphere is positioned on the left side of the page. It is composed of numerous thin lines that form a grid-like structure, giving it a three-dimensional appearance. The sphere is centered vertically and horizontally on the left half of the page.

Hull Standards

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Hull Standards

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III 199/50000/- --- 199/52510/-	1-2

1 Introduction

1.1 General about this Standard Handbook

By tradition the Hull application contains an extensive in-built hull design standard on different levels, from micro standard to standard for rather complex design elements like brackets with stiffening, cutouts, etc. This standard is available to all users of the application, after installation without any need for customisation.

This Standard Handbook describes the individual standards delivered with the standard Hull application.

A definition: The term "VIS" occurring in this document is a reference to a Swedish national standard for shipbuilding.

1.2 Overview of In-built Standards

The standard types that make up the in-built standards are

- Profile Standards
- Edge Preparation
- Profile Cutouts
- Holes
- Endcuts of Profiles
- End Connections of Profiles
- Brackets
- Clips/Collars

1.3 Customisation of Standards

For advanced users the system has support for customisation of the standard types described above and support for additional customer defined standards such as standards for flanging, swedging, knuckling, etc.

A full description of customisation of standards can be found in different sections of the guide "Hull Setup and Customisation".

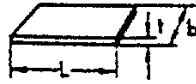
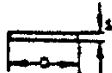
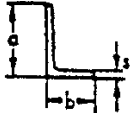
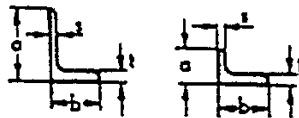
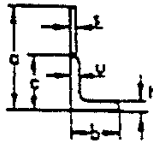
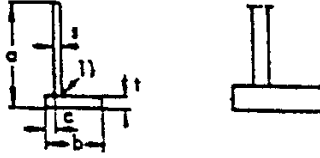
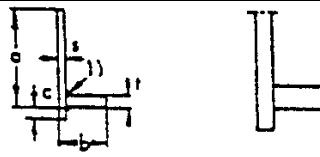
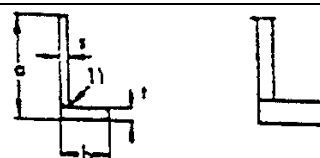
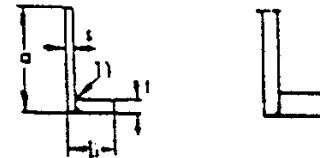
Hull Standards

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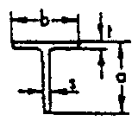
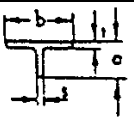
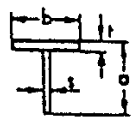
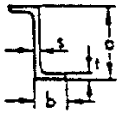
Profile Standard

Type codes, shape symbols and indication of dimensions for plate and bar products.

Type code	Designation	Figure	Shape symbol	Dimension directions
1	Flat plate		P	t x b x L
2	Floor plate			
3	Perforated plate			
Bar				
10	Flat bar and strip		- (pls)	a x s - L
20	Bulb bar, single		BE	a x s - L
30	Angle bar (equal thickness in flange and web)		L	a x b x s - L
31	Angle bar (unequal thickness in flange and web)		L	a x b x s x t - L
33	Unequal angles welded from profiles 10 and 31		L	a x b x s x t x c x u - L
35	Angle bar welded from profile 10		L	a x b x s x t x c - L
36				
37				a x b x s x t - L
38				

Profile Standard

Type codes, form symbols and indication of dimension for plate and bar products.

Type code	Designation	Figure	Shape symbol	Dimension directions
40	T-bar		T	a x b x s x t - L
				
43	T-bar welded from profile 10		T	a x b x s x t - L
50	U-bar		U	a - L
51	I-bar		I	a - L
52	IPE-bar		IPE	a - L
53	HE-A-bar		HE	a ¹⁾ A - L
54	HE-B-bar		HE	a B - L
55	HE-M-bar		HE	a ¹⁾ M - L
56*	Z-bar		Z	a x b x s x t - L
60*	Round bar and wire		Ø	a - L
61*	Half-round bar		HR	a x s - L
62*	Half-oval bar (segment bar)			
63*	Square bar		4K	a - L
64*	Hexagon bar		6K	a - L
65*	Octagon bar		8K	a - L
70*	Tube, seamless		Rör	a x s - L
71*	Tube, welded			
72*	Tube, cast			
73*	Square tube		RHS	a x b x s - L
74*	Rectangular tube			

Drawing Principles

Symbols for end-marking of profiles

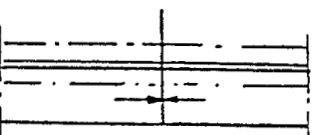
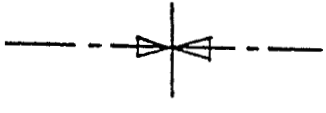
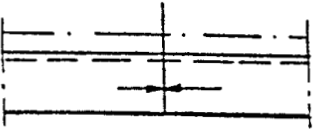
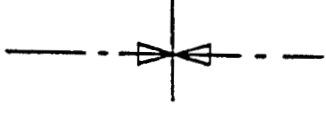
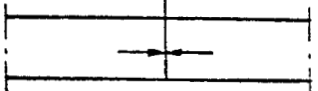
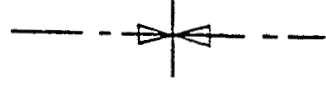
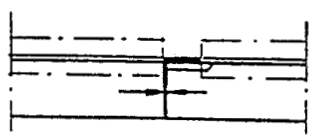
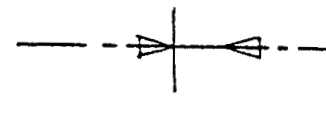
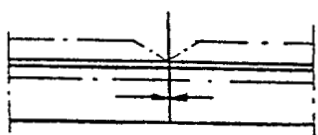
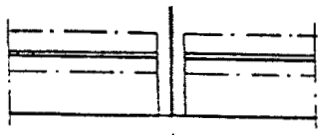
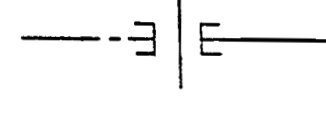
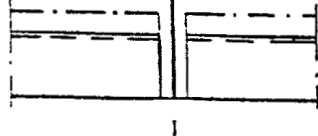
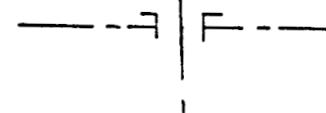
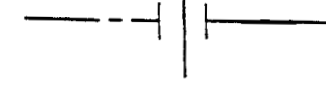
General The symbols are intended for ships hull drawings. Profiles placed on the near side are indicated by chaindot lines. Profiles placed on the far side are indicated by dashed lines. The line shows the position of the mould line and the thick line shows how the material is positioned.

For line marking, see also standard sheet KS89.

No.	Side view (elevation)	Form symbol	Description	Symbols on drawings
1		T L —	Throughgoing profile with connection at e.g. block joint.	
2		T L	Webs cut straight and welded.	
3		T L	Webs cut straight and welded. Flange bevel cut. Cutout in web.	

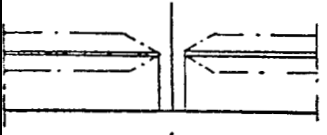
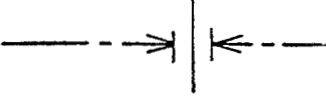
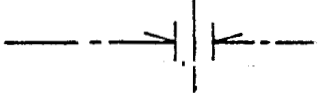
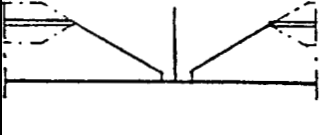
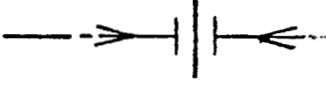
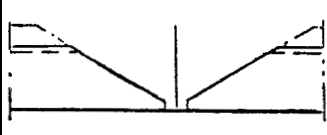
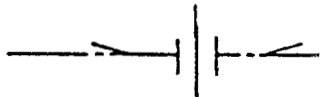
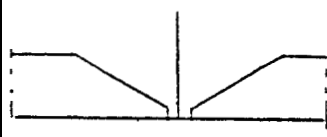
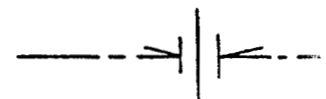
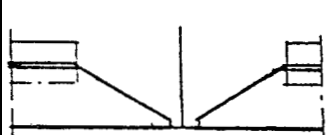
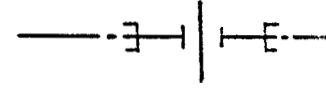
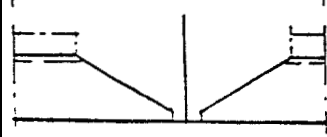
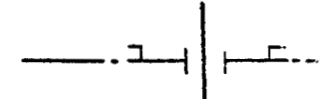
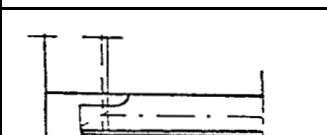
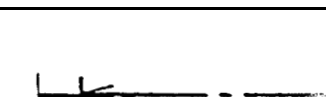
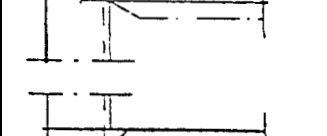
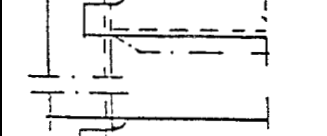
Drawing Principles

Symbols for end-marking of profiles

No.	Side view (elevation)	Form symbol	Description	Symbols on drawings
4		T	Web and flange cut straight and welded.	
		L		
		—		
5		T	Web and flange cut straight and welded. Web cutouts.	
6		T	Web and half the flange straight cut and welded, other half of the flange bevel cut.	
7		T	Web and flange cut straight, stops clear.	
		L		
		—		

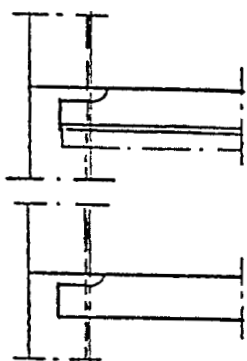
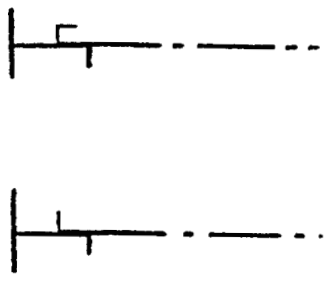
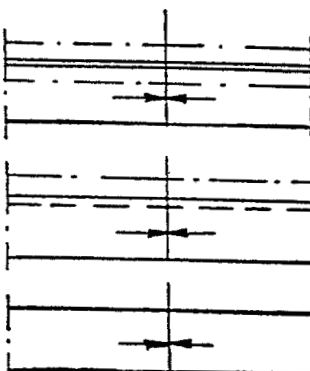
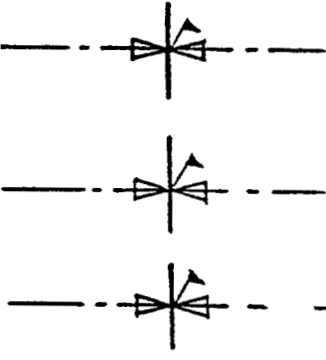
Drawing Principles

Symbols for end-marking of profiles

No.	Side view (elevation)	Form symbol	Description	Symbols on drawings
8		T	Web cut straight. Flange bevel cut, stops clear.	
		L		
9		T	Web and flange bevel cut, stops clear.	
		L		
		—		
10		T	Web bevel cut, flange cut straight, stops clear.	
		L		
11		T	Web cut straight, flange bevel cut.	
		L		
		L		

Drawing Principles

Symbols for end-marking of profiles

No.	Side view (elevation)	Form symbol	Description	Symbols on drawings
12			Web and flange cut straight.	
13			Web and flange slotted.	

Edge Preparations

Extract from VIS 444 *)

(NB. Page 6 is valid only for SLT mat.)

1. Purpose

This standard provides a numerical code for indicating edge preparation and an index over the corresponding bevel number to be used, and it is intended to be used when specifying machined plates and bars.

2. Construction of the code.

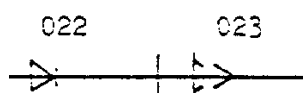
The code for edge preparation consists of 6 digits grouped as follows:

2.1 Bevel No.

The bevel No. used in the series 001-899 can be seen in the index on pages 2 - 4.

The bevel numbering has been made so that non-symmetrical bevels will get different numbers, depending upon the position of the bevel. Bevels on this side will get even numbers and the same bevel on the other side will get the nearest higher uneven no.

Example:



Bevel numbers in the series 900-999 indicate edge preparation according to separate document. In this case the type of machining and e-dimension are omitted.

2.2 Type of machining.

The following types are used:

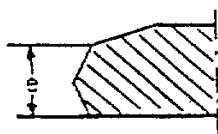
- | | |
|---------------------------------------|--|
| 0 -Basic type for a particular | 4 -Excess thickness, other side bevel |
| 1 -Machined bevel 18° this side | 5 -Machined bevel, 18° as well excess thick- |
| 2 -Machined bevel 18° other side | ness, this side |
| 3 -Excess thickness, this side excess | 6 -Machined bevel, 18° as well excess thick- |
| | ness, other side |

2.3 e-dimension.

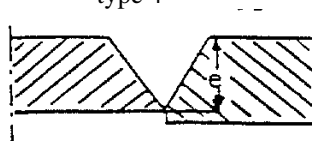
The dimension is given as 00 when the type of machining is 0. This occurs because the e-dimension is identical with the plate thickness. For basic types of machining the three last digits are thus noughts and may be omitted.

For other types of machining, the e-dimension gives that part (in mm) of the plate thickness within which the basic bevel type is placed.

Example: X - bevel, type 1

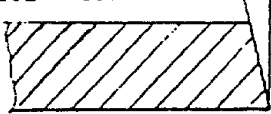
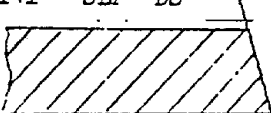
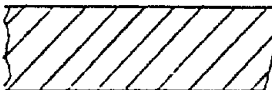
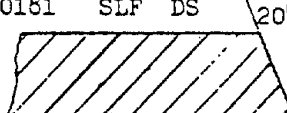
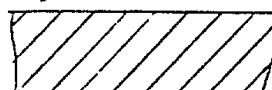
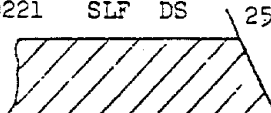
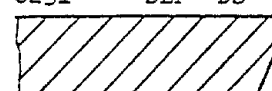


V - bevel
type 0



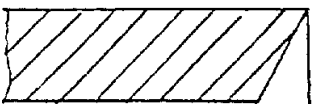
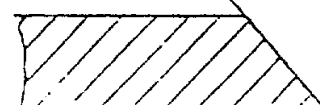
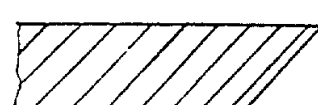
*)The standards regarding the edge preparation described in this document are used mainly for drawing purposes in SBH.

Edge Preparations

Type	Type	Remark
0021 SLF DS <u>002</u>  0022 SLF AS F1		
0101 SLF DS 10° <u>010</u>  0102 SLF AS	0111 SLF DS <u>011</u>  0112 SLF AS 10°	
0141 SLF DS 15° <u>014</u>  0142 SLF AS F54	0151 SLF DS <u>015</u>  0152 SLF AS 15° F55	
0181 SLF DS 20° <u>018</u>  0182 SLF AS F21	0191 SLF DS <u>019</u>  0192 SLF AS 20° F22	
0221 SLF DS 25° <u>022</u>  0222 SLF AS F2	0231 SLF DS <u>023</u>  0232 SLF AS 25° F3	

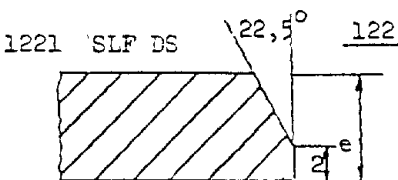
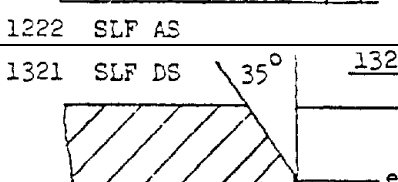
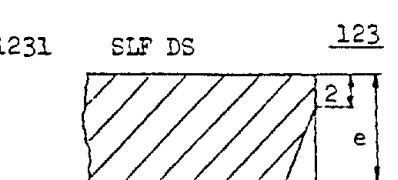
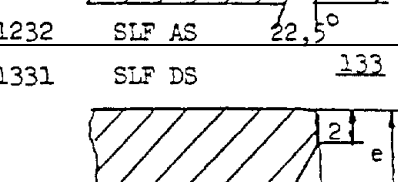
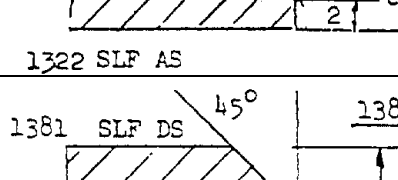
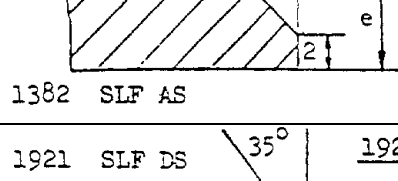
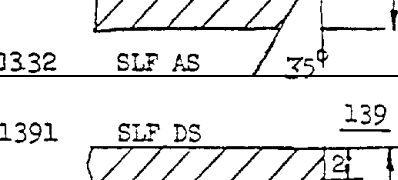
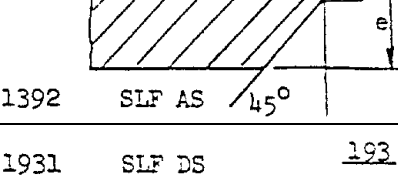
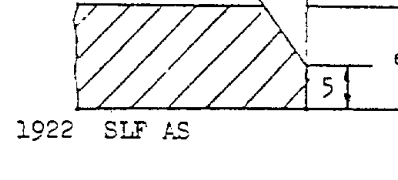
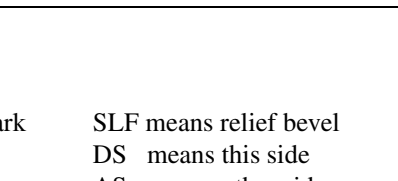
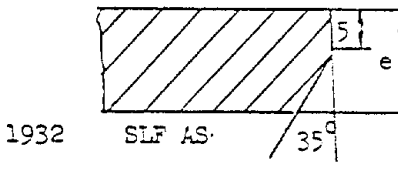
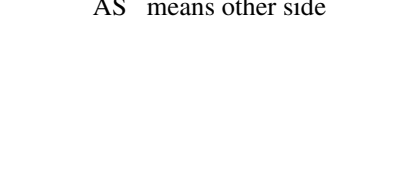
Remark SLF means relief bevel
DS means this side
AS means other side

Edge Preparations

Type	Type	Remark
0261 SLF DS 30°  0262 SLF AS	0271 SLF DS 027  0272 SLF AS 30°	
0301 SLF DS 35°  0302 SLF AS	0311 SLF DS 031  0312 SLF AS 35°	
0341 SLF DS 40°  0342 SLF AS	0351 SLF DS 035  0352 SLF AS 40°	
0381 SLF DS 45°  0382 SLF AS	0391 SLF DS 039  0392 SLF AS 45°	

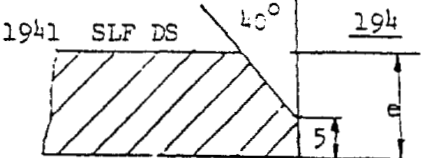
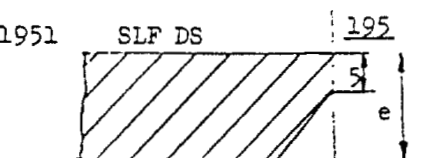
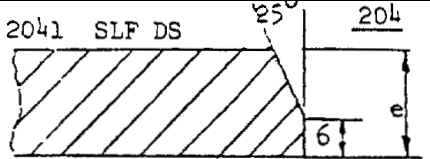
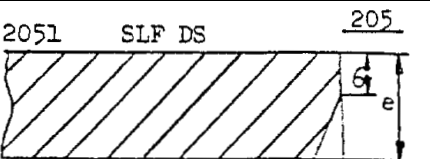
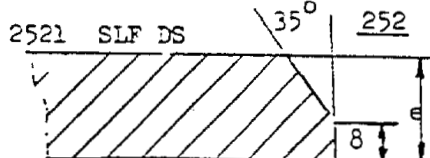
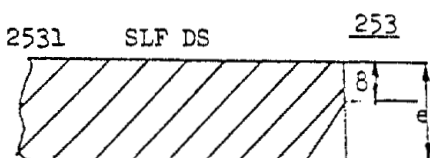
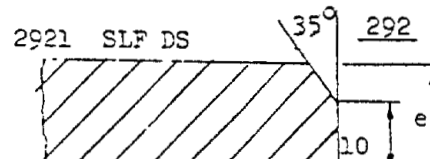
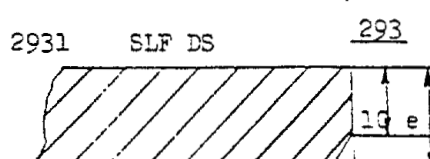
Remark SLF means relief bevel
DS means this side
AS means other side

Edge Preparations

Type	Type	Remark
1221 SLF DS  1222 SLF AS 	1231 SLF DS  1232 SLF AS 	
1321 SLF DS  1322 SLF AS 	1331 SLF DS  1332 SLF AS 	
1381 SLF DS  1382 SLF AS 	1391 SLF DS  1392 SLF AS 	
1921 SLF DS  1922 SLF AS 	1931 SLF DS  1932 SLF AS 	

Remark SLF means relief bevel
DS means this side
AS means other side

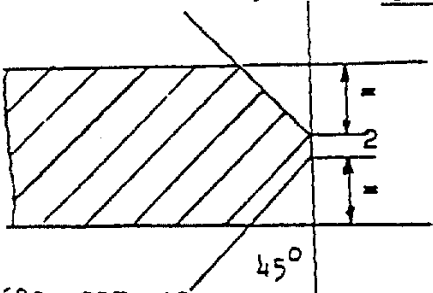
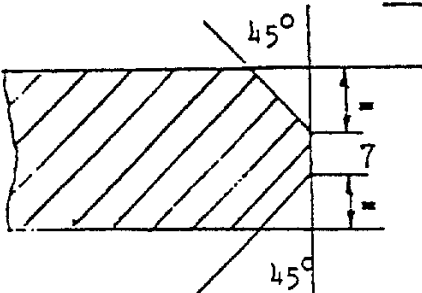
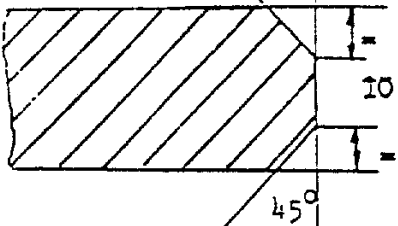
Edge Preparations

Type	Type	Remark
 1941 SLF DS 1942 SLF AS	 1951 SLF DS 1952 SLF AS	
 2041 SLF DS 2042 SLF AS F26	 2051 SLF DS 2052 SLF AS F27	e = 15, -28
 2521 SLF DS 2522 SLF AS	 2531 SLF DS 2532 SLF AS	Machine weld joint 12.5 < e ≤ 18
 2921 SLF DS 2922 SLF AS	 2931 SLF DS 2932 SLF AS	FL 2 e=15,1 -20,0

Remark

- SLF means relief bevel
- DS means this side
- AS means other side
- FL2 means panel line No.2

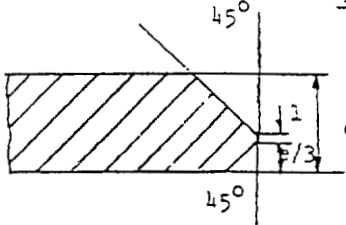
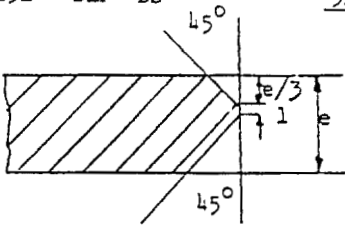
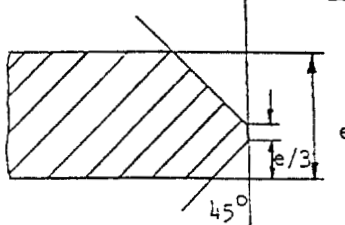
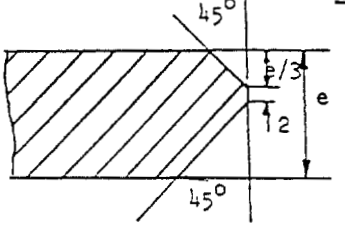
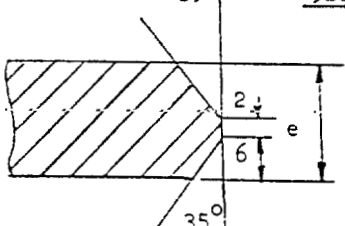
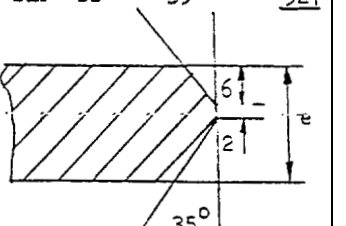
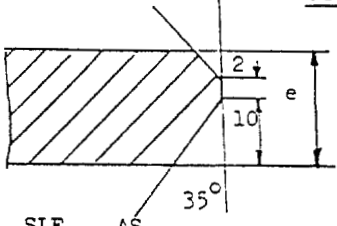
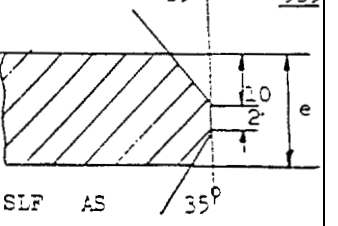
Edge Preparations

Type	Type	Remark
3681 SLF DS 45° 368  3682 SLF AS 45°		t ≥ approx 15 mm *)
4181 SLF DS 45° 418  4182 SLF AS 45°		t ≥ approx 20 mm *)
4481 SLF DS 45° 448  4482 SLF AS 45°		t ≥ approx 25 mm *)

Remark SLF means relief bevel
DS means this side
AS means other side

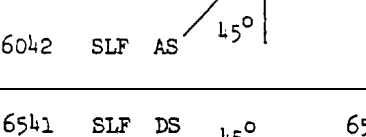
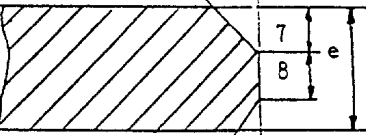
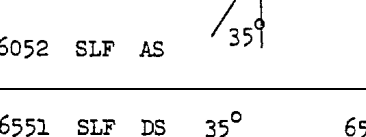
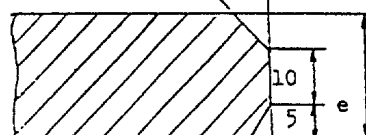
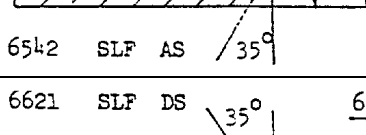
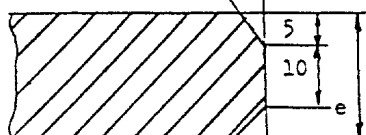
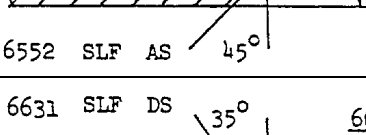
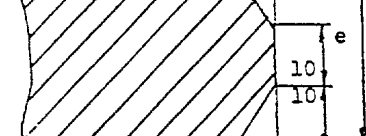
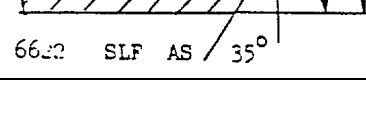
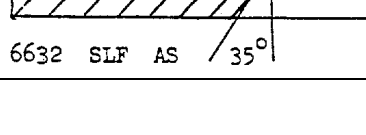
* Will not give full penetration, if clearance is missing.

Edge Preparations

Type	Type	Remark
5141 SLF DS 45° 514  5142 SLF AS	5151 SLF DS 45° 515  5152 SLF AS	
5241 SLF DS 45° 524  5242 SLF AS	5251 SLF DS 45° 525  5252 SLF AS	
5261 SLF DS 35° 526  5262 SLF AS	5271 SLF DS 35° 527  5272 SLF AS	
5341 SLF DS 35° 534  5342 SLF AS	5351 SLF DS 35° 535  5352 SLF AS	

Remark
 SLF means relief bevel
 DS means this side
 AS means other side

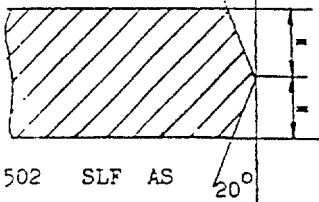
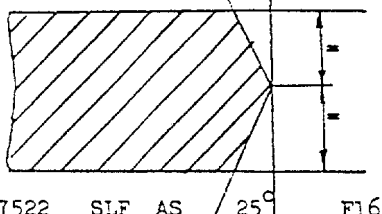
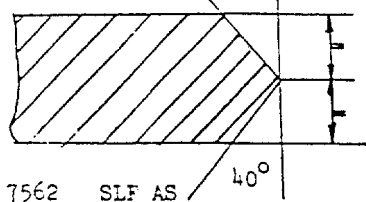
Edge Preparations

Type	Type	Remark
6041 SLF DS 35° <u>604</u>  6042 SLF AS 45° 	6051 SLF DS 45° <u>605</u>  6052 SLF AS 35° 	FL.2 e=20, 1-24,0
6541 SLF DS 45° <u>654</u>  6542 SLF AS 35° 	6551 SLF DS 35° <u>655</u>  6552 SLF AS 45° 	e=19,1-26,0
6621 SLF DS 35° <u>662</u>  6622 SLF AS 35° 	6631 SLF DS 35° <u>663</u>  6632 SLF AS 35° 	FL.2 e=28, 1-32,0

Remark

- SLF means relief bevel
- DS means this side
- AS means other side
- FL.2 means panel line No.2

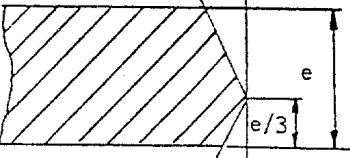
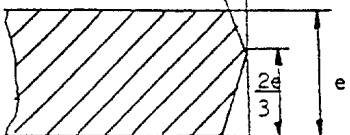
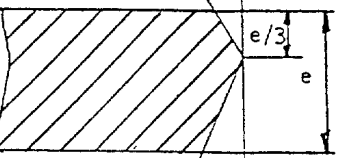
Edge Preparations

Type	Type	Remark
7501 SLF DS 20° 750  7502 SLF AS 20°		*)
7521 SLF DS 25° 752  7522 SLF AS 25° F16		*)
7561 SLF DS 40° 756  7562 SLF AS 40°		*)

Remark SLF means relief bevel
 DS means this side
 AS means other side

*) Choose joint with unsymmetric bevel, if full penetration is necessary.

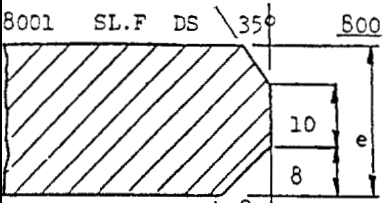
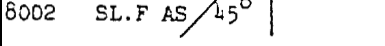
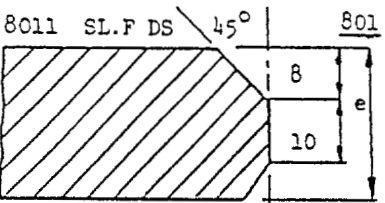
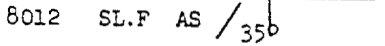
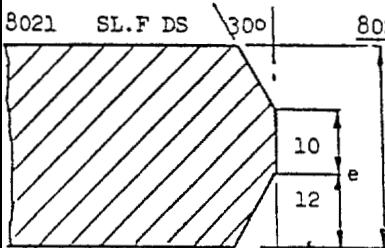
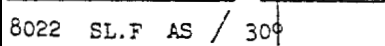
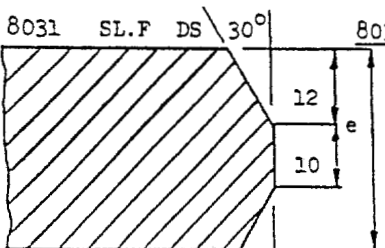
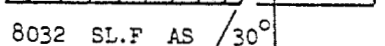
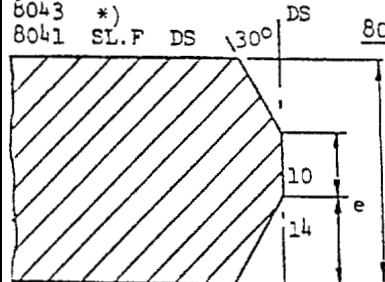
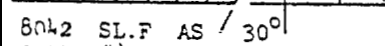
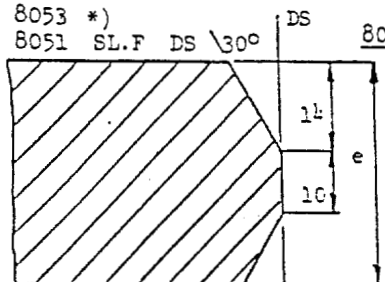
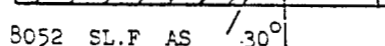
Edge Preparations

Type	Type	Remark
7723 *) DS 772 7721 SLF DS  7722 SLF AS 7724 *) AS 7761 SLF DS 25° 776  7762 SLF AS 30°	7733 *) DS 773 7731 SLF DS  7732 SLF AS 7734 *) AS 7771 SLF DS 30° 777  7772 SLF AS 25°	

Remark SLF means relief bevel
DS means this side
AS means other side

*) Thickness excess.

Edge Preparations

Type	Type	Remark
8001 SL.F DS 35° 800  8002 SL.F AS 45° 	8011 SL.F DS 45° 801  8012 SL.F AS 35° 	FL.2 e=28.1-32.0
8021 SL.F DS 30° 802  8022 SL.F AS 30° 	8031 SL.F DS 30° 803  8032 SL.F AS 30° 	e=32.1-36.0
8043 *) DS 804 8041 SL.F DS 30°  8042 SL.F AS 30° 8044 *) AS 	8053 *) DS 805 8051 SL.F DS 30°  8052 SL.F AS 30° 8054 *) AS 	e=36.1-40.0

Remark

SLF means relief bevel

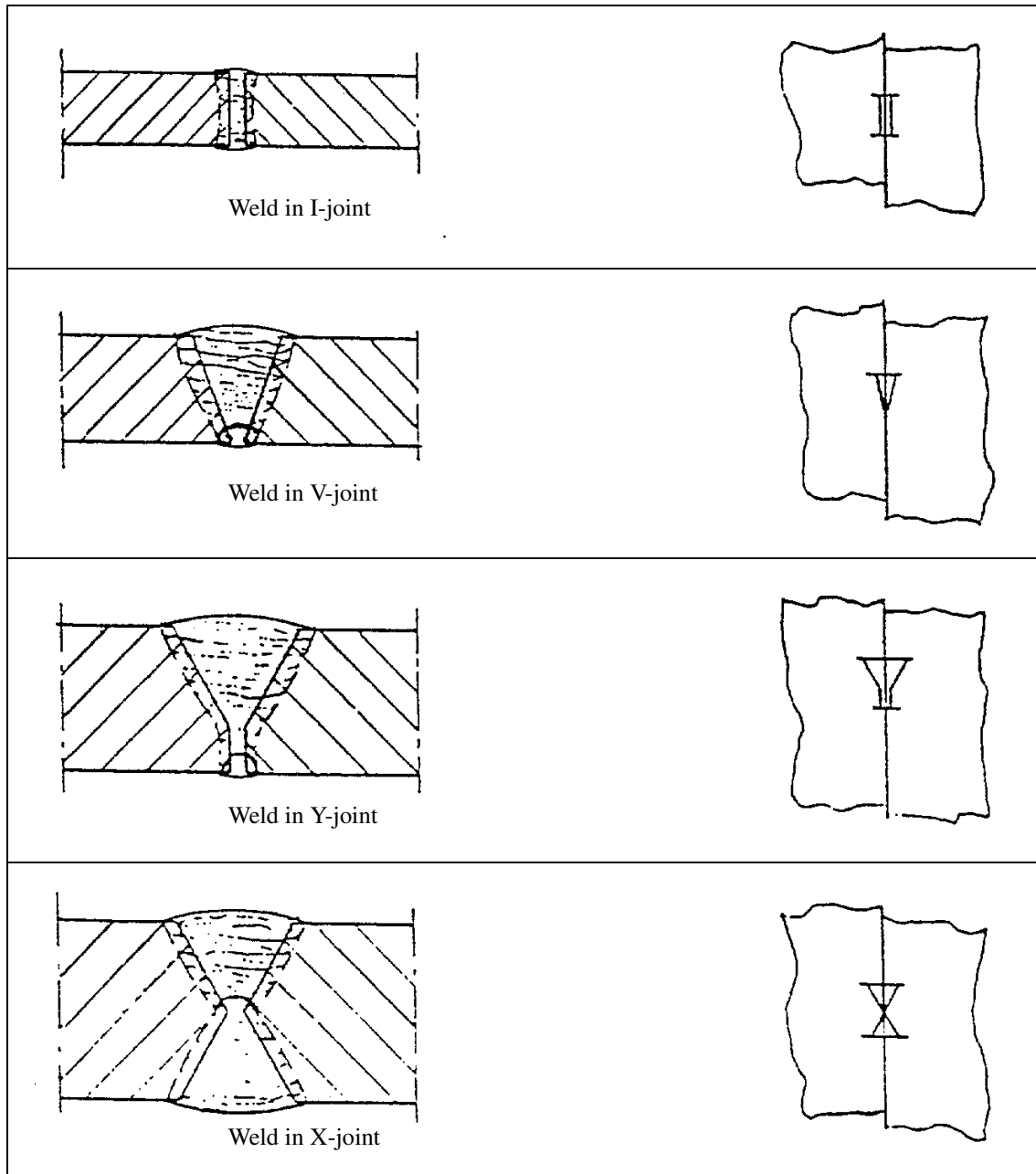
DS means this side

AS means other side

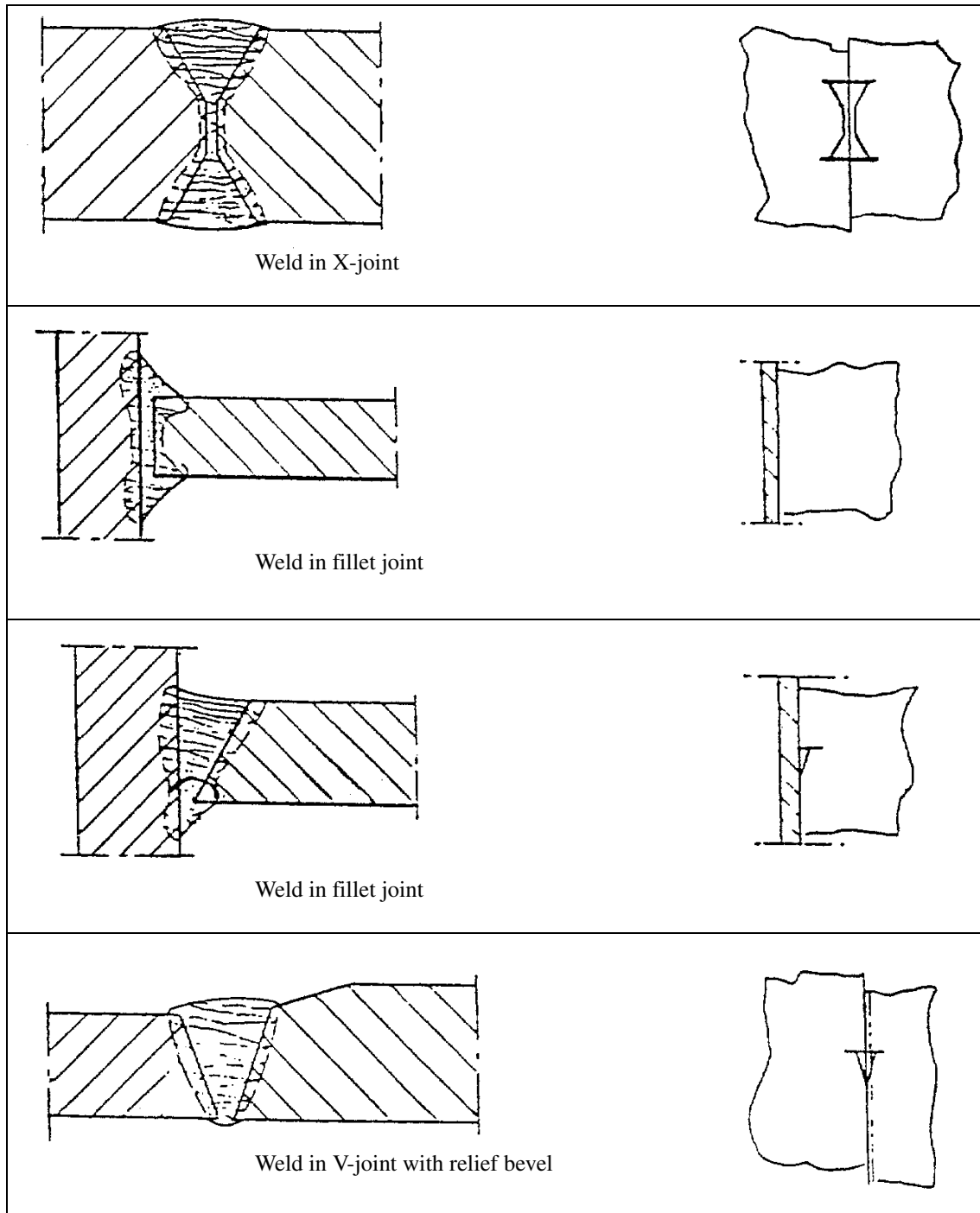
FL.2 means panel line No.2

*) Excess thickness.

Detail drawing of weld joints



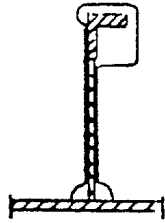
Detail drawing of weld joints



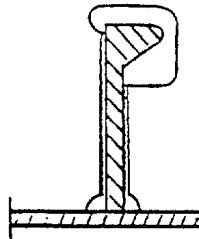
Cutouts, Survey *)



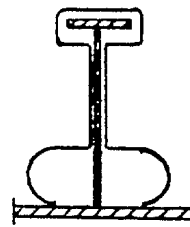
Type 1
190/20001/-



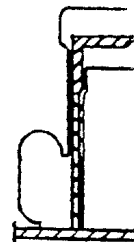
Type 2, 201
190/20002/-



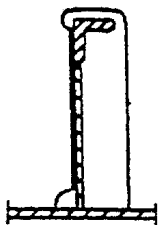
Type 202, 203,
204
190/20002/-



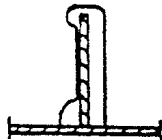
Type 21
190/20002/-



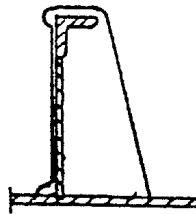
Type 22
190/20003/-



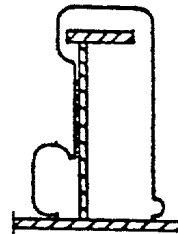
Type 3, 31, 32
301, 311 o 312
190/20003/-



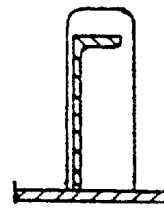
Type 33
190/20003/-



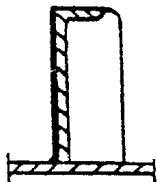
Type 34, 35,
341, 351
190/20003/-



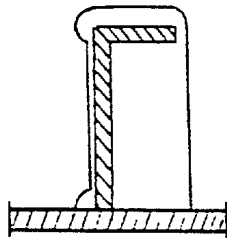
Type 36
190/20003/-



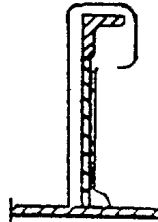
Type 37
190/20003/-



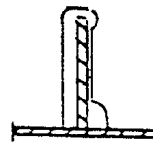
Type 38
190/20003/-



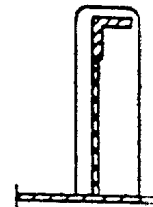
Type 302,
303, 308,
309, 322, 323
190/20003/-



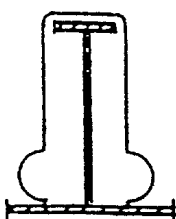
Type 4, 41, 401,
411
190/20004/-



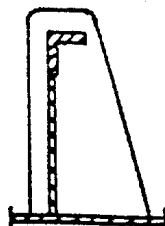
Type 42
190/20004/-



Type 5, 51
190/20005/-



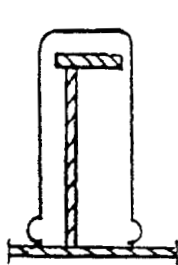
Type 52, 53
190/20005/-



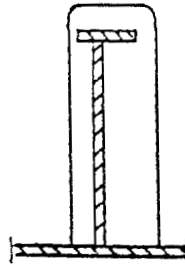
Type 54, 55
190/20005/-

*) Normally, a given cutout type is valid only for certain profile types.

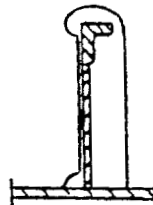
Cutouts, Survey



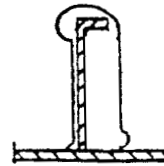
Type 56
190/20005/-



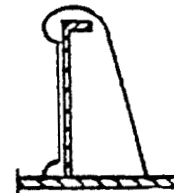
Type 57, 58
190/20005/-



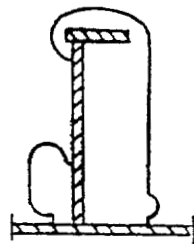
Type 6, 601
190/20006/-



Type 602
190/20006/-



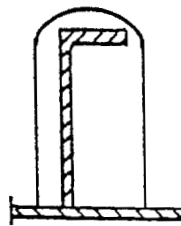
Type 62, 6
190/20006/-



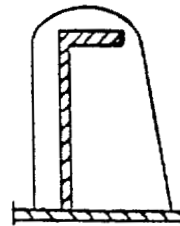
Type 65
190/20006/-



Type 101, 102, 103
190/20007/-



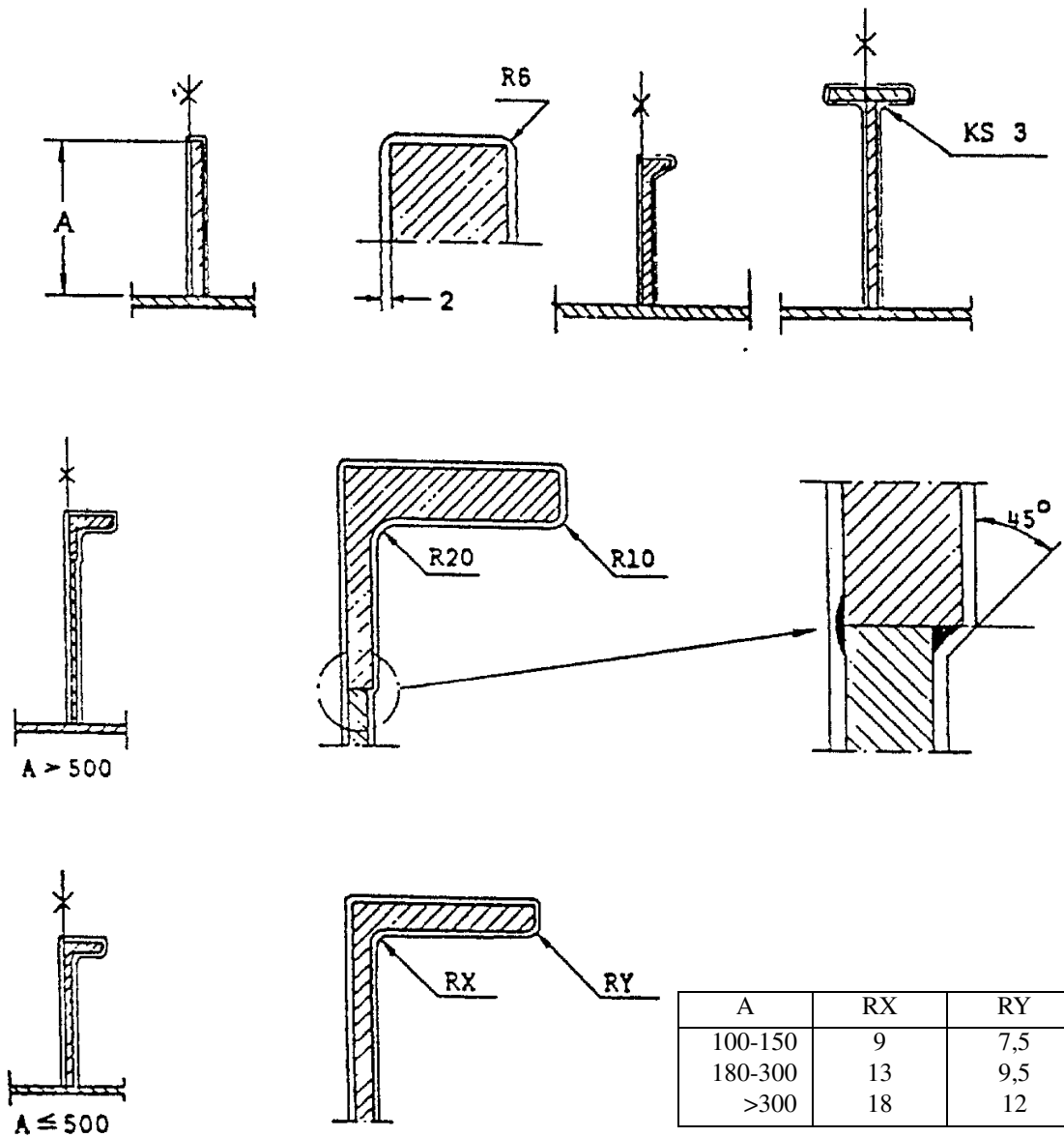
Type 7
190/20008/-



Type 72
190/20008/-

Cutouts, Slots

Type 1

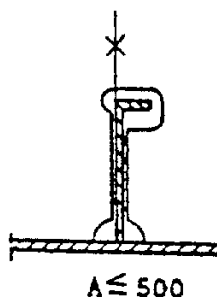
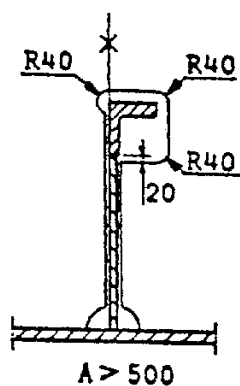
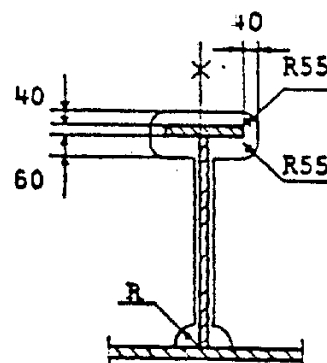
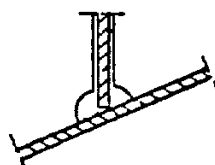
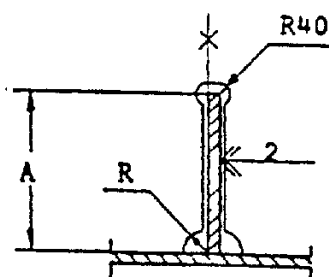


Cutouts with stub on both sides

Type 2 R = R75

Type 201 R = KS10

Type 205 R = R50



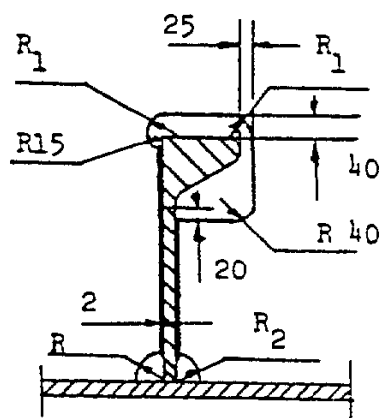
$R_1 = 40$ if $H \geq 200$ else $R_1 = 30$

Type 202 R and $R_2 = KS10$

Type 203 R = KS10

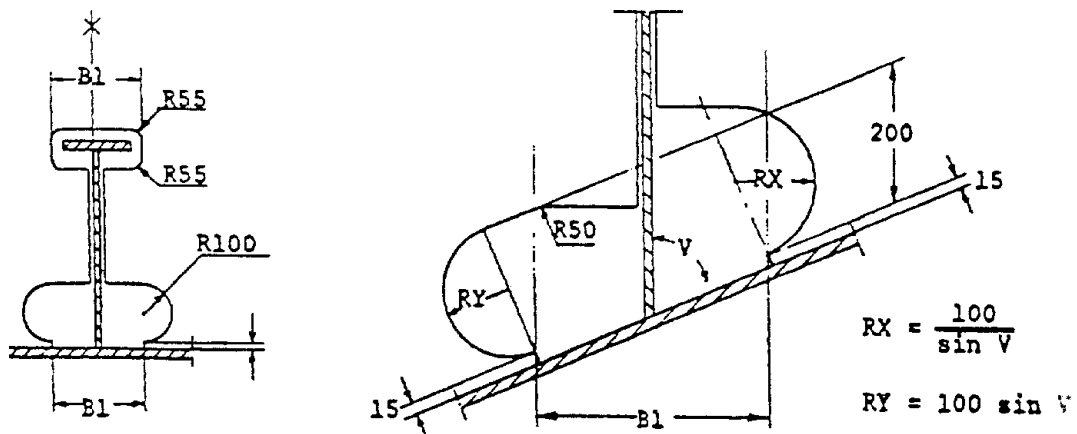
$R_2 = 75$ if $H \geq 300$, else $R = 50$

Type 204 R and $R_2 = 75$ if $H \geq 300$ else $R = 50$

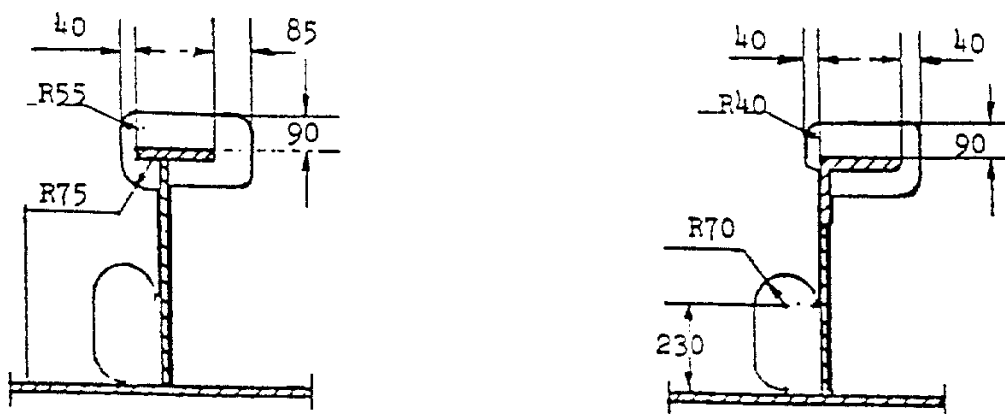


Cutouts with stub on both sides

Type 21

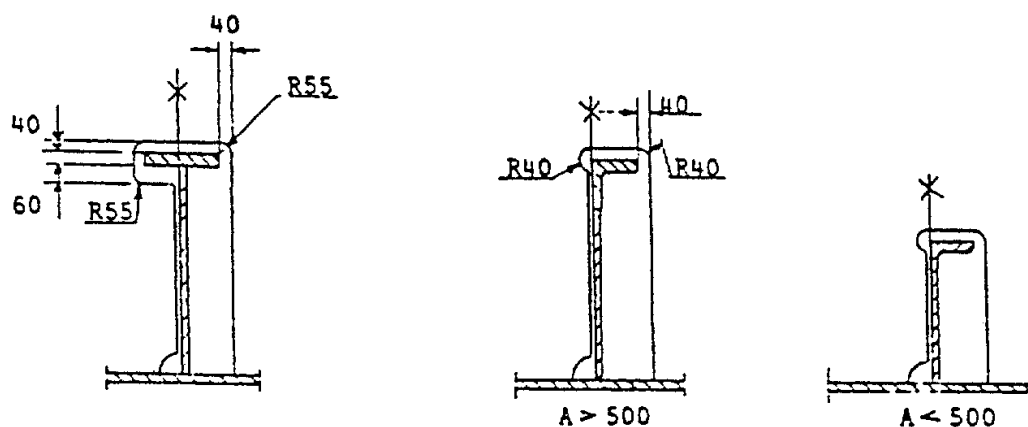
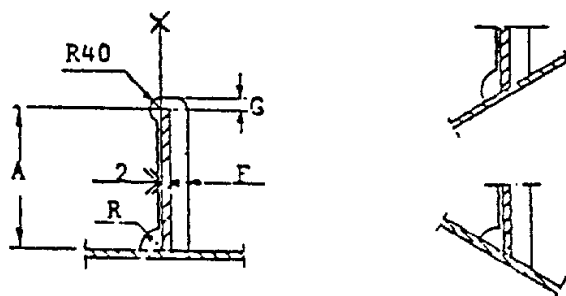


Type 22



Cutouts with stub on mould line side

<u>Type 3</u>	$F = 60, G = 40, R = R75$, if $A > 300$ else $R50$
<u>Type 31</u>	Flat bar: $F = 90, G = 40, R = R75$, if $A > 300$ else $R50$
<u>Type 32</u>	L-bar: $G = 90, R = R75$ if $A > 300$ else $R50$
<u>Type 301</u>	$F = 60, G = 40, R = KS8$
<u>Type 311</u>	Flat bar: $F = 90, C = 40, R = KS8$
<u>Type 321</u>	L-bar: $G = 90, R = KS8$

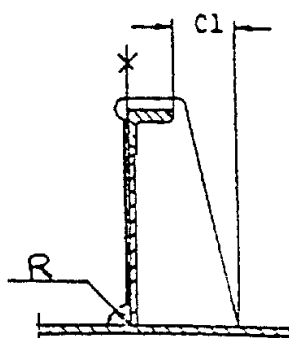
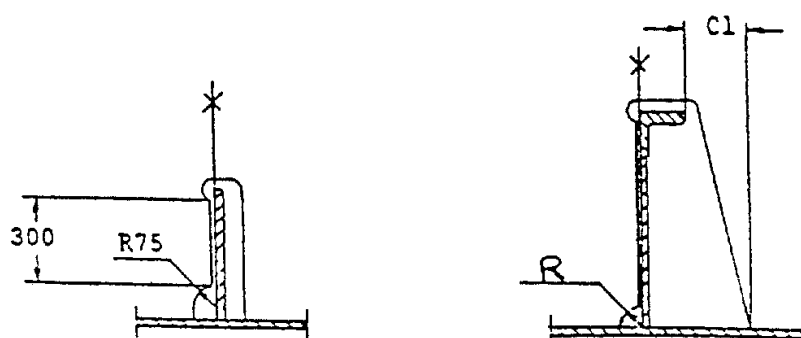


Valid for L- and BE-profiles

Type 33 $F = 60, G = 40$

Type 34 $G = 40, R = R75$ if $A > 300$ else $R50$

Type 35 $G = 90, R = R75$ if $A > 300$ else $R50$



Type 341 $G = 40, R = KS8$

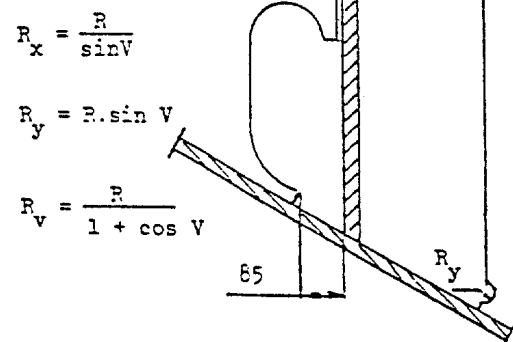
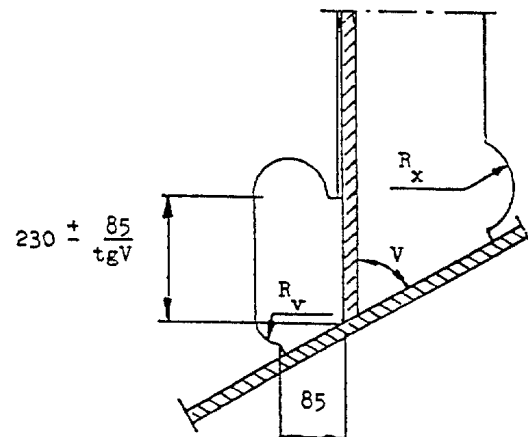
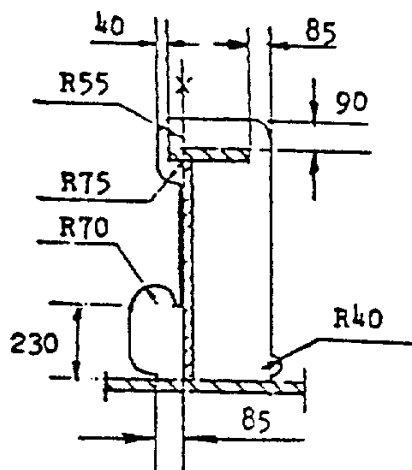
Type 351 $G = 90, R = KS8$

Valid for L- and BE-profiles

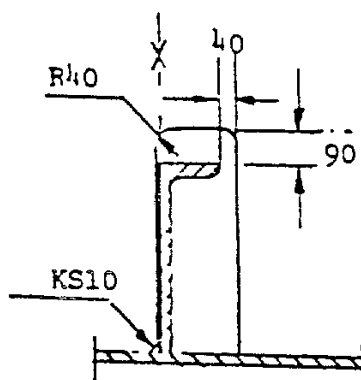
C1 given as parameter

Cutouts with stub on mould line side

Type 36

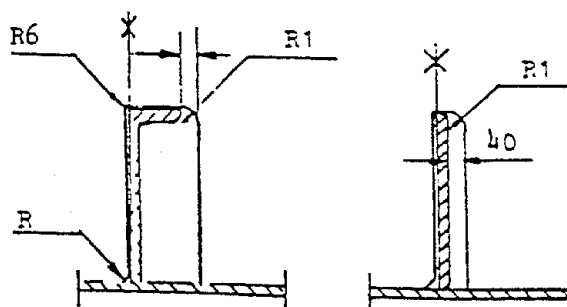


Type 37



Valid for L- and BE-profiles

- Type 38 F = 40, R = KS 10, R1 = R40
Type 381 F = 40, R = R50, R1 = R40
Type 382 F = 40, R = R75, R1 = R40
Type 386 F = 25, R = KS 10, R1 = R25

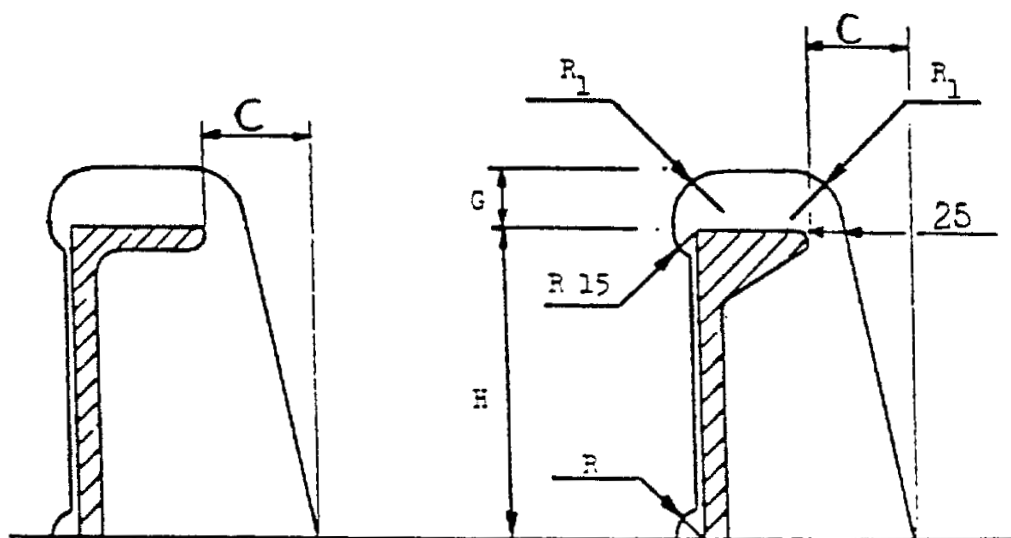


Valid for L- and BE-profiles and flat bars

Cutouts with stub on mould line side

<u>Type 308</u>	$G = 40, R = KS10$
<u>Type 309</u>	$G = 40, R = R75, \text{ if } H \geq 360 \text{ else } R50$
<u>Type 322</u>	$G = 90, R = KS10$
<u>Type 323</u>	$G = 90, R = R75, \text{ if } H \geq 300 \text{ else } R50$

RE-profile:	$R_1 = R40, \text{ if } H \geq 220 \text{ else } R30$
L-profile:	$R_1 = R40$



$C = 25$ if nothing else is given

Cutouts with stub on mould line side

Type 302

$G = 50$, $R = KS10$

Type 303

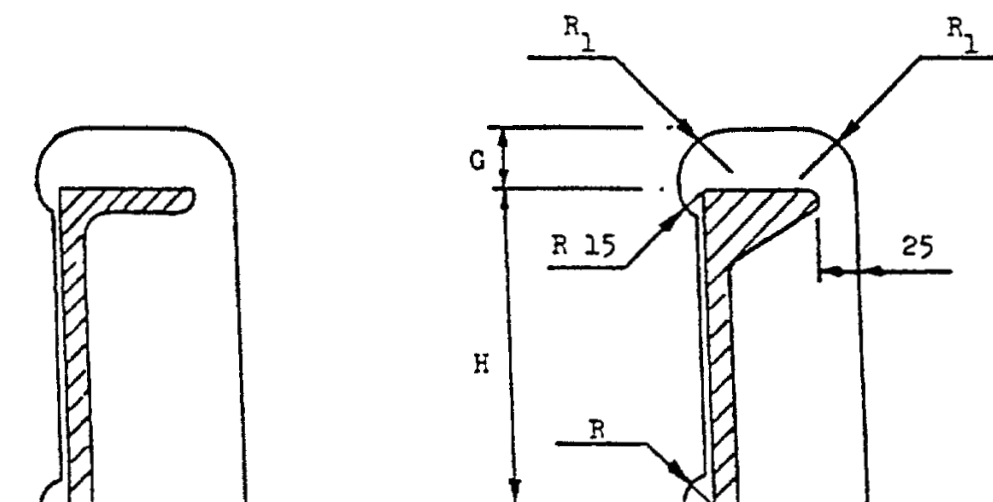
$G = 50$, $R = R75$, if $H \geq 300$ else $R50$

RE-profile:

$R_1 = R40$, if $H \geq 220$ else $R30$

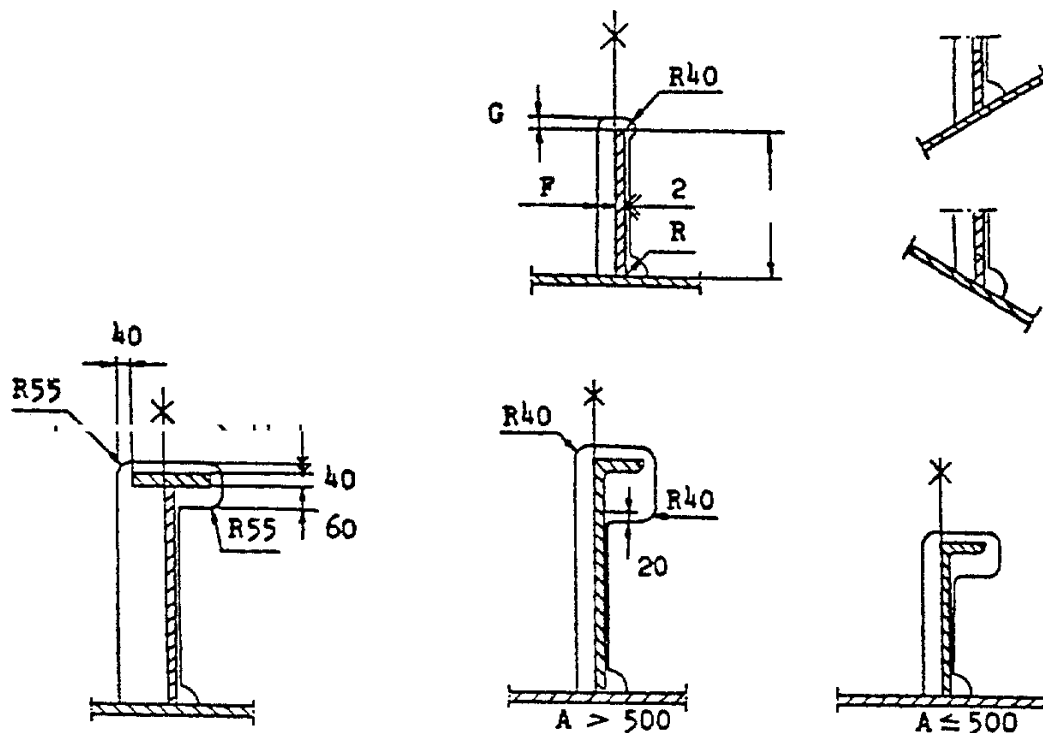
L-profile:

$R_1 = R40$

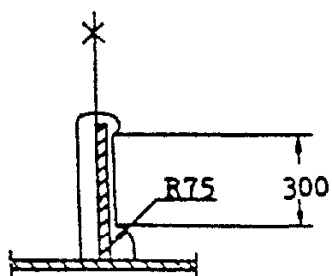


Cutouts with stub at other than mould line side

<u>Type 4</u>	$F = 60, G = 40, R = R75$
<u>Type 41</u>	Flat bar: $F = 90, G = 40, R = R75$ L-bar: $F = 90, G = 90, R = R75$
<u>Type 401</u>	$F = 60, G = 40, R = KS8$
<u>Type 411</u>	Flat bar: $F = 90, G = 40, R = KS8$ L-bar: $F = 90, G = 90, R = KS8$

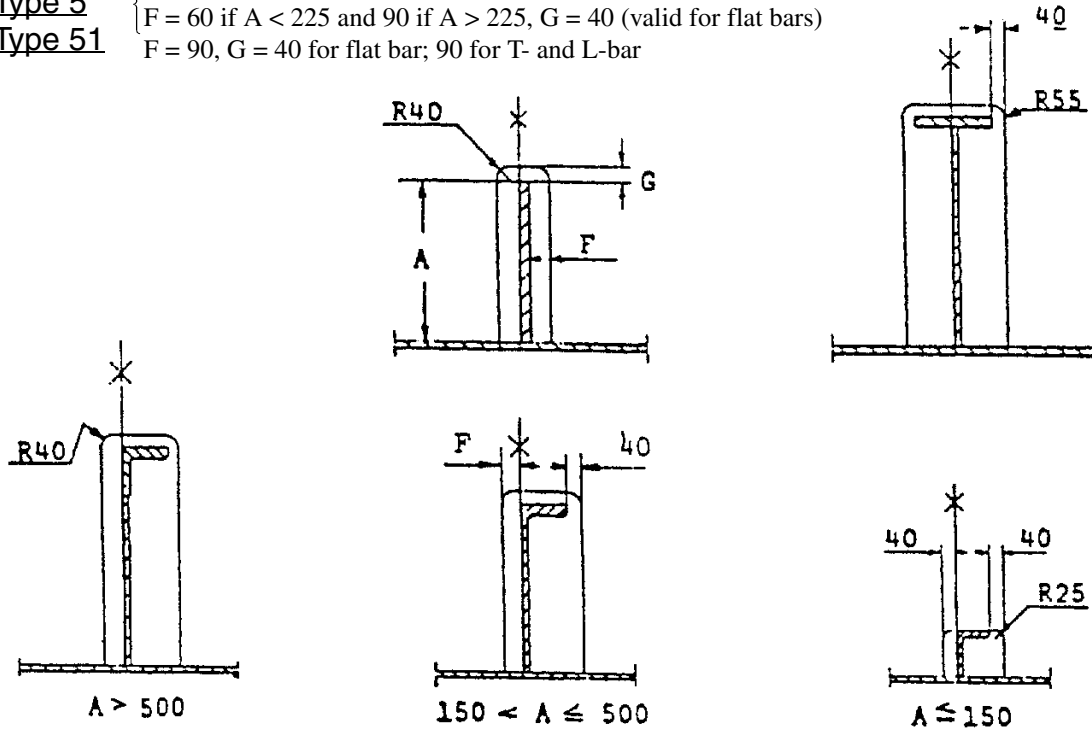


Type 42 $F = 60, G = 40$



Cutouts for fastening with or without clips

Type 5 { F = 60 if A < 300 and 90 if A ≥ 300, G = 40 (valid for L- and BE-bars)
Type 51 { F = 60 if A < 225 and 90 if A > 225, G = 40 (valid for flat bars)
 F = 90, G = 40 for flat bar; 90 for T- and L-bar



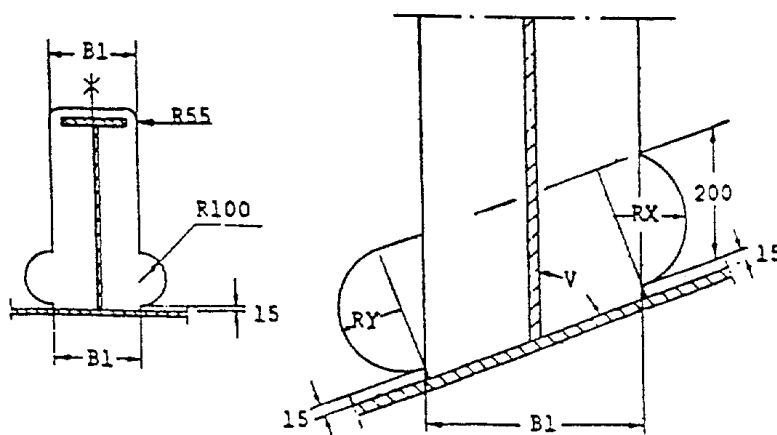
Valid for L-and BE-profiles

Type 52 G = 40
Type 53 G = 90

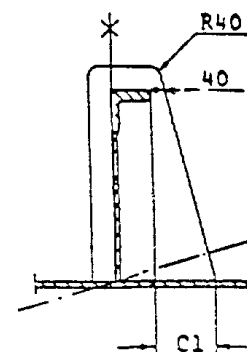
$$RX = \frac{100}{\sin V}$$

$$RY = 100 \sin V$$

Type 54 F = 60, G = 40
Type 55 F = 90, G = 90



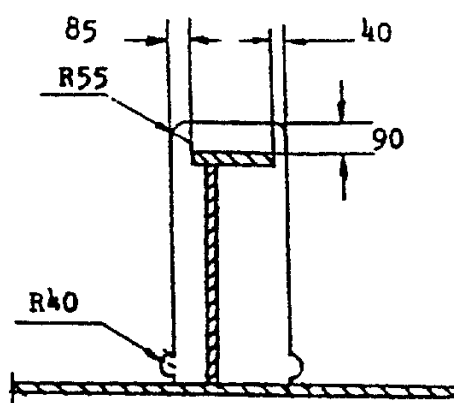
RX is always assumed to lie at the side with the acute angle



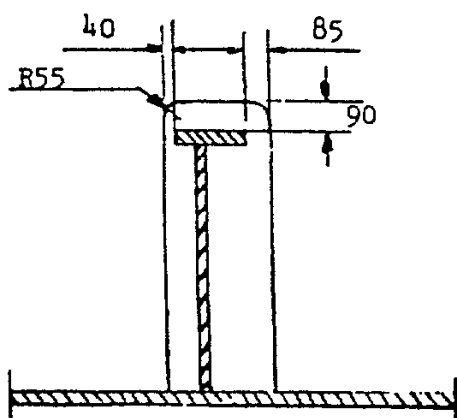
Valid for L-and BE-profiles
 C1 given as parameter

Cutouts for fastening with or without clips

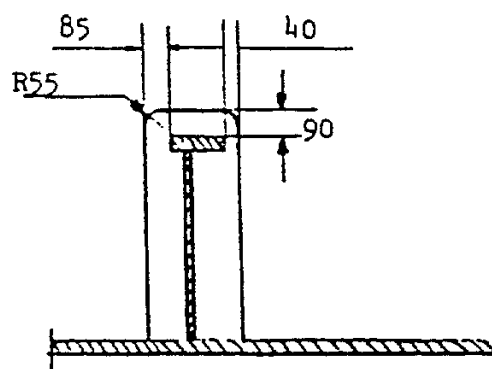
Type 56



Type 58

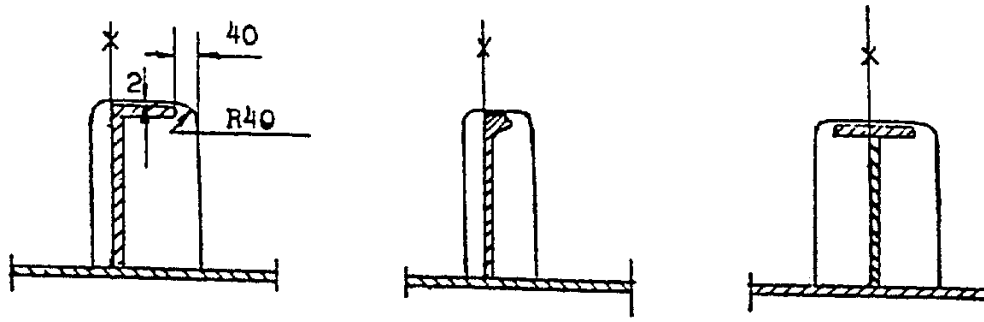


Type 57



Cutouts for fastening with or without clips

Type 511

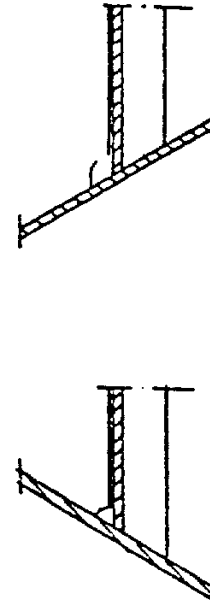
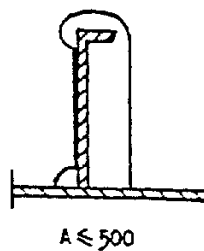
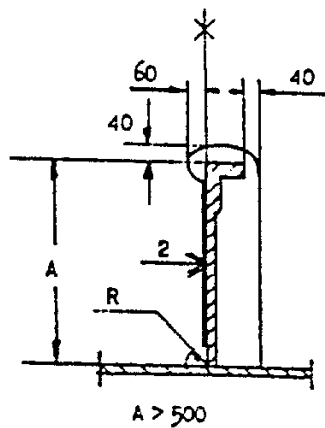


Valid for L-, BE- and T-profiles

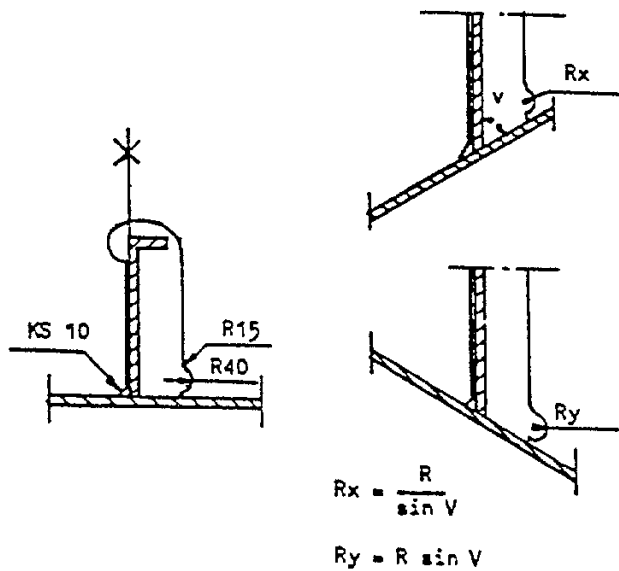
Cutouts, elliptical shapes

Type 6 R = R75

Type 601 R = KS8

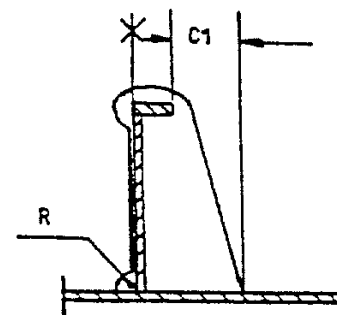


Type 602 (Not implented in SBH)



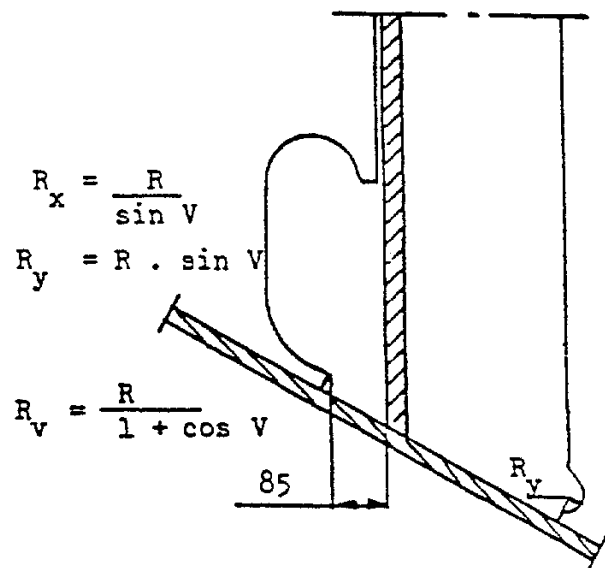
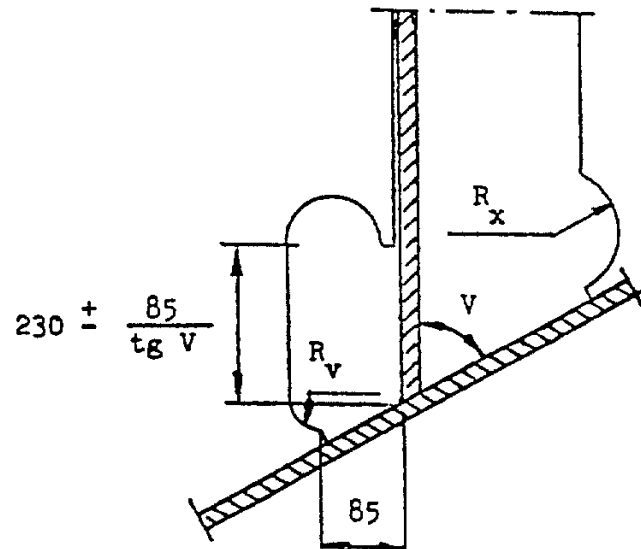
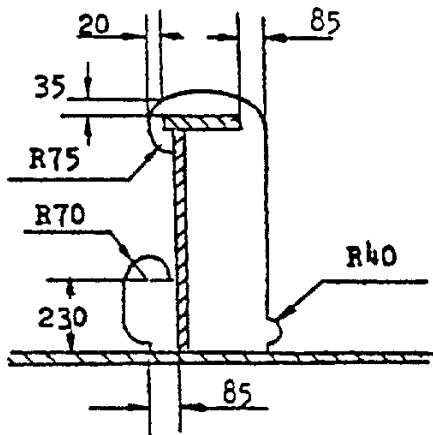
Type 62 R = R75

Type 621 R = KS8



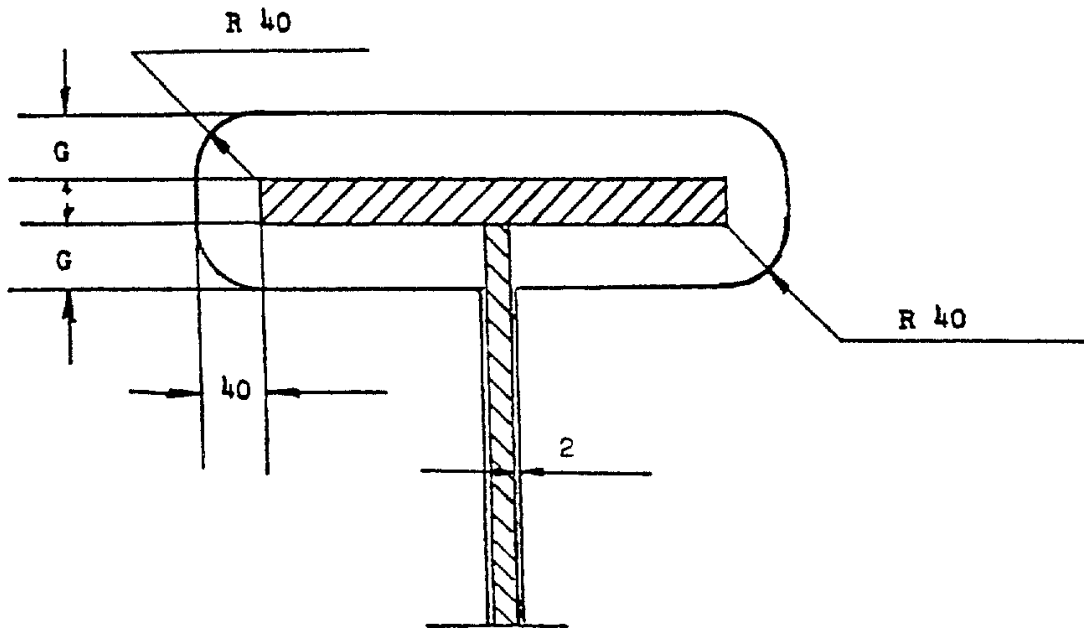
Cutouts, elliptical shapes

Type 65



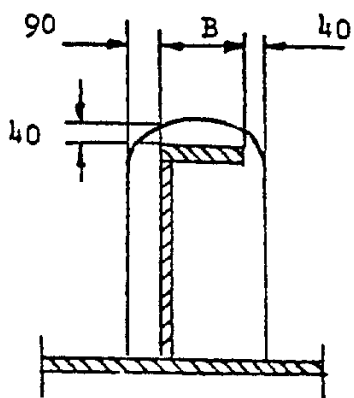
Cutouts, for girder flange

Type 101	G = 40
Type 102	G = 60
Type 103	G = 90

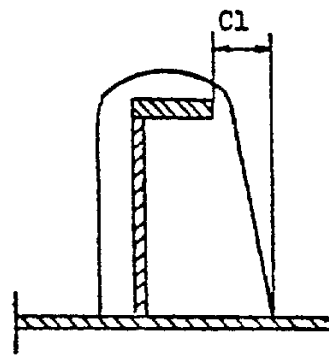


Cutouts, elliptical shapes

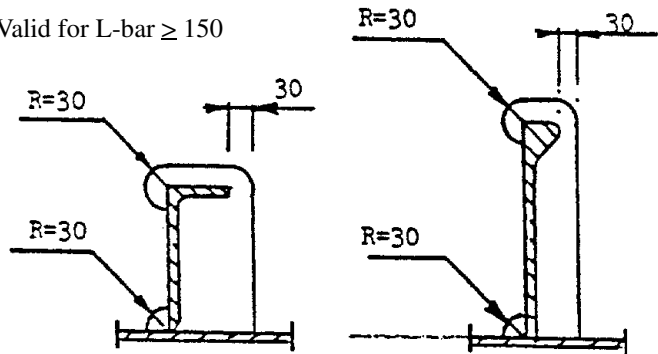
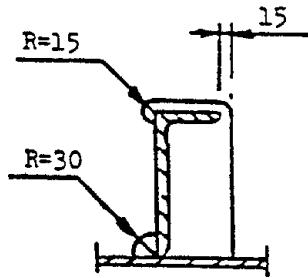
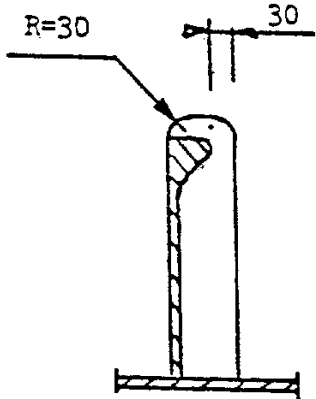
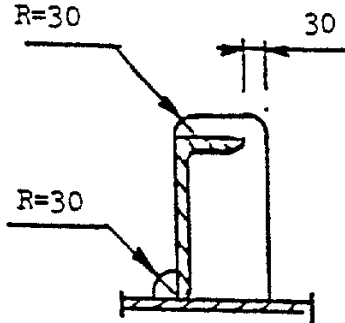
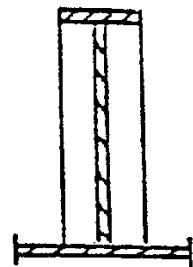
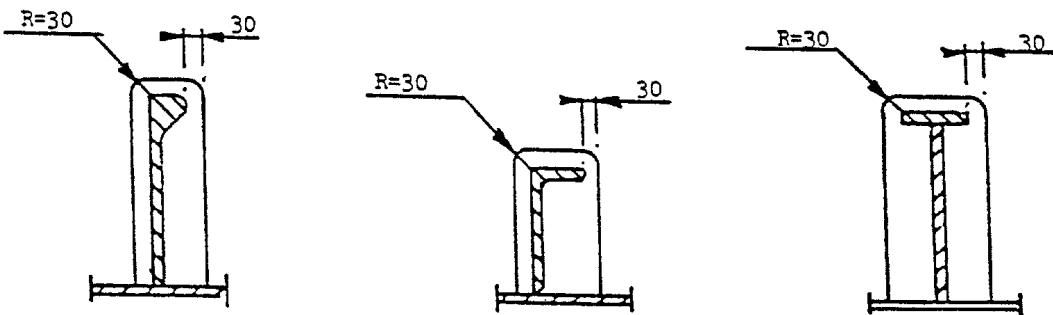
Type 7



Type 72



Cutouts, mine-layer

<u>Type 305</u> Valid for L-bar ≥ 150 	<u>Type 306</u> Valid for L-bar ≤ 130 	
<u>Type 375</u> 	<u>Type 376</u> 	<u>Type 385</u> 
<u>Type 505</u> 		

Special Cutouts

Ellipse:

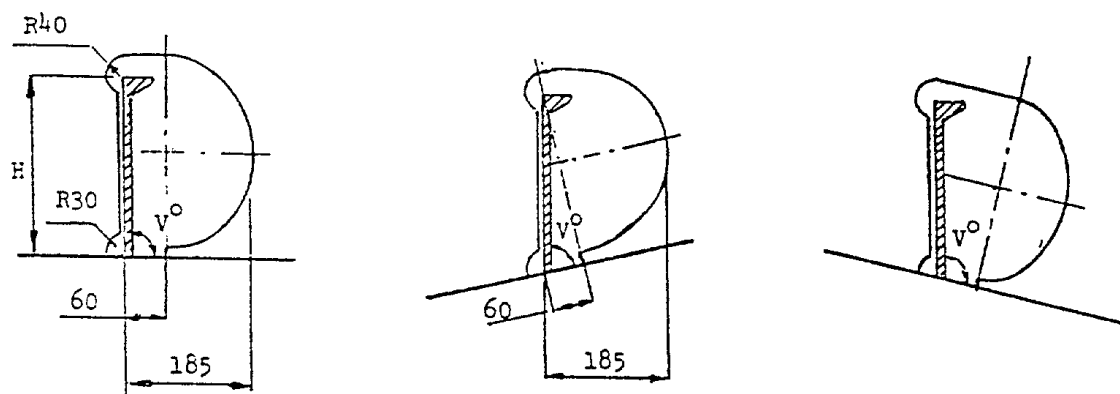
A x B

$A = H \sin V + 25$

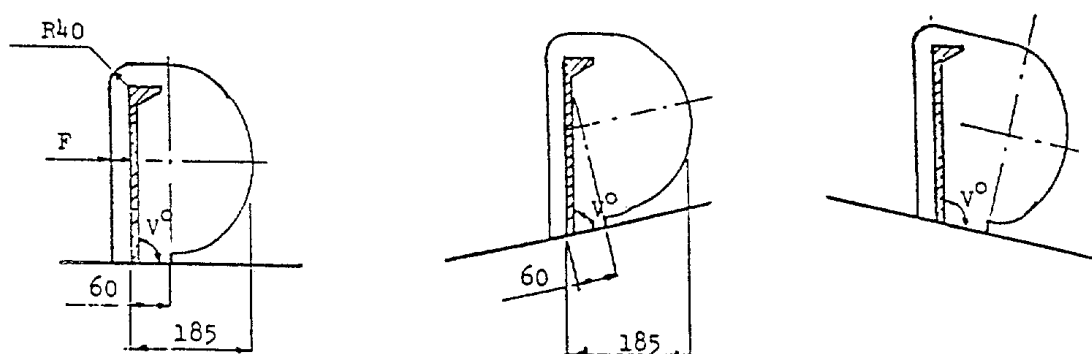
$B = \frac{370}{\sin V} + \cos V \left(H + \frac{55}{\sin V} \right) - 120$

$\cos V$ must be $< \frac{60}{H}$

Type 307

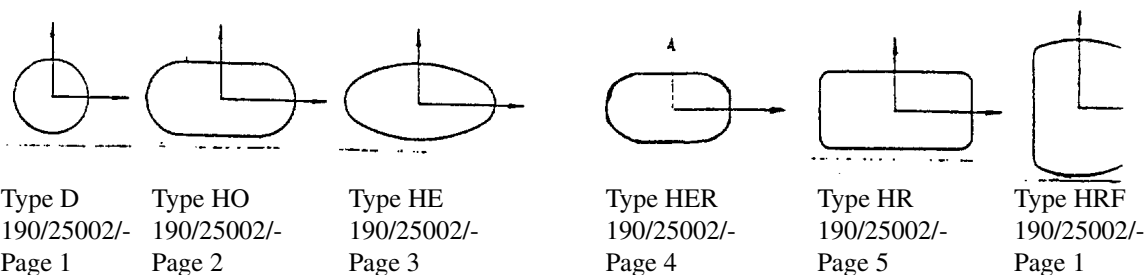


Type 507 $F = 60$ if $H < 300$, 90 if $H \geq 300$

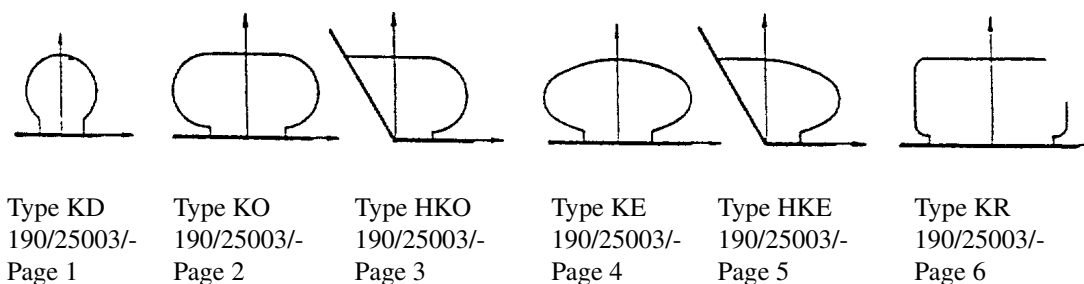


Survey of holes and cutouts

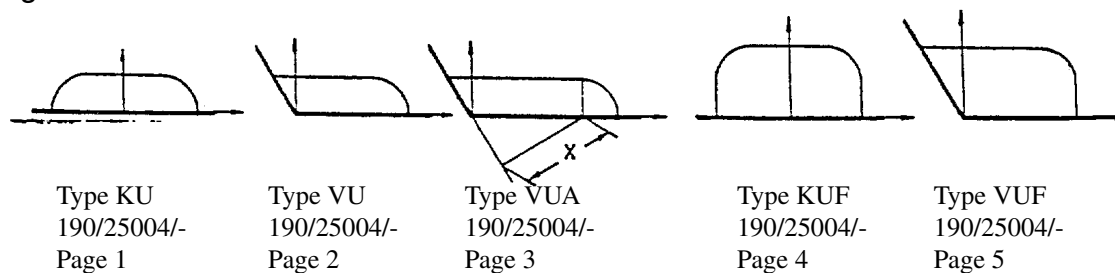
Holes



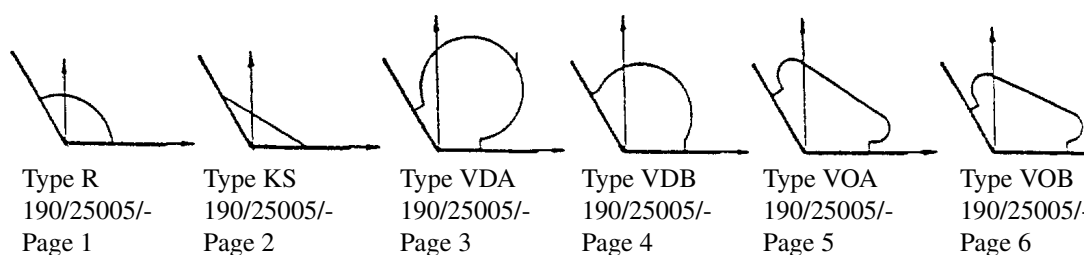
Edge holes



Edge cutouts



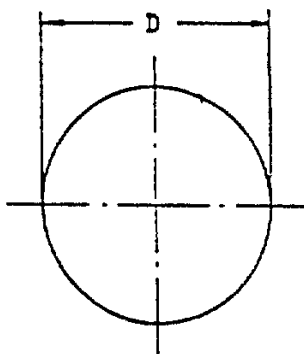
Corner cutouts



Holes

Round

Type D



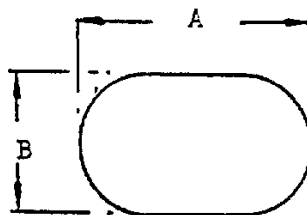
Examples

Designation	Dimens. in mm Diam.	Area m ²	Circumf m	Massa in kg at thickness			
				10	12	14	16
D 25	25	0,0005	0,08	0,039	0,047	0,055	0,063
D 40	40	0,001	0,13	0,079	0,094	0,11	0,13
D 50	50	0,002	0,16	0,16	0,19	0,22	0,25
D 75	75	0,004	0,24	0,31	0,38	0,44	0,50
D 100	100	0,008	0,31	0,63	0,75	0,88	1,00
D 150	150	0,018	0,47	1,41	1,70	1,98	2,26
D 200	200	0,031	0,63	2,43	2,92	3,41	3,89
D 250	250	0,049	0,79	3,85	4,62	5,39	6,15
D 300	300	0,071	0,94	5,57	6,69	7,80	8,92
D 350	350	0,096	1,10	7,54	9,04	10,6	12,1
D 400	400	0,126	1,26	9,89	11,9	13,8	15,8
D 450	450	0,159	1,41	12,5	15,0	17,5	20,0
D 500	500	0,196	1,57	15,4	18,5	21,5	24,6
D 550	550	0,238	1,73	18,6	22,4	26,0	29,8
D 600	600	0,283	1,88	22,2	26,6	31,1	35,5
D 650	650	0,332	2,04	26,0	31,2	36,4	41,6
D 700	700	0,385	2,20	30,2	36,2	42,3	48,3
D 750	750	0,442	2,36	34,6	41,6	48,5	55,4
D 800	800	0,503	2,51	39,4	47,3	55,2	63,1
D 850	850	0,567	2,67	44,5	53,4	62,3	71,2
D 900	900	0,636	2,83	49,9	59,9	69,9	79,9
D 950	950	0,709	2,98	55,6	66,7	77,8	89,0
D 1000	1000	0,785	3,14	61,6	73,9	86,3	98,6

Holes

Oval

Type HO



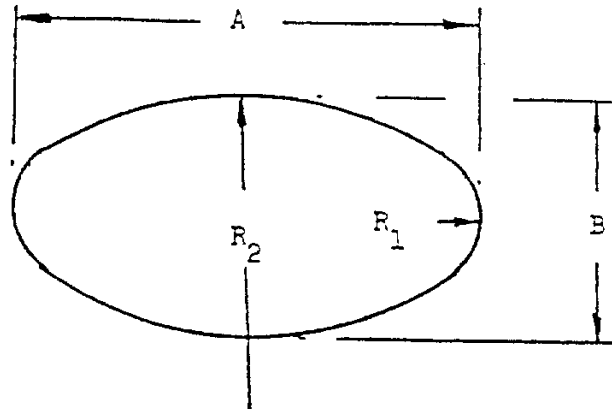
Examples

Designation	Dimensions in mm		Area m ²	Circumf. m	Mass in kg at thickness			
	A	B			10	12	14	16
HO 100x50	100	50	0,004	0,26	0,31	0,38	0,45	0,50
HO 100x75	100	75	0,006	0,29	0,39	0,47	0,55	0,63
HO 130x35	130	35	0,004	0,30	0,31	0,38	0,45	0,50
HO 130x40	130	40	0,005	0,31	0,39	0,47	0,55	0,63
HO 150x50	150	50	0,007	0,36	0,55	0,66	0,77	0,88
HO 150x75	150	75	0,009	0,39	0,71	0,85	0,99	1,13
HO 200x75	200	75	0,013	0,49	1,02	1,22	1,43	1,63
HO 200x100	200	100	0,018	0,51	1,41	1,70	1,98	2,26
HO 250x75	250	75	0,017	0,59	1,33	1,60	1,87	2,13
HO 250x100	250	100	0,023	0,61	1,81	2,17	2,53	2,89
HO 300x75	300	75	0,021	0,69	1,65	1,98	2,31	2,64
HO 300x100	300	100	0,028	0,71	2,20	2,64	3,08	3,52
HO 300x150	300	150	0,040	0,77	3,14	3,77	4,40	5,02
HO 350x75	350	75	0,025	0,79	1,96	2,35	2,74	3,14
HO 350x100	350	100	0,033	0,81	2,59	3,11	3,63	4,14
HO 400x75	400	75	0,029	0,89	2,28	2,73	3,19	3,64
HO 400x100	400	100	0,038	0,91	2,98	3,58	4,18	4,77
HO 400x150	400	150	0,055	0,97	4,32	5,18	6,04	6,91
HO 400x200	400	200	0,071	1,03	5,57	6,69	7,80	8,92
HO 450x100	450	100	0,043	1,01	3,38	4,05	4,73	5,40
HO 500x100	500	100	0,048	1,11	3,77	4,52	5,28	6,03
HO 500x200	500	200	0,091	1,23	7,14	8,57	10,00	11,4
HO 500x350	500	350	0,149	1,40	11,70	14,00	16,40	18,7
HO 600x150	600	150	0,085	1,37	6,67	8,01	9,34	10,7
HO 600x200	600	200	0,111	1,43	8,71	10,5	12,2	13,9
HO 600x300	600	300	0,161	1,54	12,60	15,2	17,7	20,2
HO 600x400	600	400	0,206	1,66	16,20	19,4	22,6	25,9
HO 700x150	700	150	0,100	1,57	7,85	9,42	11,0	12,6
HO 700x400	700	400	0,246	1,86	19,30	23,2	27,0	30,9
HO 800x400	800	400	0,286	2,06	22,50	26,9	31,4	35,9
HO 800x600	800	600	0,403	2,28	31,60	38,0	44,3	50,6
HO 900x600	900	600	0,463	2,48	36,30	43,6	50,9	58,2
HO 1500x950	1500	950	1,23	4,08	96,60	115,9	135,2	154,5

Holes

Elliptical

Type HE



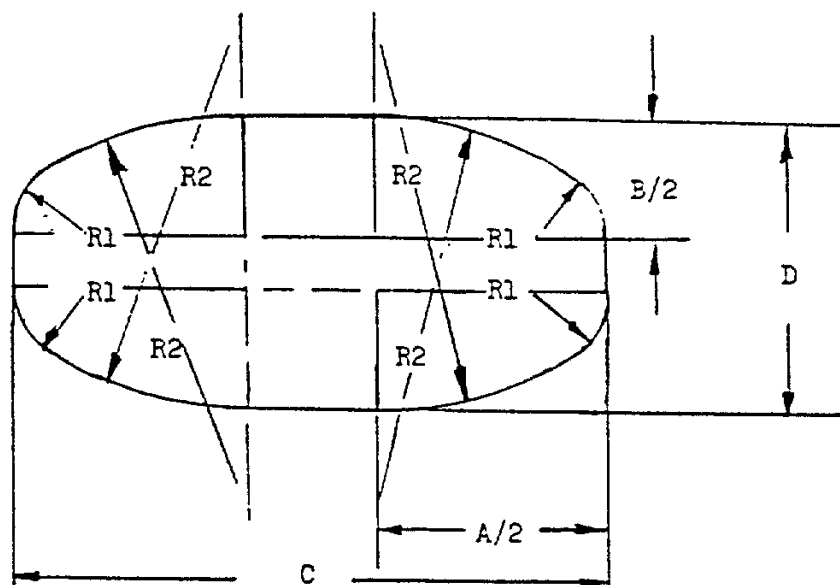
Examples

Designation	Dimensions in mm				B : A	Area m ²	Cir- cumf. m	Mass in kg at thickness			
	A	B	R1	R2				10	12	14	1-
HE 125x25	125	25	8	340	1:5	0,003	0,27	0,24	0,28	0,33	0,-
HE 100x50	100	50	15	82	1:2	0,004	0,24	0,31	0,38	0,44	0,-
HE 150x50	150	50	11	166	1:3	0,006	0,33	0,47	0,57	0,66	0,-
HE 150x75	150	75	23	123	1:2	0,009	0,36	0,71	0,85	0,99	1-
HE 200x75	200	75	18	203	1:2,7	0,012	0,45	0,94	1,13	1,32	1-
HE 200x100	200	100	30	164	1:2	0,016	0,48	1,26	1,51	1,76	2-
HE 200x120	200	120	42	144	1:1,67	0,019	0,51	1,49	1,79	2,09	2-
HE 250x100	250	100	26	242	1:2,5	0,020	0,58	1,57	1,88	2,20	2-
HE 250x125	250	125	38	205	1:2	0,025	0,60	1,96	2,35	2,75	3-
HE 300x150	300	150	46	246	1:2	0,035	0,73	2,75	3,30	3,85	4-
HE 350x175	350	175	53	287	1:2	0,048	0,84	3,77	4,52	5,28	6-
HE 318x255	318	255	109	186	1:1,25	0,064	0,90	5,02	6,03	7,03	8-
HE 400x200	400	200	61	328	1:2	0,063	0,97	4,95	5,93	6,92	7-
HE 500x250	500	250	76	410	1:2	0,098	1,21	7,96	9,23	10,8	12-
HE 600x300	600	300	91	492	1:2	0,141	1,45	11,1	13,3	15,5	17-
HE 700x350	700	350	107	574	1:2	0,192	1,69	1,51	18,1	21,1	24-
HE 730x400	730	400	130	560	1:1,83	0,227	1,81	1,68	20,2	23,5	26-
HE 800x400	800	400	122	656	1:2	0,251	1,94	1,97	23,6	27,6	31-
HE 900x450	900	450	137	738	1:2	0,318	2,18	2,50	30,0	34,9	39-
HE 900x650	900	650	258	567	1:1,38	0,46	2,63	3,61	43,4	50,6	57-
HE 1000x500	1000	500	152	820	1:2	0,354	2,42	2,78	33,3	38,9	44-
HE 1100x550	1100	550	168	902	1:2	0,469	2,66	3,68	44,7	51,5	58-
HE 1200x600	1200	600	183	984	1:2	0,558	2,90	4,38	52,6	61,3	70-
HE 1470x770	1470	770	243	1166	1:1,91	0,879	3,60	6,90	82,8	96,6	110-
HE 1500x870	1500	870	295	1105	1:1,72	1,017	3,79	79,8	95,8	111,8	127,-
HE 1500x950	1500	950	343	1037	1:1,58	1,113	3,90	87,4	104,8	122,3	139,-
HE 1500x650	1300	650	198	1066	1:2	0,655	3,15	51,4	61,7	72,0	8,-

Holes

Elliptical - Rectangular

Type HER



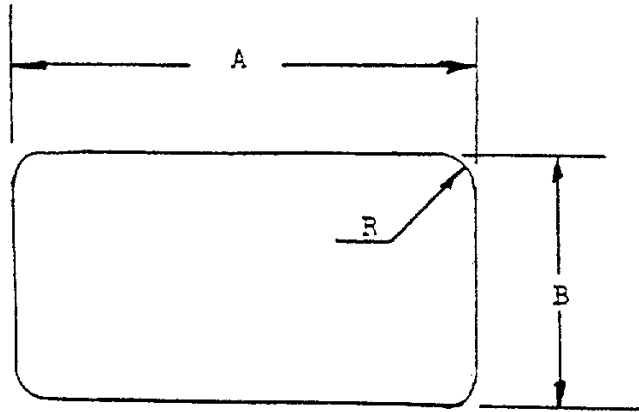
Examples

Designation	Dimensions in mm						Area m ²	Circumf m
	C	D	A	B	R1	R2		
HER 2100 x 1250 x 1500 x 950	2100	1250	1500	950	343	1037	2.3127	5.70
HER 2100 x 950 x 1500 x 950	2100	950	1500	950	343	1037	1.6827	5.10

Holes

Rectangular

Type HR



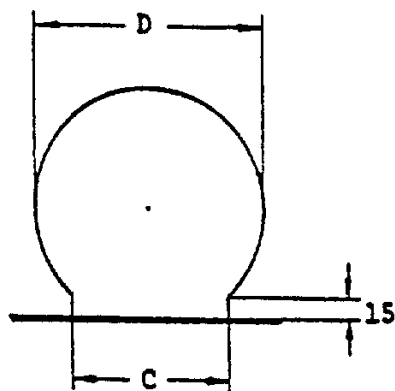
Examples

Designation	Dimensions in mm			Mass in kg at thickness				Used as
	A	B	R	10	12	14	16	
HR 600x600x75	600	600	75	28,3	33,9	39,6	45,2	Temporary passage
HR 750x750x75	750	750	75	44,1	53,0	61,8	70,6	Temporary passage
HR 1200x700x75	1200	700	75	65,9	79,1	92,3	105,5	Temporary passage
HR 1480x680x40	1480	680	40	79,0	94,8	110,6	126,4	Hole in plate for door
HR 1480x830x40	1480	830	40	96,4	115,7	135,0	154,3	Hole in plate for door
HR 1670x670x15	1670	670	15	87,8	105,4	123,0	140,5	Hole in plate for door
HR 1670x870x15	1670	870	15	114,1	136,9	159,7	182,5	Hole in plate for door
HR 1680x680x40	1680	680	40	89,7	107,6	125,5	143,5	Hole in plate for door
HR 1680x830x40	1680	830	40	109,5	131,4	153,2	175,1	Hole in plate for door
HR 1820x670x15	1820	670	15	95,7	114,9	134,0	153,2	Hole in plate for door
HR 1820x870x15	1820	870	15	124,3	149,2	174,0	198,9	Hole in plate for door
HR 1880x680x40	1880	680	40	100,4	120,4	140,5	160,6	Hole in plate for door
HR 1880x830x40	1880	830	40	122,5	147,0	171,5	196,0	Hole in plate for door
HR 1990x670x15	1990	670	15	104,7	125,6	146,5	167,5	Hole in plate for door
HR 1990x870x15	1990	870	15	135,9	163,1	190,3	217,5	Hole in plate for door
HR 681x501x125	681	501	125	26,2	31,5	36,7	42,0	Hole in plate for window
HR 761x551x125	761	551	125	32,4	39,0	45,5	52,0	Hole in plate for window
HR 851x611x125	851	611	125	40,5	48,6	56,7	64,9	Hole in plate for window
HR 1051x501x125	1051	501	125	41,1	49,2	57,3	65,7	Hole in plate for window
HR 1051x801x125	1051	801	125	66,2	79,5	92,8	106,1	Hole in plate for window
HR 1001x1051x125	1101	1051	125	91,5	109,8	128,1	146,4	Hole in plate for window
HR1401x1051x125	1401	1051	125	116,7	140,1	163,4	186,7	Hole in plate for window
HR 1701x1051x125	1701	1051	125	141,9	170,3	198,7	227,1	Hole in plate for window

Edge holes

Round

Type KD



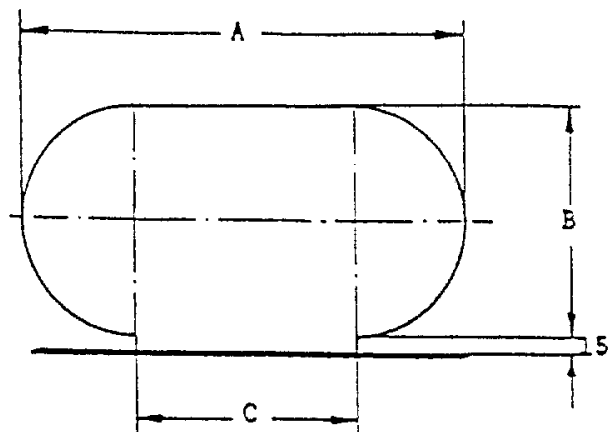
Examples

Designation	Dimensions in mm	
	D	C
KD 75x60	75	60
KD 100x70	100	70
KD 150x90	150	90
KD 200x105	200	105
KD 250x120	250	120
KD 300x130	300	130

Edge holes

Oval

Type KO



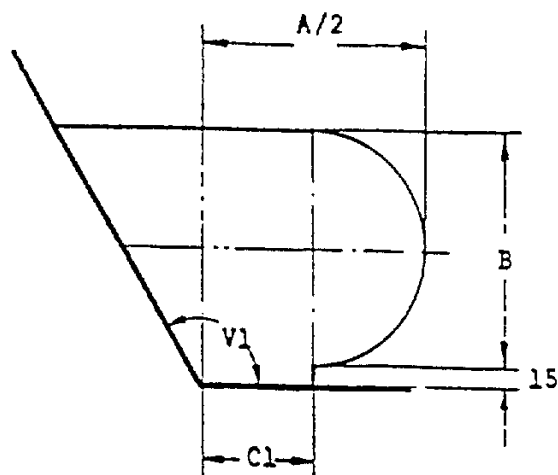
Examples

Designation	Dimensions in mm			Area m ²	Circumf ¹⁾
	A	B	C = A - B		
KO 100x50	100	50	50	0,005	0,24
KO 100x75	100	75	25	0,006	0,30
KO 150x50	150	50	100	0,009	0,29
KO 150x50	150	75	75	0,010	0,35
KO 200x75	200	75	125	0,015	0,40
KO 200x100	200	100	100	0,020	0,44
KO 250x 75	250	75	175	0,020	0,45
KO 250x100	250	100	150	0,025	0,49
KO 300x 75	300	75	225	0,024	0,50
KO 300x100	300	100	200	0,031	0,54
KO 300x150	300	150	150	0,042	0,65
KO 350x 75	350	75	275	0,029	0,55
KO 350x100	350	100	250	0,037	0,59
KO 400x 75	400	75	325	0,034	0,60
KO 400x100	400	100	300	0,043	0,64
KO 400x150	400	150	250	0,059	0,75
KO 400x200	400	200	200	0,074	0,86
KO 450x100	450	100	350	0,048	0,69
KO 500x100	500	100	400	0,054	0,74
KO 500x200	500	200	300	0,096	0,95
KO 500x350	500	350	150	0,151	1,28
KO 600x200	600	200	400	0,117	1,06
KO 600x300	600	300	300	0,166	1,27
KO 600x400	600	400	200	0,209	1,49
KO 700x400	700	400	300	0,251	1,59
KO 800x400	800	400	400	0,292	1,69
KO 1000x500	1000	500	500	0,454	2,1

Edge holes

Oval at corners

Type HKO

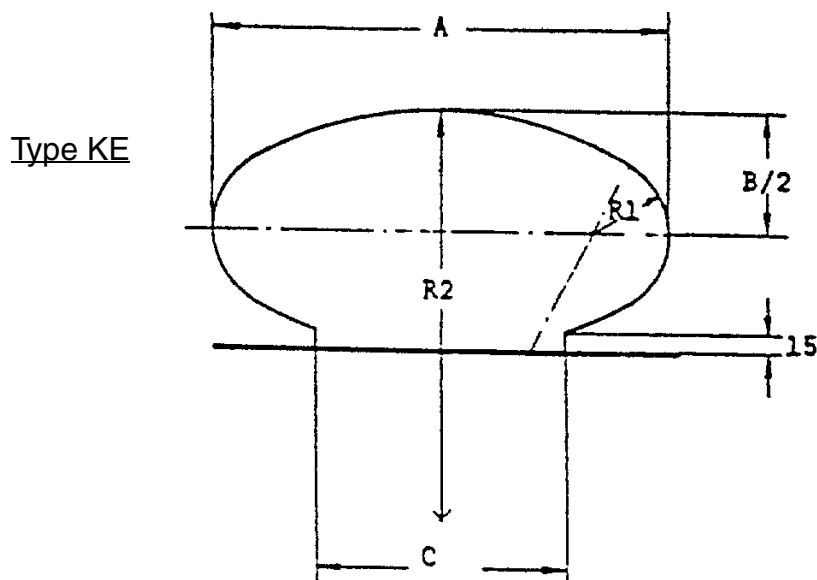


Examples

Designation	Dimensions in mm		
	A	B	C ¹
HKO 90x40	90	40	25
HKO 100x 50	100	50	25
HKO 100x 75	100	75	12,5
HKO 150x 50	150	50	50
HKO 150x 75	150	75	35,7
HKO 200x 75	200	75	62,5
HKO 200x100	200	100	50
HKO 250x 75	250	75	87,5
HKO 250x100	250	100	75
HKO 300x 75	300	75	112,5
HKO 300x100	300	100	150
HKO 300x150	300	150	75
HKO 350x 75	350	75	137,5
HKO 350x100	350	100	125
HKO 400x 75	400	75	162,5
HKO 400x100	400	100	150
HKO 400x150	400	150	125
HKO 400x200	400	200	100
HKO 450x100	450	100	175
HKO 500x100	500	100	200
HKO 500x200	500	200	150
HKO 500x350	500	350	75
HKO 600x200	600	200	200
HKO 600x300	600	300	150
HKO 600x400	600	400	100
HKO 700x400	700	400	150
HKO 800x400	800	400	200
HKO 1000x500	1000	500	250

Edge holes

Elliptical



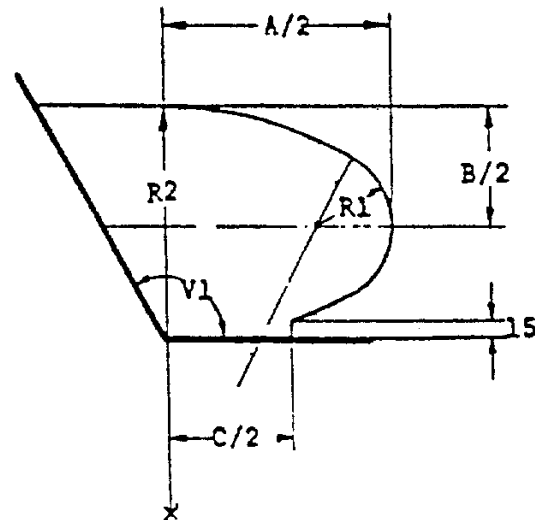
Examples

Designation	Dimensions in mm					Area m ²	Cir- cumf m
	A	B	C	R1	R2		
KE 150x50x100	150	20	100	11	166	0,007	0,17
KE 150x75x100	150	75	100	23	123	0,010	0,29
KE 200x75x100	200	75	100	18	203	0,013	0,38
KE 200x100x100	200	100	100	30	164	0,017	0,41
KE 250x100x100	250	100	100	26	242	0,021	0,51
KE 250x125x100	250	125	100	38	205	0,026	0,53
KE 300x150x100	300	150	100	46	246	0,036	0,66
KE 350x175x100	350	175	100	53	287	0,049	0,77
KE 400x200x100	400	200	100	61	328	0,064	0,90

Edge holes

Elliptical at corners

Type HKE



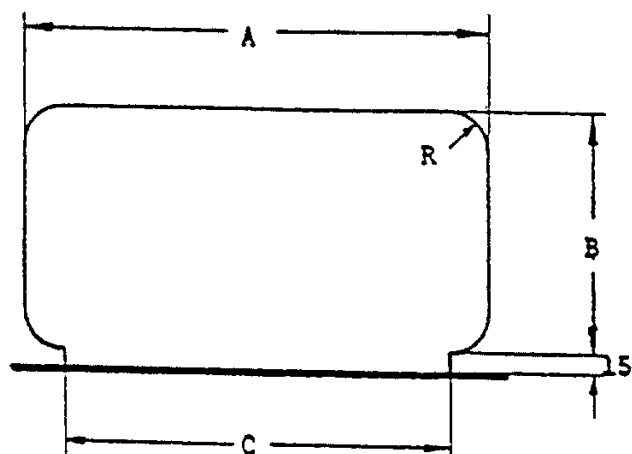
Examples

Designations	Dimensions in mm				
	A	B	C	R1	R2
HKE 150x 50x100	150	50	100	11	166
HKE 150x 75x100	150	75	100	23	123
HKE 200x 75x100	200	75	100	18	203
HKE 200x100x100	200	100	100	30	164
HKE 250x100x100	250	100	100	26	242
HKE 250x125x100	250	125	100	38	205
HKE 300x150x100	300	150	100	46	246
HKE 400x200x100	400	200	100	61	328

Edge holes

Rectangular

Type KR



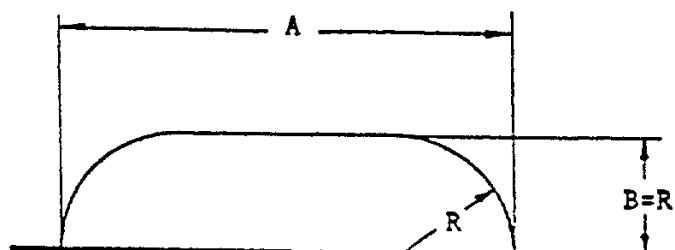
Examples

Designations	Dimensions in mm			
	A	B	C	R
KR 600x600x 500x50	600	600	500	50
KR 750x750x 650x50	750	750	650	50
KR 1200x700x1100x50	1200	700	1100	50
KR 1670x680x1590x40	1670	680	1590	40

Edge cutouts

Half oval

Type KU



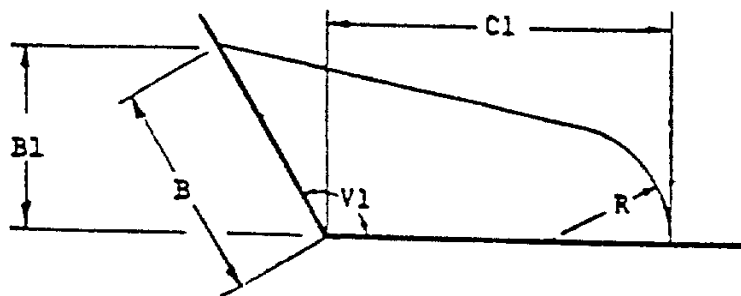
Examples

Designations	Dimensions in mm	
	A	B
KU 150x 25	150	25
KU 150x 50	150	50
KU 150x 60	150	60
KU 200x 50	200	50
KU 250x 50	250	50
KU 300x 50	300	50
KU 300x 75	300	75
KU 350x 50	350	50
KU 400x 50	400	50
KU 400x 75	400	75
KU 400x100	400	100

Edge cutouts

Half oval at corners

Type VU



Designation: VU C1xBxR or VU C1xR (when B1=R)

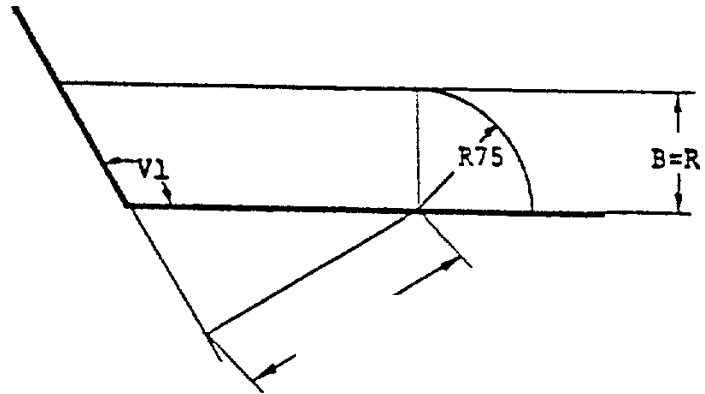
If $V1=90^\circ$ then $B=B1$

Dimensions in mm	
C1	R
60	25
60	50
75	60
100	50
125	25
125	75
150	50
150	75
200	50
225	75
250	50
300	50
300	100
325	75
350	50

Edge cutouts

Half oval at corners

Type VUA



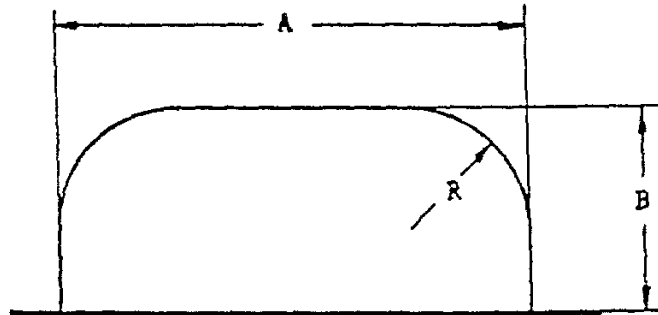
Examples

Designations	X
VUA 50	50
VUA 100	100
VUA 165	165
VUA 200	200
VUA 300	300

Edge cutouts

Half oval, deep type

Type KUF

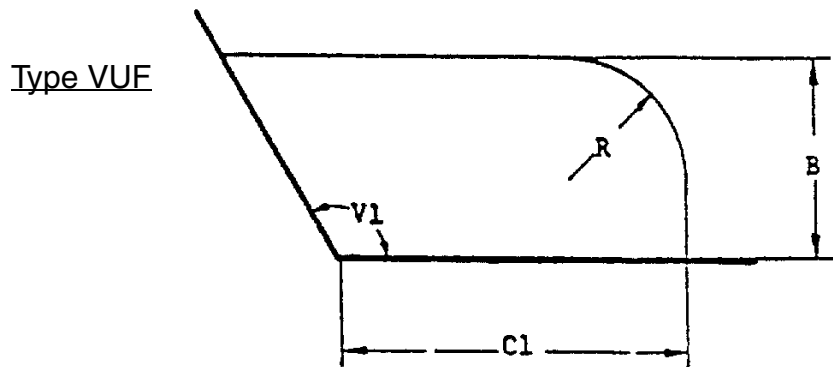


Designation KUF A x B x R

A and R first of all chosen from the series for type KU.

Edge cutouts

Half oval at corners, deep type

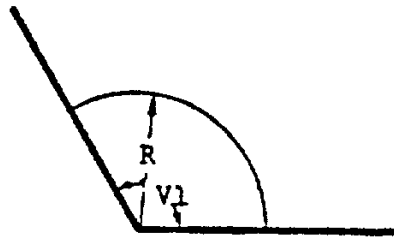


Designation VUF $C1 \times B \times R$
 $C1$ and R chosen from series for type VU.

Corner cutouts

Radii

Type R



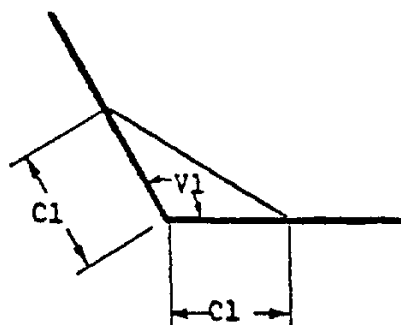
Examples

Designations	R
R 35	35
R 50	50
R 75	75
R 100	100
R 150	150

Corner cutouts

Bevel cut

Type KS



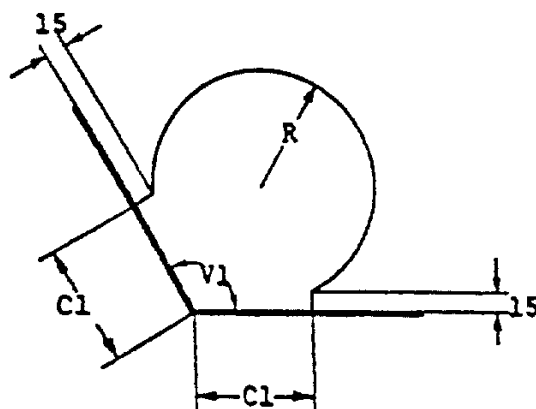
Examples

Designations	X
KS 3	3
KS 5	5
KS 8	8
KS 10	10
KS 15	15
KS 25	25
KS 50	50
KS 75	75

Corner cutouts

Round

Type VDA



$$C1 = X (1 + \cot V1) \text{ for } V1 \leq 90^\circ$$

$$C1 = X \text{ for } V1 > 90^\circ$$

$$R = C1 \tan \frac{V1}{2} - 15 \text{ however, max } X$$

Examples

For VDA 75 $X = 75$

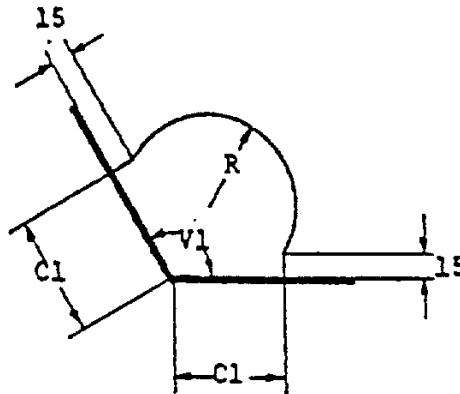
For VDA 100 $X = 100$

For VDA 150 $X = 150$

Corner cutouts

Round

Type VDB



$$C1 = X (1 + \cot V1) \text{ for } V1 \leq 90^\circ$$

$$C1 = X \text{ for } V1 > 90^\circ$$

$$R = \cos \frac{V1}{2} (C1 \tan \frac{V1}{2} - 15)$$

Examples

For VDB 75 $X = 75$

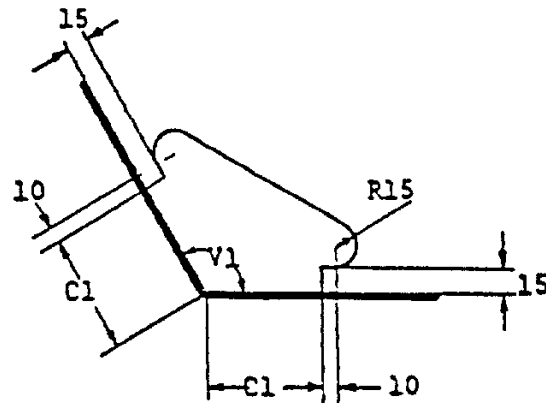
For VDB 100 $X = 100$

For VDB 150 $X = 150$

Corner cutouts

Oval symmetrical

Type VOA



$$C1 = X (1 + \cot V1) \text{ for } V1 \leq 90^\circ$$

$$C1 = X \text{ for } V1 > 90^\circ$$

Examples

For VOA 75 $X = 75$

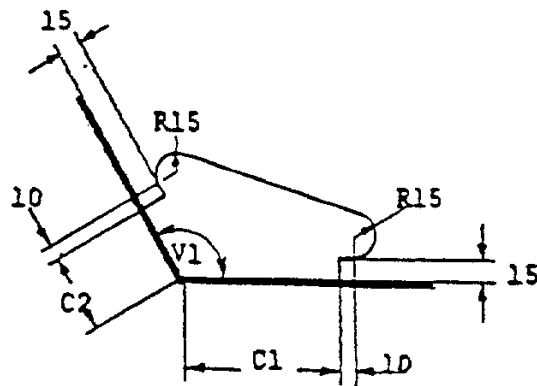
For VOA 100 $X = 100$

For VOA 150 $X = 150$

Corner cutouts

Oval non-symmetrical

Type VOB



$$C2 = \frac{C1}{2} (1 + \cot V1) \text{ for } V1 \leq 90^\circ$$

$$C2 = \frac{C1}{2} \text{ for } V1 > 90^\circ$$

Examples

For VOB 100 C1 = 100

For VOB 150 C1 = 150

For VOB 200 C1 = 200

End cutting of profiles

Code designations and dimensions

List of Contents

- 0 Purpose
- 0.1 General
- 0.2 Construction of the code.
- 0.3 Example
- 1 Flat and universal bars
- 2 Angle and bulb bars
- 3 T-bars

0. Purpose

0.1 General

This standard deals with end cutting of profiles and types and code designations for them. The standard is mainly intended for use when specifying machined profiles.

It is assumed that the end cutting is made at right angles to the profiles longitudinal axis if no other special instructions are given. Dimensions in mm and degrees when not stated otherwise.

Associated standards:

Material specifications for machined profiles are under preparation. Material specifications for plates and bars, edge preparations, VIS 444, Class 0541.21.

0.2 Construction of the code

The code for end cutting consists of four digits divided into groups as follows:



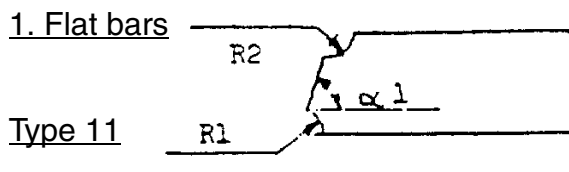
The code for variations, the 3rd and 4th digits, is built up according to the following directives:

- | | |
|--------------------------------|---|
| 3rd digit 0 for R1 = 0 | 4th digit for R2 = 0:0 if V = 90°, 1 if V = 30° |
| 1 for R1 = 35 | for R2 = 35:2 if V = 90°, 3 if V = 30° |
| 2 for R1 = 50 | for R2 = 50:4 if V = 90°, 5 if V = 30° |
| 3 for R1 = 75 | for R2 = 75:6 if V = 90°, 7 if V = 30° |
| 4 for R1 = 60 | |
| 9 if R1 is replaced by a bevel | |

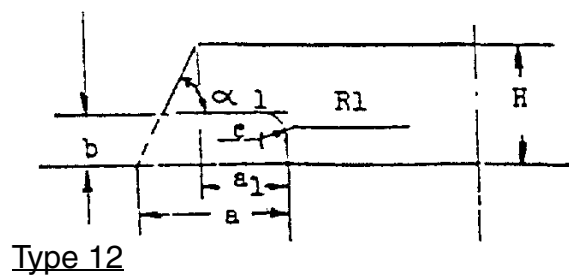
In addition to the basic code there may be variable dimensional designations. In these cases the dimensions are given for those directly connected to the basic code. If a and b are variable, the complete code designation will be as follows: 1 2 3 4, a b.

End Cutting of Profiles

1. Flat bars

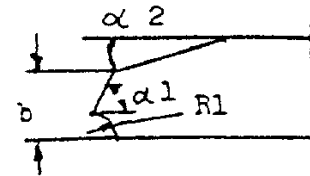


Code designation	R ₁	R ₂
1100, α 1	0	0
1102, α 1	0	35
1104, α 1	0	50
1105, α 1	0	60
1106, α 1	0	75
1110, α 1	35	0
1112, α 1	35	35
1114, α 1	35	50
1115, α 1	35	60
1116, α 1	35	75
1120, α 1	50	0
1122, α 1	50	35
1124, α 1	50	50
1125, α 1	50	60
1126, α 1	50	75
1130, α 1	75	0
1132, α 1	75	35
1134, α 1	75	50
1135, α 1	75	60
1136, α 1	75	75
1140, α 1	60	0
1142, α 1	60	35
1144, α 1	60	50
1145, α 1	60	60
1146, α 1	60	75
1190, α 1	KS10	0



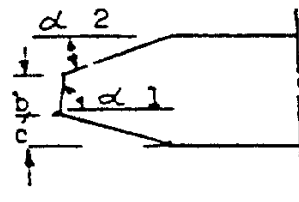
Code designation	R ₁
1210, a,b α 1	35
1220, a,b α 1	50
1230, a,b α 1	75
1240, a,b α 1	60

Type 13



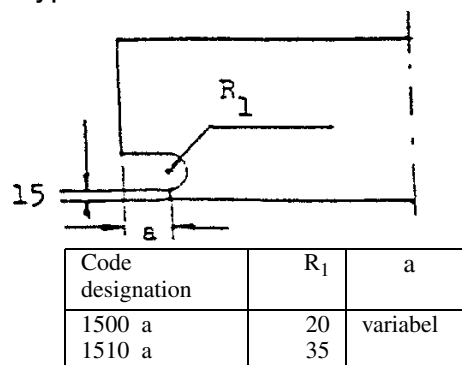
Code designation	R ₁	α 2
1301, b, α 1	0	18
1302, b, α 1	0	30
1303, b, α 1	0	45
1311, b, α 1	35	18
1312, b, α 1	35	30
1313, b, α 1	35	45
1321, b, α 1	50	18
1322, b, α 1	50	30
1323, b, α 1	50	45
1331, b, α 1	75	18
1332, b, α 1	75	30
1333, b, α 1	75	45
1341, b, α 1	60	18
1342, b, α 1	60	30
1343, b, α 1	60	45
1391, b, α 1	KS10	18
1392, b, α 1	KS10	30
1393, b, α 1	KS10	45

Type 14



Code designation	α 2
1401, b,c α 1	18
1402, b,c α 1	30
1403, b, c α 1	60

Type 15



Code designation	R ₁	a
1500 a	20	variabel
1510 a	35	

If $b = R_1$, then b need to be given in the code
 If $b > H/2$, then a1 will be given instead of a
 c will be given manually, but not if $c = 0$

End cutting of profiles

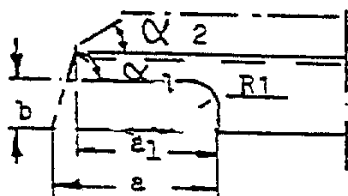
2. L- and bulb bars

Type 21



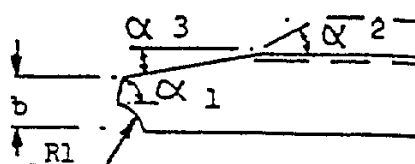
Code designation	R ₁	α ₂
2100, α 1, α 2	0	90
2101, α 1,	0	30
2110, α 1, α 2	35	90
2111, α 1,	35	30
2120, α 1, α 2	50	90
2121, α 1,	50	30
2130, α 1,	75	90
2131, α 1,	75	30
2140, α 1, α 2	60	90
2141, α 1,	60	30
2190, α 1, α 2	KS10	90
2191, α 1,	KS10	30

Type 22



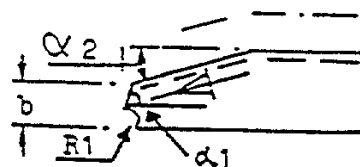
Code designation See also type 12	R ₁	α ₂
2210, a, b, α 1, α 2	35	90
2211, a, b, α 1	35	30
2220, a, b, α 1, α 2	50	90
2221, a, b, α 1,	50	30
2230, a, b, α 1, α 2	75	90
2231, a, b, α 1	75	30
2240, a, b, α 1, α 2	60	90
2241, a, b, α 1	60	30

Type 23



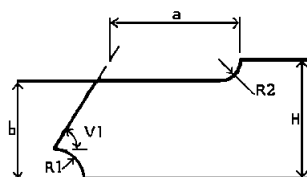
Code designation	R ₁	α ₂	α ₃
2301, b, α 1	0	30	18
2302, b, α 1	0	90	18
2303, b, α 1	0	30	30
2304, b, α 1	0	90	30
2311, b, α 1	35	30	18
2312, b, α 1	35	90	18
2313, b, α 1	35	30	30
2314, b, α 1	35	90	30
2321, b, α 1	50	30	18
2322, b, α 1	50	90	18
2323, b, α 1	50	30	30
2331, b, α 1	75	30	18
2332, b, α 1	75	90	18
2333, b, α 1	75	30	30
2341, b, α 1	60	30	18
2342, b, α 1	60	90	18
2343, b, α 1	60	30	30
2391, b, α 1	KS10	30	18
2392, b, α 1	KS10	90	18
2393, b, α 1	KS10	30	30
2394, b, α 1	KS10	90	30

Type 24



Code designation	R ₁	α ₂
2401, b, α 1	0	18
2411, b, α 1	35	18
2421, b, α 1	50	18
2431, b, α 1	75	18
2441, b, α 1	60	18
2491, b, α 1	KS10	18

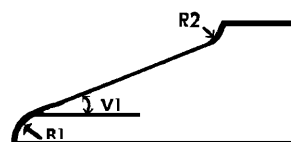
Type 112



Code, parameters	R1	R2
11210, a [b,] V1	0	35
11220, a [b,] V1	0	50
11230, a [b,] V1	0	75

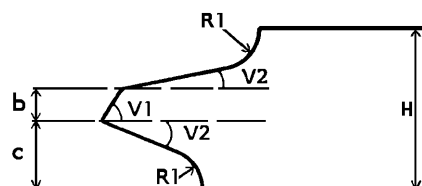
If $b = 0$ (not given) then
 $b = H - R2$

Type 113



Code	R1	R2	V2
11301	25	25	18
11302	25	25	30
11303	25	25	45

Type 114

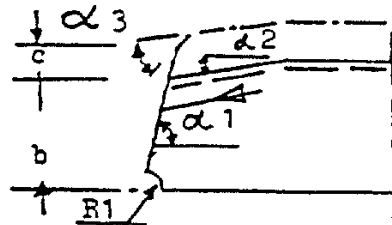


Code, parameters	R1	V2
11401, b [c,] V1	25	18
11402, b [c,] V1	25	30
11403, b [c,] V1	25	60

If $c = 0$ (not given) then
 $c = (H - b) / 2$

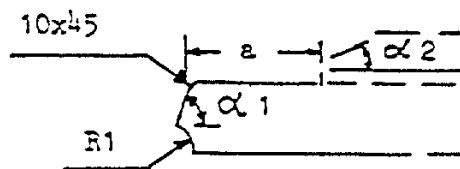
End cutting of profiles

Type 25



Code designation	R_1	$\alpha 2$	$\alpha 3$
2501, b, c, $\alpha 1$	0	18	18
2511, b, c, $\alpha 1$	35	18	18
2521, b, c, $\alpha 1$	50	18	18
2531, b, c, $\alpha 1$	75	18	18
2541, b, c, $\alpha 1$	60	18	18
2591, b, c, $\alpha 1$	KS10	18	18

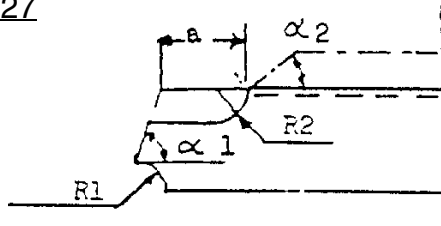
Type 26



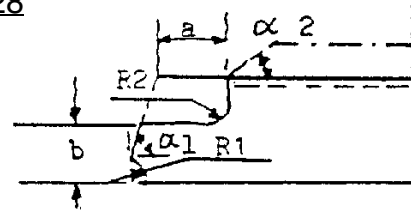
Code designation	R_1	$\alpha 2$
2600, a, $\alpha 1$, $\alpha 2$	0	90
2601, a, $\alpha 1$,	0	30
2610, a, $\alpha 1$, $\alpha 2$	35	90
2611, a, $\alpha 1$,	35	30
2620, a, $\alpha 1$, $\alpha 2$	50	90
2621, a, $\alpha 1$,	50	30
2630, a, $\alpha 1$, $\alpha 2$	75	90
2631, a, $\alpha 1$,	75	30
2640, a, $\alpha 1$, $\alpha 2$	60	90
2641, a, $\alpha 1$,	60	30

End Cutting of Profiles

Type 27



Type 28



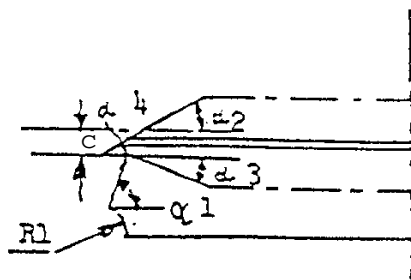
Code Designation	R ₁	R ₂	$\alpha 2$
2700, a, $\alpha 1$, $\alpha 2$	0	0	90
2701, a, $\alpha 1$	0	0	30
2702, a, $\alpha 1$, $\alpha 2$	0	35	90
2703, a, $\alpha 1$	0	35	30
2704, a, $\alpha 1$, $\alpha 2$	0	50	90
2705, a, $\alpha 1$	0	50	30
2706, a, $\alpha 1$, $\alpha 2$	0	75	90
2707, a, $\alpha 1$	0	75	30
2708, a, $\alpha 1$, $\alpha 2$	0	60	90
2709, a, $\alpha 1$	0	60	30
2710, a, $\alpha 1$, $\alpha 2$	35	0	90
2711, a, $\alpha 1$	35	0	30
2712, a, $\alpha 1$, $\alpha 2$	35	35	90
2713, a, $\alpha 1$	35	35	30
2714, a, $\alpha 1$, $\alpha 2$	35	50	90
2715, a, $\alpha 1$	35	50	30
2716, a, $\alpha 1$, $\alpha 2$	35	75	90
2717, a, $\alpha 1$	35	75	30
2718, a, $\alpha 1$, $\alpha 2$	35	60	90
2719, a, $\alpha 1$	35	60	30
2720, a, $\alpha 1$, $\alpha 2$	50	0	90
2721, a, $\alpha 1$	50	0	30
2722, a, $\alpha 1$, $\alpha 2$	50	35	90
2723, a, $\alpha 1$	50	35	30
2724, a, $\alpha 1$, $\alpha 2$	50	50	90
2725, a, $\alpha 1$	50	50	30
2726, a, $\alpha 1$, $\alpha 2$	50	75	90
2727, a, $\alpha 1$	50	75	30
2728, a, $\alpha 1$, $\alpha 2$	50	60	90
2729, a, $\alpha 1$	50	60	30
2730, a, $\alpha 1$, $\alpha 2$	75	0	90
2731, a, $\alpha 1$	75	0	30
2732, a, $\alpha 1$, $\alpha 2$	75	35	90
2733, a, $\alpha 1$	75	35	30
2734, a, $\alpha 1$, $\alpha 2$	75	50	90
2735, a, $\alpha 1$	75	50	30
2736, a, $\alpha 1$, $\alpha 2$	75	75	90
2737, a, $\alpha 1$	75	75	30
2738, a, $\alpha 1$, $\alpha 2$	75	60	90
2739, a, $\alpha 1$	75	60	30
2740, a, $\alpha 1$, $\alpha 2$	60	0	90
2741, a, $\alpha 1$	60	0	30
2742, a, $\alpha 1$, $\alpha 2$	60	35	90
2743, a, $\alpha 1$	60	35	30
2744, a, $\alpha 1$, $\alpha 2$	60	50	90
2745, a, $\alpha 1$	60	50	30
2746, a, $\alpha 1$, $\alpha 2$	60	75	90
2747, a, $\alpha 1$	60	75	30

Code Designation	R ₁	R ₂	$\alpha 2$
2802, a, b, $\alpha 1$, $\alpha 2$	0	35	90
2803, a, b, $\alpha 1$	0	35	30
2804, a, b, $\alpha 1$, $\alpha 2$	0	50	90
2805, a, b, $\alpha 1$	0	50	30
2806, a, b, $\alpha 1$, $\alpha 2$	0	75	90
2807, a, b, $\alpha 1$	0	75	30
2808, a, b, $\alpha 1$, $\alpha 2$	0	60	90
2809, a, b, $\alpha 1$	0	60	30
2812, a, b, $\alpha 1$, $\alpha 2$	35	35	90
2813, a, b, $\alpha 1$	35	35	30
2814, a, b, $\alpha 1$, $\alpha 2$	35	50	90
2815, a, b, $\alpha 1$	35	50	30
2816, a, b, $\alpha 1$, $\alpha 2$	35	75	90
2817, a, b, $\alpha 1$	35	75	30
2818, a, b, $\alpha 1$, $\alpha 2$	35	60	90
2819, a, b, $\alpha 1$	35	60	30
2822, a, b, $\alpha 1$, $\alpha 2$	50	35	90
2823, a, b, $\alpha 1$	50	35	30
2824, a, b, $\alpha 1$, $\alpha 2$	50	50	90
2825, a, b, $\alpha 1$	50	50	30
2826, a, b, $\alpha 1$, $\alpha 2$	50	75	90
2827, a, b, $\alpha 1$	50	75	30
2828, a, b, $\alpha 1$, $\alpha 2$	50	60	90
2829, a, b, $\alpha 1$	50	60	30
2832, a, b, $\alpha 1$, $\alpha 2$	75	35	90
2833, a, b, $\alpha 1$	75	35	30
2834, a, b, $\alpha 1$, $\alpha 2$	75	50	90
2835, a, b, $\alpha 1$	75	50	30
2836, a, b, $\alpha 1$, $\alpha 2$	75	75	90
2837, a, b, $\alpha 1$	75	75	30
2838, a, b, $\alpha 1$, $\alpha 2$	75	60	90
2839, a, b, $\alpha 1$	75	60	30
2842, a, b, $\alpha 1$, $\alpha 2$	60	35	90
2843, a, b, $\alpha 1$	60	35	30
2844, a, b, $\alpha 1$, $\alpha 2$	60	50	90
2845, a, b, $\alpha 1$	60	50	30
2846, a, b, $\alpha 1$, $\alpha 2$	60	75	90
2847, a, b, $\alpha 1$	60	75	30
2848, a, b, $\alpha 1$, $\alpha 2$	60	60	90
2849, a, b, $\alpha 1$	60	60	30

End Cutting of Profiles

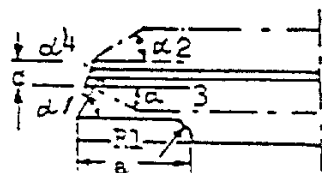
3. T-bars

Type 31



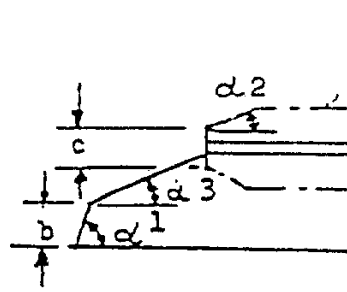
Code designation	R ₁	α ₂	α ₃
3100, α ₁ , α ₄	0	90	90
3101, c, α ₁ , α ₃ , α ₄	0	30	90
3102, c, α ₁ , α ₄	0	30	30
3103, c, α ₁ , α ₃ , α ₄	0	18	90
3104, c, α ₁ , α ₄	0	18	18
3110, α ₁ , α ₄	35	90	90
3111, c, α ₁ , α ₃ , α ₄	35	30	90
3112, c, α ₁ , α ₄	35	30	30
3113, c, α ₁ , α ₃ , α ₄	35	18	90
3114, c, α ₁ , α ₄	35	18	18
3120, α ₁ , α ₄	50	90	90
3121, c, α ₁ , α ₃ , α ₄	50	30	90
3122, c, α ₁ , α ₄	50	30	30
3123, c, α ₁ , α ₃ , α ₄	50	18	90
3124, c, α ₁ , α ₄	50	18	18
3130, α ₁ , α ₄	75	90	90
3131, c, α ₁ , α ₃ , α ₄	75	30	90
3132, c, α ₁ , α ₄	75	30	30
3133, c, α ₁ , α ₃ , α ₄	75	18	90
3134, c, α ₁ , α ₄	75	18	18
3140, α ₁ , α ₄	60	90	90
3141, c, α ₁ , α ₃ , α ₄	60	30	90
3142, c, α ₁ , α ₄	60	30	30
3143, c, α ₁ , α ₃ , α ₄	60	18	90
3144, c, α ₁ , α ₄	60	18	18

Type 32



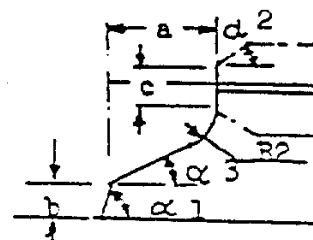
Code designation	R ₁	α ₂ α ₃
3210, a, α ₁ , α ₄	35	90
3211, a, c, α ₁ , α ₄	35	30
3212, a, c, α ₁ , α ₄	35	18
3220, a, α ₁ , α ₄	50	90
3221, a, c, α ₁ , α ₄	50	30
3222, a, c, α ₁ , α ₄	50	18
3230, a, α ₁ , α ₄	75	90
3231, a, c, α ₁ , α ₄	75	30
3232, a, c, α ₁ , α ₄	75	18
3240, a, α ₁ , α ₄	60	90
3241, a, c, α ₁ , α ₄	60	30
3242, a, c, α ₁ , α ₄	60	18

Type 33



Code designation	α ₂	α ₃
3311, b, c, α ₁	30	30
3312, b, c, α ₁	18	30
3321, b, c, α ₁	30	18
3322, b, c, α ₁	18	18

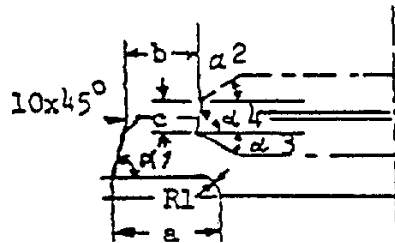
Type 34



Code designation	R ₁	α ₂	α ₃
3411, a, b, c, α ₁	35	30	30
3412, a, b, c, α ₁	35	18	30
3413, a, b, c, α ₁	35	30	18
3414, a, b, c, α ₁	35	18	18
3421, a, b, c, α ₁	50	30	30
3422, a, b, c, α ₁	50	18	30
3423, a, b, c, α ₁	50	30	18
3424, a, b, c, α ₁	50	18	18
3431, a, b, c, α ₁	75	30	30
3432, a, b, c, α ₁	75	18	30
3433, a, b, c, α ₁	75	30	18
3434, a, b, c, α ₁	75	18	18
3441, a, b, c, α ₁	60	30	30
3442, a, b, c, α ₁	60	18	30
3443, a, b, c, α ₁	60	30	18
3444, a, b, c, α ₁	60	18	18

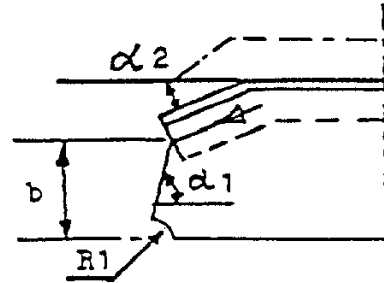
End cutting of profiles

Type 35



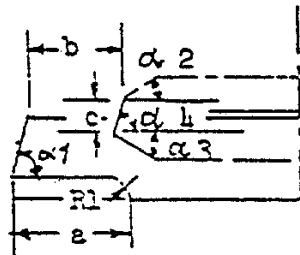
Code designation	R ₁	α2 α3
3510, a, b, α1, α4	35	90
3511, a, b, c, α1, α4	35	30
3512, a, b, c, α1, α4	35	18
3520, a, b, α1, α4	50	90
3521, a, b, c, α1, α4	50	30
3522, a, b, c, α1, α4	50	18
3530, a, b, α1, α4	75	90
3531, a, b, c, α1, α4	75	30
3532, a, b, c, α1, α4	75	18
3540, a, b, α1, α4	60	90
3541, a, b, c, α1, α4	60	30
3542, a, b, c, α1, α4	60	18

Type 37



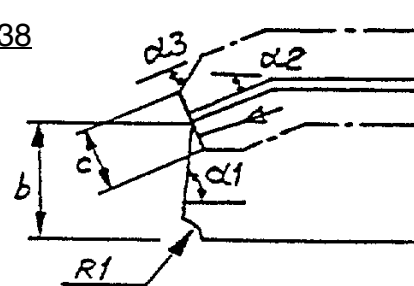
Code designation	R ₁	2
3701, b, α1	0	18
3711, b, α1	35	18
3721, b, α1	50	18
3731, b, α1	75	18
3741, b, α1	60	18
3791, b, α1	KS10	18

Type 36



Code designation	R ₁	α2 α3
3610, a, b, α1, α4	35	90
3611, a, b, c, α1, α4	35	30
3612, a, b, c, α1, α4	35	18
3620, a, b, α1, α4	50	90
3621, a, b, c, α1, α4	50	30
3622, a, b, c, α1, α4	50	18
3630, a, b, α1, α4	75	90
3631, a, b, c, α1, α4	75	30
3632, a, b, c, α1, α4	75	18
3640, a, b, α1, α4	60	90
3641, a, b, c, α1, α4	60	30
3642, a, b, c, α1, α4	60	18

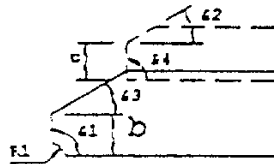
Type 38



Code designation	R ₁	α2	α3
3801, b, c, α1	0	18	18
3811, b, c, α1	35	18	18
3821, b, c, α1	50	18	18
3831, b, c, α1	75	18	18
3841, b, c, α1	60	18	18
3891, b, c, α1	KS10	18	18

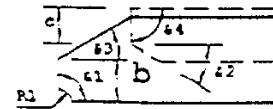
End cutting of profiles

Type 40



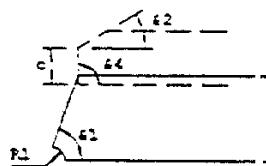
Code designation	R_1	α_2	α_3
4001, b, c, α_1 , α_4	0	30	30
4002, b, c, α_1 , α_4	0	30	18
4011, b, c, α_1 , α_4	35	30	30
4012, b, c, α_1 , α_4	35	30	18
4031, b, c, α_1 , α_4	75	30	30
4032, b, c, α_1 , α_4	75	30	18
4041, b, c, α_1 , α_4	60	30	30
4042, b, c, α_1 , α_4	60	30	18

Type 41



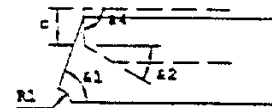
Code designation	R_1	α_2	α_3
4101, b, c, α_1 , α_4	0	30	30
4102, b, c, α_1 , α_4	0	30	18
4111, b, c, α_1 , α_4	35	30	30
4112, b, c, α_1 , α_4	35	30	18
4131, b, c, α_1 , α_4	75	30	30
4132, b, c, α_1 , α_4	75	30	18
4141, b, c, α_1 , α_4	60	30	30
4142, b, c, α_1 , α_4	60	30	18

Type 42



Code designation	R_1	α_2
4201, c, α_1 , α_4	0	30
4211, c, α_1 , α_4	35	30
4231, c, α_1 , α_4	75	30
4241, c, α_1 , α_4	60	30

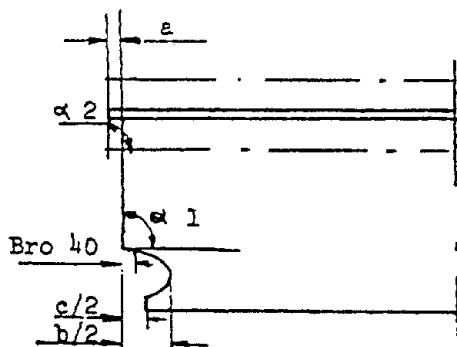
Type 43



Code designation	R_1	α_2
4301, c, α_1 , α_4	0	30
4311, c, α_1 , α_4	35	30
4331, c, α_1 , α_4	75	30
4341, c, α_1 , α_4	60	30

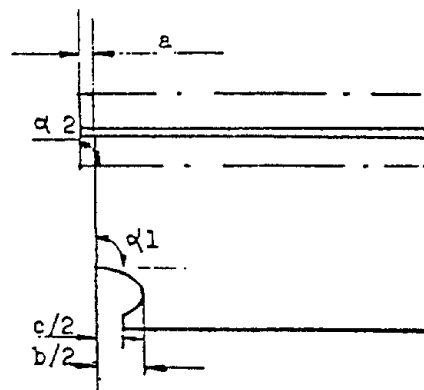
End cutting of profiles

Type 45



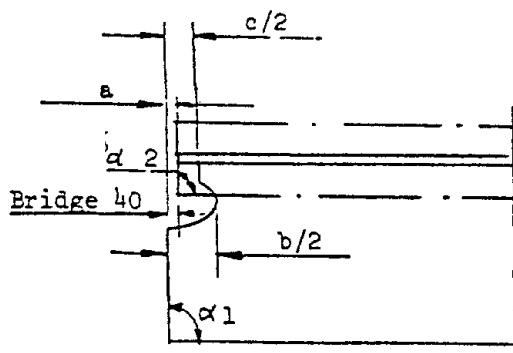
Code designation	$\alpha 2$	Relation major axis/ minor axis
4560, a, b, c, $\alpha 1$	90	1/0,5
4580, a, b, c, $\alpha 1$	90	1/0,6

Type 47



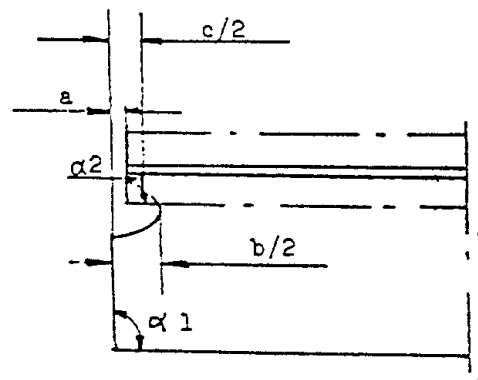
Code designation	$\alpha 2$	Relation major axis/ minor axis
4760, a, b, c, $\alpha 1$	90	1/0,5
4780, a, b, c, $\alpha 1$	90	1/0,6

Type 46



Code designation	$\alpha 2$	Relation major axis/ minor axis
4660, a, b, c, $\alpha 1$	90	1/0,5
4680, a, b, c, $\alpha 1$	90	1/0,6

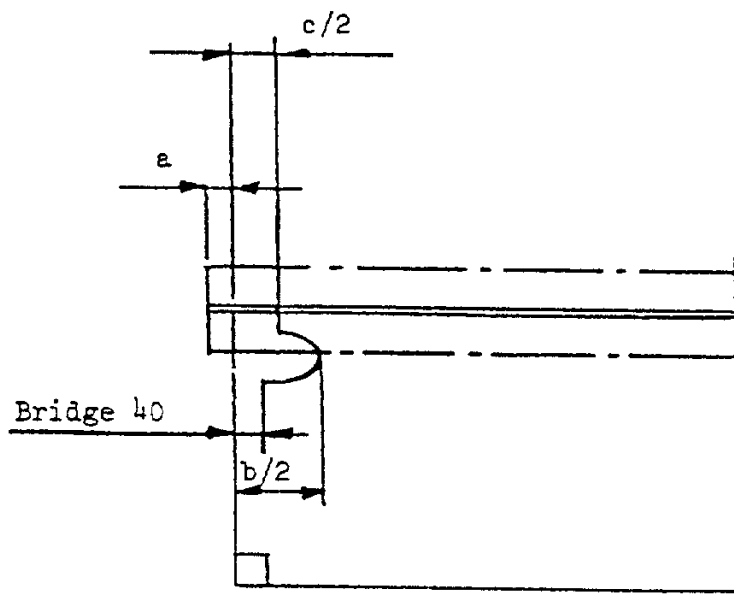
Type 48



Code designation	$\alpha 2$	Relation major axis/ minor axis
4760, a, b, c, $\alpha 1$	90	1/0,5
4780, a, b, c, $\alpha 1$	90	1/0,6

End cutting of profiles

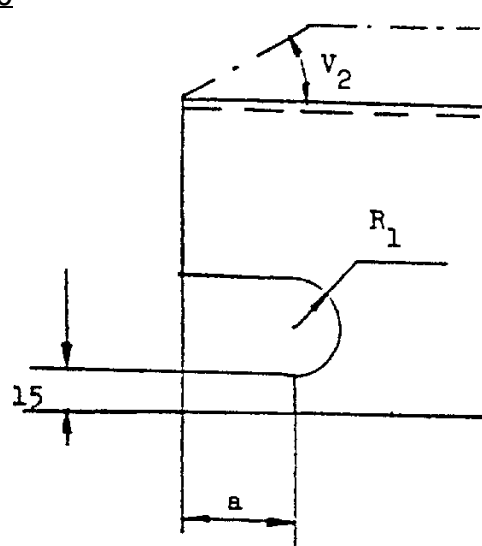
Type 50



Code designation	Relation major axis/ minor axis
5060, a, b, c	1/0,5

End cutting of profiles

Type 60

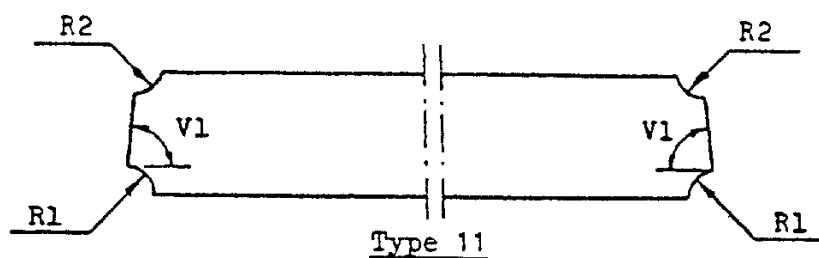


Code designation	R_1	V_2	a
6000 a	20	90	variable
6001 a	20	30	variable
6010 a	35	90	variable
6011 a	35	30	variable

End cutting of profiles

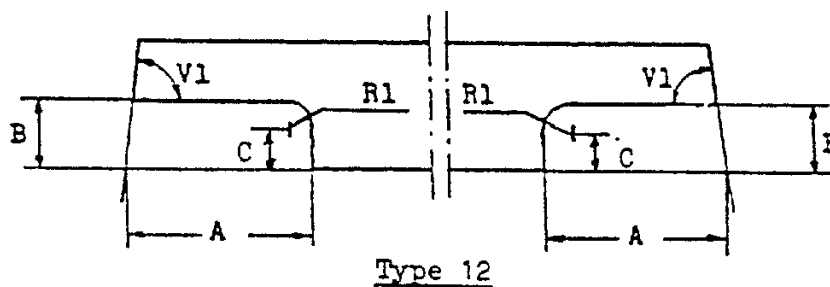
Marking-up list for profiles

Flat bars



code x 1 x 11

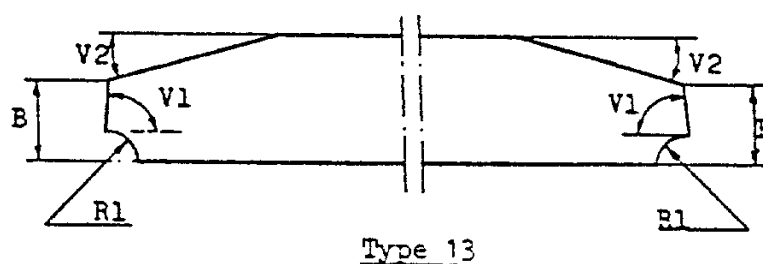
code x 2 x 11



code x 1 x 12

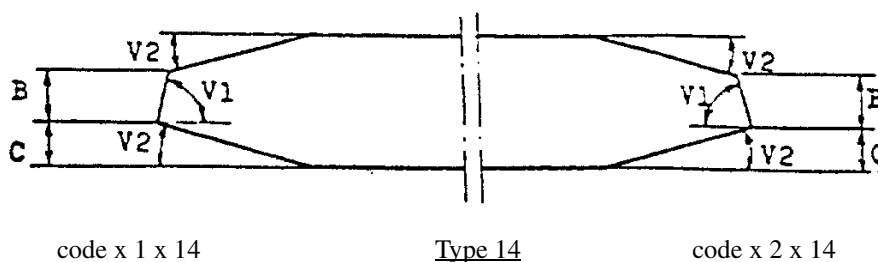
code x 2 x 12

C will be given manually, but not if C = 0



code x 1 x 13

code x 2 x 13



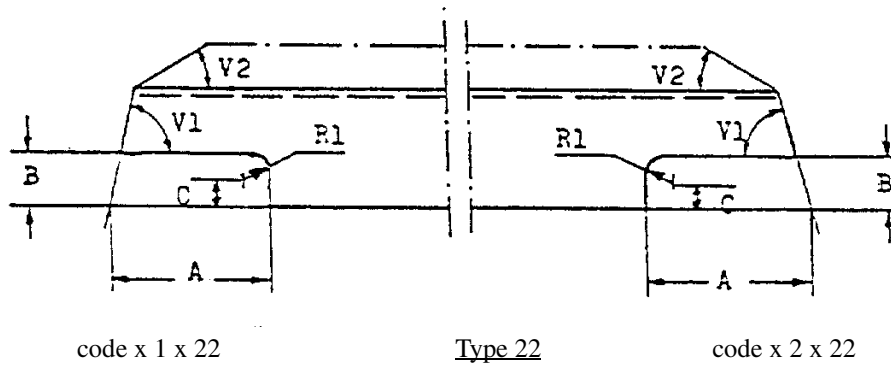
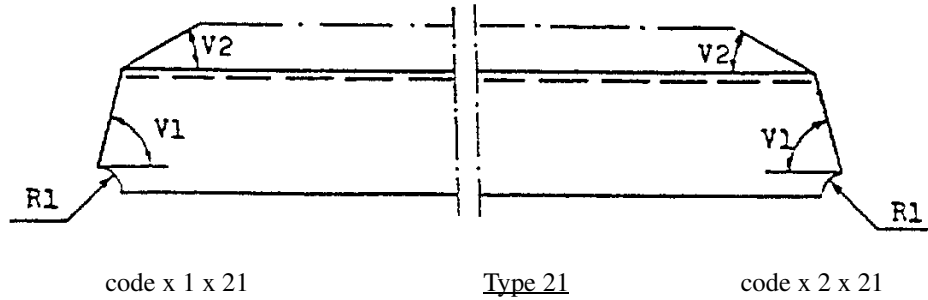
code x 1 x 14

code x 2 x 14

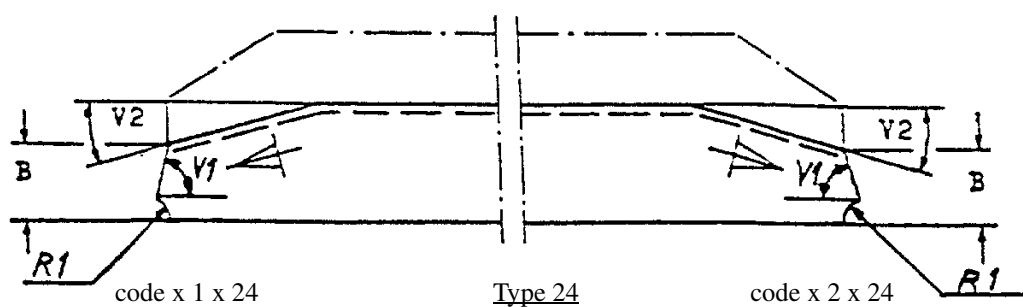
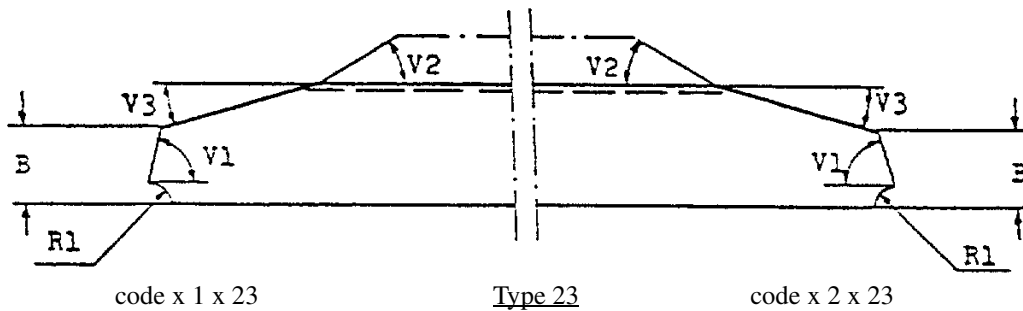
End cutting of profiles

Marking-up list for profiles

L-bars

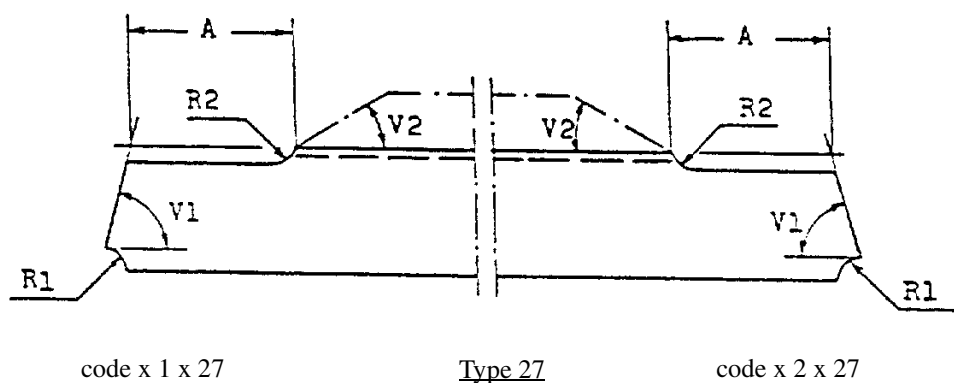
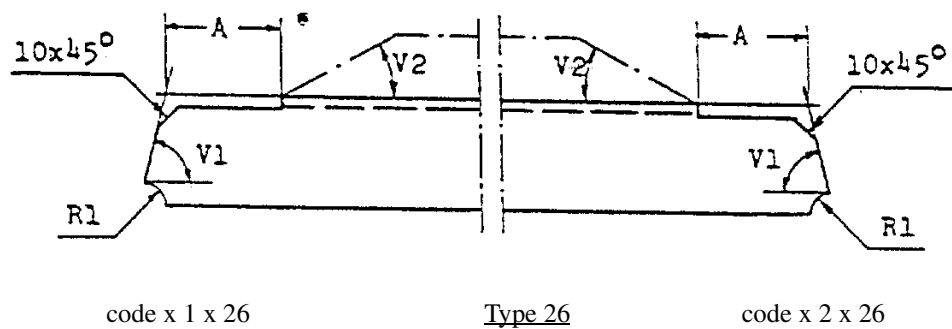
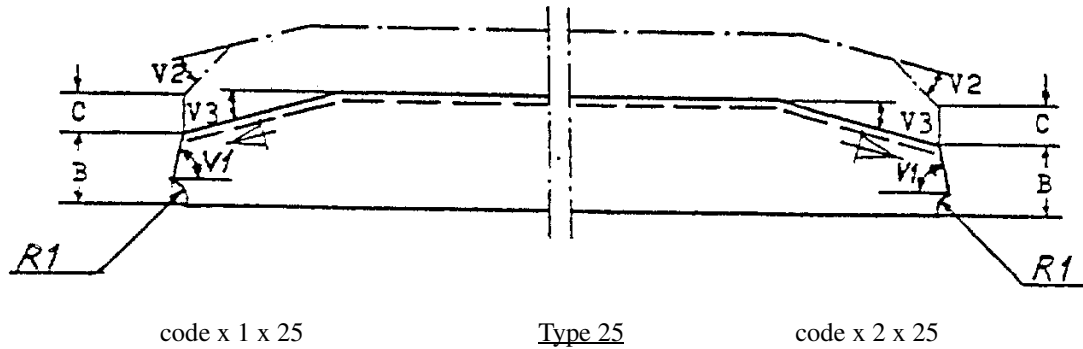


C will be given manually, but not if C = 0



End cutting of profiles

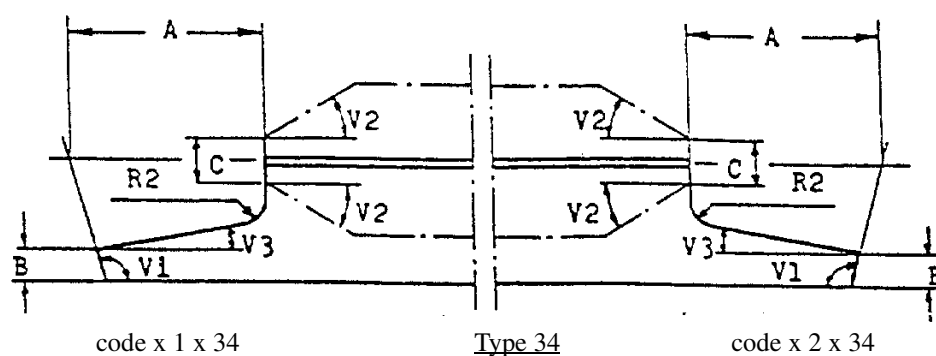
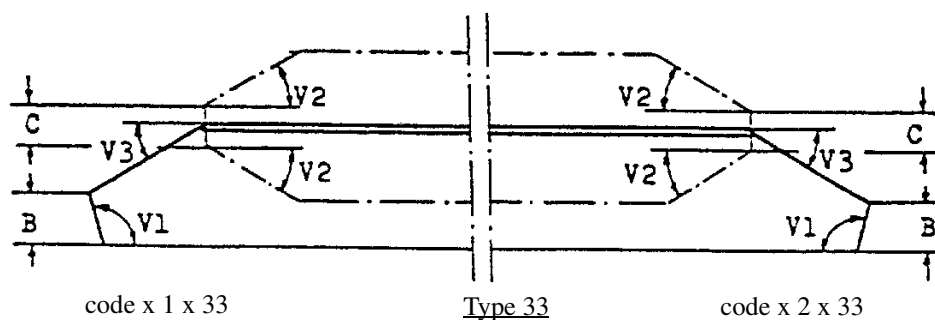
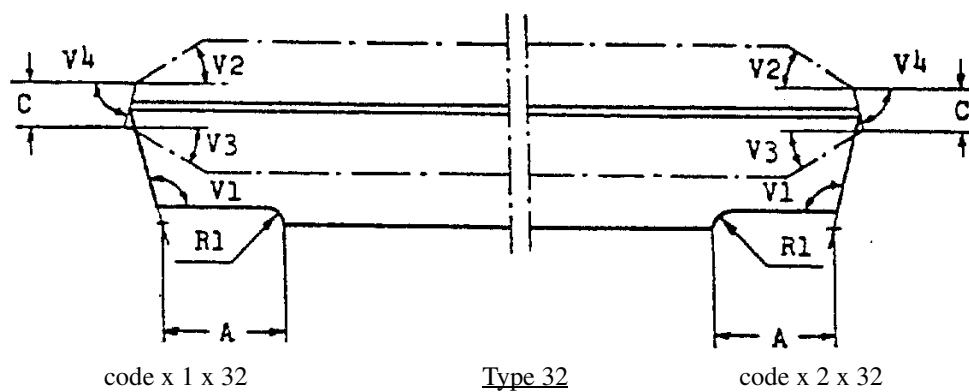
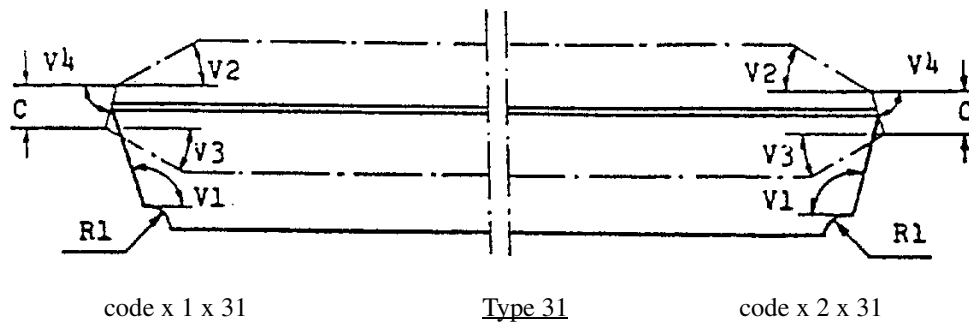
Marking-up list for profiles



End cutting of profiles

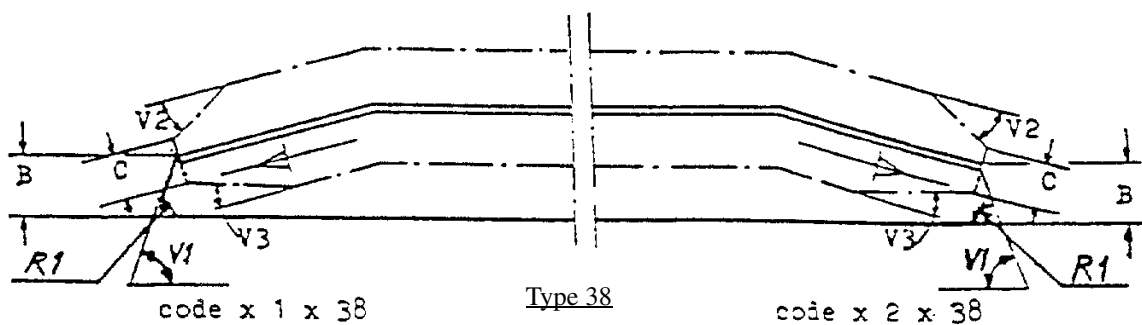
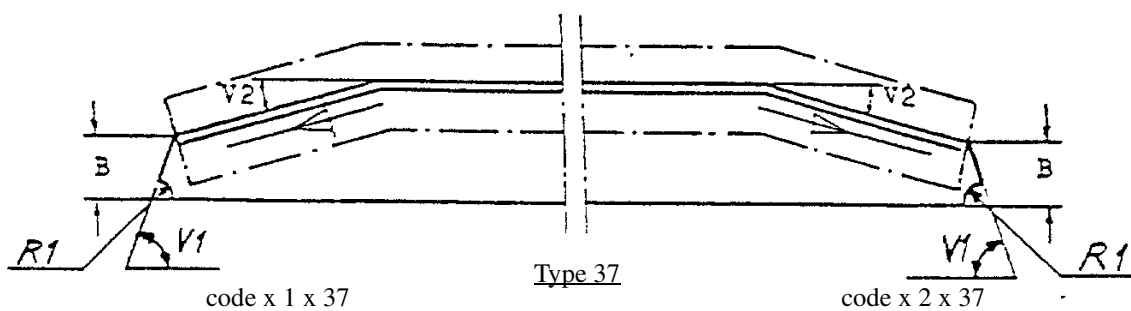
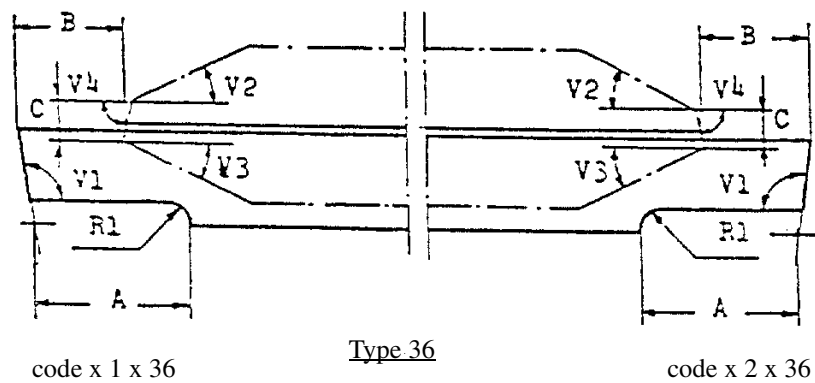
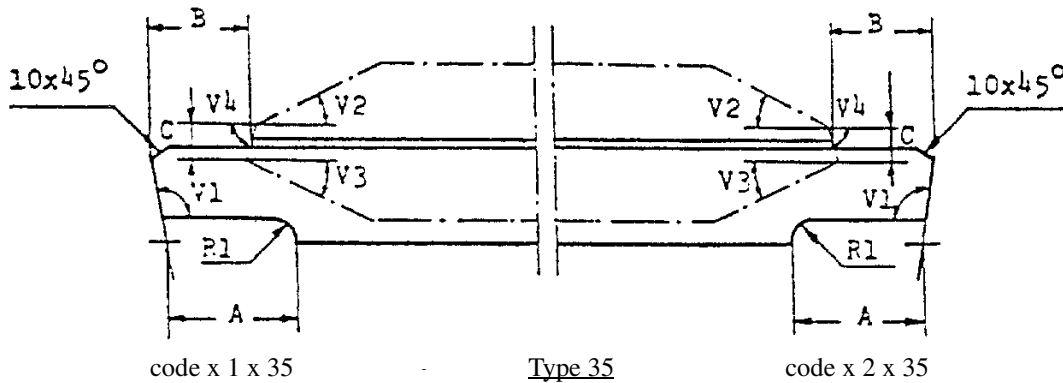
Marking-up list for profiles

T-bars



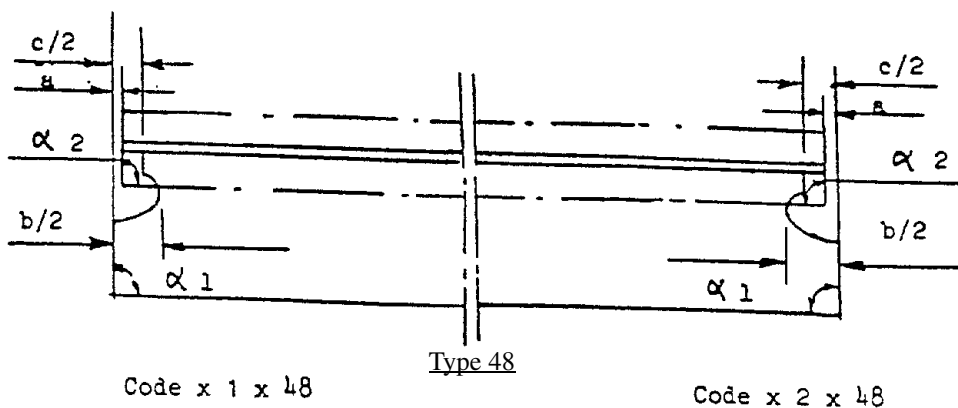
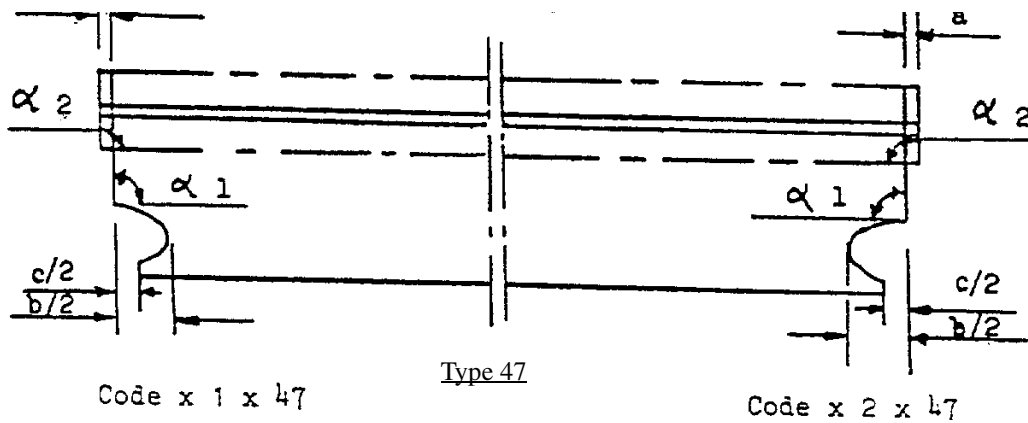
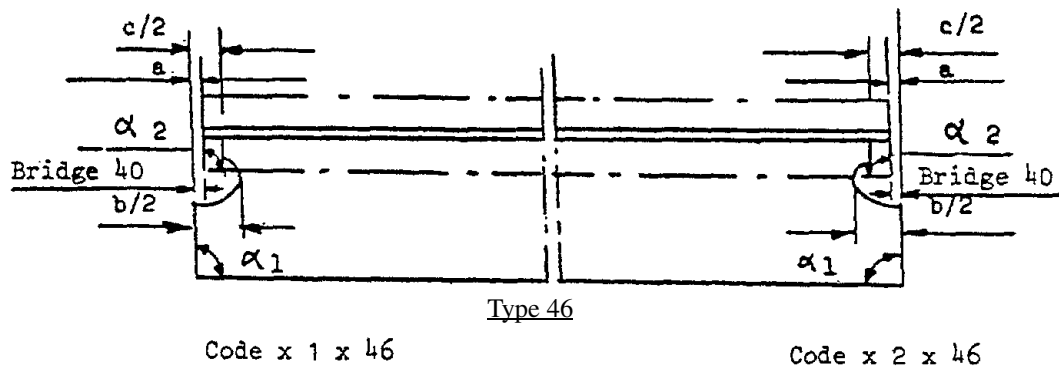
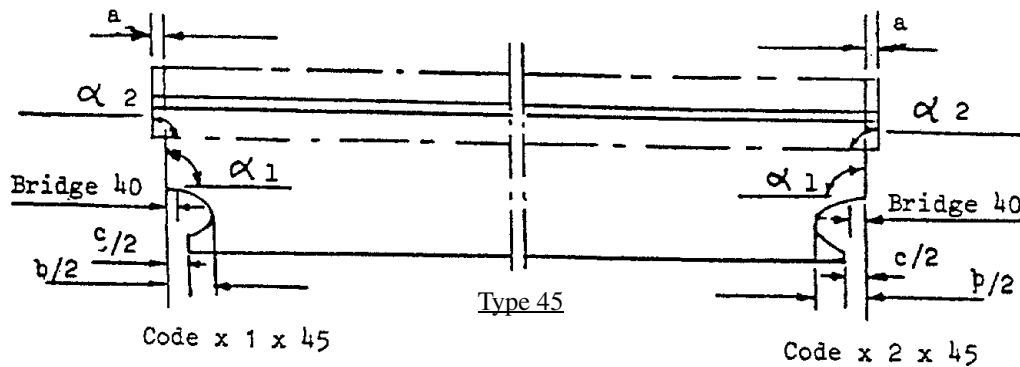
End cutting of profiles

Marking-up list for profiles



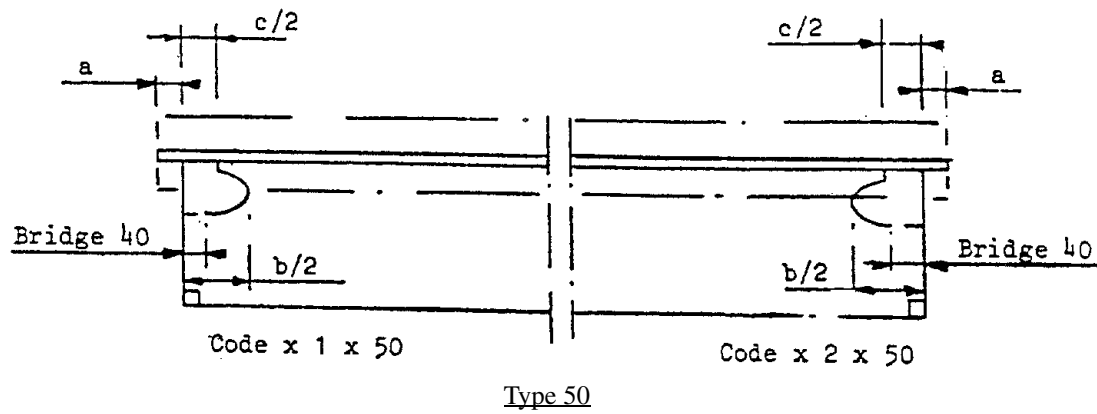
End cutting of profiles

Marking-up list for profiles



End cutting of profiles

Marking-up list for profiles



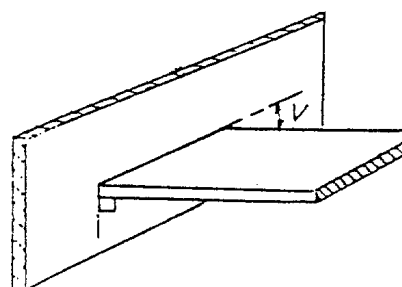
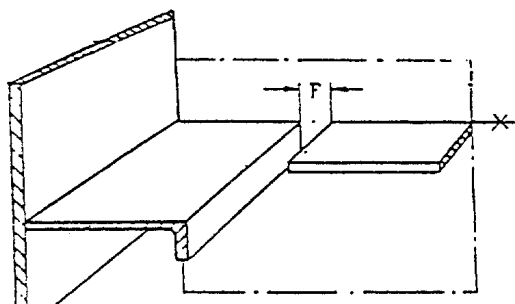
Clearance

Positioning of profile end relative to another profile on another panel

F=0, Type 4
F=20, Type 80
F=25, Type 44
F=30, Type 83
F=40, Type 32
F=50, Type 3
F=60, Type 35

F=75, Type 2
F=85, Type 36
F=100, Type 1
F=125, Type 37
F=150, Type 38
F=175, Type 39

F=200, Type 41
F=225, Type 42
F=250, Type 43



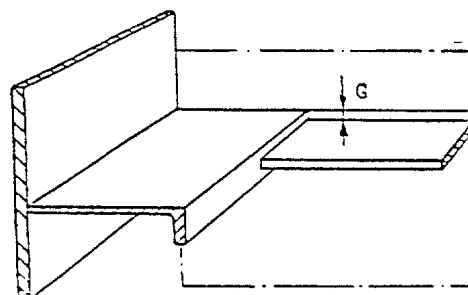
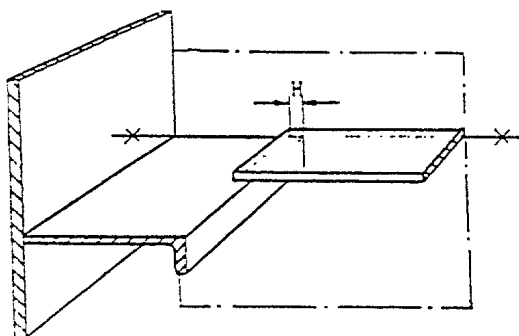
H=30, Type 81
H=40, Type 31
H=50, Type 13
H=70, Type 82
H=100, Type 20
H=150, Type 21

H=200, Type 22
H=250, Type 23
H=300, Type 24
H=350, Type 25

H=400, Type 26
H=450, Type 27
H=500, Type 28
H=550, Type 29
H=600, Type 30
H=65, Type 46
H=75, Type 47

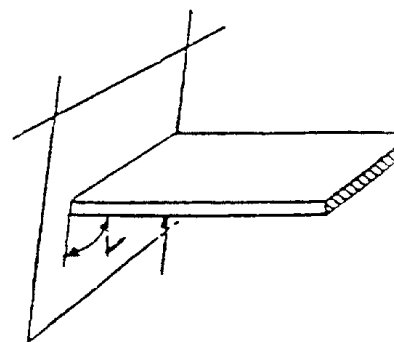
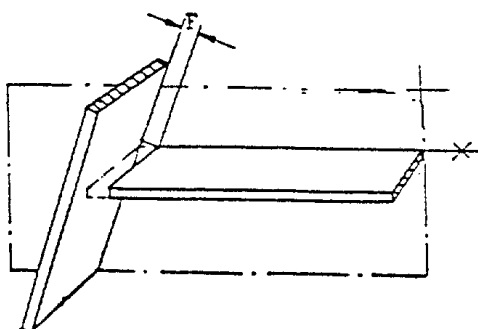
Only for stiffeners, no flangers

G=5, Type 5
G=8, Type 6
G=10, Type 7
G=12, Type 8
G=15, Type 9



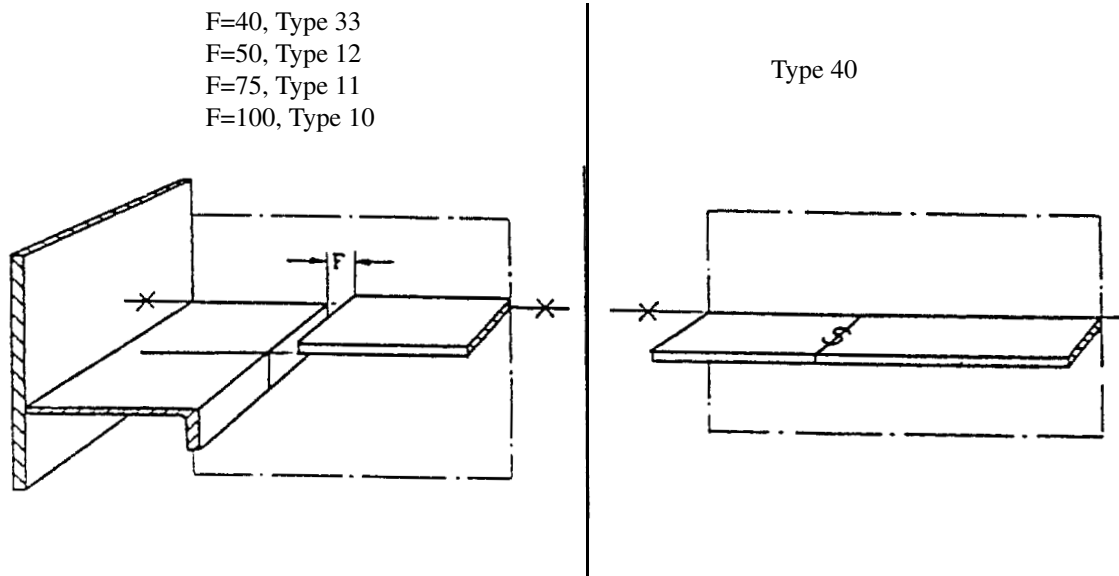
Only for stiffenes on the same panel.

F=0, Type 15
F=10, Type 85
F=15, Type 86
F=20, Type 87
F=25, Type 45
F=30, Type 88
F=40, Type 19
F=50, Type 14
F=75, Type 16
F=85, Type 49
F=100, Type 17
F=125, Type 34
F=150, Type 18



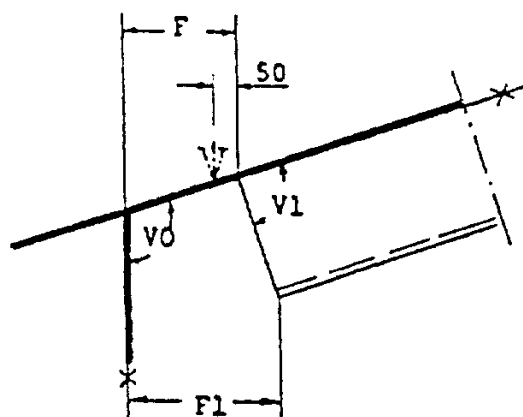
Clearance

Positioning of profile end relative to another profile



Clearance

Positioning of profile end relative to a plate



Code	Clearance	Recommended use
51	F= 600	In the case of extra large clearance, for instance at a seam. If F1 at $V1 = 90^\circ$ is less than 100 make $V1 = V0$
52	55	
53	500	
54	450	
55	400	
56	350	
57	300	For brackets connected with large overlap and profile connections: if $V0 < 60^\circ$ make $V1 = V0$
58	250	
59	200	
61	150	
62	100	For brackets connected with large overlap and profile connections: $60^\circ < V0 < 90^\circ$ make $V1 = V0$
63	75	For brackets connected with large overlap: if $V0 > 90^\circ$ make $V1 = 90^\circ$
64	75	For connection of profiles: if $V0 > 90^\circ$ make $V1 = V0$
65	50	Clearance at normal overlap: Make $V1 = V0$
70	0	When butting (Note: Automatically corrected for thivkness): Make $V1 = V0$



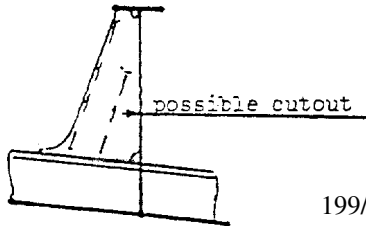
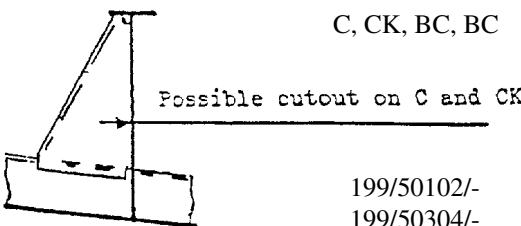
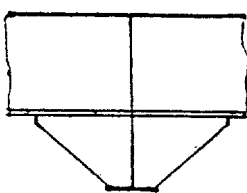
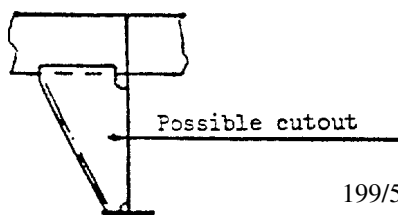
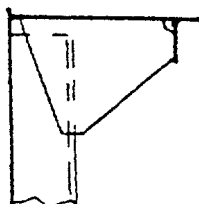
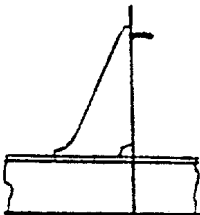
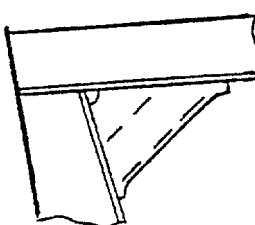
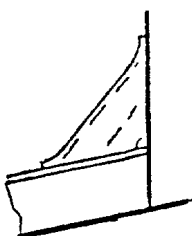
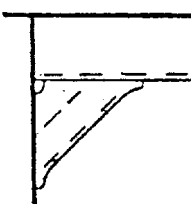
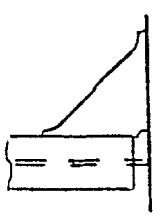
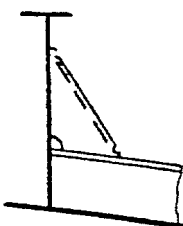
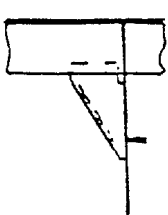
Hull Standards

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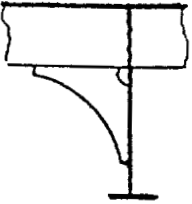
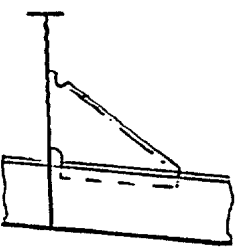
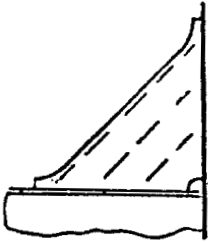
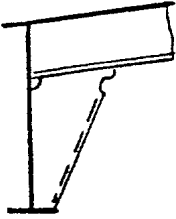
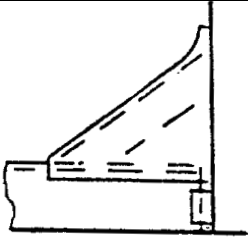
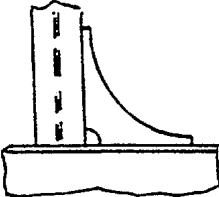
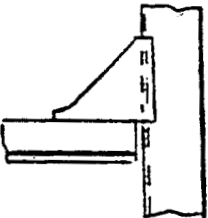
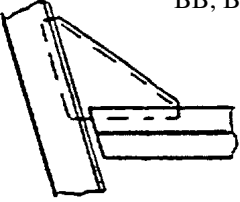
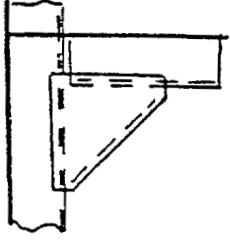
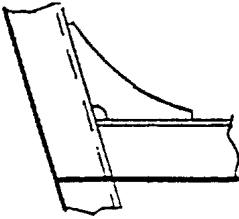
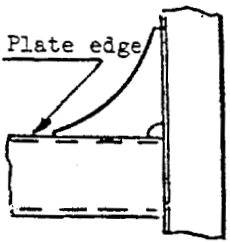
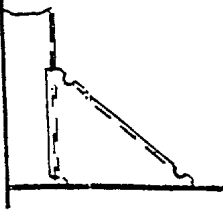
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199/52510/-	Positioning of Clips	2:145

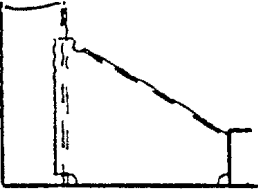
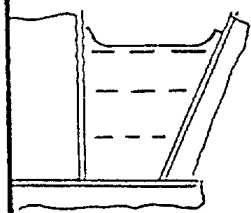
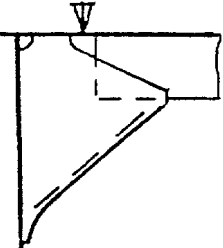
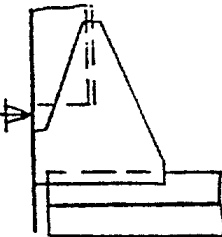
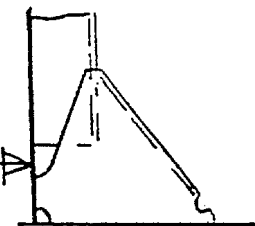
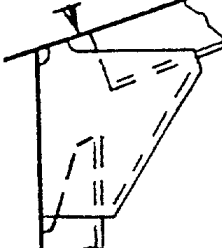
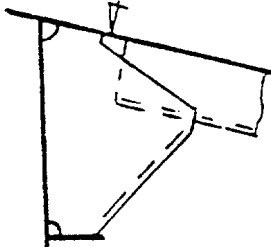
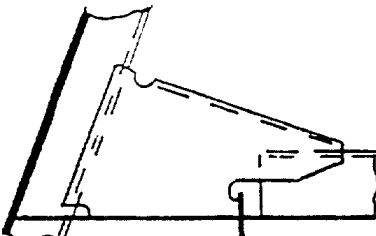
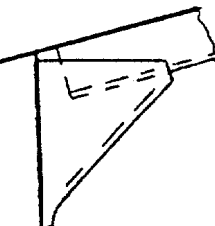
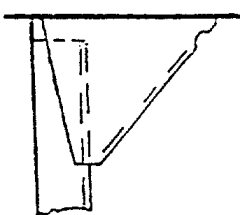
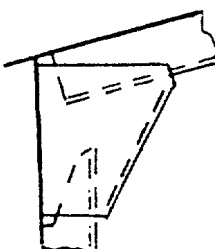
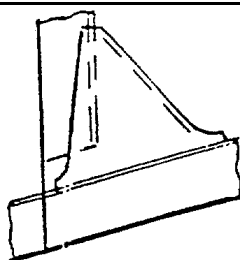
Bracket Connections, Survey

 A possible cutout 199/50101/-	 C, CK, BC, BC Possible cutout on C and CK 199/50102/- 199/50304/-
 BCB 199/50311/-	 CK Possible cutout 199/50102/-
 GC, GCK 199/50412/-	 LS 199/50103/-
 K, KC, KD KBA, KCA, KP 199/502--/-	 B, K 199/50301/-
 B, BS 199/50301/-	 BS 199/50301/- 199/50201/-
 BK 199/50301/-	 BA 199/50302/-

Bracket Connections, Survey

 B, BRS 199/50301/- 199/50301/-	 BAK 199/50302/-
 B, BS, KB 199/50311/- -	 BGK 199/50308/-
 KPA 199/50210/-	 BR 199/50203/-
 BA 199/50302/-	 BB, BBS, BBK, BBKS, BDK 199/50303/- 199/50305/-
 BBS, BBKS 199/50301/-	 BR, BRS 199/50203/-
 B, BR 199/50301/- 199/50203/-	 BEK 199/50306/-

Bracket Connections, Survey

 BKF 199/50307/-	 KPV 199/50204/-
 EA 199/50401/-	 EB 199/50402/-
 EAK 199/50401/-	 EC 199/50403/-
 EC 199/50403/-	 EGK 199/50404/-
 GA 199/50410/-	 GAK 199/50410/-
 GC 199/50411/-	 GB 199/50411/-

Bracket Survey

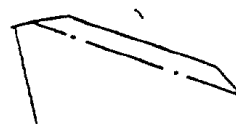
Edge brackets



Type A = 01
199/501 01/-



Type C = 02
199/501 02/-



Type CK = 03
199/501 02/-



Type L = 04
199/501 03/-

Strengthening brackets



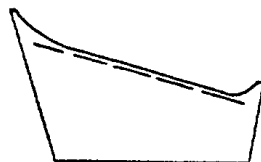
Type K = 10
199/502 01/-



Type KP = 11
199/502 02/-



Type BR=12
199/502 03/-



Type KPV = 13
199/502 04/-



Type KB = 14
199/502 05/-



Type KC = 15
199/502 06/-



Type KD = 16
199/502 07/-



Type KBA = 17
199/502 08/-



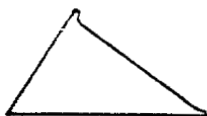
Type KCA = 18
199/502 09/-



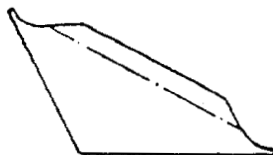
Type KPA = 19
199/502 10/-

Bracket Survey

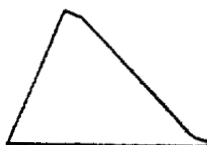
Angle brackets



Type B = 20
199/503 01/-



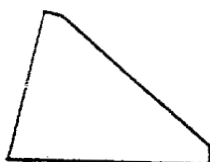
Type BK = 21
199/503 01/-



Type BA = 22
199/503 02/-



Type BAK = 23
199/503 02/-



Type BB = 24
199/503 03/-



Type BBK = 25
199/503 03/-



Type BC = 26
199/503 04/-



Type BCK = 27
199/503 04/-



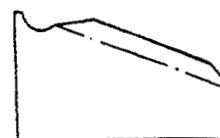
Type BDK = 28
199/503 05/-



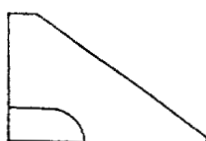
Type BEK = 29
199/503 06/-



Type BFH = 30
199/503 07/-



Type BCK = 31
199/503 08/-



Type BAA = 32
199/50309/-



Type BCB = 33
199/50311/-



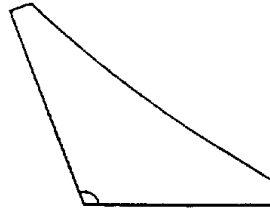
Type BAB = 34
199/50316/-

Bracket Survey

Angle brackets



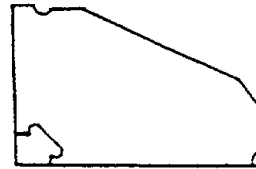
Type BBA = 35
199/50312/-



Type EBR = 36
199/50313/-



Type BCA = 37
199/50314/-



Type BHK = 38
199/50315/-

Bracket Survey

Connection brackets



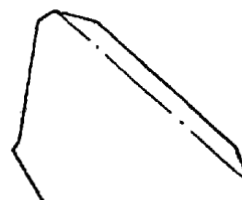
Type EA = 40
199/504 01/-



Type EAK = 41
199/504 02/-



Type EB = 42
199/504 02/-



Type EBK = 43
199/504 02/-



Type EC = 44
199/504 03/-



Type ECK = 45
199/504 03/-



Type ECK = 46
199/504 05/-



Type GA = 50
199/504 10/-



Type GAK = 51
199/504 10/-



Type GB = 52
199/504 11/-



Type GC = 53
199/504 12/-



Type GCK = 54
199/504 12/-

Marking of Brackets and Lugs *)

1. Type standard brackets

In this group are brackets which follow the bracket description but are not included in the recommended selection of brackets. They deviate from "pure standard brackets" by for instance:

- bracket angle
- length of leg
- size of stiffener
- extra holes or cut-outs

This group can be divided into two subgroups, namely:

- A. Brackets which occur in relatively large numbers.
The position numbering of such brackets is not necessary, since these must be completely interchangeable with each other.
It is, however, necessary to have a specific marking of these brackets, consisting of:

Bracket designation thickness - quality.

The bracket designation used, when generating the bracket consists of:

Bracket type - size - variant, where:

The bracket type will be seen from the bracket survey.

The size generally consists of the leg lengths of the bracket in cm.

The variant consists of a combination of max three digits having the following meaning:

*) The rules in this document are in some details special to the Kockum Shipyard, but they are mainly used in SBH.

Marking of Brackets and Lugs

First digit:
Geometric alternatives, e.g.
corner cut-out as shown on the
type standard bracket sketch.

1 = no corner cut out
2
3
4
5
6
7
8
9

Second digit:
a) Positioning of flanges and
flat bars.

0
1 = this side
2 = other side
3 = flanges both sides,
others this side
4 = flange both sides, others
other side
5

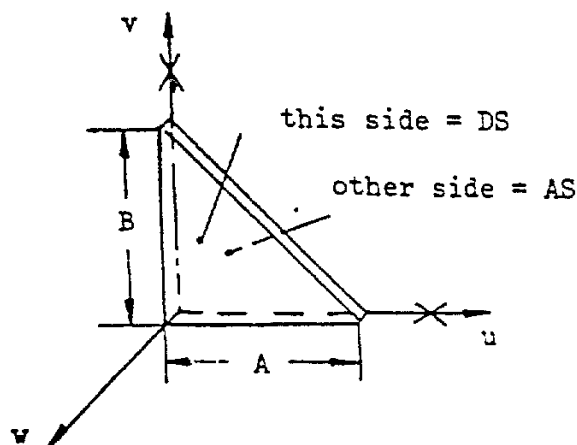
b) Running number to be
combined

6
7
8
9

with

Third digit:
The third digit gives running
number for finer subdivisions
of variants.

0 = pure standard
1
2
3
4
5
6
7
8
9 = individual bracket



Marking of Brackets and Lugs

Example: K 210/135 - 122

K 210/210 - 122

This bracket designation must, in order to be quite clear, have the thickness in mm including one decimal and also the quality coded in accordance with VIS 441 with the VIS code's 2nd to 4th digits included. The VIS code's 1st digit, which indicates Classification Society, is not included.

The 2nd digit indicates basic quality as below, 3rd and 4th digit means yield limit in kp/cm² (00 = no yield limit required - "mild steel"):

Code for basic quality:

0 = A

1 = B

2 = C

3 = D

4 = E

5 = CS (ABS), W (NV)

6 = R

The yield limit for "mild steel" (00) may be omitted.

Example:

16.0 - 336 quality D yield limit 36 kp/cm²

18.5 - 3 quality D "mild steel"

16.0 - 036 quality A yield limit 36 kp/cm²

18.5 A "mild steel"

The complete marking will then be:

K 210/135 -122 - 16.0 - 366

in its longest form.

B. Brackets which occur in small numbers.

These brackets are to be marked with

Location No. - Type No. - Position No.

similar to other items in the vessel, but in addition the position No. must start with a B.

Example:

461 - 71 - 3601 in its longest form.

Marking of Brackets and Lugs

2. Standard brackets

In this group are brackets which follow the bracket description and are included in the recommended bracket selection. These brackets have an S in the type code.

Position numbering of such brackets is not necessary, since these are to be completely interchangeable with each other.

Marking of Brackets and Lugs

On the other hand it is necessary to have a definite marking of these brackets consisting of:

Bracket designation - thickness - quality

The differences between standard and type standard brackets are:

Bracket type and variant number

The bracket type has previously been mentioned and should contain an S.

The variant number consists of a combination of a max of three digits with a meaning similar to the one for type standard brackets except the second digit in group b.

Example: KS 210/135 - 120 - 16.0 - 336.

3. OTHER BRACKETS

In this group are brackets which do not belong to any of the previous groups. These brackets are to be marked with

Location No. - type No. - position No.

similar to group 1B above and other items in the vessel.

On the drawing all measurements must be given.

Note. Brackets which are fabricated in Helsingborg are to have an E added to its marking.

In order to make it easier to sort the brackets, the marking must be placed on the outside of flat bar A or flange.

4. TYPE STANDARD LUGS

In this group are lugs which are based on the rectangular shape, but which deviate by e.g.

bracket angle
length and width
cut-outs

Marking of Brackets and Lugs

The marking of these lugs consists of:

Lug designation - thickness - quality

The lug designation consist of:

Type of lug - size - variant where the type of lug is KLIPS, the size consists of the width and length in cm and the variant consists of a combination of a max. of two digits.

Example: KLIPS 22/35 - 21

This lug designation must in order to be quite clear be completed with the thickness in mm including one decimal and also quality coded in the same way as is mentioned under "type standard brackets" above.

Example:

16.0 - 036 quality A yield limit 36 kp/cm²
18.5 quality "mild steel"

The complete marking will then be:

KLIPS 22/35 - 21 - 16.0 - 036
in its longest form.

5. STANDARD LUGS

This group includes lugs which belong to the recommended selection of lugs.

The marking of these lugs consists of:

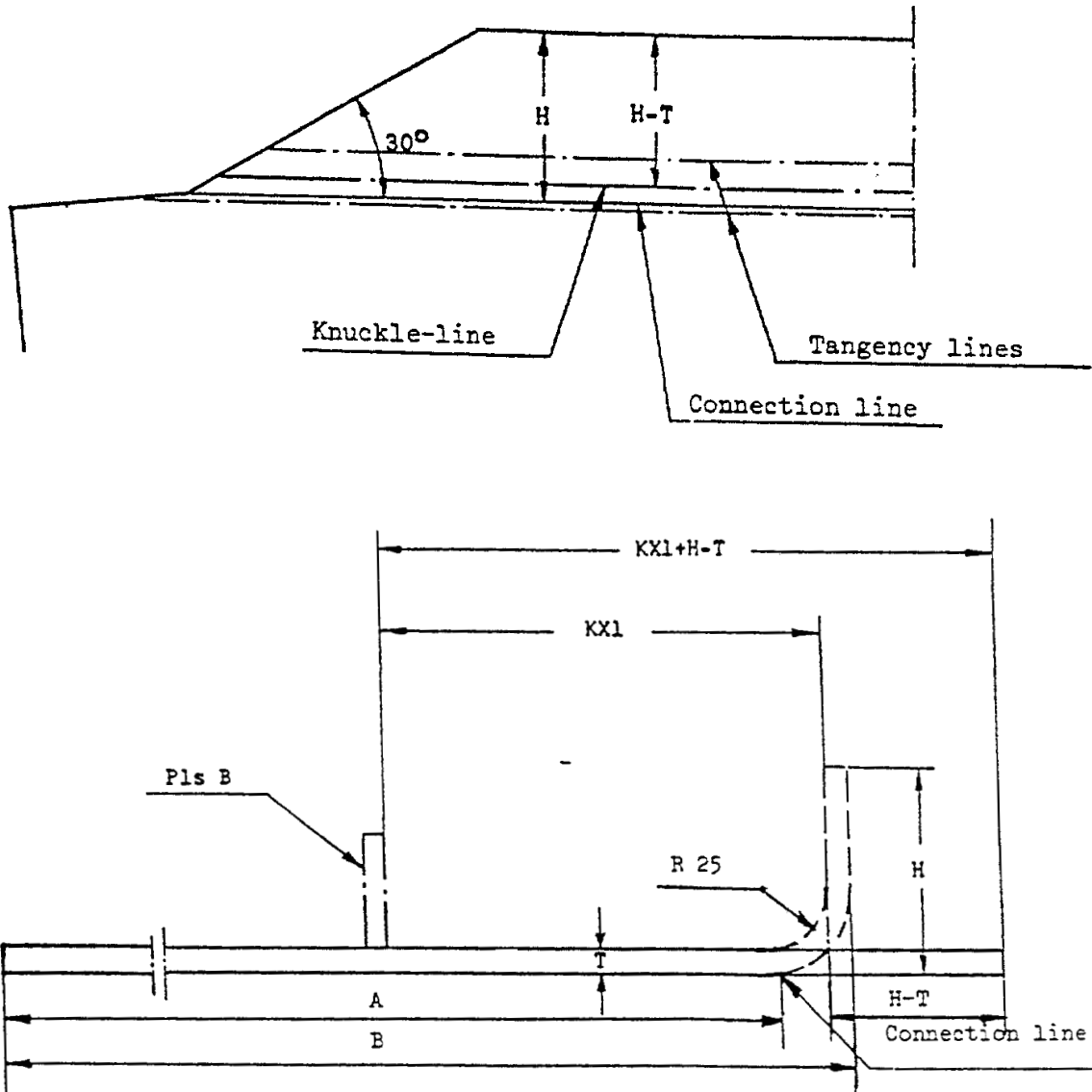
Lug designation - thickness - quality

The difference between standard and type standard lugs is the variant.

The type of lug has already been mentioned above.
The variant is omitted.

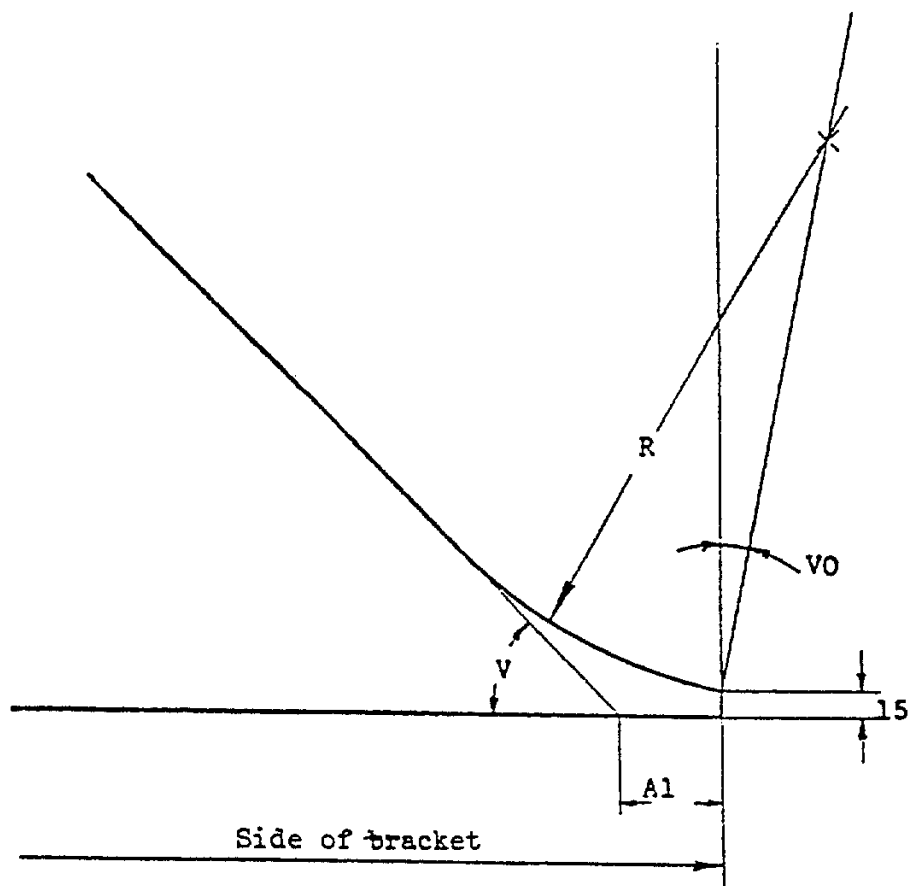
Example: KLIPS 22/35 - 16.0 - 036.

Flanging of Brackets and Positioning of Flat Bars B



If the breadth of a bracket is given as input to the structure generation, the A-measure shall be given. $A=B-R$, which can be rounded off to the nearest multiple of 5. The connection line denotes the outer geometry before "flanging"

Principles for the Shape of Bracket Toes



For all brackets toes it applies that the radius R is calculated according to the formula

$$R = \frac{A1 \times \tan V + 15}{\tan \frac{V - V0}{2} (\cos V0 \times \tan - \sin V0)}$$

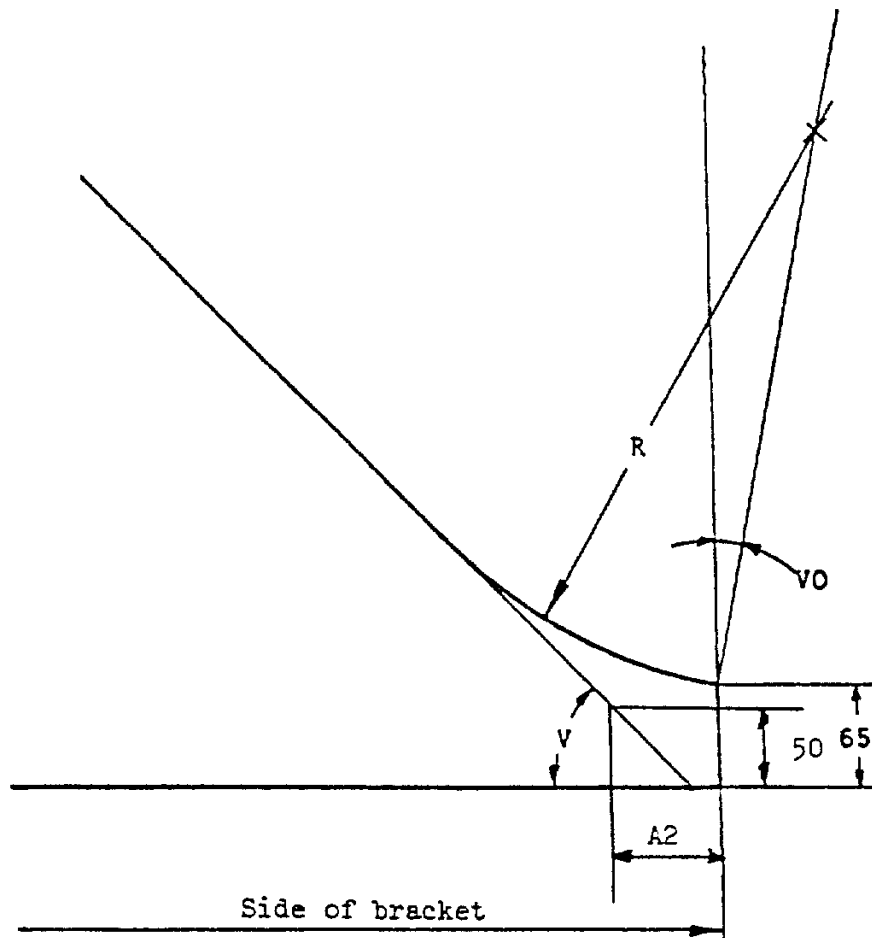
For toe 1 (T1) it applies that $A1 = 0.175 \times \text{side of bracket}$
 $V0 = 15^\circ$

For toe 2 (T2) it applies that $A1 = 100$
 $V0 = 10^\circ$

For toe 3 (T3) it applies that $A1 = 300$
 $V0 = 10^\circ$

For toe 4 (T4) it applies that $A1 = 35$
 $V0 = 0^\circ$

Principles for the Shaping of Bracket Toes



For all brackets toes it applies that the radius R is calculated according to the formula

$$R = \frac{A_2 \times \tan V + 15}{\tan \frac{V - V_0}{2} (\cos V_0 \times \tan - \sin V_0)}$$

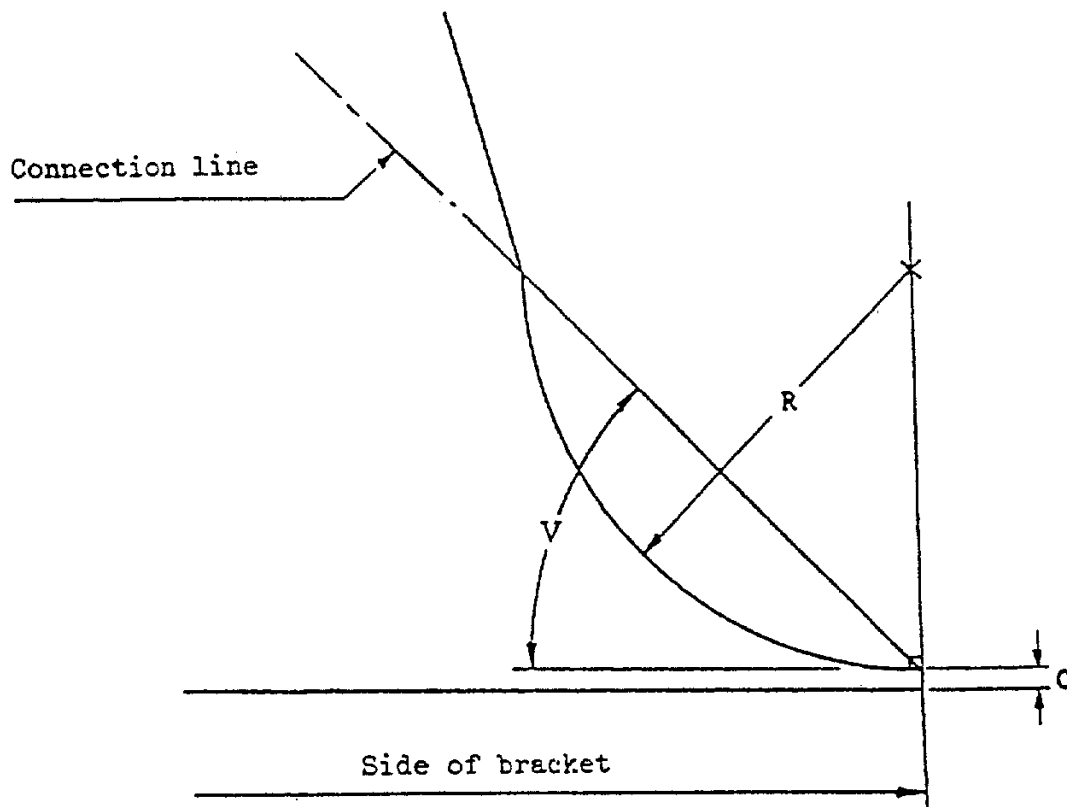
For toe 31 (T31) it applies that $A_1 = 0.175 \times \text{side of bracket}$
 $V_0 = 15^\circ$

For toe 32 (T32) it applies that $A_1 = 100$
 $V_0 = 10^\circ$

For toe 33 (T33) it applies that $A_1 = 300$
 $V_0 = 10^\circ$

For toe 34 (T34) it applies that $A_1 = 35$
 $V_0 = 0^\circ$

Principles for the Shaping of Bracket Toes and Brackets with Flanges



For all brackets toes it applies that the radius R is calculated according to the formula

$$R = \frac{R1}{\sqrt{2} \sin V}$$

For toe 11 (T11) it applies that R1 = 50, C = 15

For toe 12 (T12) it applies that R1 = 75, C = 15

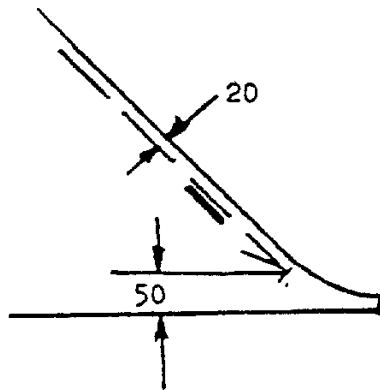
For toe 13 (T13) it applies that R1 = 100, C = 15

For toe 21 (T21) it applies that R1 = 50, C = 65

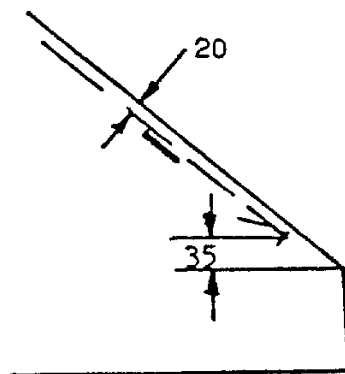
For toe 22 (T22) it applies that R1 = 75, C = 65

For toe 23 (T23) it applies that R1 = 100, C = 65

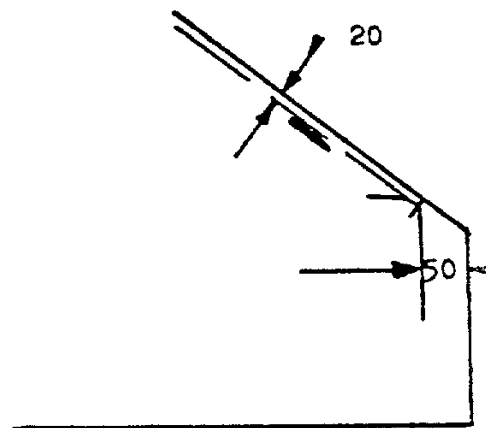
Principles Position of Stiffeners on some Types of Bracket



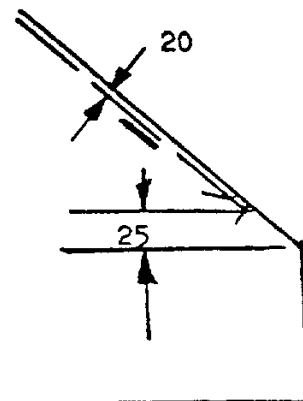
Type B, BA, EA, GA, GB



Type BB

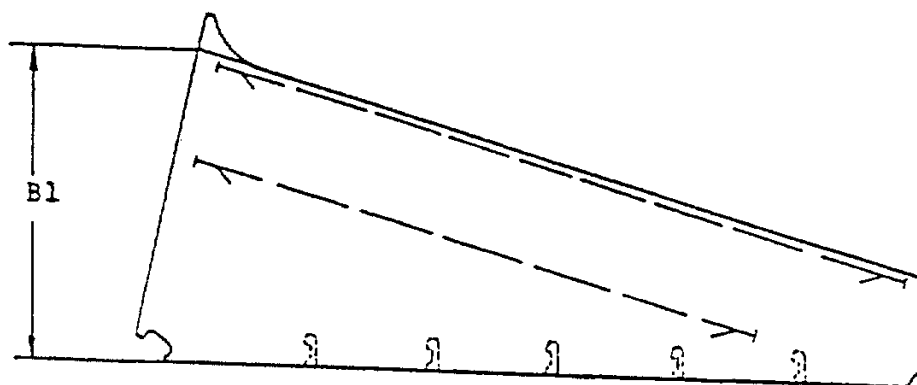
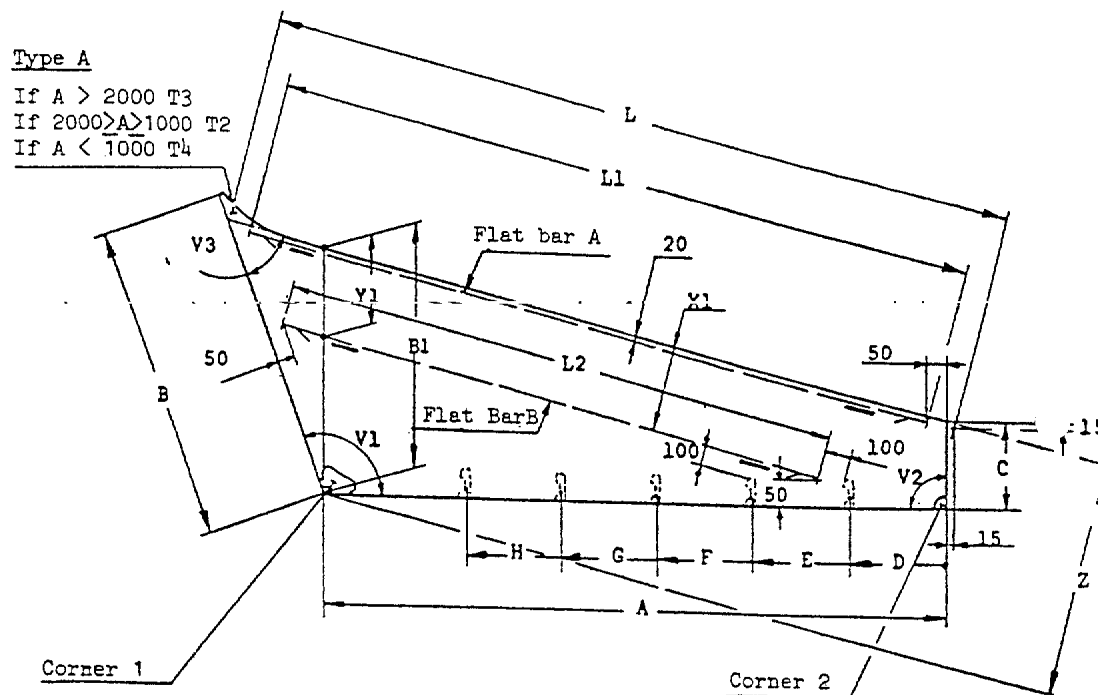


Type BC, GC



Type EB, EC

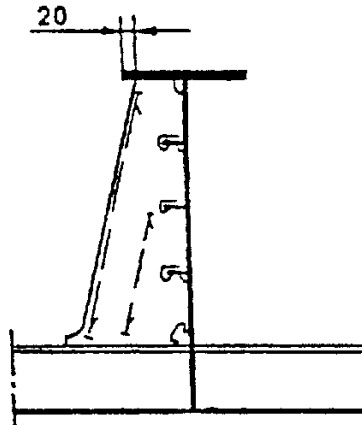
Bracket Type A



Cutout	Alt 1	Alt 2	Alt 3	Alt 4	Alt 5	Alt 6	Alt 7	Alt 8	Alt 9
Corner 1	No cutout	No cutout	No cutout	VOB100	VOB100	VOB100	R75	R75	KS10
Corner 2	KS10	R50	R75	KS10	R50	R75	KS10	R50	KS10
		To be avoided			To be avoided				

Bracket Type A

Example for Fitting



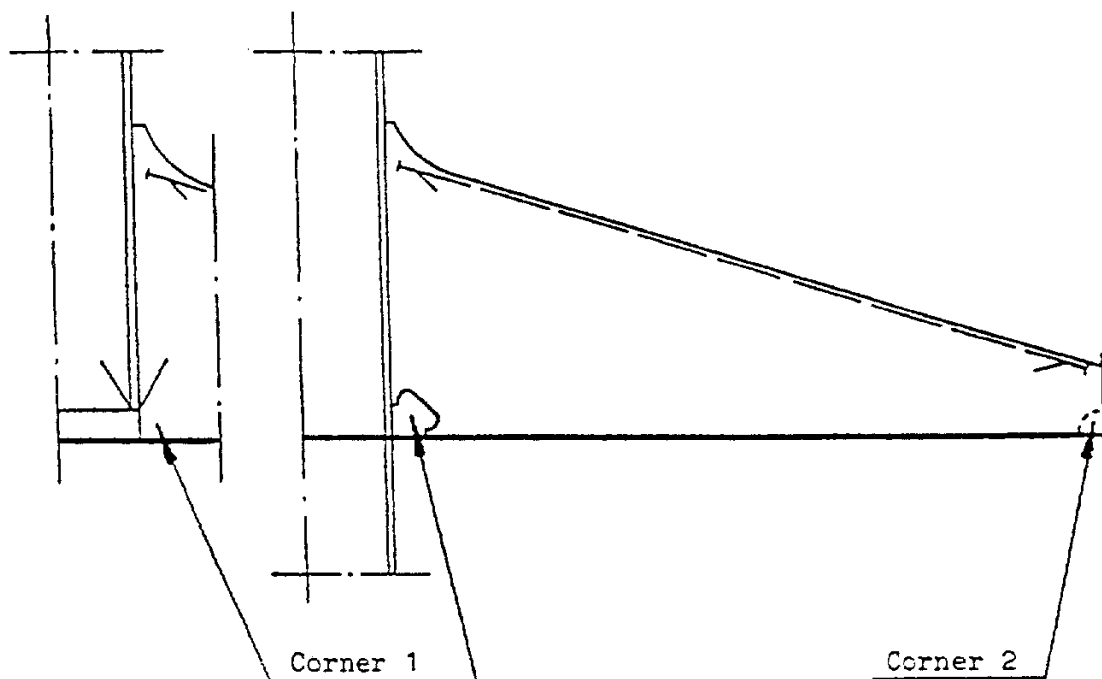
Bracket Type A

Description

- 1 A-dimension determined by depth of web - profile height.
- 2 B1-dimension = $0.4 \times A$, if nothing else is given, min is, however, 300 mm.
- 3 C-dimension is variable.
- 4 The bracket may occur both with and without cut-out for stiffener.
- 5 If $L < 800$ mm, no flat bar is required.
- 6 If $L > 800$ mm and $Z \leq 1100$, one flat bar is required.
The area of this flat bar to be $0.1 \times L \text{ cm}^2$.
- 7 If $L > 800$ mm and $1100 < Z \leq 1850$, then two flat bar are required.
The area of flat bar A to be $0.6 \times 0.1 \times L \text{ cm}^2$
The area of flat bar B to be $0.4 \times 0.1 \times L \text{ cm}^2$
For positioning of flat bar B set $X1 = 0.4 \times Z$.
- 8 If $L > 800$ mm and $1850 < Z \leq 2700$, then three flat bars are required.
The area of flat bar A to be $0.6 \times 0.1 \times L \text{ cm}^2$
The area of flat bar B to be $0.4 \times 0.1 \times L \text{ cm}^2$
The area of flat bar C to be $0.1 \times L \text{ cm}^2$
For positioning of flat bars B and C set
 $X1 = 0.28 \times Z$ and $X2 = 0.59 \times Z$
- 9 The above areas are to be rounded off to the nearest recommended standard dimension, lowest flat bar value is, however, 120 x 12.
The relationship width - thickness should be a max of 18/1.
- 10 All welds to be  4 when nothing else is given on the class drawing.
- 11 From the class drawing is also obtained:
Plate thickness, quality of material erection welding.

Bracket Type A

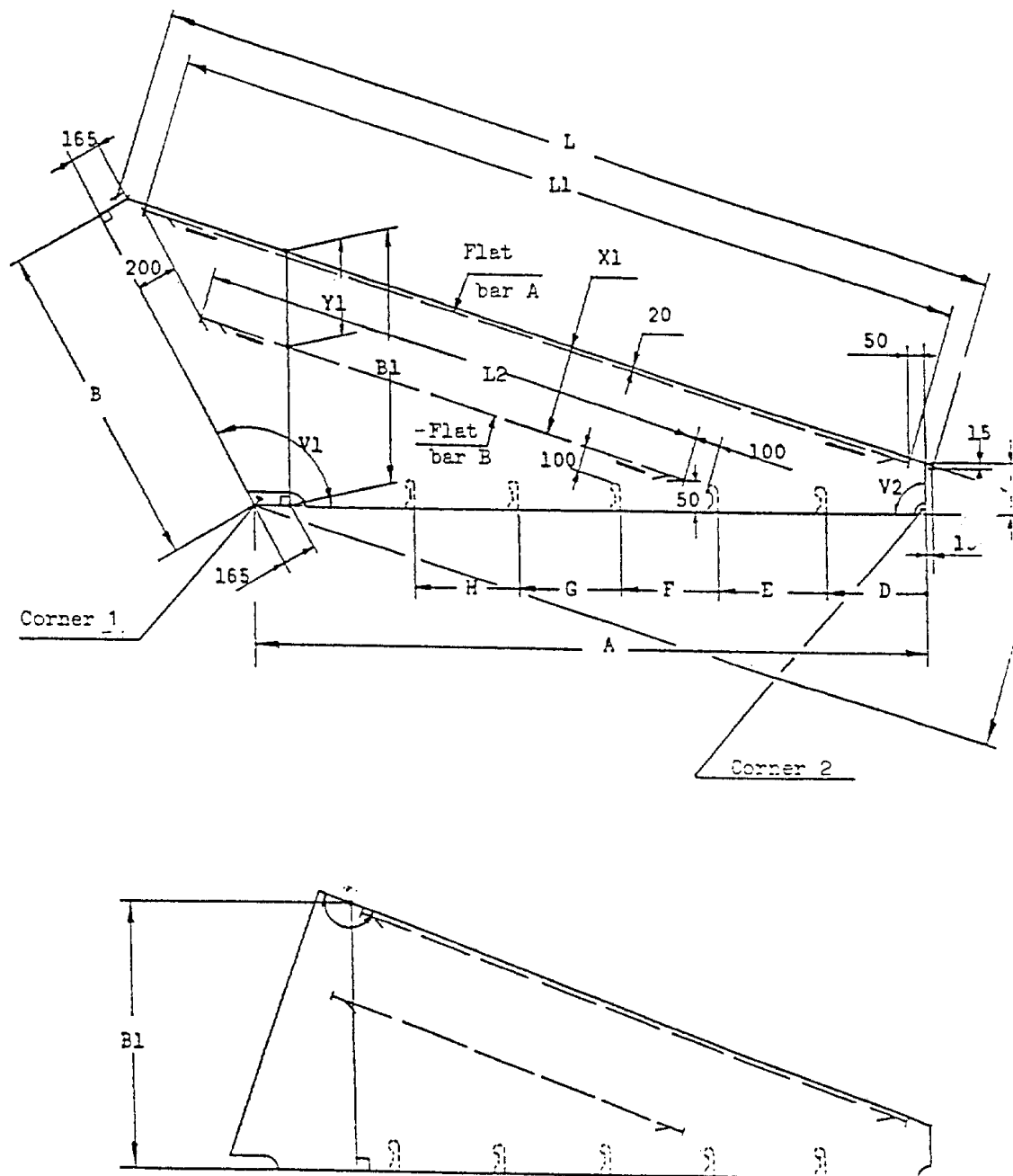
Connections for Type A (01)



Connection	Cut-out		Variant	End out	Clearance
	Corner 1	Corner 2			
011	No cut-out	KS10	10-29	3102	62
012	No cut-out	R50	10-29	3102	62
013	No cut-out	R75	10-29	3102	62
014	VOB 100	KS10	10-29	3100	70
015	VOB 100	R50	10-29	3100	70
016	VOB 100	R75	10-29	3100	70
017	R75	KS10	10-29	3100	70
018	R75	R50	10-29	3100	70
019	KS10	KS10	10-29	3100	70

Bracket Type C, CK

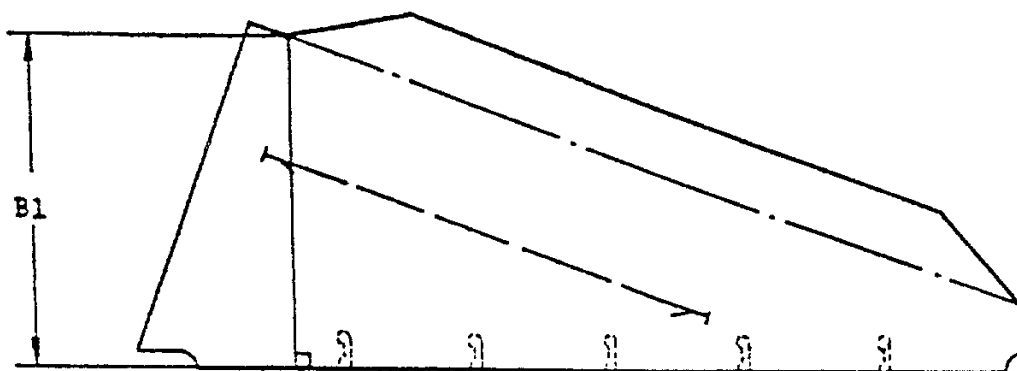
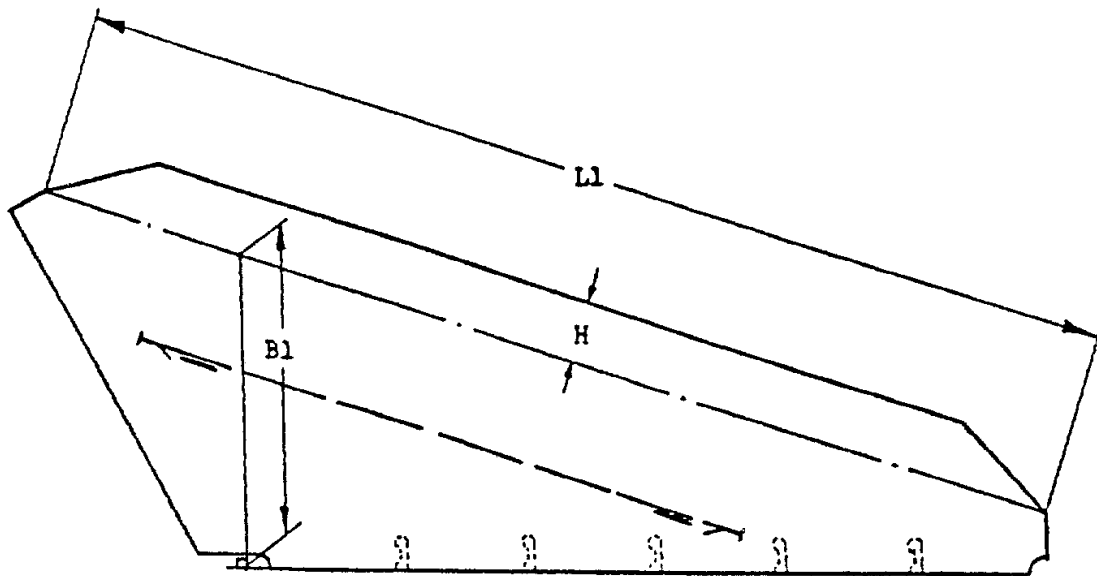
Type C



Cut-out	Alt 1	Alt 2	Alt 3	Alt 4	Alt 5	Alt 6
Corner 1	No cut-out	No cut-out	No cut-out	VUA165	VUA165	VUA165
Corner 2	KS10	R50	R75	KS10	R50	R75
		To be avoided			To be avoided	

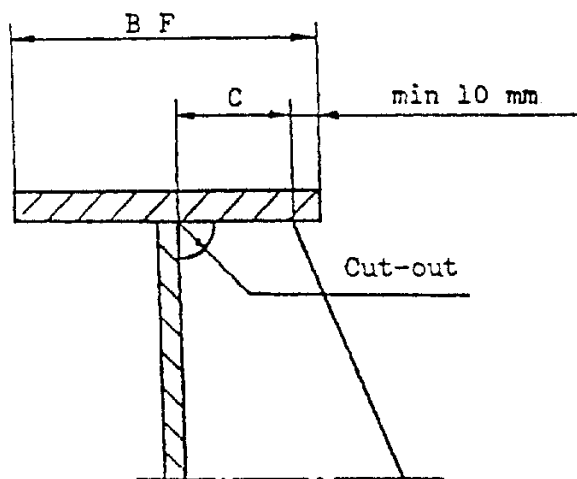
Bracket Type C, CK

Type CK



For dimensions not shown see page 1.

Bracket Type C, CK

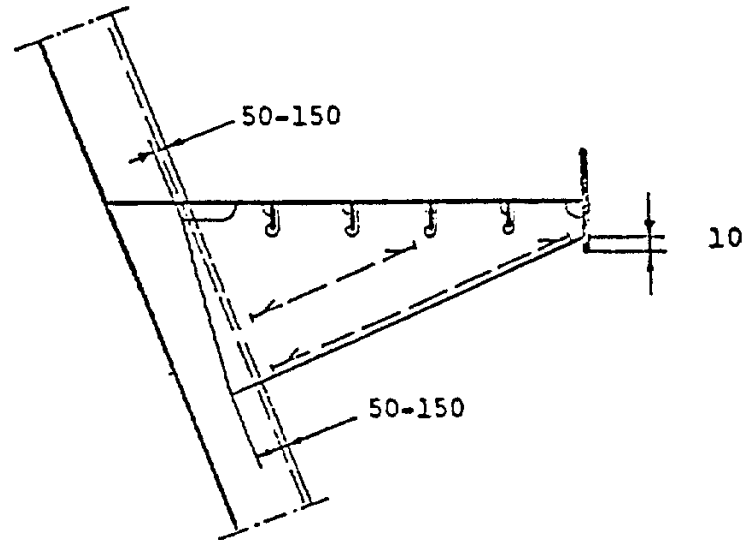


Example

B F	C	Recommended Cut-out
200	80	KS10
250	100	KS10
300	100	KS10
350	150	KS10
400	150	KS10
450	200	R75
500	200	R75
550	200	R75
600	200	R75
650	300	R75
700	300	R75

B F	C	Recommended Cut-out
750	300	R75
800	300	R75
850	400	R75
900	400	R75
950	400	R75
1000	400	R75
1050	500	R75
1100	500	R75
1150	500	R75
1200	500	R75

Bracket Type C, CK



Bracket Type C, CK

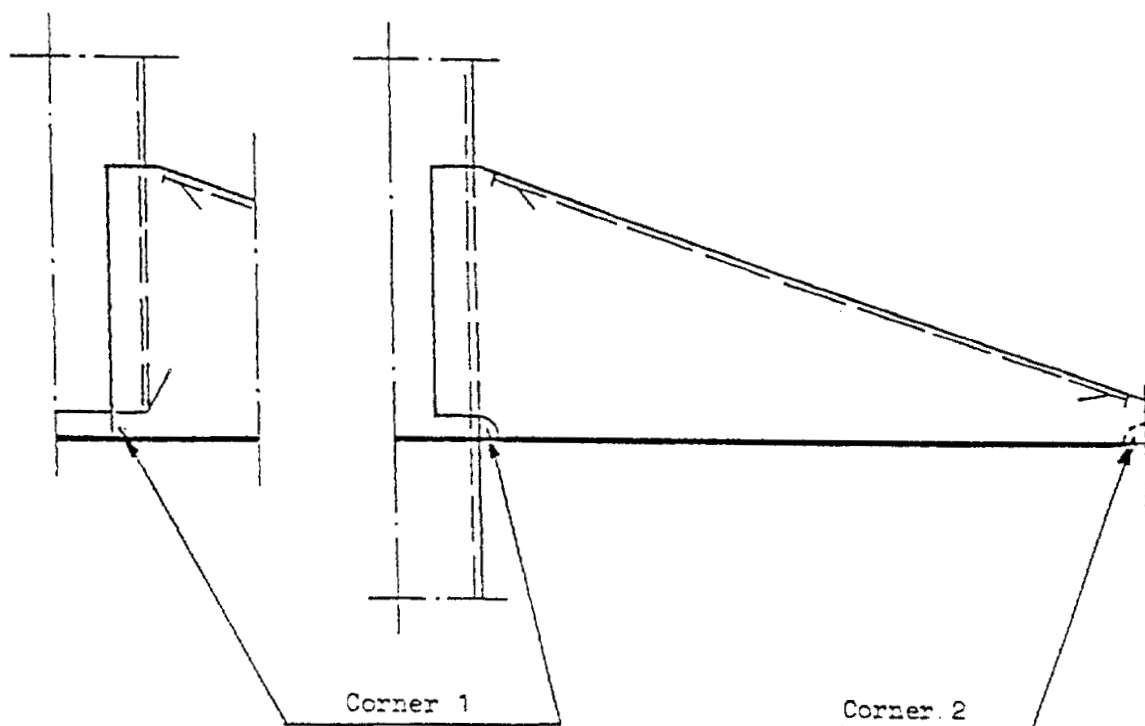
Description

- 1 A-dimension determined by depth of web - profile height + 50 mm and rounded up to the nearest 100 mm if nothing else is given.
- 2 B1-dimension = $0.4 \times A$, if nothing else is given, min is, however, 300 mm.
- 3 C-dimension is variable in the program but values according to the table is recommended.
- 4 The bracket may occur both with and without cut-out for stiffener.
- 5 If $L < 800$ mm, no flat bar is required.
- 6 If $L > 800$ mm and $Z \leq 1300$, one flat bar is required.
The area of this flat bar to be $0.1 \times L \text{ cm}^2$.
- 7 If $L > 800$ mm and $1300 < Z \leq 2180$, then two flat bar are required.
The area of flat bar A to be $0.1 \times L \text{ cm}^2$
The area of flat bar B to be $0.1 \times L2 \text{ cm}^2$
For positioning of flat bar B set $X1 = 0.4 \times Z$.
- 8 If $L > 800$ mm and $2180 < Z \leq 3200$, then three flat bars are required.
The area of flat bar A to be $0.1 \times L \text{ cm}^2$
The area of flat bar B to be $0.1 \times L2 \text{ cm}^2$
The area of flat bar C to be $0.1 \times L3 \text{ cm}^2$
For positioning of flat bars B and C set
 $X1 = 0.28 \times Z$ and $X2 = 0.59 \times Z$
- 9 The above areas are to be rounded off to the nearest recommended standard dimension, lowest flat bar value is, however, 120 x 12.
The relationship width - thickness should be a max of 18/1.
- 10 When flanging then corresponding flange areas are used, at $t = 10$ mm.
If $Z \leq 1300$ mm then the widths of the flanges will be as follows:

FL 100 if $80 \text{ cm}^2 < \text{flange area} \leq 130 \text{ cm}^2$
FL 160 if $130 \text{ cm}^2 < \text{flange area} \leq 190 \text{ cm}^2$
FL 220 if $190 \text{ cm}^2 < \text{flange area} \leq 235 \text{ cm}^2$
FL 250 if $235 \text{ cm}^2 < \text{flange area} \leq 400 \text{ cm}^2$
- 11 All welds to be  4 when nothing else is given on the class drawing.
- 12 From the class drawing is also obtained:
Plate thickness, quality of material erection welding.

Bracket Type C, CK

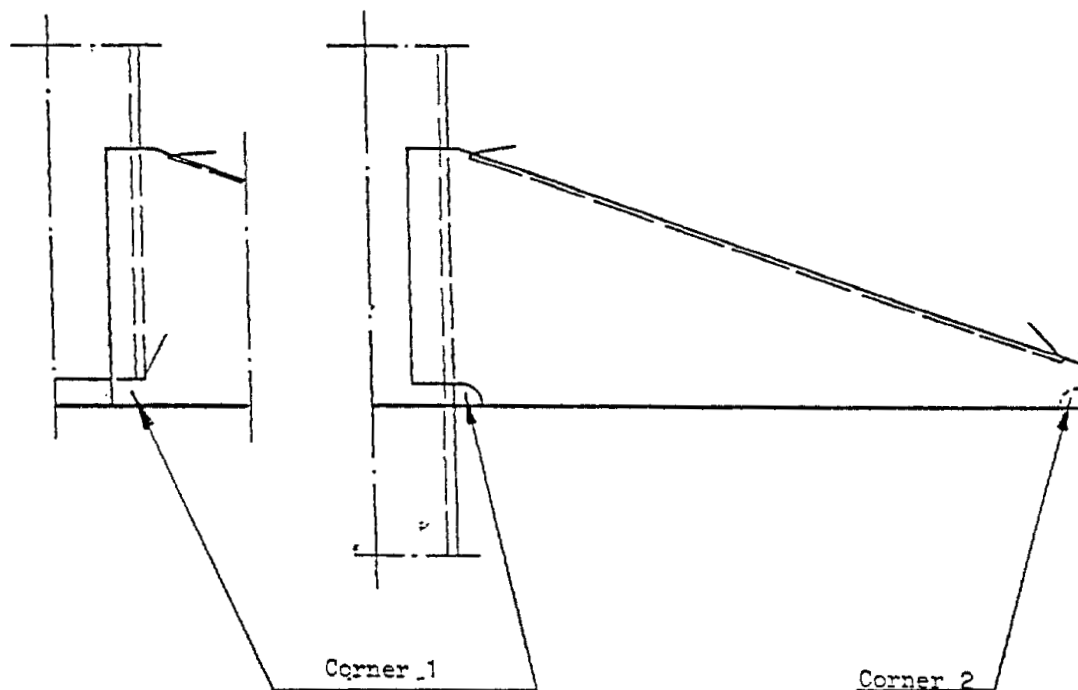
Connections for Type C (02)



Alt.	Cut-outs	
	Corner 1	Corner 2
1	No cut-out	KS10
2	No cut-out	R50
3	No cut-out	R75
4	VUA 165	KS10
5	VUA 165	R50
6	VUA 165	R75

Bracket Type C, CK

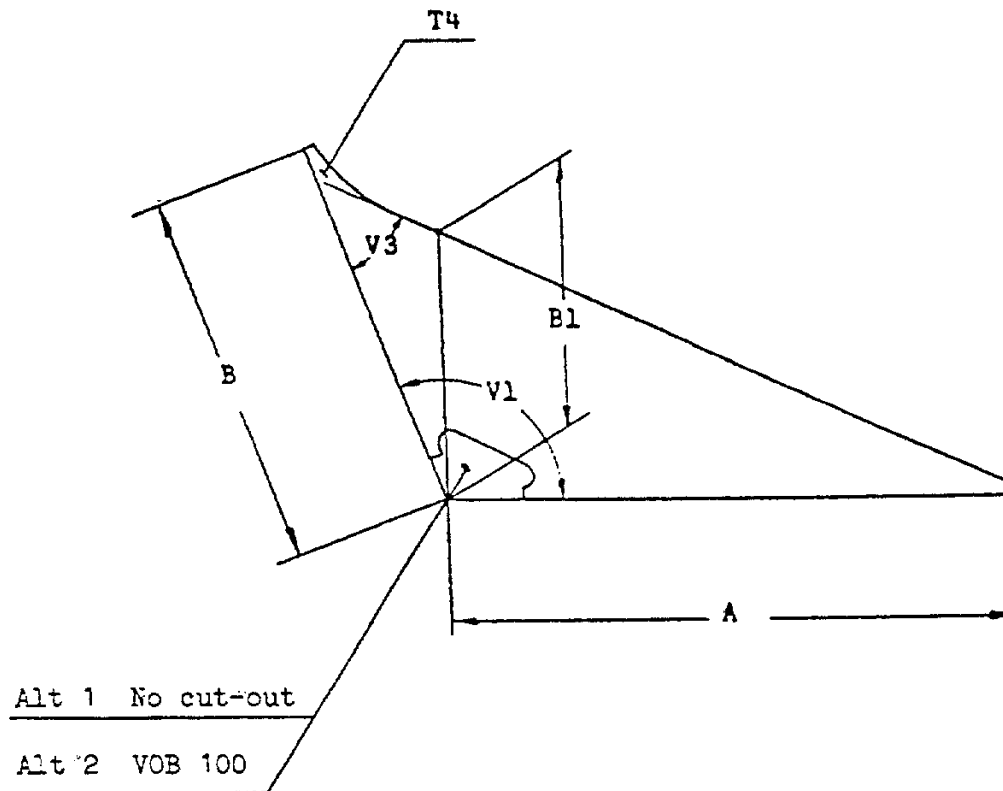
Connections for Type CK (03)



Alt.	Cut-outs	
	Corner 1	Corner 2
1	No cut-out	KS10
2	No cut-out	R50
3	No cut-out	R75
4	VUA 165	KS10
5	VUA 165	R50
6	VUA 165	R75

Bracket Type L

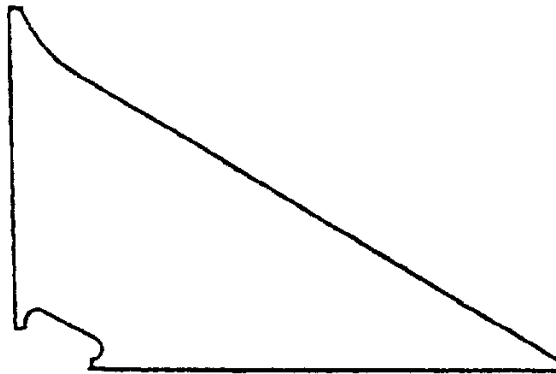
Type L



Length of legs A and B to be chosen according to page 2.

Bracket Type L

Type L



A to be selected from series 600-1200 with intervals of 20

B to be selected from series 350-600 with intervals of 50

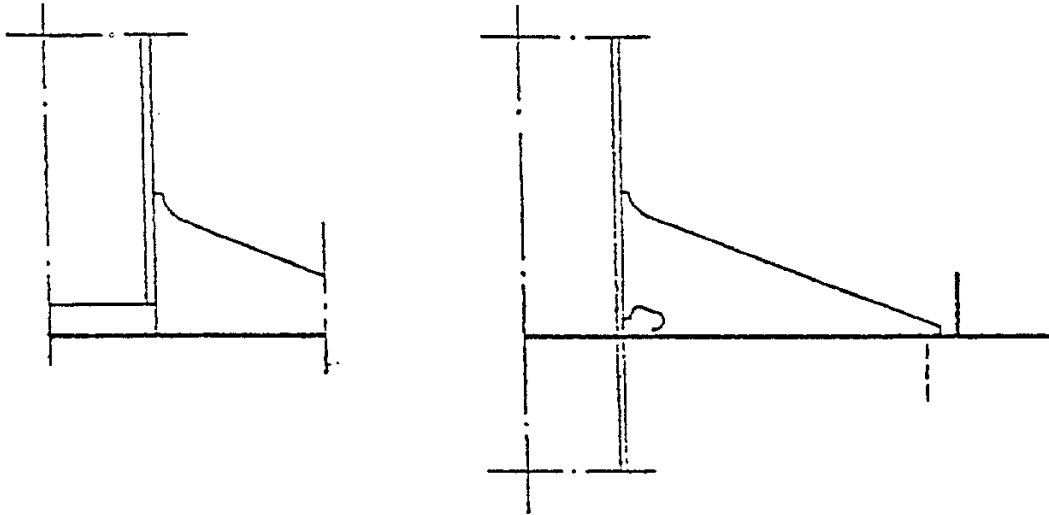
The relationship between A and B1 to lie within the limits obtained from formula

$$0.4 \times A \leq B1 \leq 0.6 \times A$$

For dimensions not given see page 1.

Bracket Type L

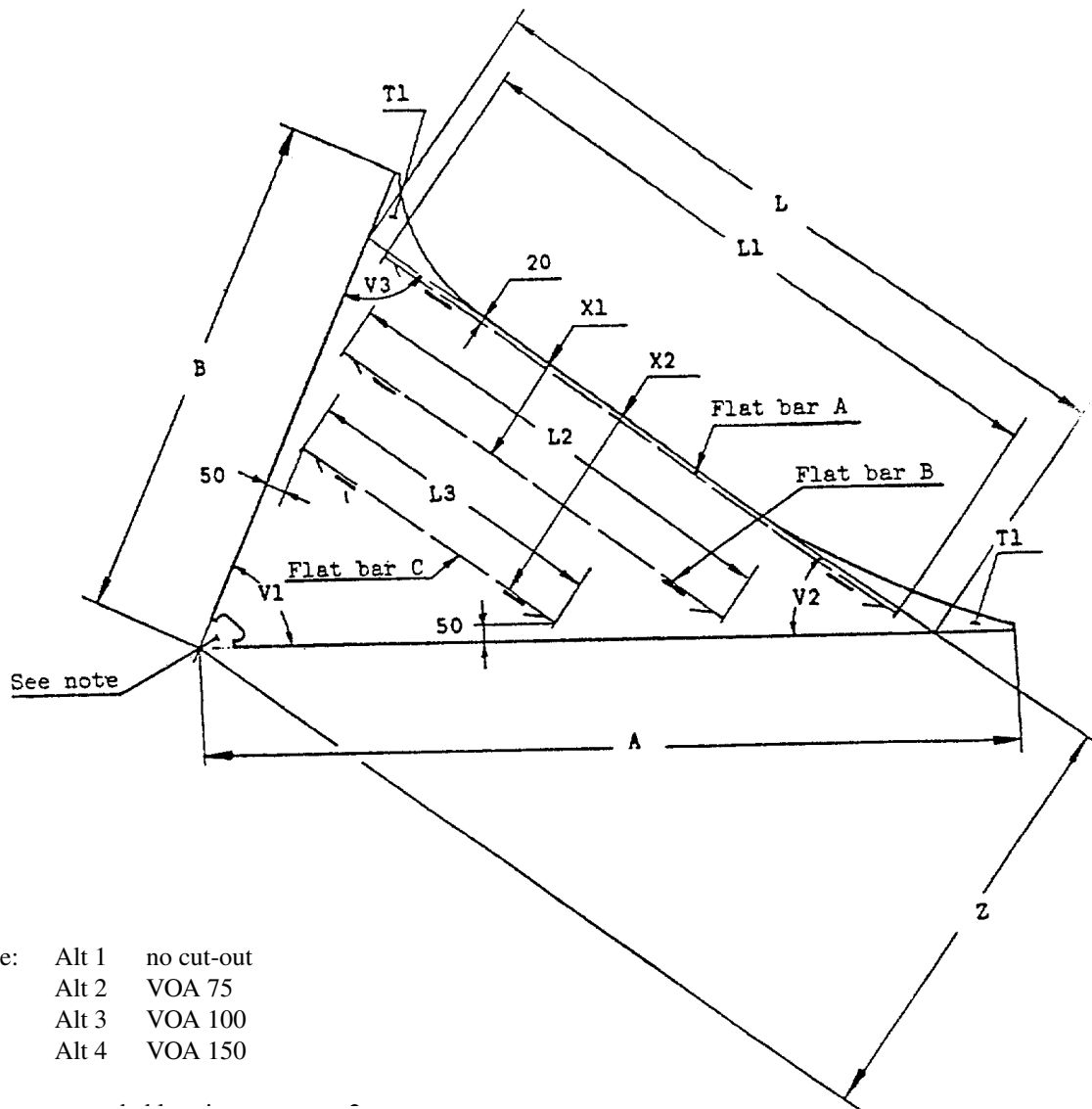
Connections for Type L (04)



Alt.	Cut-out
1	No cut-out
2	VOB 100

Bracket Type K

Type K



Note: Alt 1 no cut-out
 Alt 2 VOA 75
 Alt 3 VOA 100
 Alt 4 VOA 150

Bracket Type K

Description

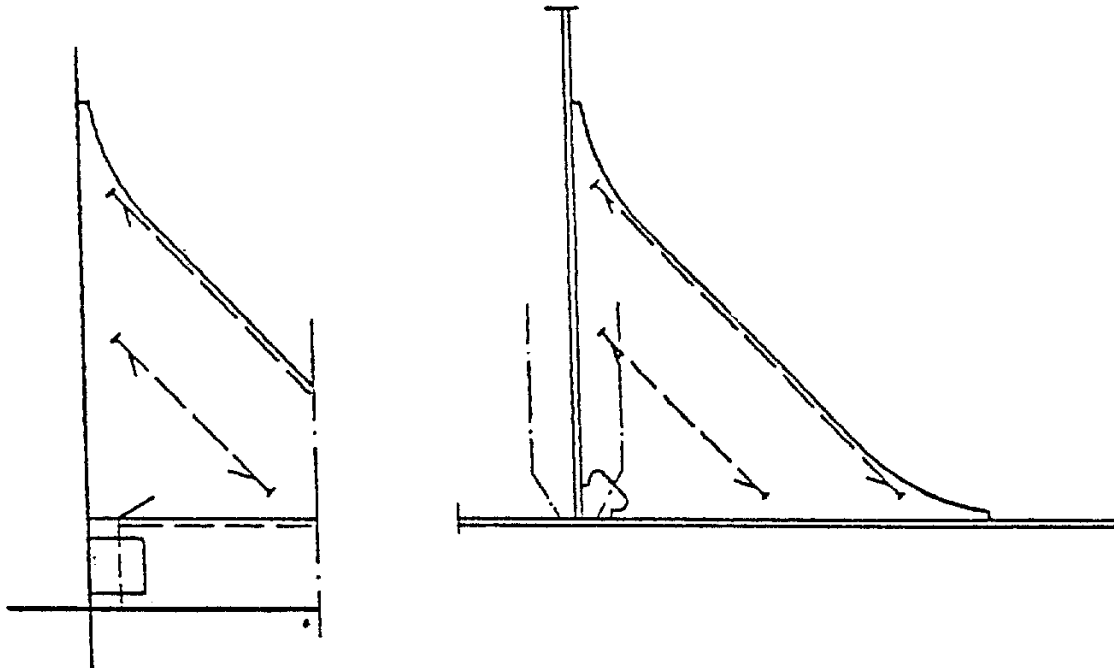
- 1 For $Z \leq 750$ mm one flat bar is required, but if $L < 800$ mm it may be left out.
The area of this flat bar to be $0.14 \times L \text{ cm}^2$.
- 2 For $750 < Z < 1850$ mm two flat bar are required.
The area of flat bar A to be $0.6 \times 0.14 \times L \text{ cm}^2$
The area of flat bar B to be $0.4 \times 0.14 \times L \text{ cm}^2$
For positioning of flat bar B set $X1 = 0.4 \times Z$.
- 3 For $1850 < Z < 2700$ mm three flat bars are required.
The area of flat bar A to be $0.6 \times 0.14 \times L \text{ cm}^2$
The area of flat bar B to be $0.4 \times 0.14 \times L \text{ cm}^2$
The area of flat bar C to be $0.1 \times L^3 \text{ cm}^2$
For positioning of flat bars B and C set
 $X1 = 0.28 \times Z$ and $X2 = 0.59 \times Z$
- 4 For $2700 < Z < 3550$ mm four flat bars are required.
The area of flat bar A to be $0.6 \times 0.14 \times L \text{ cm}^2$
The area of flat bar B to be $0.4 \times 0.14 \times L \text{ cm}^2$
The area of flat bar C to be $0.1 \times L^3 \text{ cm}^2$
The area of flat bar D to be $0.1 \times L^4 \text{ cm}^2$
For positioning of flat bars B, C and D set
 $X1 = 0.21 \times Z$, $X2 = 0.45 \times Z$ and $X3 = 0.69 \times Z$.
- 5 For $3550 < Z < 4400$ mm five flat bars are required.
The area of flat bar A to be $0.6 \times 0.14 \times L \text{ cm}^2$
The area of flat bar B to be $0.4 \times 0.14 \times L \text{ cm}^2$
The area of flat bar C to be $0.1 \times L^3 \text{ cm}^2$
The area of flat bar D to be $0.1 \times L^4 \text{ cm}^2$
The area of flat bar E to be $0.1 \times L^5 \text{ cm}^2$
For positioning of flat bars B, C and D set
 $X1 = 0.17 \times Z$, $X2 = 0.36 \times Z$, $X3 = 0.55 \times Z$ and $X4 = 0.74 \times Z$

Bracket Type K

- 6 If the plate thickness of the bracket is δ 17 mm only an un-symmetrical flat bar A is required according to above.
- 7 If the plate thickness of the bracket is > 17 mm, then double flat bars A are to be used with a total area of $0.25 \times L \text{ cm}^2$.
- 8 Flat bars A and B are to be of the same quality as the plate of the bracket, while the other flat bars can be of usual quality.
- 9 The above areas are to be rounded to the nearest recommended standard dimensions, smallest bar, however, 120×12 mm.
The relationship of width/thickness should be a max. of 18/1???
- 10 If required, the flat bars F are placed as follows:
For $L1 < 2000$ mm no bar F
? For $200 \delta L1 < 4000$ mm one bar F
For $4000 \delta L1 < 6000$ mm two bars F
For $6000 \delta L1 < 8000$ mm three bars F
At intervals with equally divided spaces over L1.
- 11 All welds to be  4 when nothing else is given on the class drawing.
- 12 From the class drawing is also obtained:
Plate thickness, quality of material erection welding.

Bracket Type K

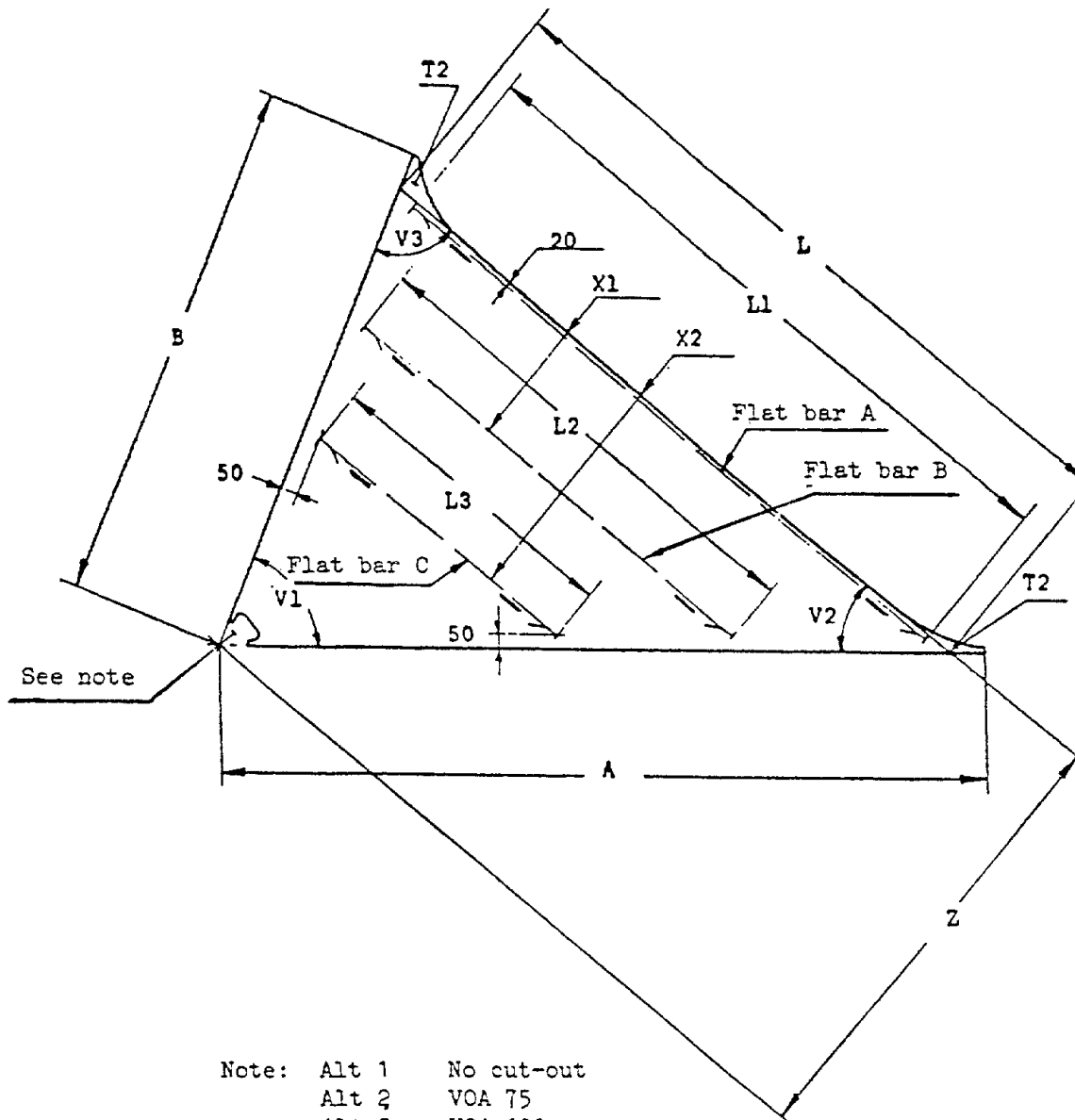
Connections for Type K



Alt.	Cut-out
1	No cut-out
2	VOA 75
3	VOA 100
4	VOA 150

Brackets Type KP

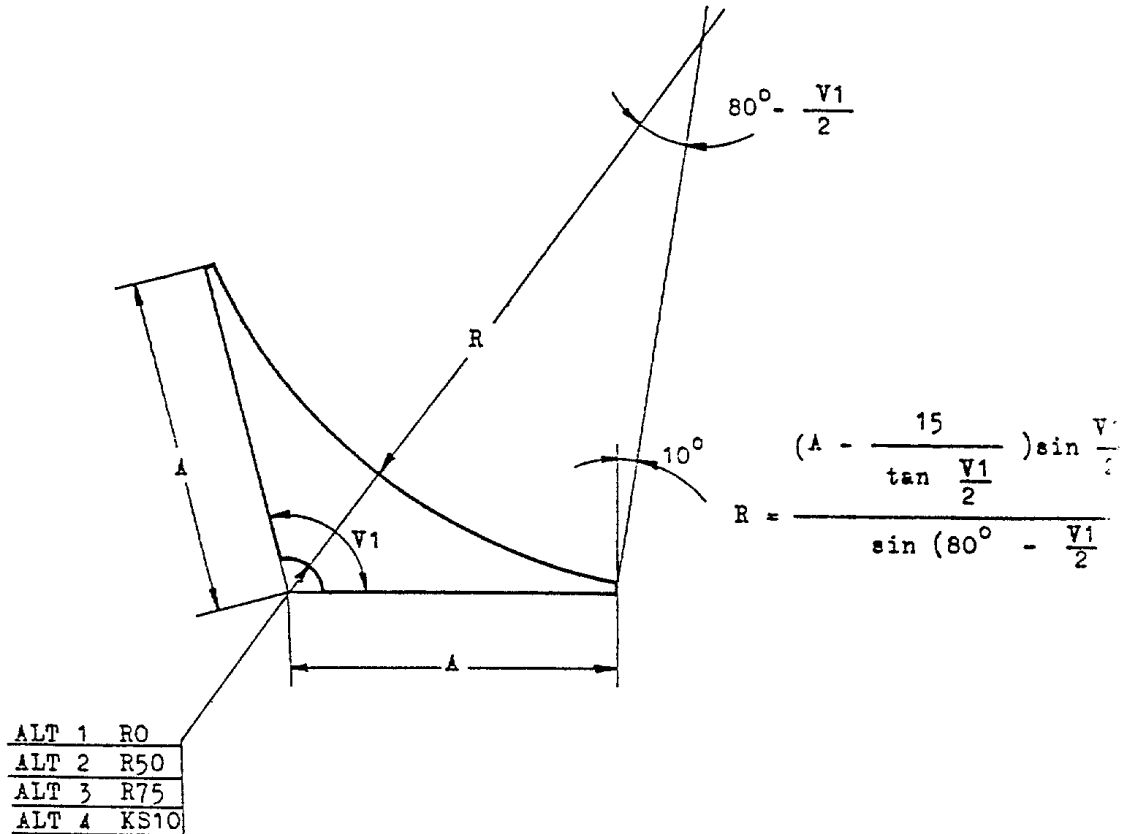
Type KP



Note: Alt 1 No cut-out
 Alt 2 VOA 75
 Alt 3 VOA 100
 Alt 4 VOA 150

Brackets Type BR

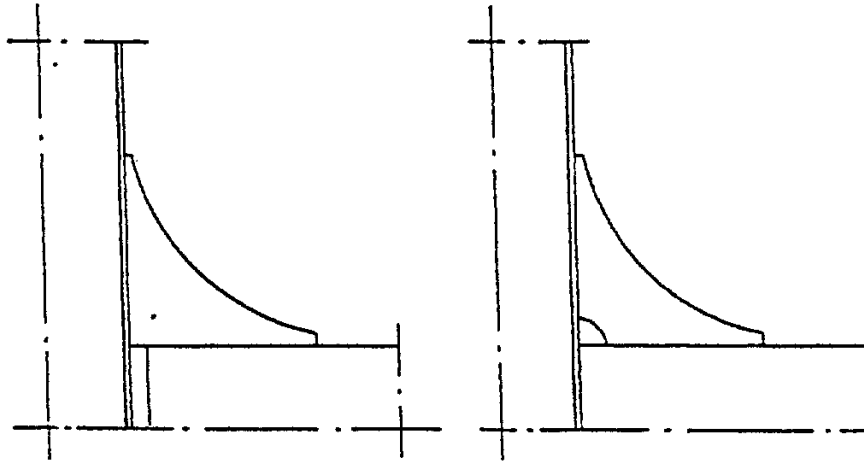
Type BR



Designation	A	R at V1		
		70°	90°	110°
BR 15/15	150	104	166	270
BR 20/20	200	145	228	367
BR 25/25	250	186	290	464
BR 30/30	300	226	352	561
BR 35/35	350	267	413	657
BR 40/40	400	307	475	754
BR 45/45	450	348	536	851
BR 50/50	500	389	598	948
BR 55/55	550	429	660	1045
BR 60/60	600	469	722	1141
BR 70/70	700	551	845	1335
BR 80/80	800	632	969	1529
BR 90/90	900	713	1092	1722
BR 100/100	1000	794	1215	1916
BR 110/110	1100	876	1339	2109
BR 120/120	1200	957	1462	2303
BR 130/130	1300	1038	1586	2497
BR 140/140	1400	1119	1709	2690
BR 150/150	1500	1200	1832	2884
BR 160/160	1600	1282	1956	3078

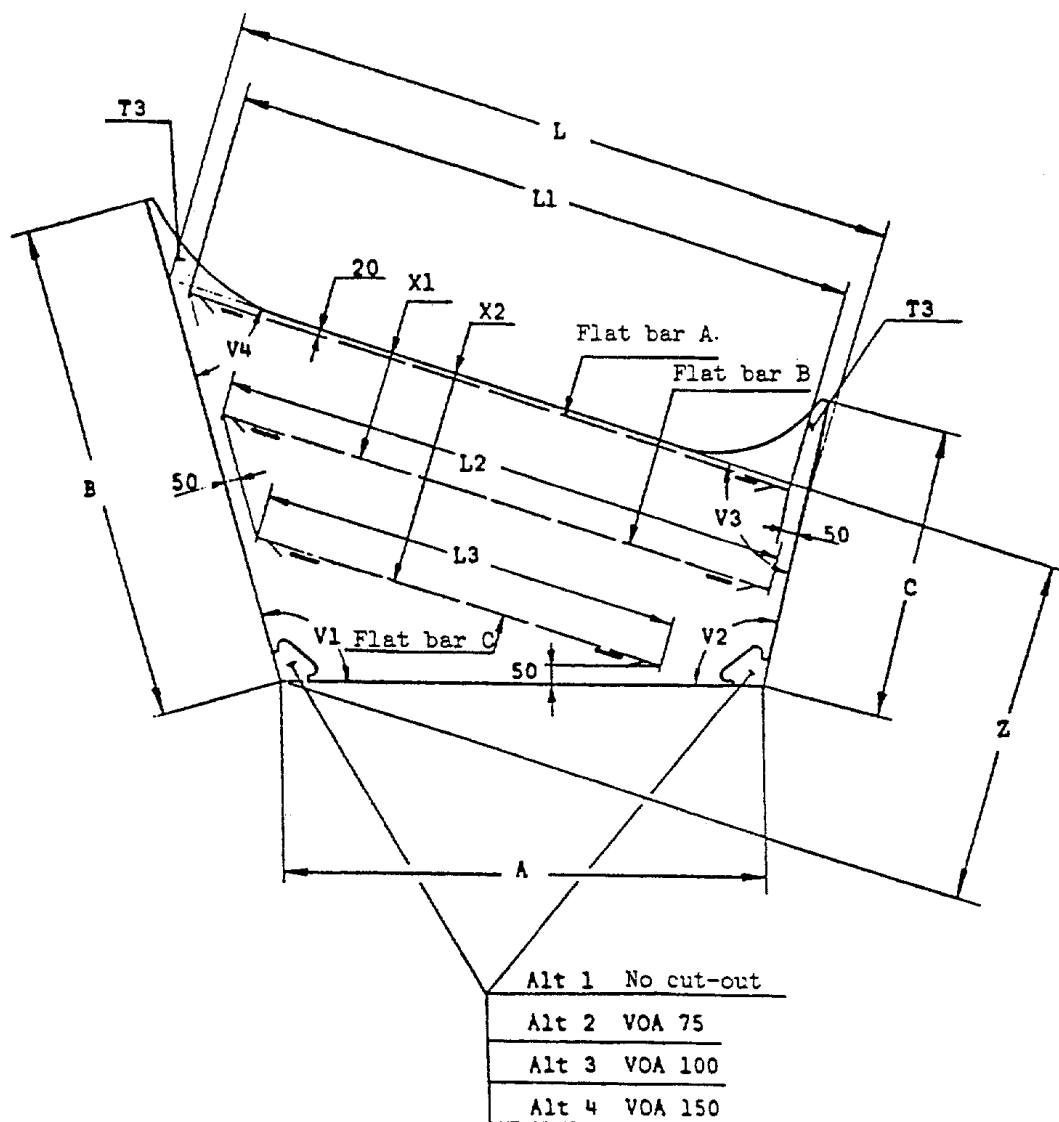
Brackets Type BR

Connection for Type BR



Alt.	Cut-out
1	R0
2	R50
3	R75

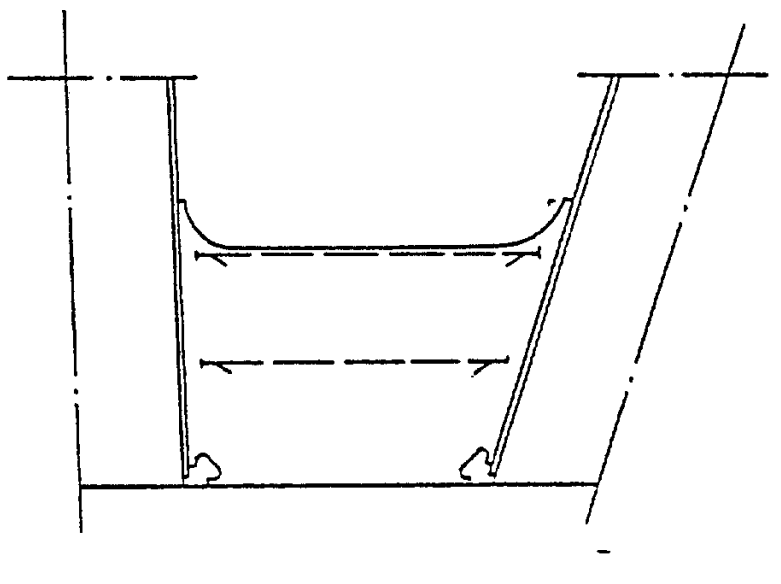
Brackets Type KPV



For description see 199/50201/- pages 3 and 4.

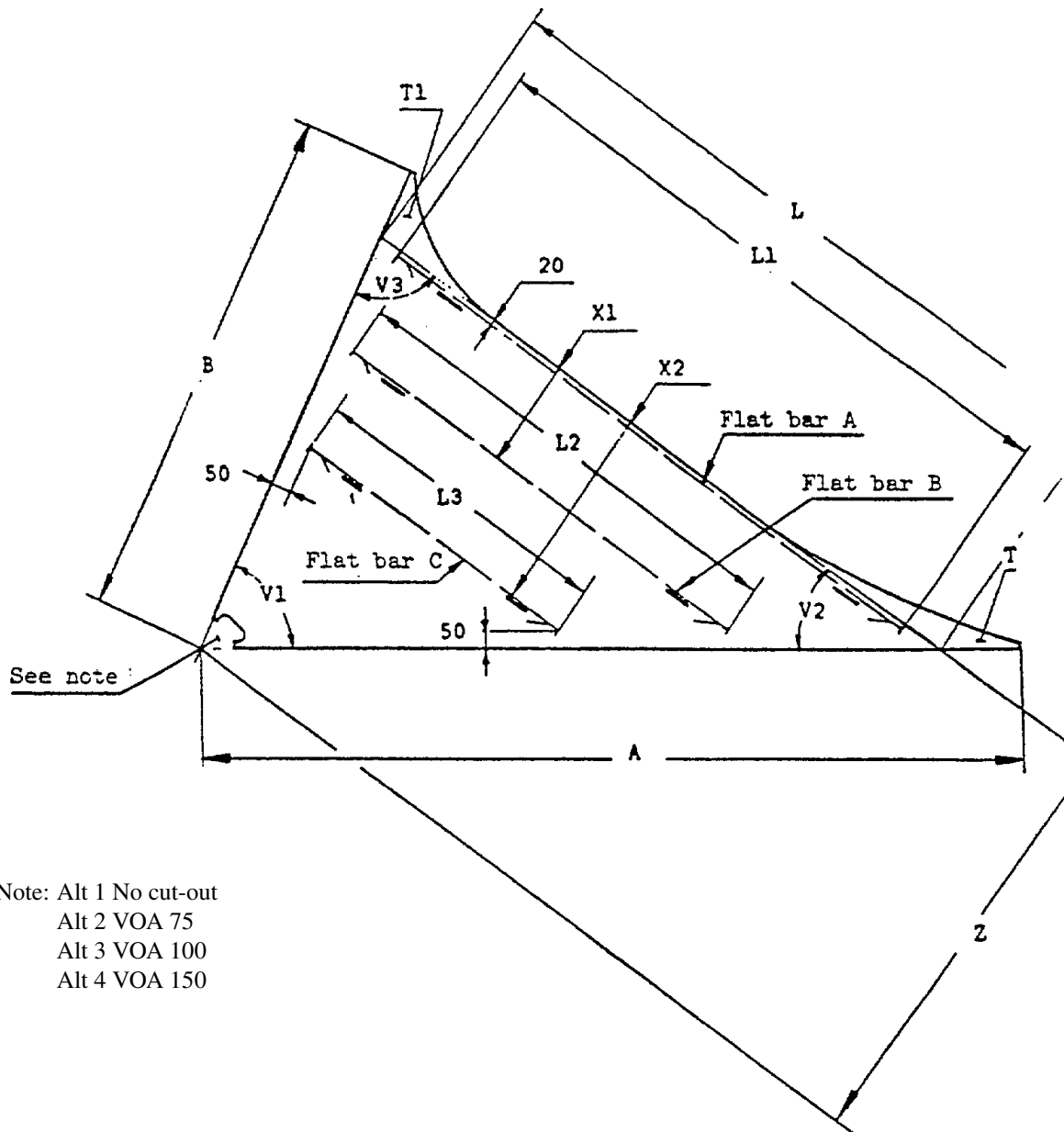
Brackets Type BR, BRS

Connection for Type KPV (13)



Alt.	Cut-out
1	No cut-out
2	VOA 75
3	VOA 100
4	VOA 150

Brackets Type KB



Note: Alt 1 No cut-out
 Alt 2 VOA 75
 Alt 3 VOA 100
 Alt 4 VOA 150

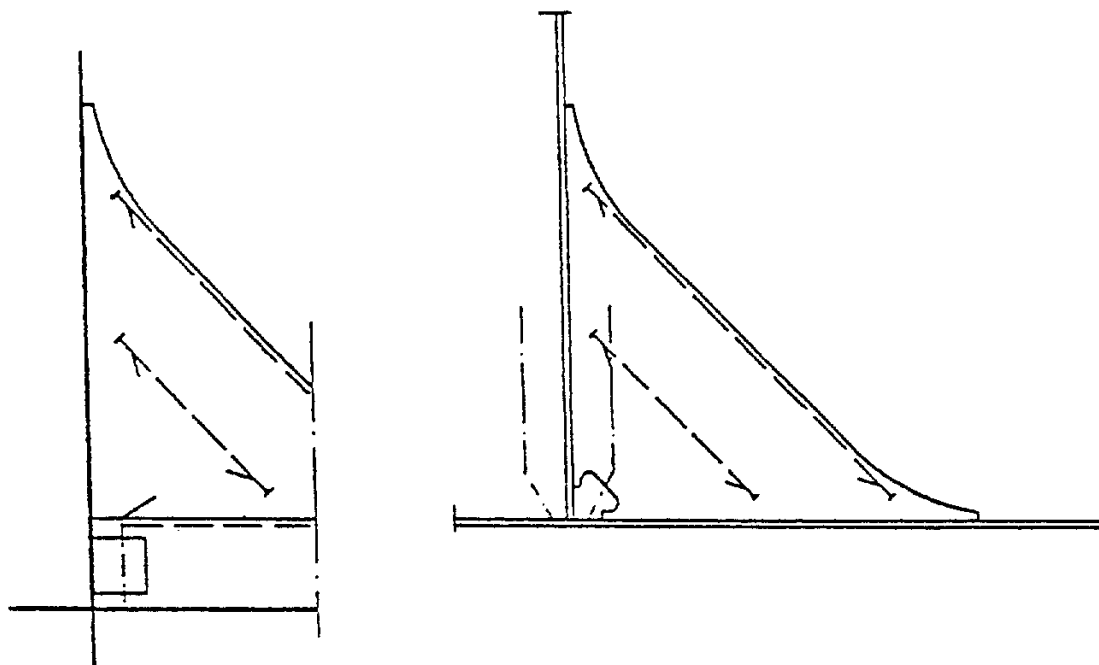
Brackets Type KB

Description

1. For $Z < 750$ mm, one flat bar is required, but if $L < 800$ mm this may be omitted.
The area of this bar must be $0.30 \times L$ cm².
2. For $750 < Z < 1850$ mm two flat bars are required.
Area of bar A to be $0.67 \times 0.30 \times L$ cm².
Area of bar B to be $0.33 \times 0.30 \times L$ cm².
For the positioning of bar B set $X1 = 0.40 \times Z$.
3. If $1850 < Z < 2700$ mm three flat bars are required.
Area of bar A to be $0.67 \times 0.30 \times L$ cm².
Area of bar B to be $0.33 \times 0.30 \times L$ cm².
Area of bar C to be $0.075 \times L^3$ cm².
For the positioning of bars C and B set
 $X1 = 0.28 \times Z$ and $X2 = 0.59 \times Z$.
4. Flat bars A and B are to be of the same quality and thickness as the bracket plate, the width being rounded to the nearest standard width.
5. The area for flat bar C to be rounded off to the nearest recommended standard dimension, smallest bar, however, 120 x 12 mm.
6. If the width for bar A exceeds 250 mm, double bars are to be used instead, each with half the area.
7. All welds to be  4 where nothing else is indicated on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

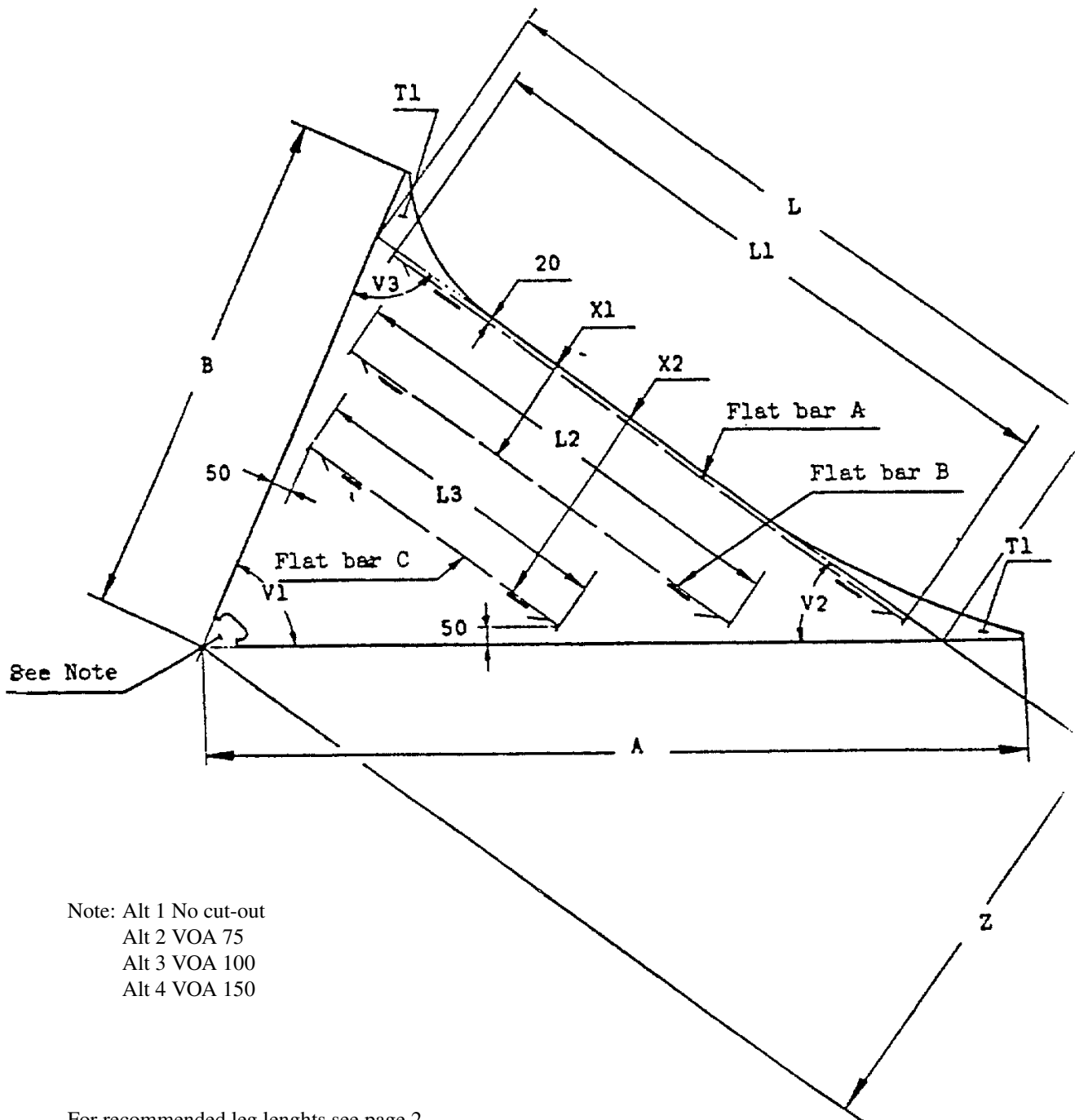
Brackets Type KB

Connection for Type KB



Alt.	Cut-out
1	No cut-out
2	VOA 75
3	VOA 100
4	VOA 150

Brackets Type KC



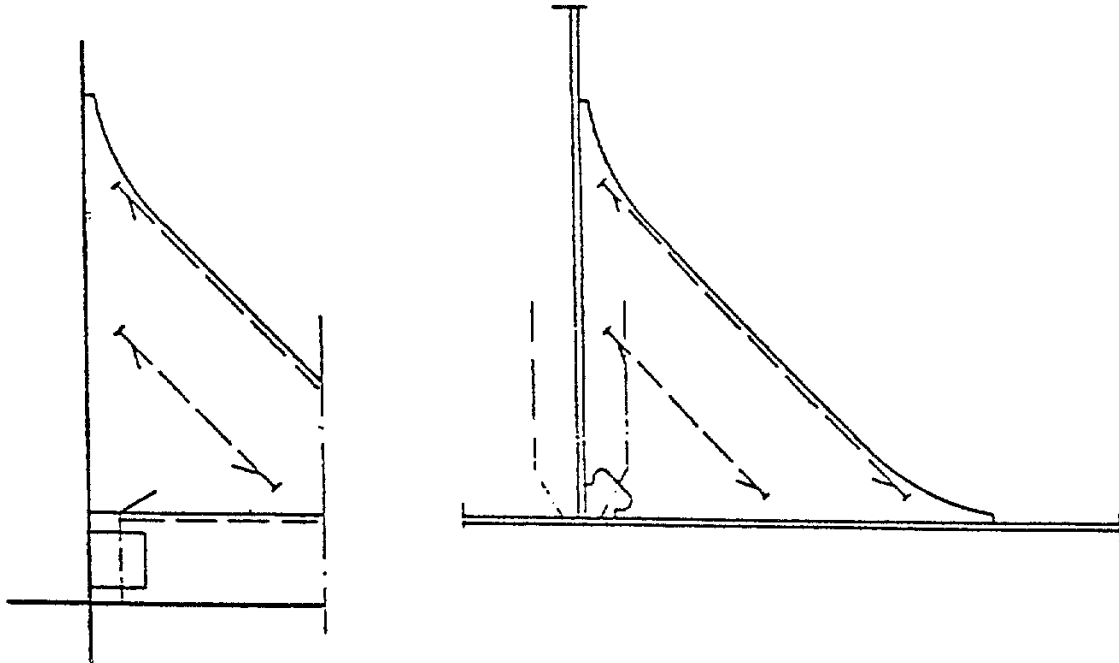
Brackets Type KC

Description

1. If $Z < 750$ mm then one flat bar is required, but if $L < 800$ mm then this may be omitted.
The area of this bar must be $0.20 \times L \text{ cm}^2$.
2. If $750 < Z < 1850$ mm, two flat bars are required.
Area of bar A to be $0.7 \times 0.20 \times L \text{ cm}^2$.
Area of bar B to be $0.3 \times 0.20 \times L \text{ cm}^2$.
For the positioning of bar B set $X1 = 0.40 \times Z$.
3. If $1850 < Z < 2700$ mm, three flat bars are required.
Area of bar A to be $0.7 \times 0.20 \times L \text{ cm}^2$.
Area of bar B to be $0.3 \times 0.20 \times L \text{ cm}^2$.
Area of bar C to be $0.01 \times L^3 \text{ cm}^2$.
For the positioning of bars B and C set $X1 = 0.28 \times Z$ and $X2 = 0.59 \times Z$.
4. Flat bar A to be of the same quality and thickness as the plate of the bracket.
The width to be rounded off to the nearest standard width.
5. The areas for flat bars B and C are to be rounded off to the nearest recommended standard dimension, smallest bar is however, 120×12 mm.
6. If the width of bar A should exceed 250 mm, then in its place choose double bars each having half the area.
7. All welds to be  where nothing else is indicated on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

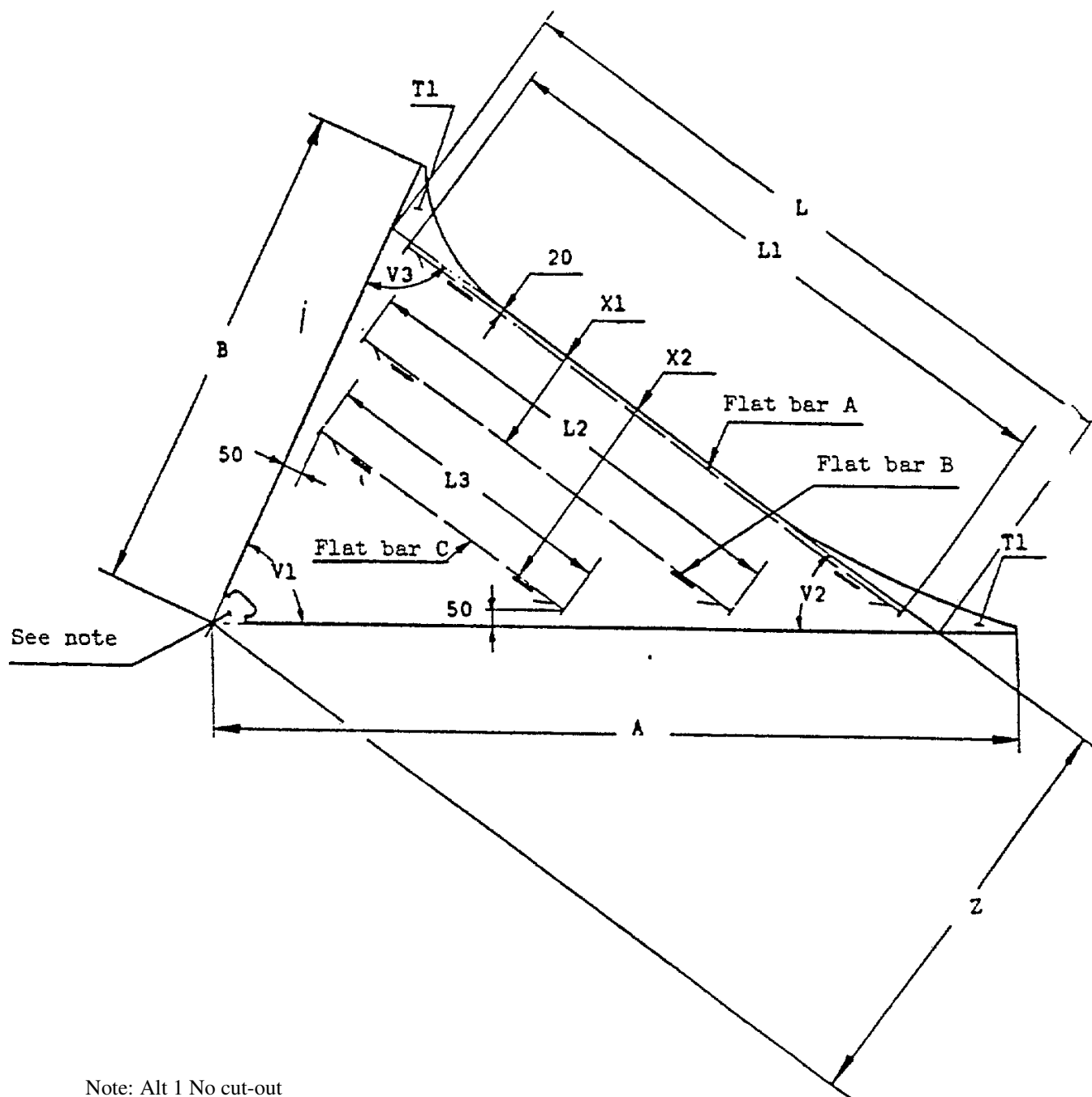
Brackets Type KC

Connection for Type KC



Alt.	Cut-out
1	No cut-out
2	VOA 75
3	VOA 100
4	VOA 150

Brackets Type KD



Note: Alt 1 No cut-out
 Alt 2 VOA 75
 Alt 3 VOA 100
 Alt 4 VOA 150

For recommended leg lengths see page 2.

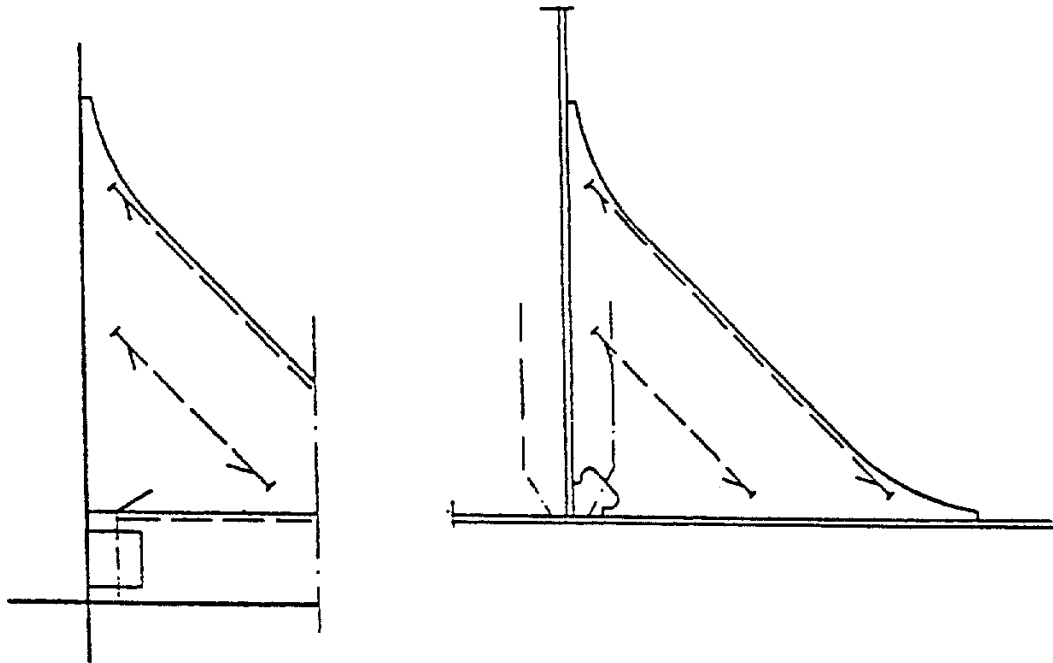
Brackets Type KD

Description

1. If $Z < 750$, one flat bar is required, but if $L < 800$ mm then this may be omitted.
The area of this bar to be $0.28 \times L \text{ cm}^2$.
2. If $750 < Z < 1850$ mm, then two bars are required.
The area of bar A to be $0.8 \times 0.28 \times L \text{ cm}^2$.
The area of bar B to be $0.2 \times 0.28 \times L \text{ cm}^2$.
For the positioning of bars B set $X1 = 0.40 \times Z$.
3. If $1850 < Z < 2700$ mm, three bars are required.
The area of bar A to be $0.8 \times 0.28 \times L \text{ cm}^2$.
The area of bar B to be $0.2 \times 0.28 \times L \text{ cm}^2$.
The area of bar C to be $0.1 \times L^3 \text{ cm}^2$.
For the positioning of bars B and C set
 $X1 = 0.28 \times Z$ and $X2 = 0.59 \times Z$.
4. Flat bars A are to be of the same quality and thickness as the plate of the bracket.
The width is rounded off to the nearest standard width.
5. The areas for B and C to be rounded off to the nearest recommended standard dimension, however, smallest bars to be 120 x 12 mm.
6. All welds to be  4 where nothing else is indicated on the class drawing.
7. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

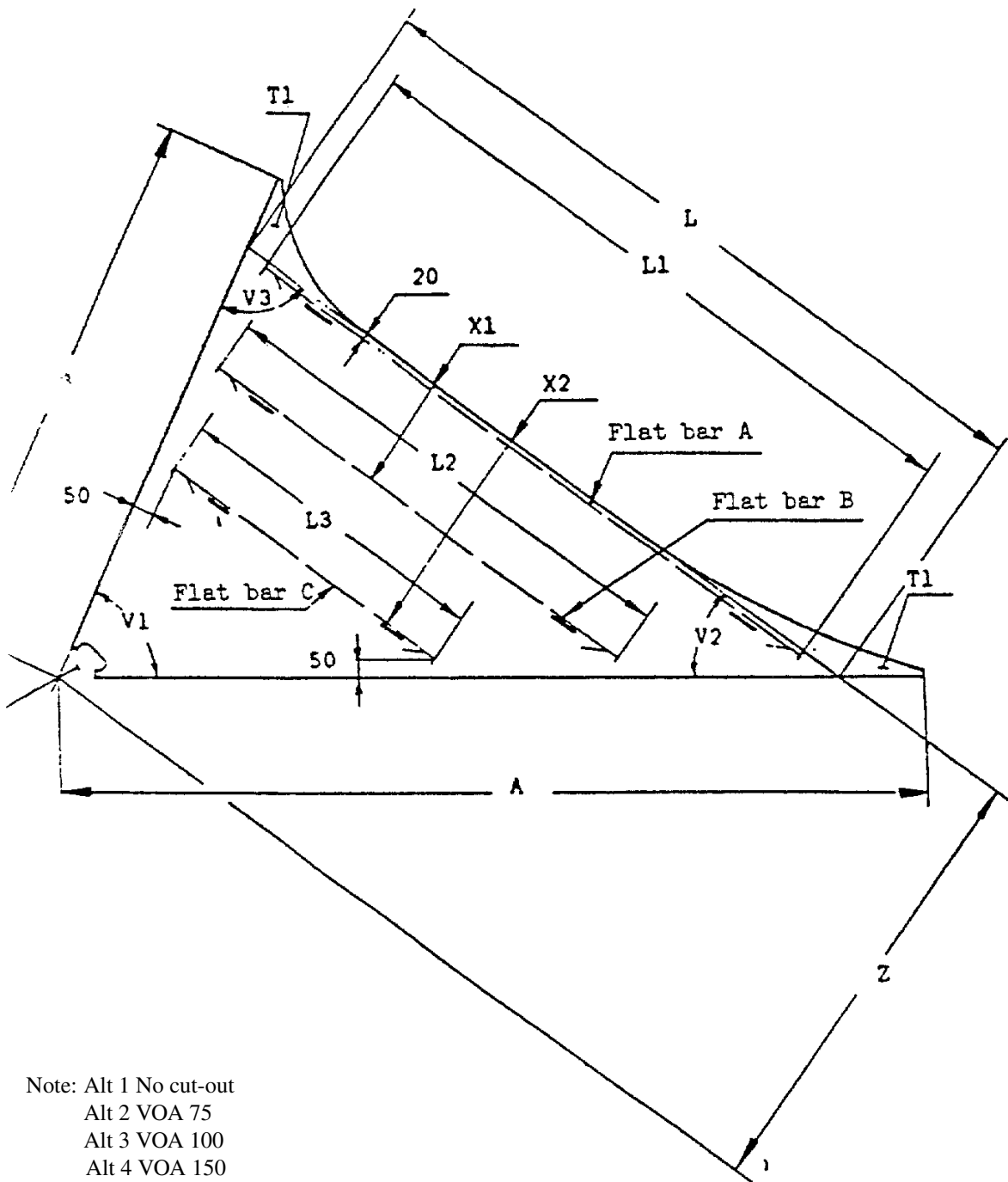
Bracket Type KD

Connections for Type KD



Alt.	Cut-out
1	No cut-out
2	VOA 75
3	VOA 100
4	VOA 150

Bracket Type KBA



Note: Alt 1 No cut-out

Alt 2 VOA 75

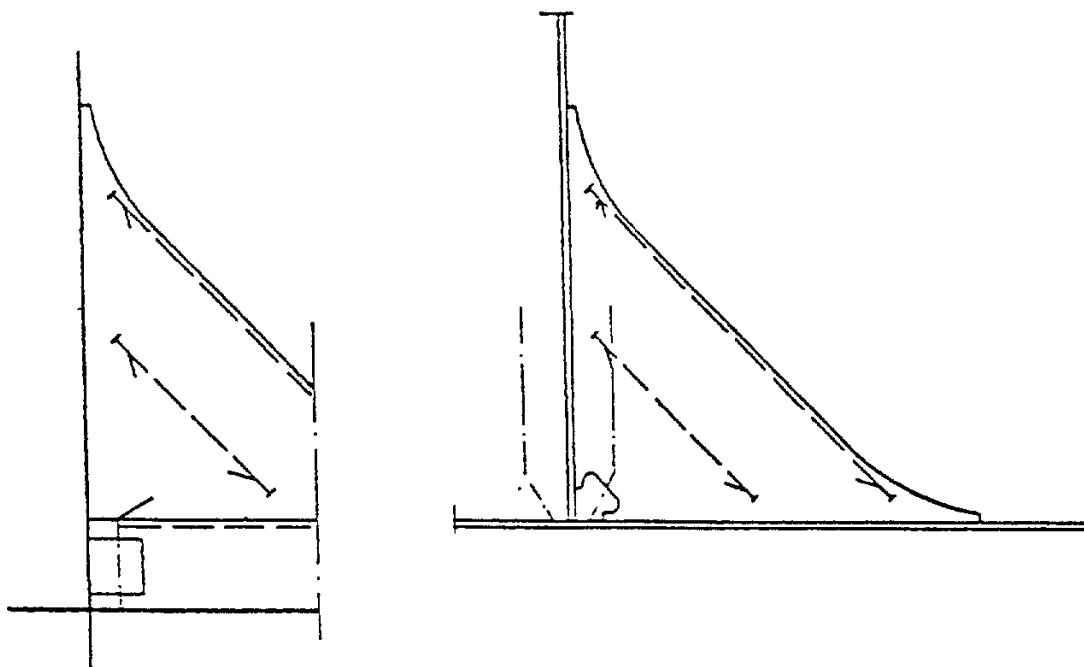
Alt 3 VOA 100

Alt 4 VOA 150

For recommended leg lengths see page 2.

Bracket Type KBA

Connections for Type KBA



Alt.	Cut-out
1	No cut-out
2	VOA 75
3	VOA 100
4	VOA 150

Technical drawing of a structural component showing three alternative views (Alt 1, Alt 2, Alt 3, Alt 4) of a cut-out. The drawing includes dimensions A, B, L, L1, L2, L3, X1, X2, V1, V2, V3, T1, and Z. Labels include 'Flat bar A', 'Flat bar B', 'Flat bar C', and 'See note'.

Note: Alt 1 No cut-out
 Alt 2 VOA 75
 Alt 3 VOA 100
 Alt 4 VOA 150

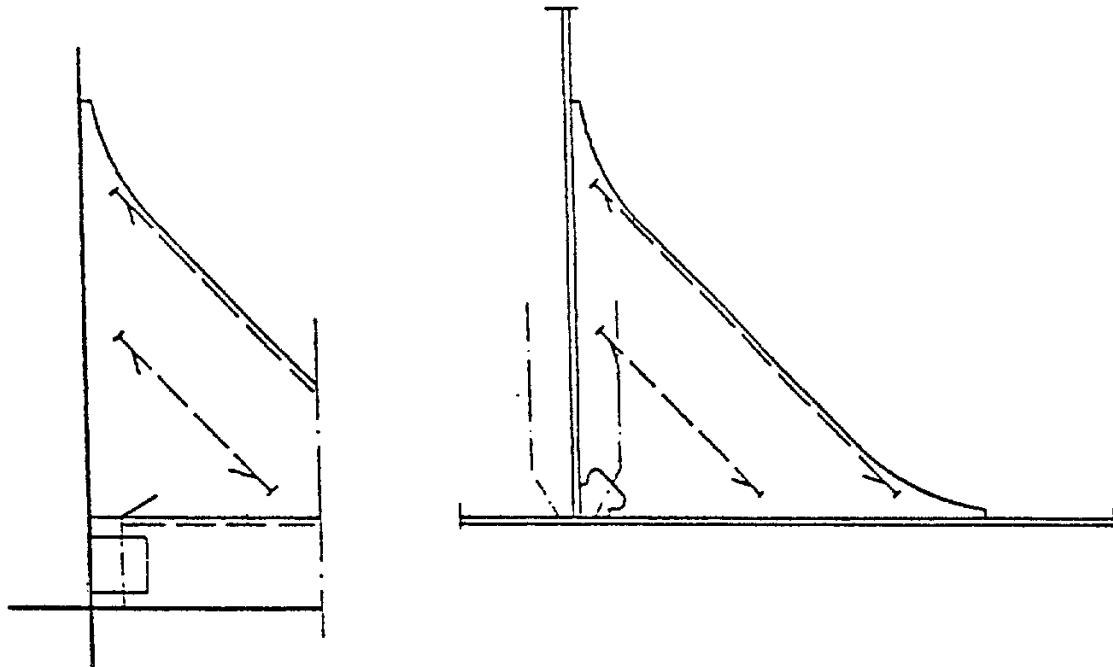
Brackets Type KCA

Description

1. If $Z < 750$ mm, then one flat bar is required, but if $L < 800$ mm this may be omitted.
The area of this bar to be $0.22 \times L \text{ cm}^2$.
2. If $750 < Z < 1850$ mm, then two bars are required.
The area of bar A to be $0.72 \times 0.22 \times L \text{ cm}^2$.
The area of bar B to be $0.28 \times 0.22 \times L \text{ cm}^2$.
For the positioning of bar B set $X1 = 0.40 \times Z$.
3. If $1850 < Z < 2700$ mm, three bars are required.
The area of bar A to be $0.72 \times 0.22 \times L \text{ cm}^2$.
The area of bar B to be $0.28 \times 0.22 \times L \text{ cm}^2$.
The area of bar C to be $0.01 \times L3 \text{ cm}^2$.
For the positioning of bars B and C set
 $X1 = 0.28 \times Z$ and $X2 = 0.59 \times Z$.
4. Bar A to be of the same quality and thickness as the plate of the bracket.
The width to be rounded off to the nearest standard width.
5. The areas for bars B and C to be rounded off to the nearest recommended standard dimension, however, smallest bar to be 120×12 .
6. If the width for bar A should exceed 250 mm, then in its place use double bars each having half the area.
7. All welds to be  where nothing else is indicated on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

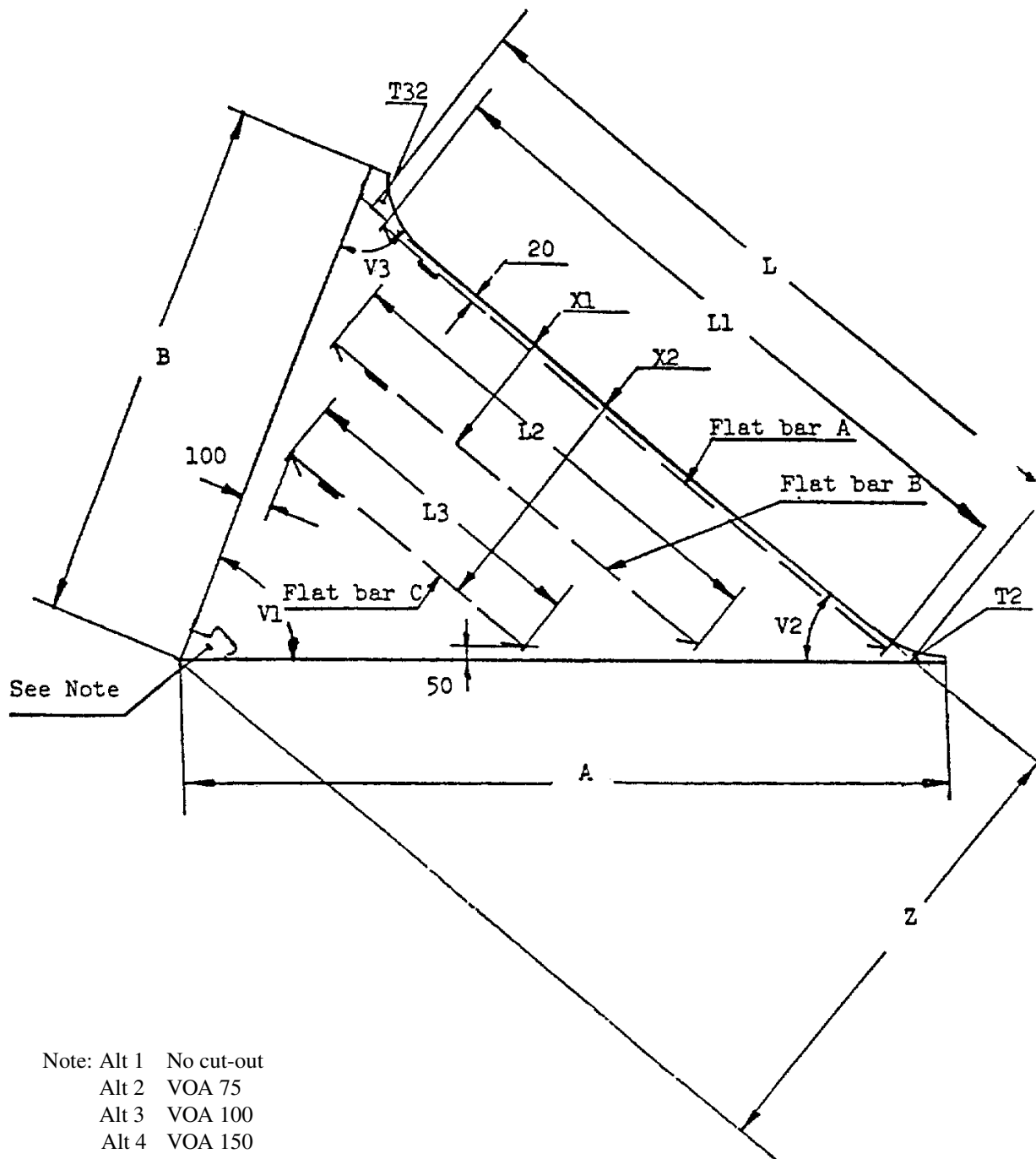
Bracket Type KCA

Connections for Type KCA



Alt.	Cut-out
1	No cut-out
2	VOA 75
3	VOA 100
4	VOA 150

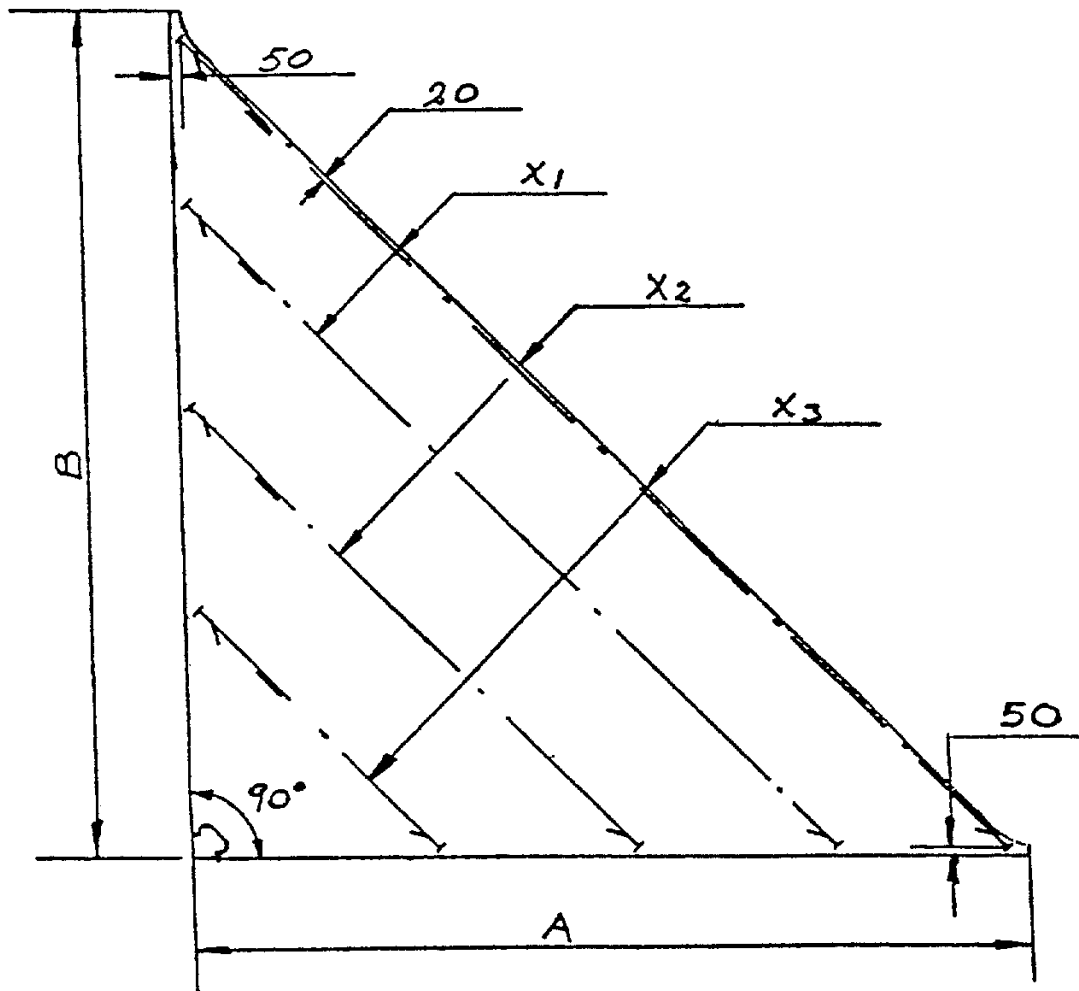
Bracket Type KPA



For recommended leg lengths see 199/50202/- page 2.

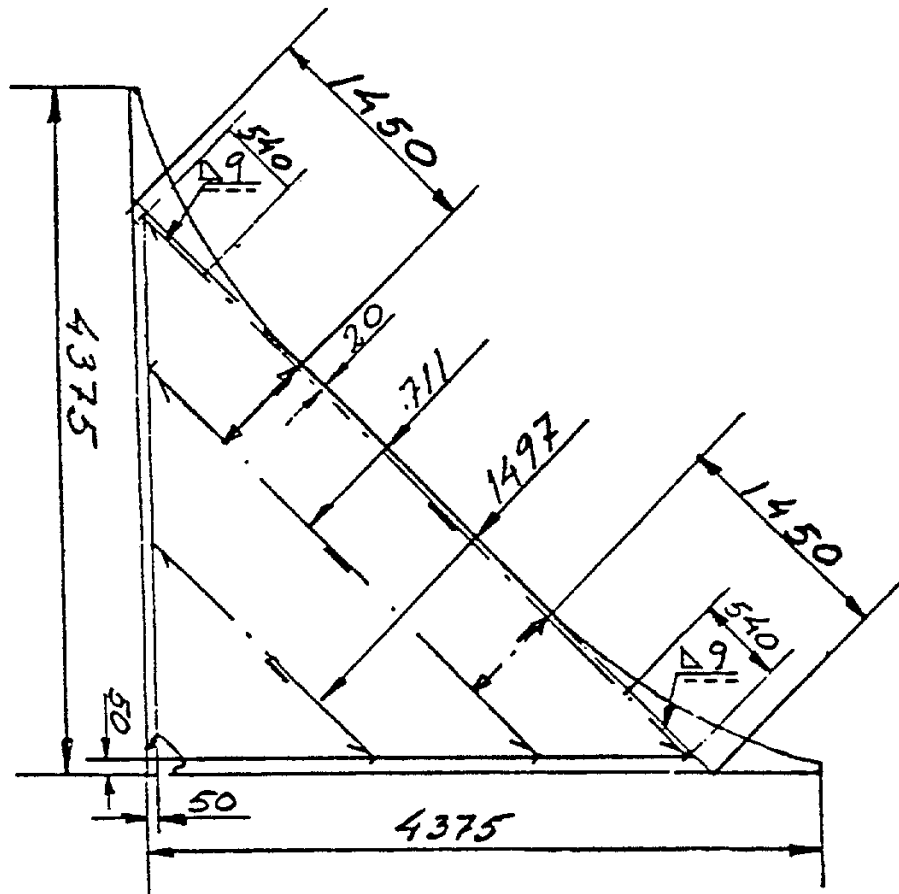
Brackets Type KP, KPA

Placing of stiffeners



Brackets Type KB with extra Stiffeners

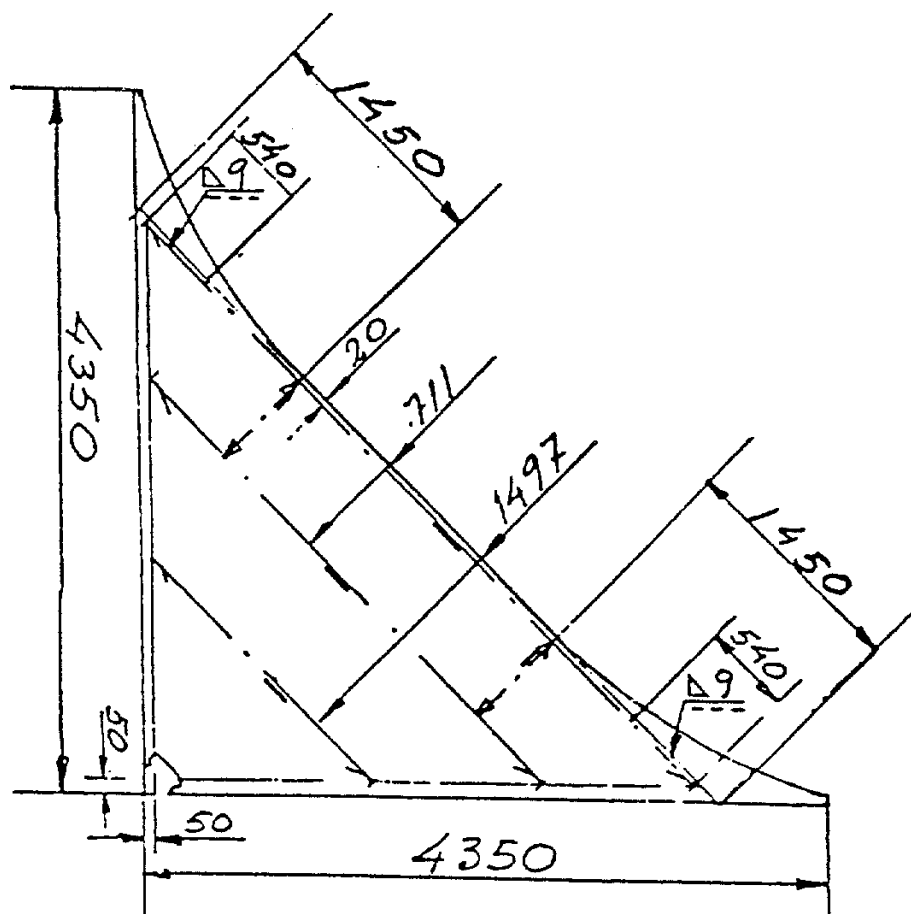
Placing of stiffeners



Weld of stiffener to be $\nabla 4$, if nothing else is given.

Brackets Type KBA with extra Stiffeners

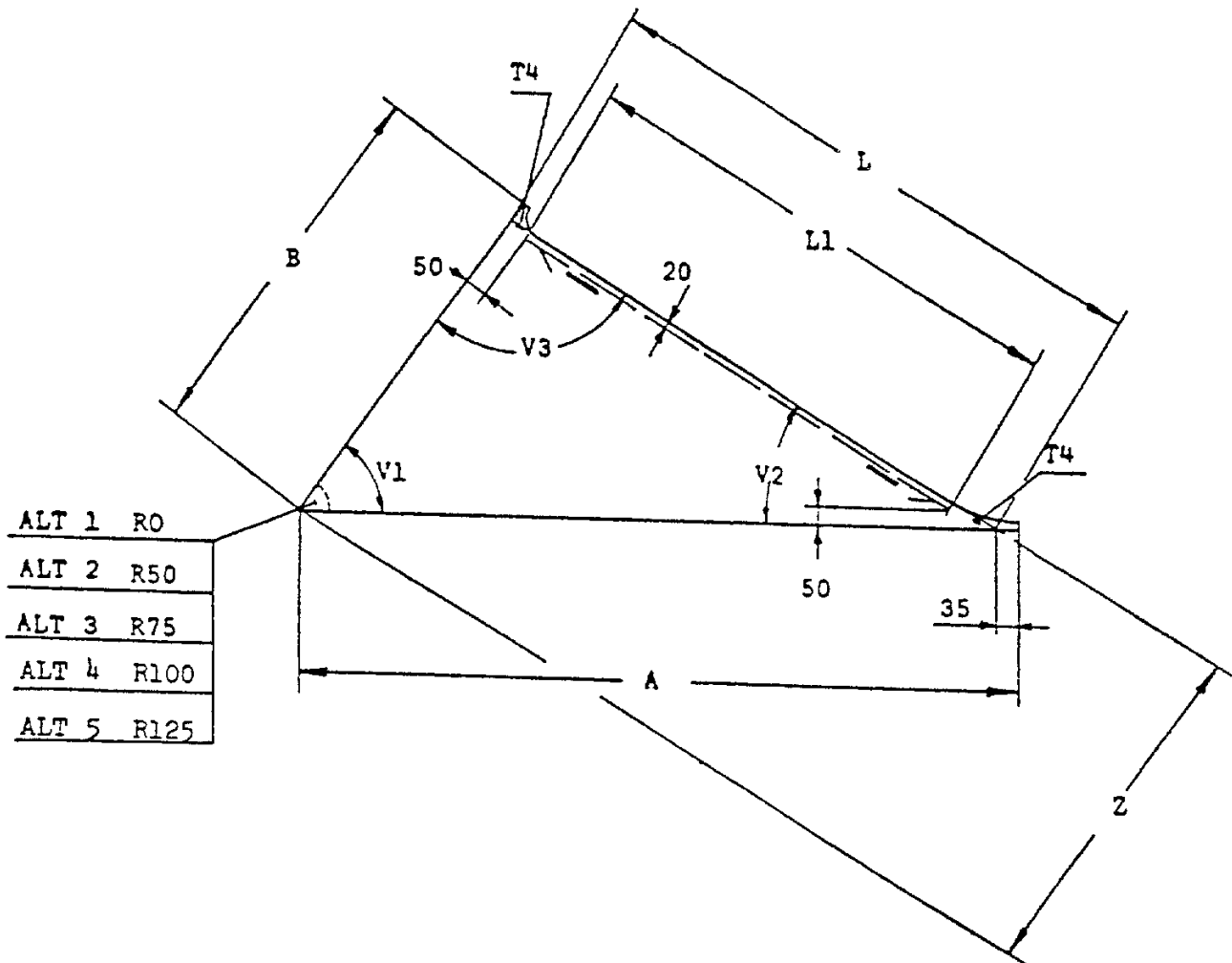
Placing of stiffeners



Weld of stiffener to be $\nabla 4$, if nothing else is given.

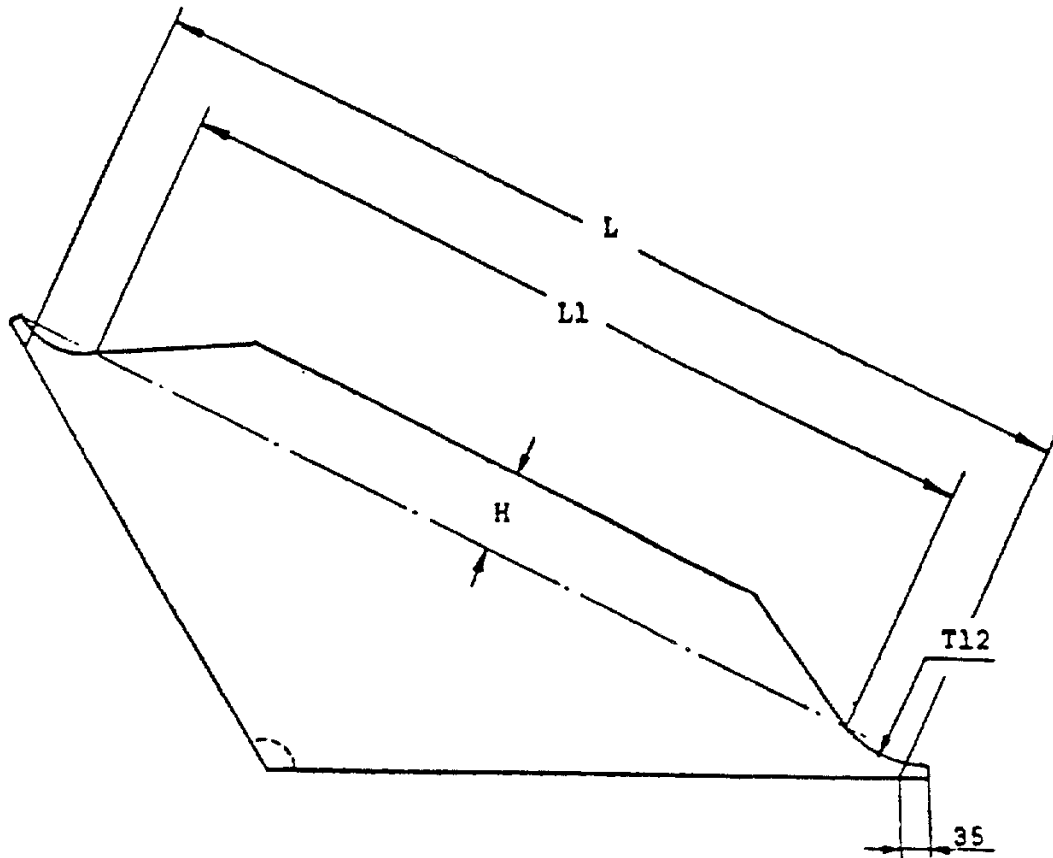
Brackets Type B, BK

Type B



Brackets Type B, BK,

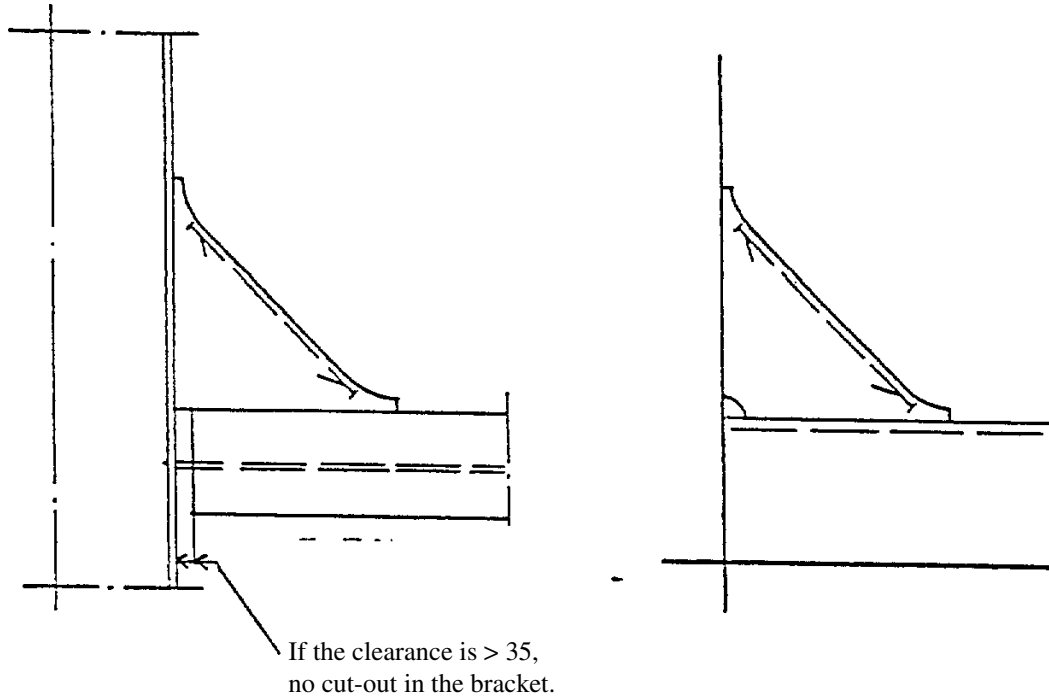
Type BK



Brackets Type B, BK

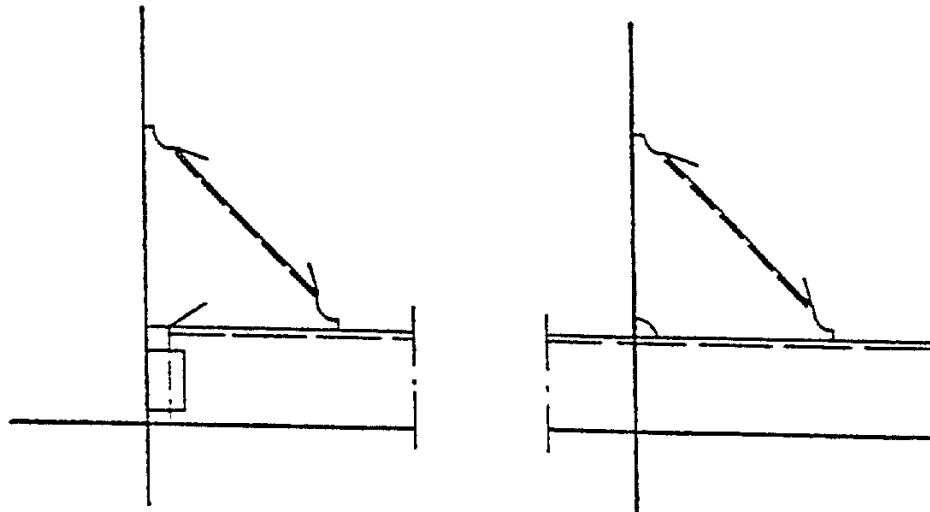
1. If $L < 800$ mm, then no flat bar is required.
2. If $L > 800$ mm, then one flat bar is required.
The area of this bar to be $0.14 \times L$ cm².
3. The flat bar to be of the same quality as the plate of the bracket
4. The above area to be rounded off downwards to a suitable recommended standard dimension, smallest bar, however, to be 90 x 12.
5. When flanging, corresponding flange areas are used, where $t = 14$ mm and is rounded off to the nearest standard flange (199/500 20/- page 2).
6. For V-brackets the Z-dimension should not be less than $0.55 \times A$, where A is the leg length at 90° for the bracket.
The Z-dimension is the distance measured perpendicular to the free edge to the root. A suitable procedure is to close up the slot or increase the length of the leg.
7. All welds to be  4 where nothing else is indicated on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Brackets Type B, BK



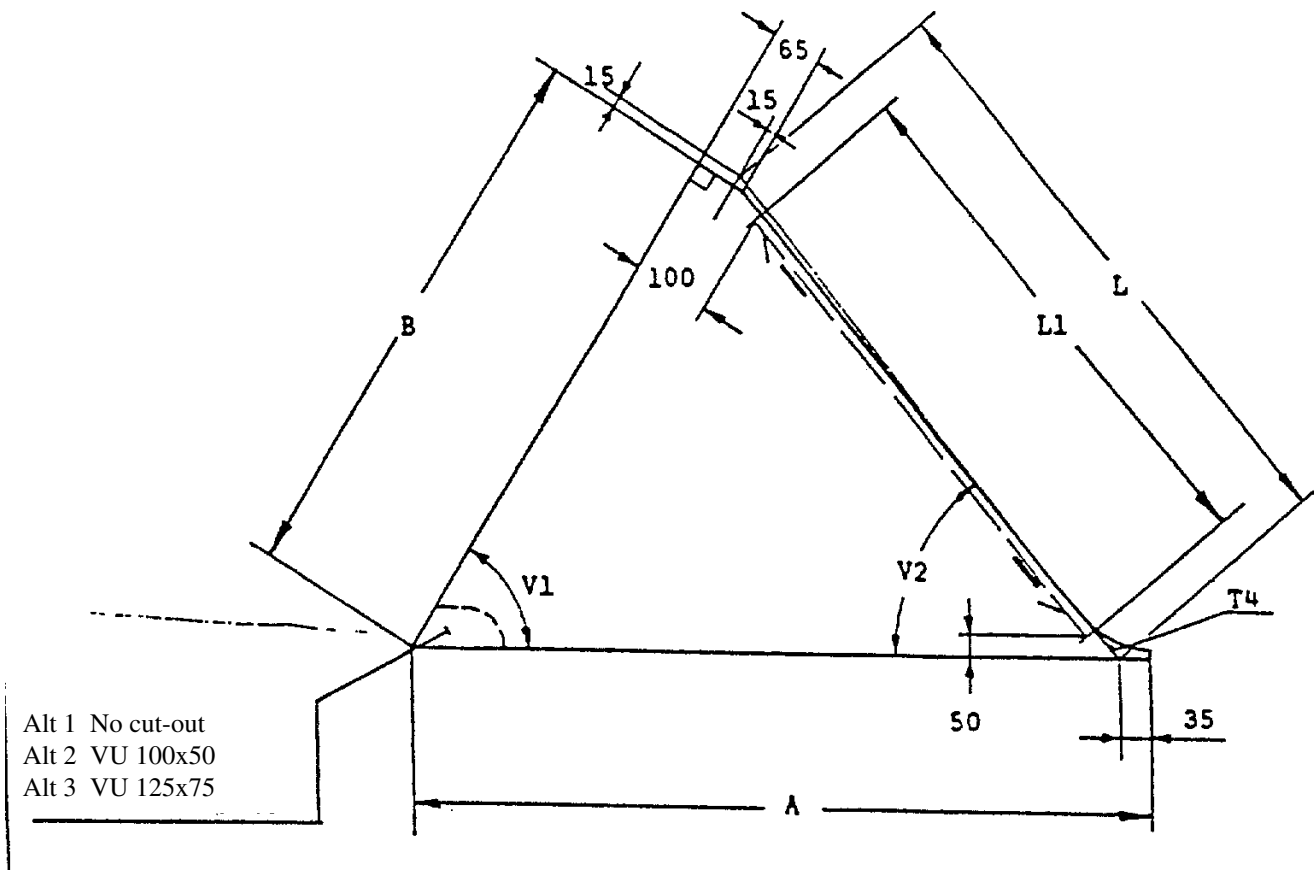
Alt.	Cut-out
1	R0
2	R50
3	R75
4	R100
5	R125

Brackets Type B, BK



Alt.	Cut-out
1	R0
2	R50
3	R75
4	R100
5	R125

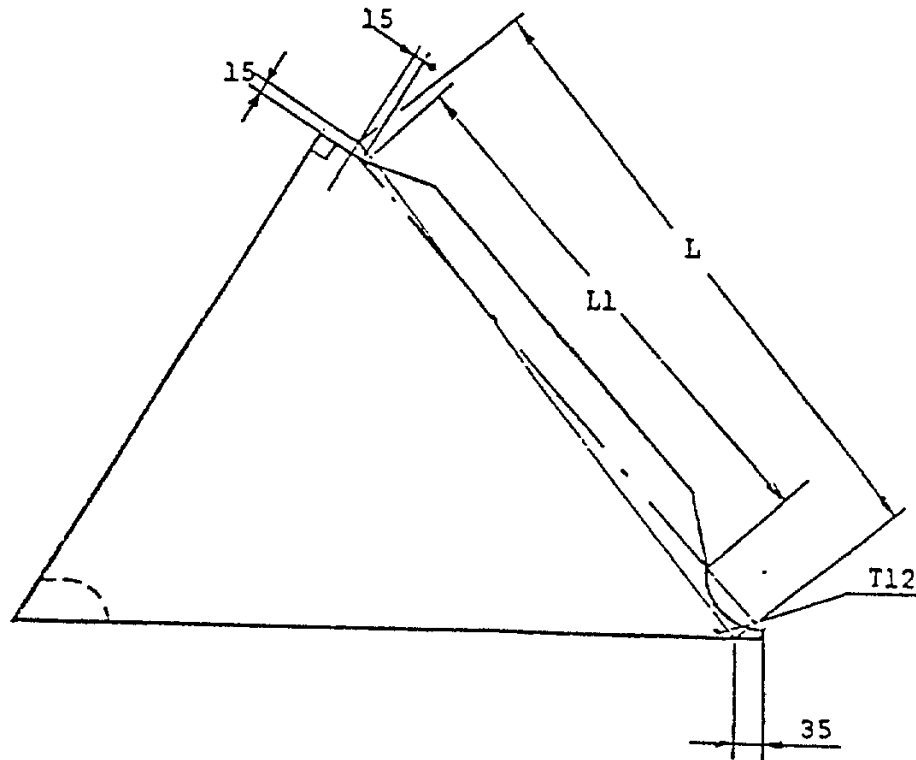
Brackets Type BA, BAK



For description see 199/50301/- page 5.

Brackets Type BA, BAK

Type BAK

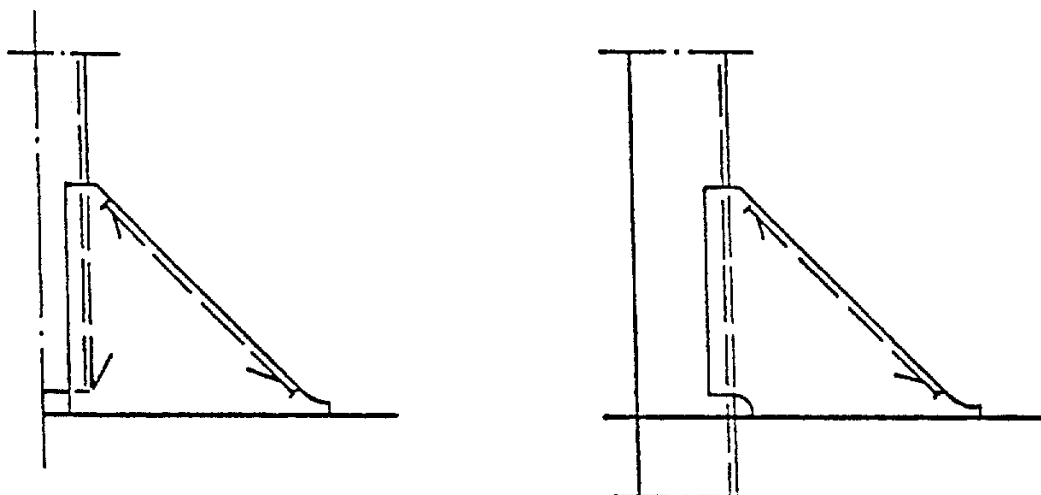


Leg lengths A and B to be selected from the A-values given in the table on page 4.

For dimensions not shown see page 1.

Brackets Type BA, BAK

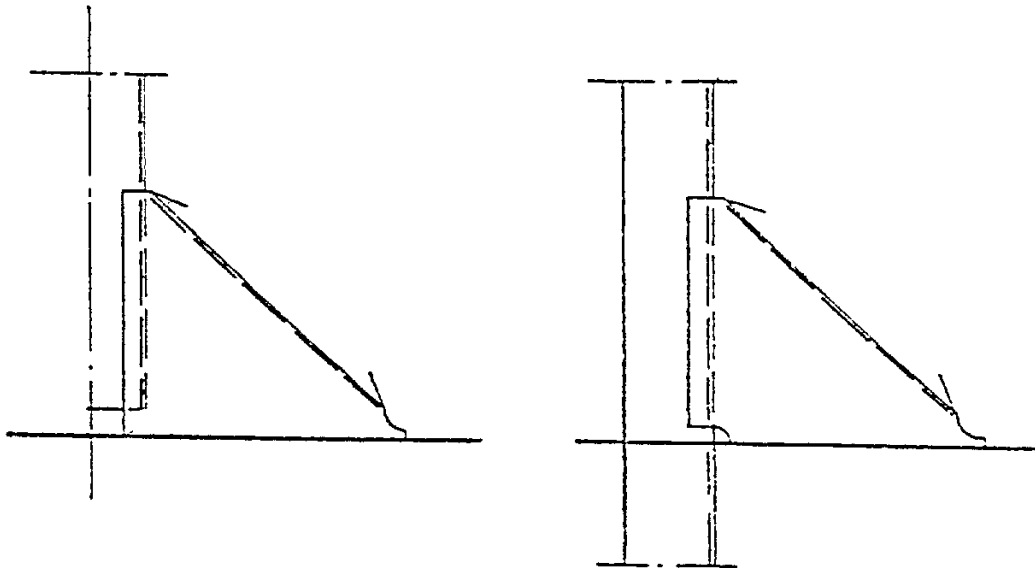
Connections for Type BA (22)



Alt.	Cut-out
1	No cut-out
2	VU 100x50
3	VU 125x75

Brackets Type BA, BAK

Connections for Type BAK)



Alt.	Cut-out
1	No cut-out
2	VU 100x50
3	VU 125x75

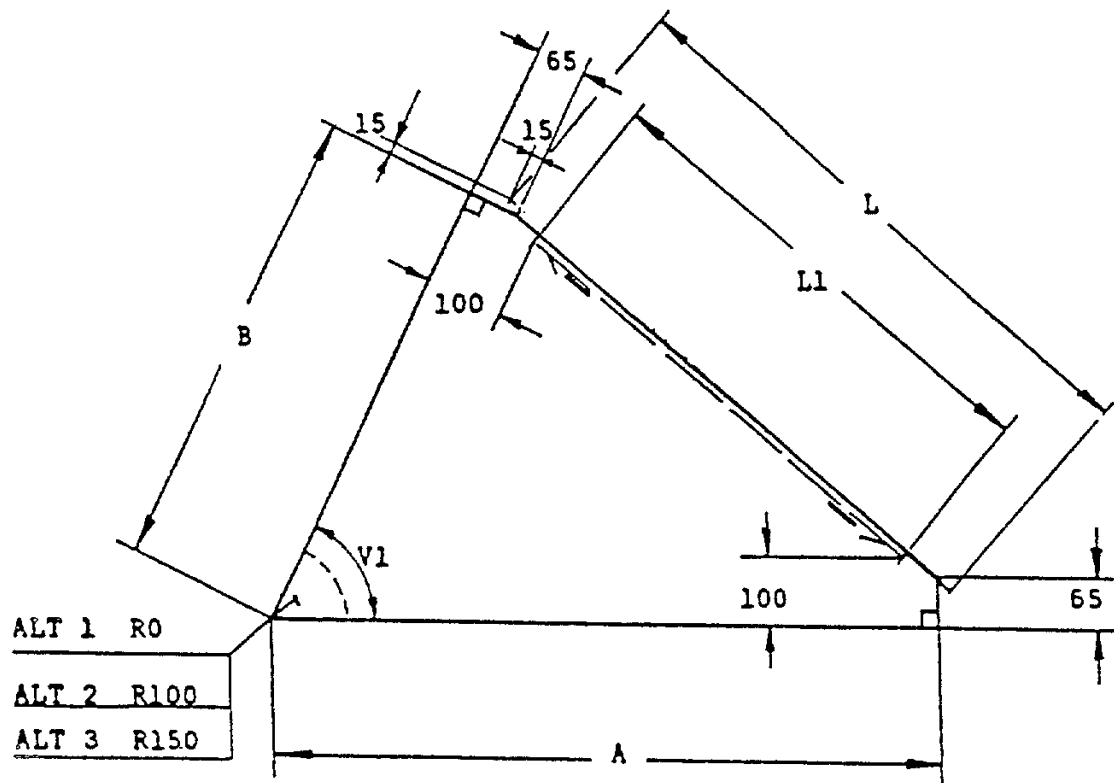
Brackets Type BA, BAK

Description

1. If $L < 800$ mm, then no flat bar will be required.
2. If $L > 800$ mm, one flat bar will be required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
3. The flat bars to be of the same quality as the plate of the bracket
4. The above areas to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, to be 90×12 .
5. When flanging, corresponding flange areas are used, where $t = 14$ mm rounded off to the nearest standard flange (199/500 20/- page 2).
6. For V-brackets the Z-dimension should not be less than $0.55 \times A$, where A is the leg length at 90° for the bracket.
The Z-dimension is the distance measured perpendicular to the free edge to the root.
A suitable procedure is to close up the slot or increase the length of the leg.
7. All welds to be  4 where nothing else is shown on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

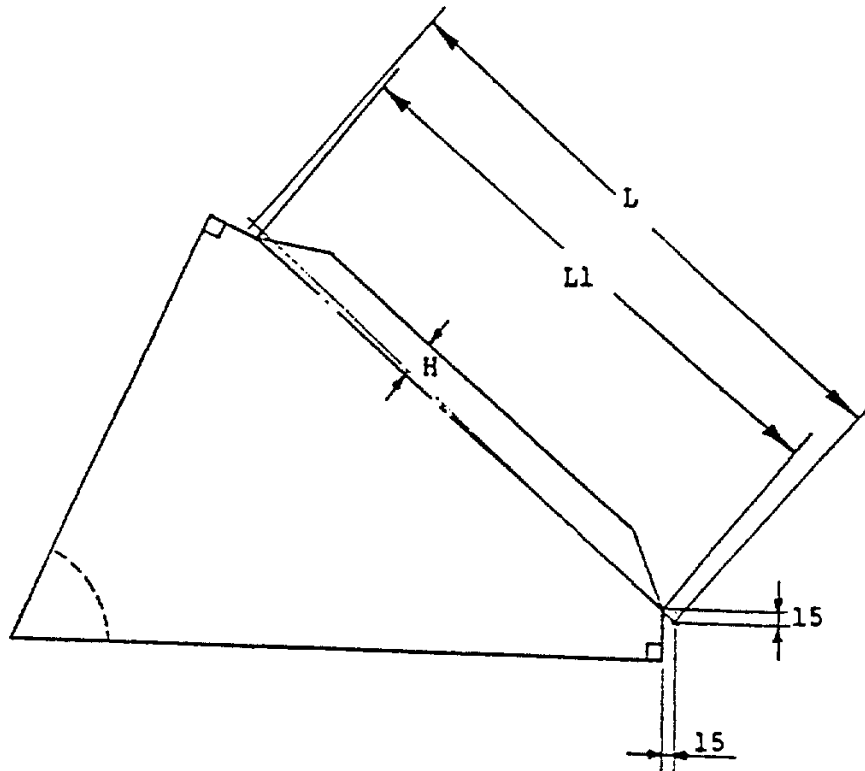
Brackets Type BB, BBK

Type BB



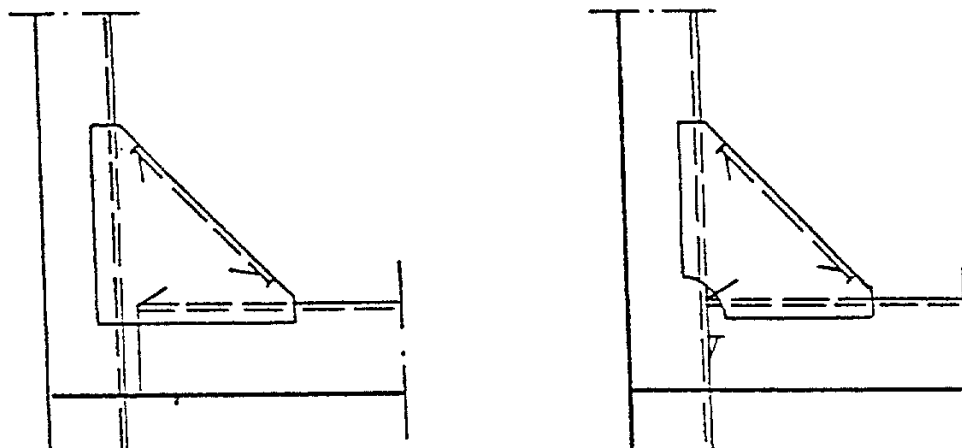
Brackets Type BB, BBK

Type BBK



Brackets Type BB, BBK

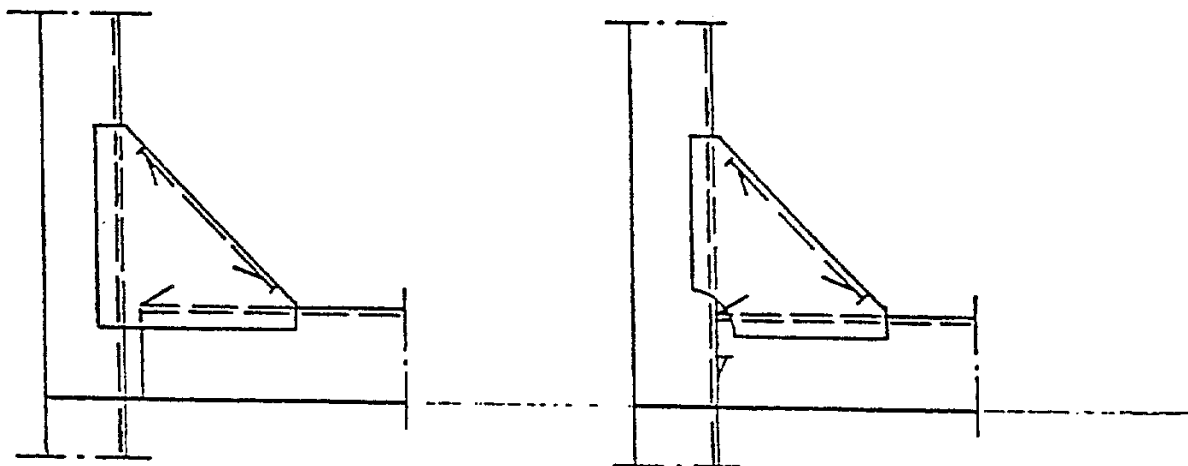
Connections for Type BE



Alt.	Cut-out
1	R0
2	R100
3	R150

Brackets Type BB, BBK

Connections for Type BBK



Alt.	Cut-out
1	R0
2	R100
3	R150

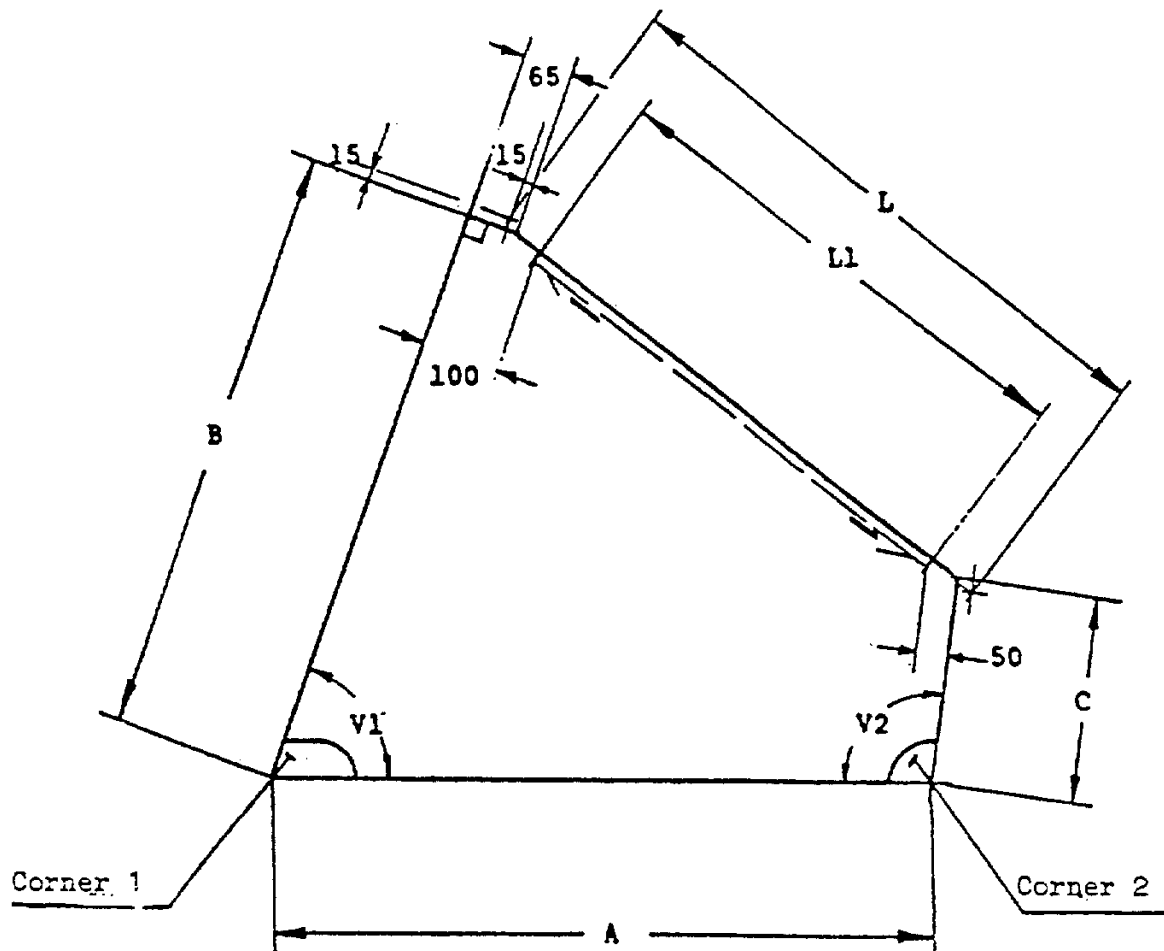
Brackets Type BB, BBK

Description

1. If $L < 800$ mm, then no flat bar will be required.
2. If $L > 800$ mm, one flat bar will be required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
3. The flat bars to be of the same quality as the plate of the bracket.
4. The above areas to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, to be 90×12 mm.
5. When flanging, corresponding flange areas is used where $t = 14$ mm rounded off to the nearest standard flange (199/500 20/- page 2).
6. For V-brackets the Z-dimension should not be less than $0.55 \times A$,
where A is the leg length at 90° for the bracket.
The Z-dimension is the distance measured perpendicular to the free edge to the root.
A suitable procedure is to close up the slot or increase the length of the leg.
7. All welds to be  4 where nothing else is shown on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Brackets Type BC, BCK

Type BC

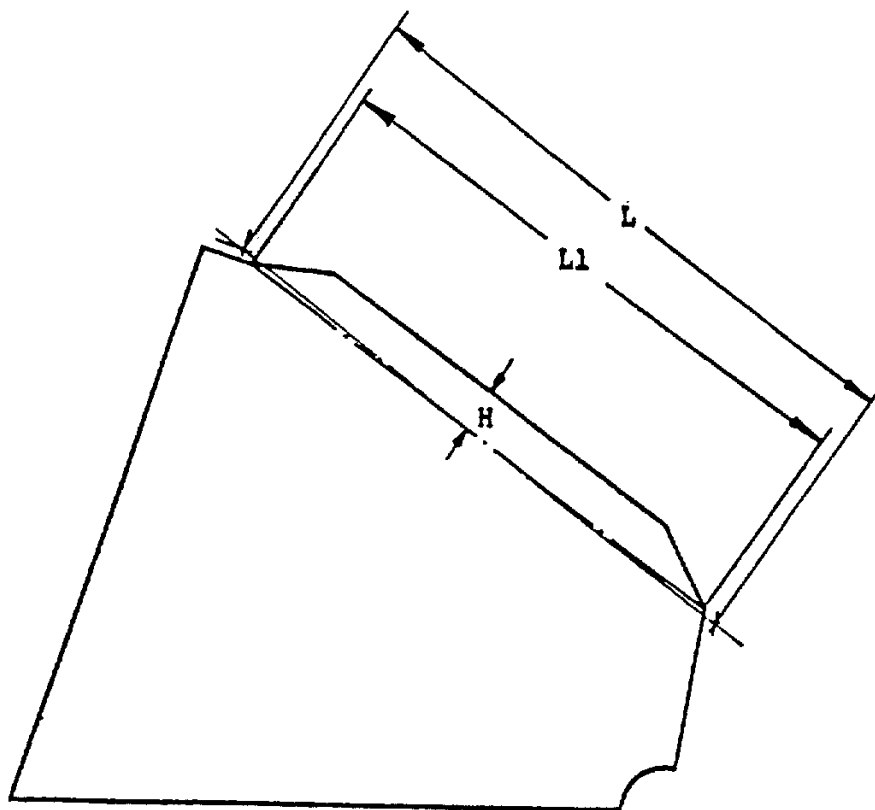


For dimension not given see page 1.

Alt	Cut-out	
	Corner 1	Corner 2
1	No cut-out	KS10
2	No cut-out	R50
3	No cut-out	R75
4	VU 100x50	KS10
5	VU 100x50	R50
6	VU 100x50	R75
7	VU 125x75	KS10
8	VU 125x75	R50
9	VU 125x75	R75

Brackets Type BC, BCK

Type BCK



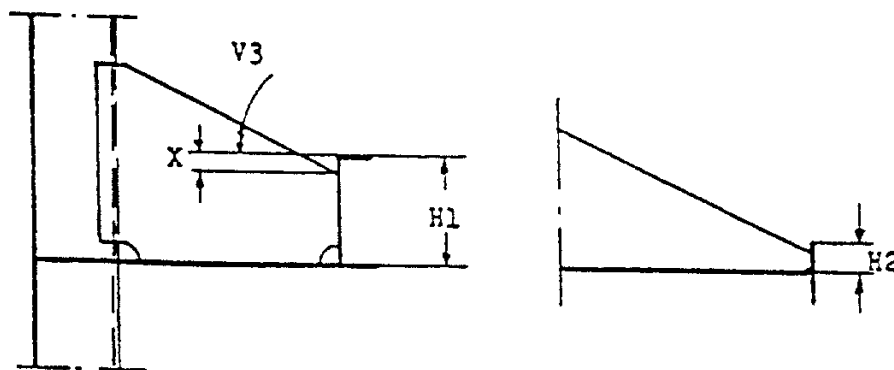
For dimension not given see page 1.

Brackets Type BC, BCK

Description

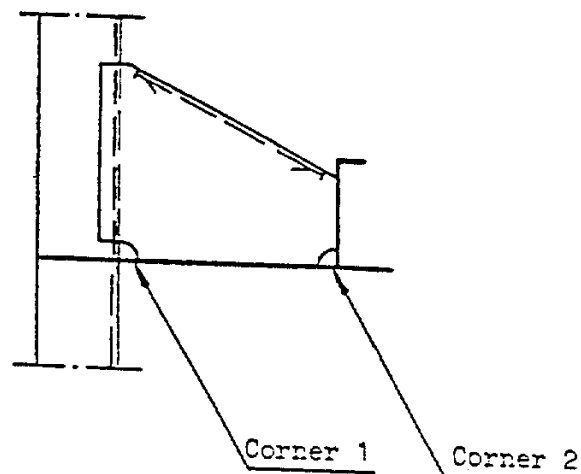
1. The A dimension is determined by the height of the profile and the depth of the web or the position of the stiffeners or by the position of the joint + 50 mm overlap.
2. The B-dimension is obtained from the class drawing.
3. The C-dimension is determined by H1 or H2.
4. If $V3 < 45^\circ$, set $X = 15$, otherwise 0.
5. If $L < 800$ mm, no flat bar is required.
6. If $L > 800$ mm, one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
7. The bar to be of the same quality as the plate of the bracket.
8. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, to be 90×12 .
9. When flanging, corresponding flange areas is used where $t = 14$ mm rounded off to the nearest standard flange (199/500 20/- page 2).
10. For V-brackets the Z-dimension should not be less than $0.55 \times A$,
where A is the leg length of the bracket at 90° .
The Z-measure is the distance from the original (lower left corner) of the bracket to free edge.
11. All welds to be  where nothing else is shown on the class drawing.
12. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Example of Assembly



Brackets Type BC, BCK

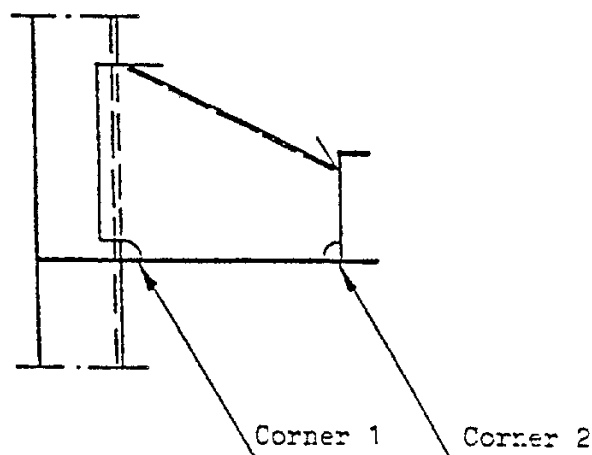
Connection for Type BC



Alt	Cut-out	
	Corner 1	Corner 2
1	No cut-out	KS10
2	No cut-out	R50
3	No cut-out	R75
4	VU 100x50	KS10
5	VU 100x50	R50
6	VU 100x50	R75
7	VU 125x75	KS10
8	VU 125x75	R50
9	VU 125x75	R75

Brackets Type BC, BCK

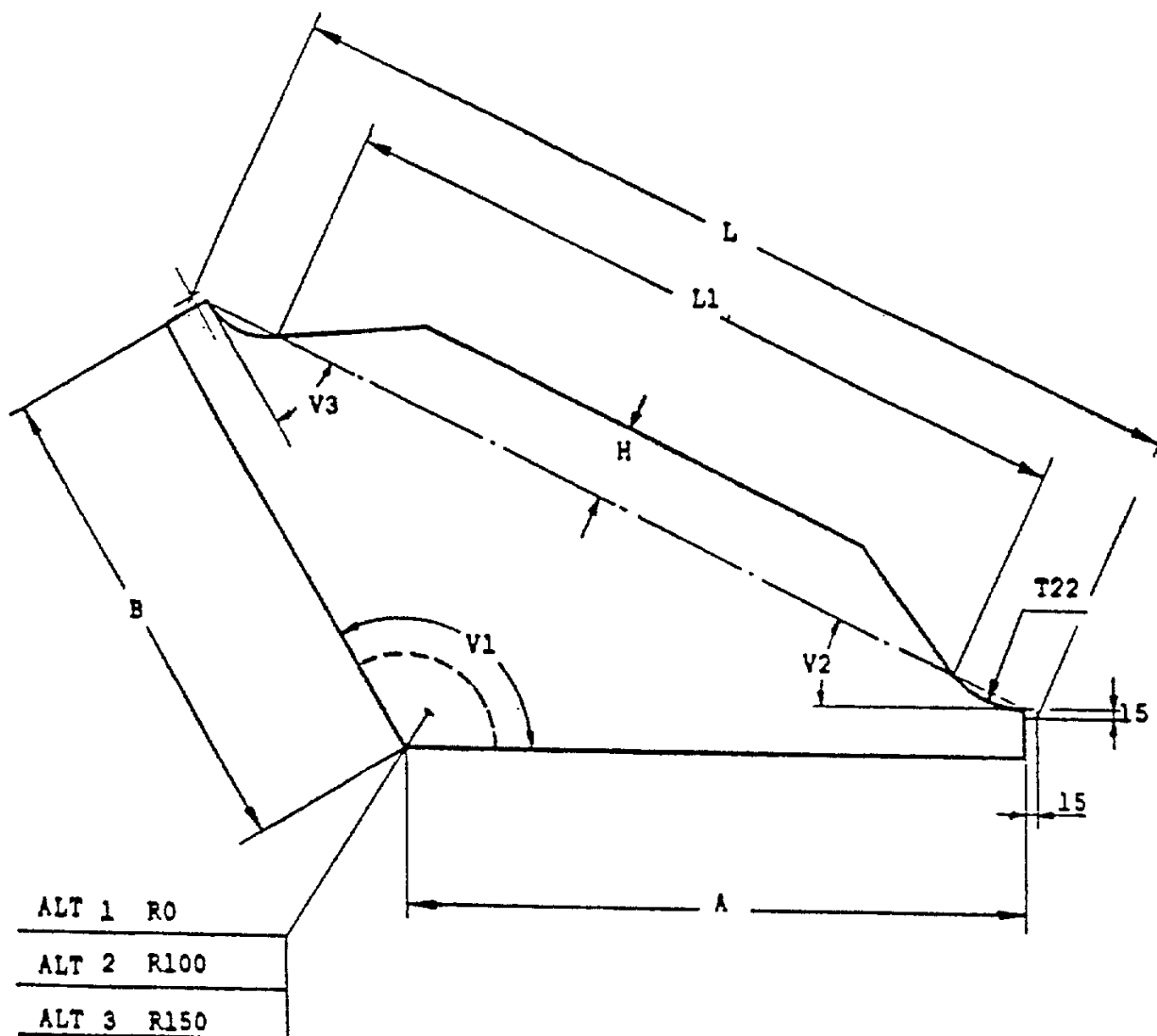
Connection for Type BCK



Alt	Cut-out	
	Corner 1	Corner 2
1	No cut-out	KS10
2	No cut-out	R50
3	No cut-out	R75
4	VU 100x50	KS10
5	VU 100x50	R50
6	VU 100x50	R75
7	VU 125x75	KS10
8	VU 125x75	R50
9	VU 125x75	R75

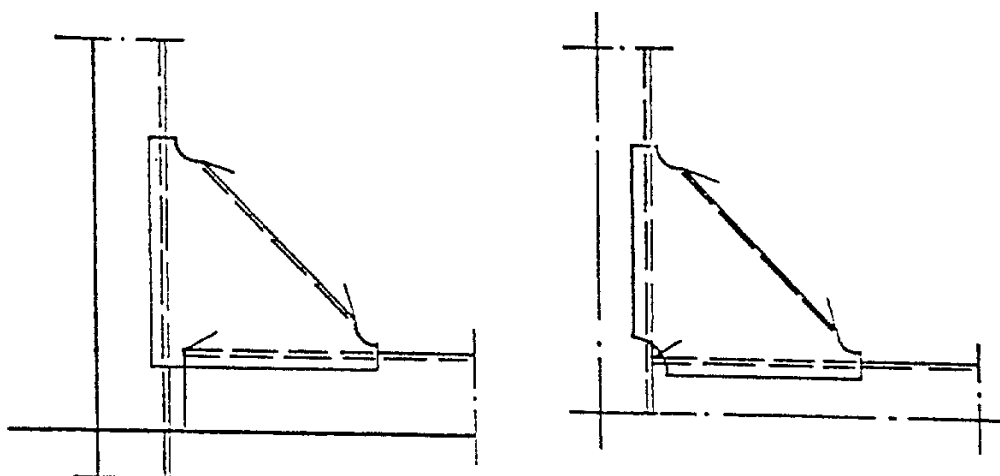
Brackets Type BDK

Type BDK



Brackets Type BDK

Connections for Type BDK



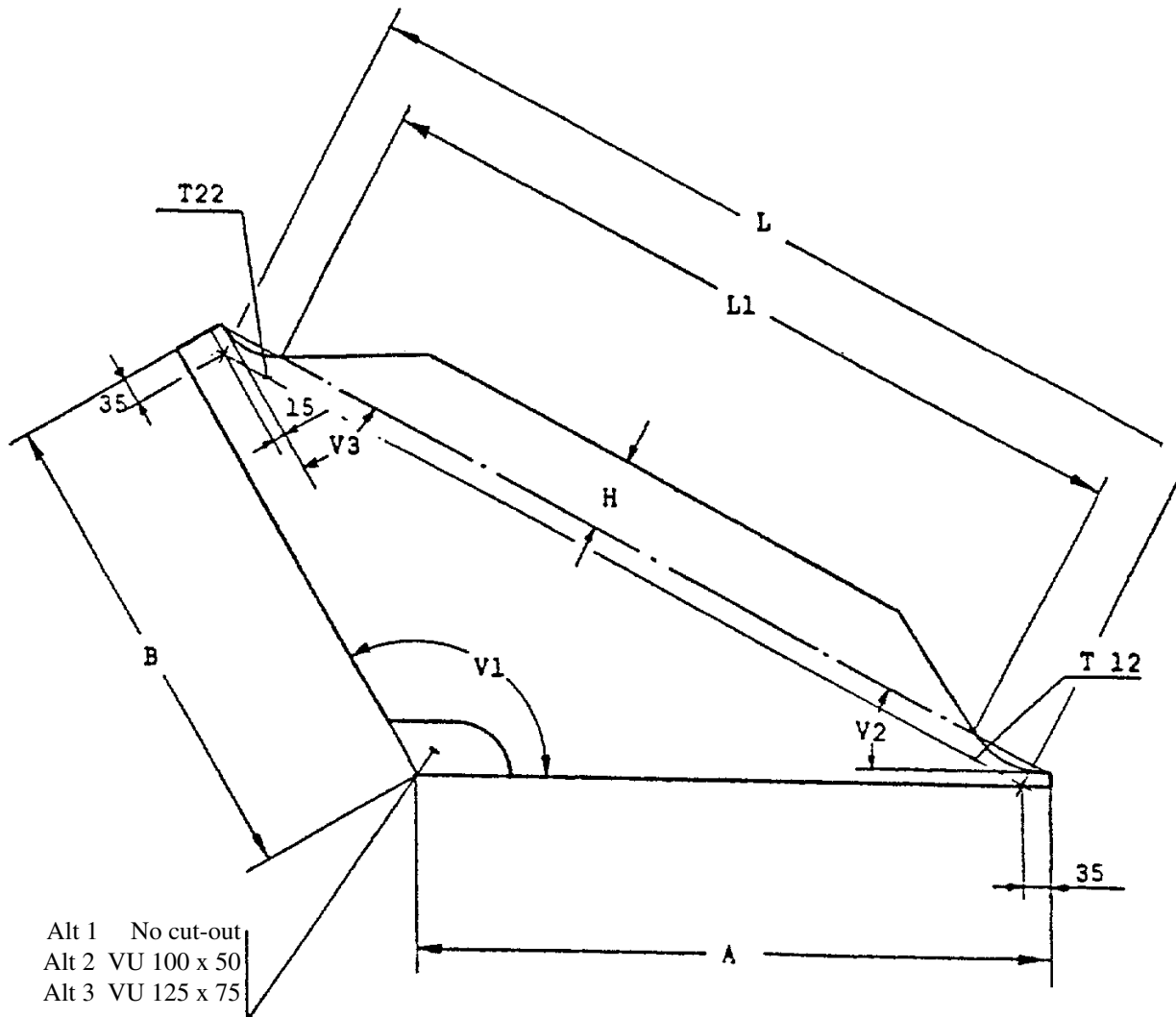
Alt.	Cut-out
1	R0
2	R100
3	R150

Brackets Type BDK

Description

1. The flange area where $t = 14 \text{ mm}$ must be $0.14 \times L \text{ cm}^2$, rounded off to the nearest standard flange (119/50020/-, page 2).
2. For V-brackets, the Z-dimension should not be less than $0.55 \times A$, where A is the leg length of the bracket at 90° .
The Z-dimension is the distance measured perpendicular to the free edge to the root.
A suitable procedure is to close-up the slot or increase the leg length.
3. All welds to be  where nothing else is shown on the class drawing.
4. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

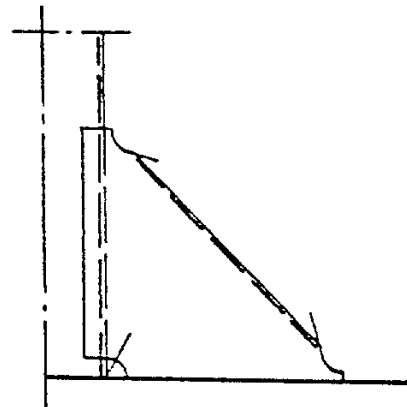
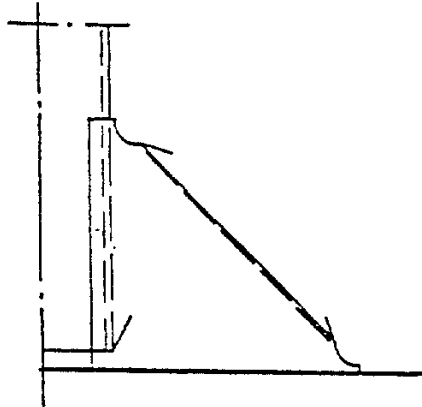
Brackets Type BEK



Leg lengths A and B to be selected from the A and B-values in the table on page 2.

Brackets Type BEK

Connections for Type BEK



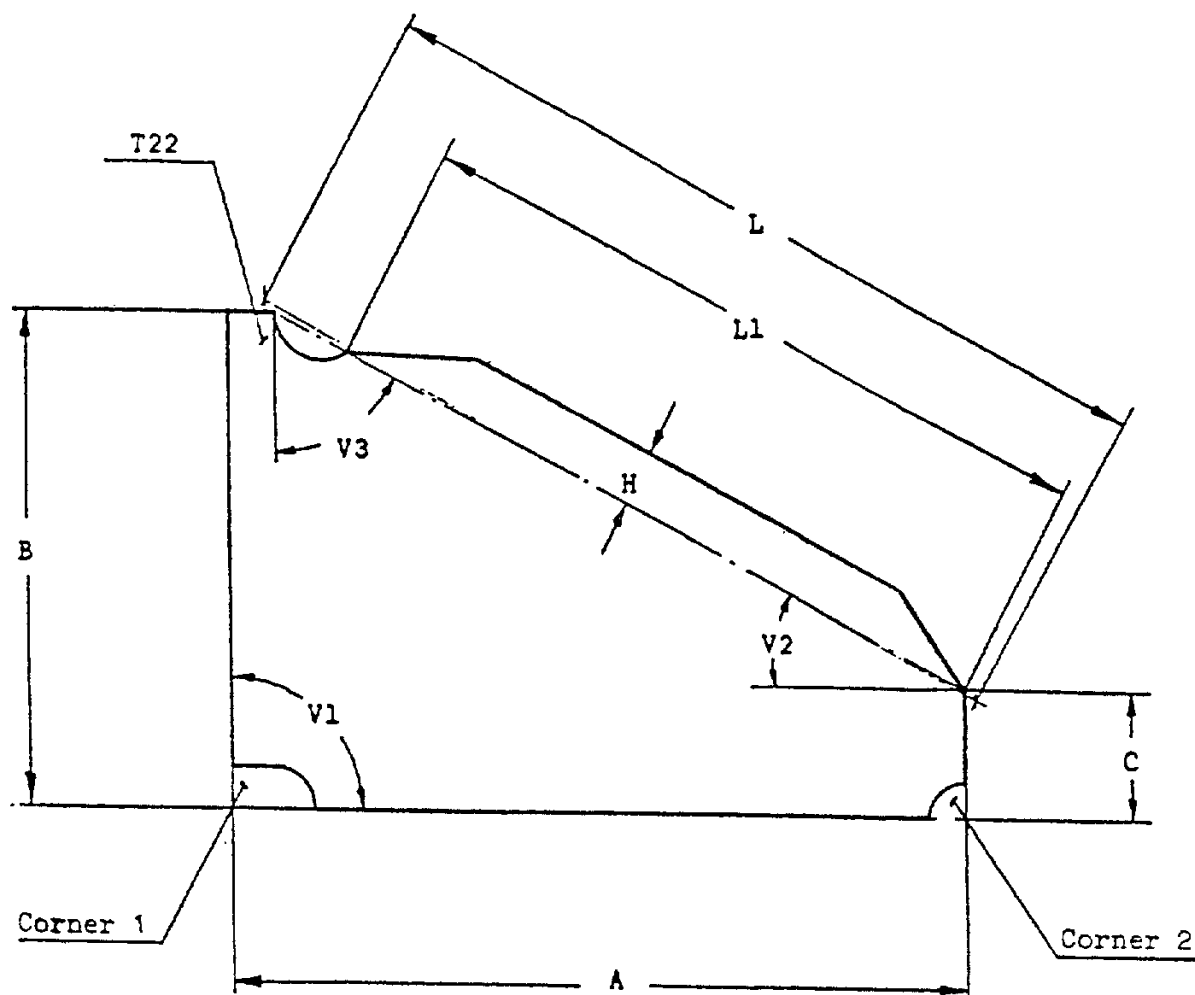
Alt.	Cut-out
1	No cut-out
2	VU 100x50
3	VU 125x75

Brackets Type BEK

Description

1. The flange area where $t = 14 \text{ mm}$ must be $0.14 \times L \text{ cm}^2$, rounded off to the nearest standard flange (119/50020/-, page 2).
2. For V-brackets, the Z-dimension should not be less than $0.55 \times A$, where A is the leg length of the bracket at 90° .
The Z-dimension is the distance measured perpendicular to the free edge to the root.
A suitable procedure is to close-up the slot or increase the leg length.
3. All welds to be  where nothing else is shown on the class drawing.
4. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Brackets Type BFK



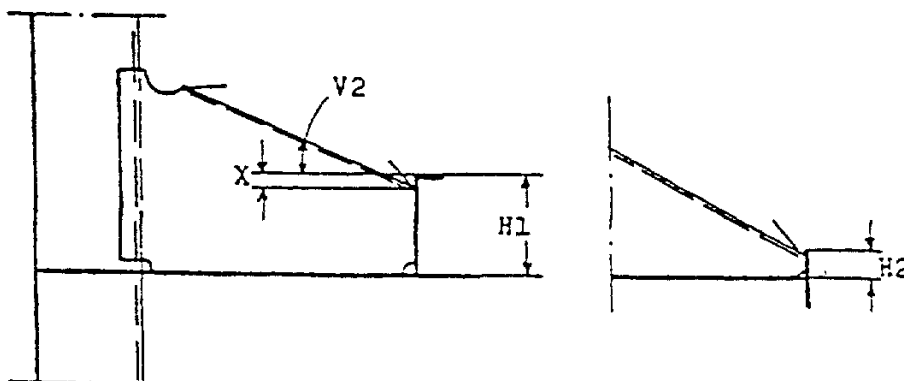
Alt	Cut-out	
	Corner 1	Corner 2
1	No cut-out	KS10
2	No cut-out	R50
3	No cut-out	R75
4	VU 100x50	KS10
5	VU 100x50	R50
6	VU 100x50	R75
7	VU 125x75	KS10
8	VU 125x75	R50
9	VU 125x75	R75

Brackets Type BFK

Description

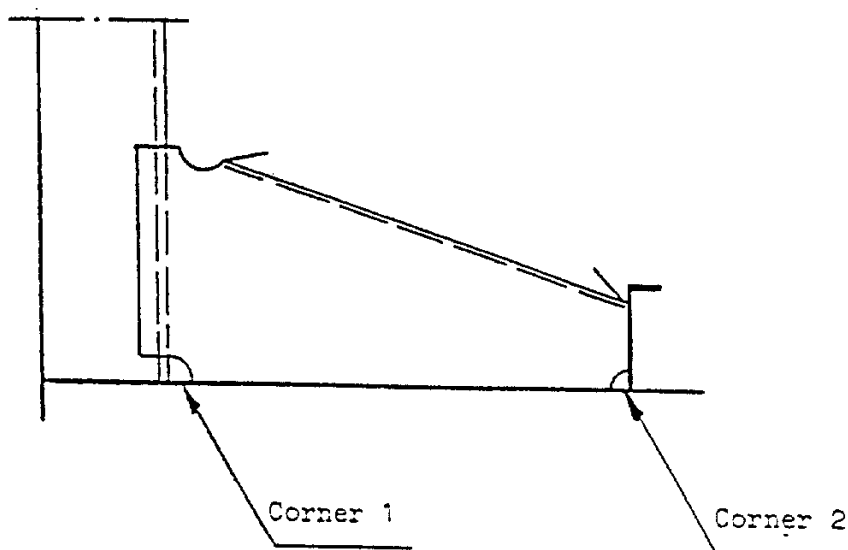
1. The A dimension is determined by the height of the profile and the depth of the web or the position of the stiffeners or by the position of the joints + 50 mm overlap.
2. The B-dimension is obtained from the class drawing.
3. The C-dimension is determined by H1 or H2.
4. If $V2 < 45$, set $X = 15$, otherwise 0.
5. The flange area where $t = 14$ mm must be $0.14 \times L$ cm², rounded off to the nearest standard flange (119/50020/-, page 2).
6. For V-brackets, the Z-dimension should not be less than $0.55 \times A$, where A is the leg length of the bracket at 90°. The Z-dimension is the distance measured perpendicular to the free edge to the root. A suitable procedure is to close-up the slot or increase the leg length.
7. All welds to be  where nothing else is shown on the class drawing.
8. From the class drawing is also obtained: plate thickness, quality of material and erection weld.

Example of Assembly



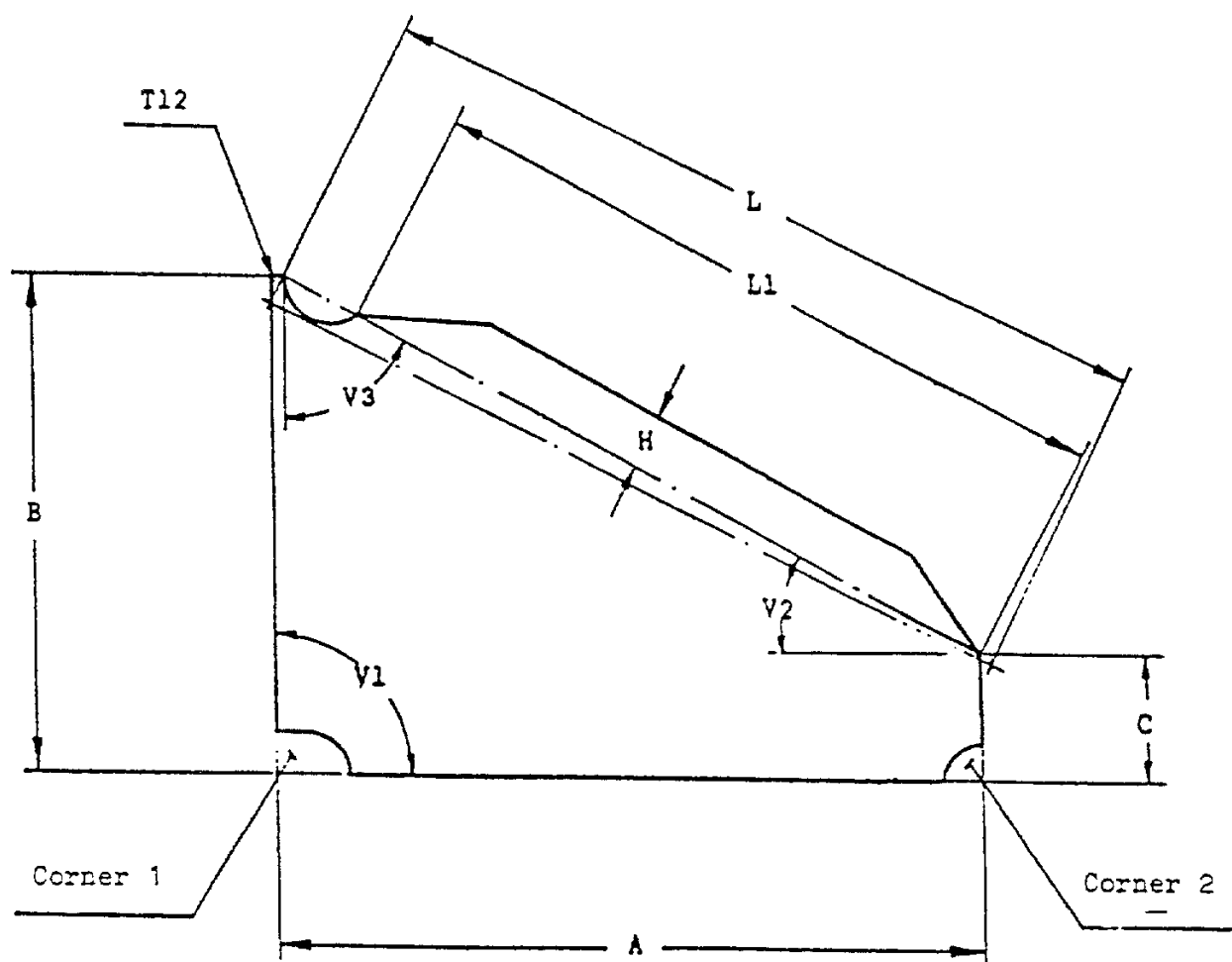
Brackets Type BFK

Connections for Type BFK



Alt	Cut-out	
	Corner 1	Corner 2
1	No cut-out	KS10
2	No cut-out	R50
3	No cut-out	R75
4	VU 100x50	KS10
5	VU 100x50	R50
6	VU 100x50	R75
7	VU 125x75	KS10
8	VU 125x75	R50
9	VU 125x75	R75

Brackets Type BGK



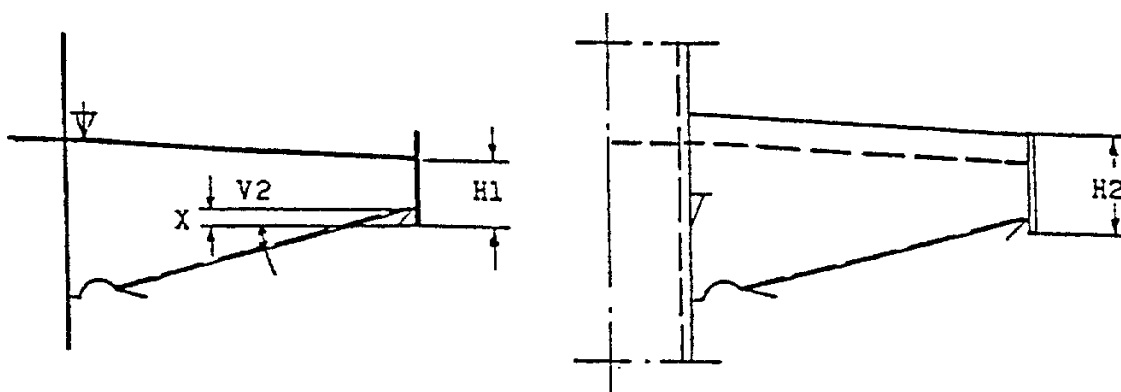
Alt	Cut-out	
	Corner 1	Corner 2
1	No cut-out	KS8
2	No cut-out	R50
3	No cut-out	R75
4	VU 100x50	KS8
5	VU 100x50	R50
6	VU 100x50	R75
7	VU 125x75	KS8
8	VU 125x75	R50
9	VU 125x75	R75

Bracket Type BGK

Description

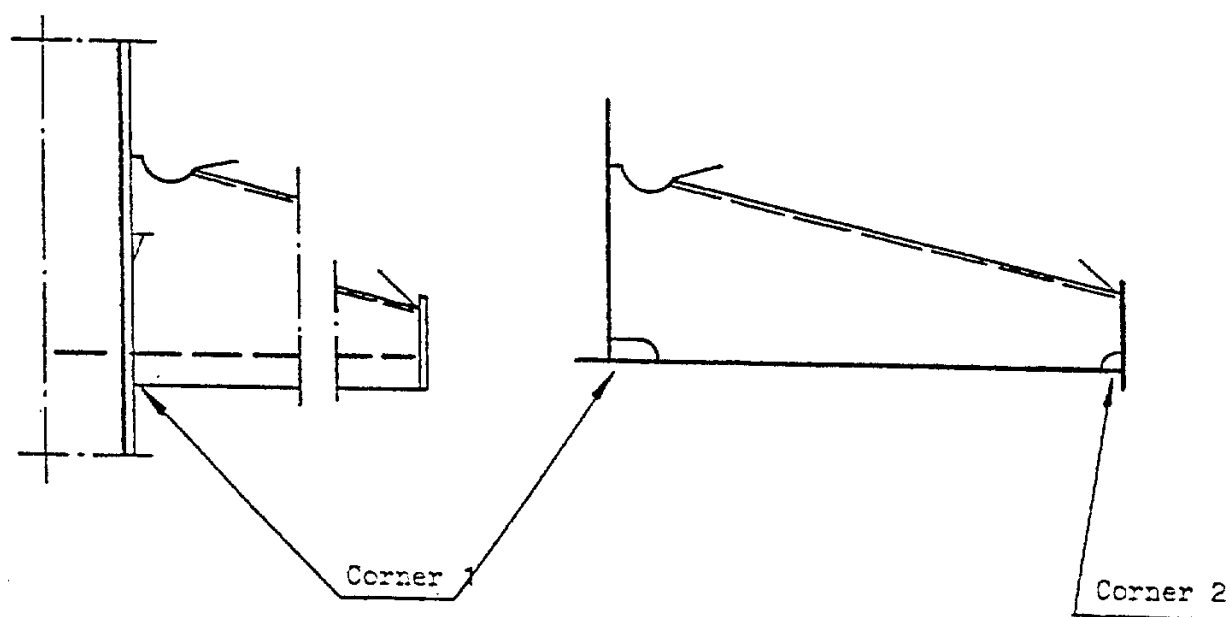
1. The A-dimension is determined by the width of the platform, minus - in certain cases - a projecting plate edge.
2. The B-dimension is obtained from the class drawing.
3. The C-dimension is determined by H1 or H2.
4. If $V2 < 45^\circ$, set $X = 15$, otherwise 0.
5. The flange areas where $t = 14$ mm must be $0.14 \times L$ cm², rounded off to the nearest standard flange (119/50020/-, page 2).
6. For V-brackets, the Z-dimension should not be less than $0.55 \times A$, where A is the leg length of the bracket at 90° .
The Z-dimension is the distance measured perpendicular to the free edge to the root.
A suitable procedure is to close-up the slot or increase the leg length.
7. All welds to be  where nothing else is shown on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Example of Assembly



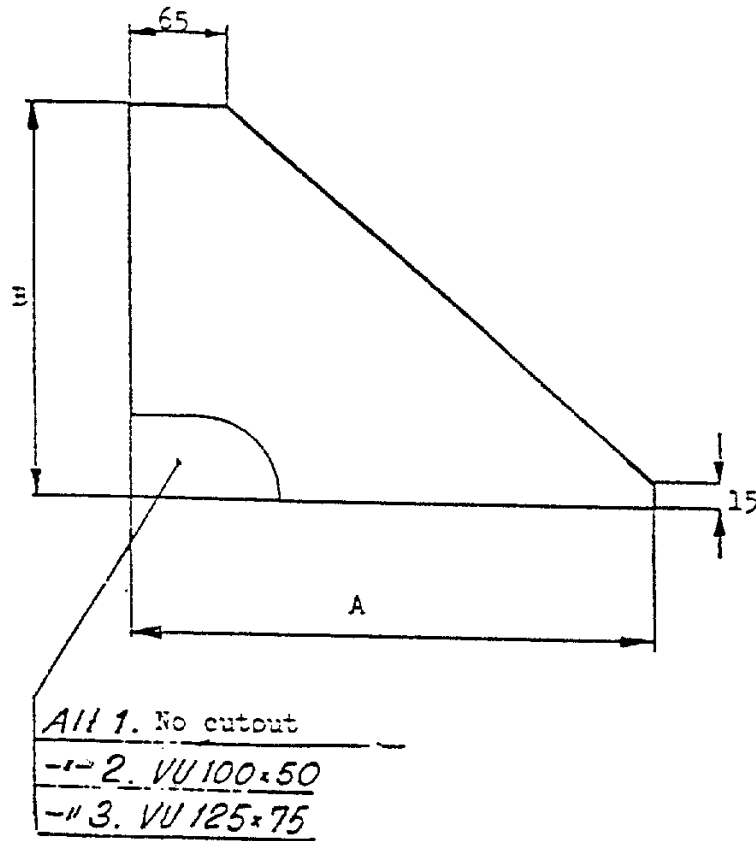
Bracket Type BGK

Connections for Type BGK

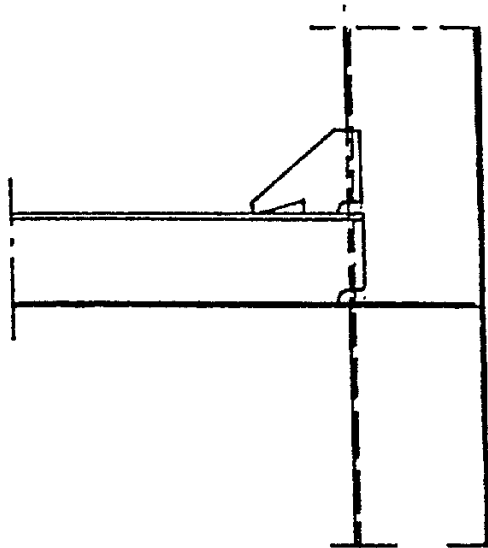


Alt	Cut-out	
	Corner 1	Corner 2
1	No cut-out	KS8
2	No cut-out	R50
3	No cut-out	R75
4	VU 100x50	KS8
5	VU 100x50	R50
6	VU 100x50	R75
7	VU 125x75	KS8
8	VU 125x75	R50
9	VU 125x75	R75

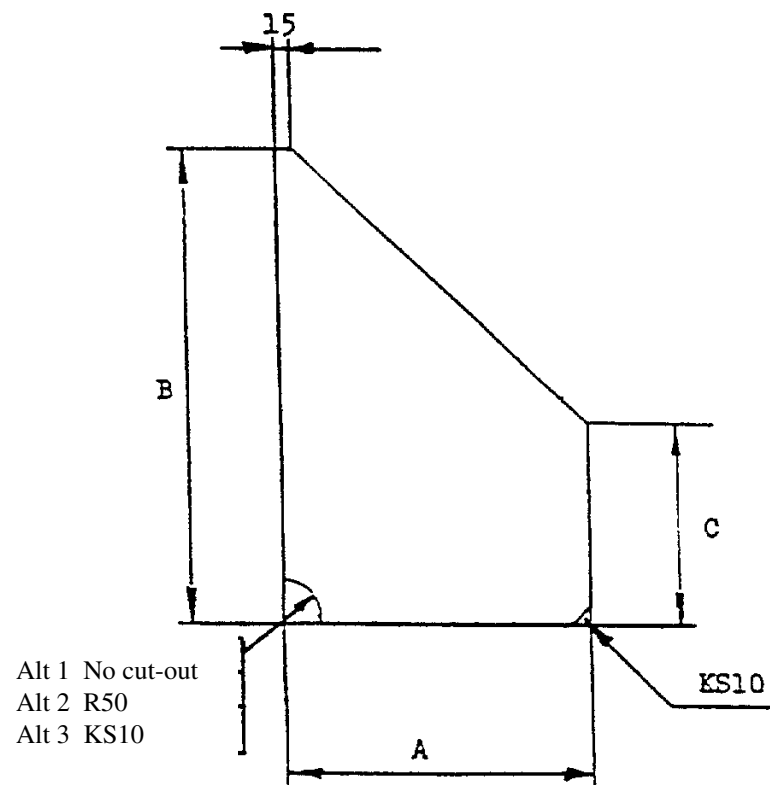
Bracket Type BAA



Bracket Type BAA



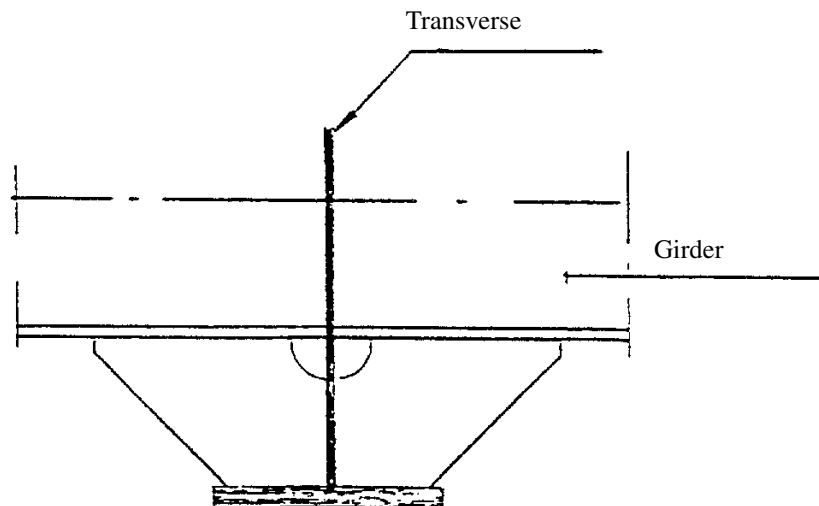
Bracket Type BCB



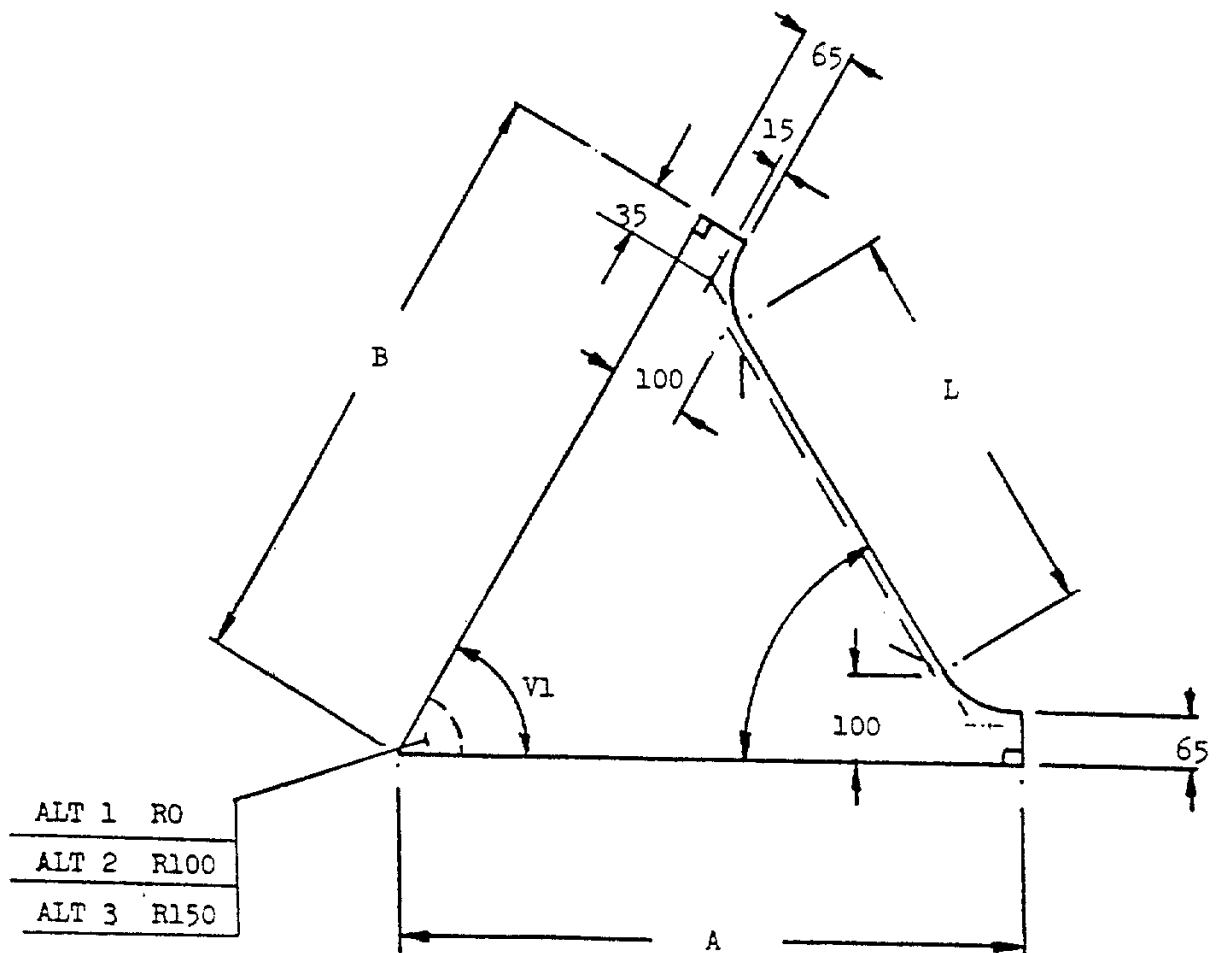
Bracket Type BCB

Description

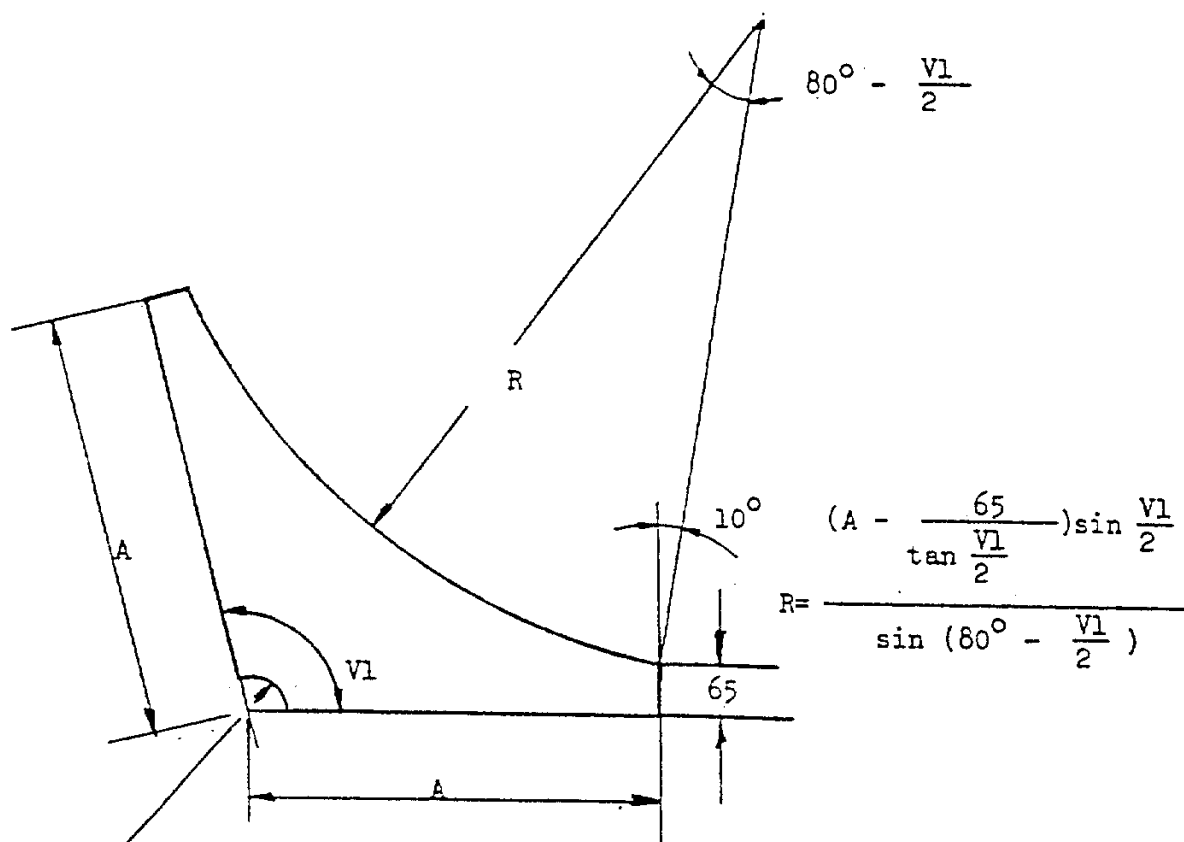
- 1 The B-measure is equal to the one given on the class drawing.
- 2 The A-measure will be equal to the difference between the depth of the web of the transverse and the total height of the girder.
- 3 The C-measure is determined by the breadth of the flange of the transverse, C is equal to half the breadth of the flange minus 30 mm.



Bracket Type BBA



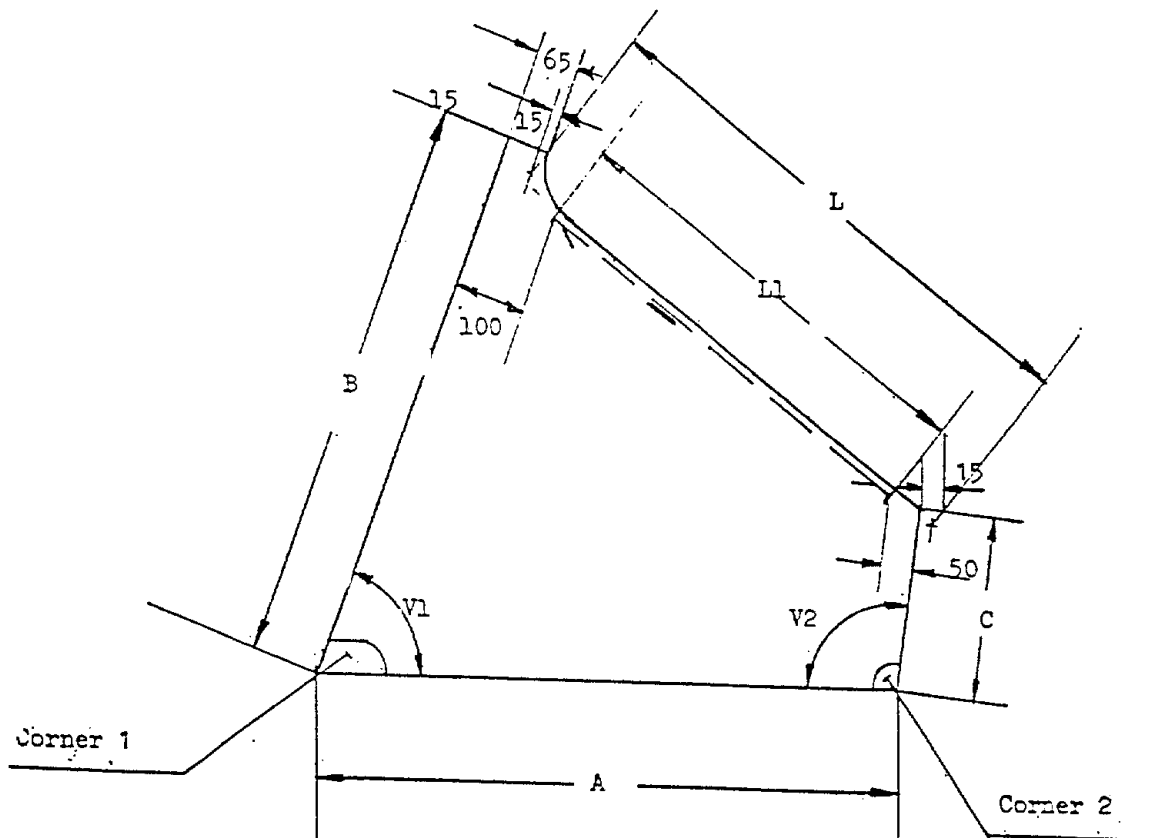
Bracket Type BBR



$$R = \frac{(A - \frac{65}{\tan \frac{V1}{2}}) \sin \frac{V1}{2}}{\sin (80^\circ - \frac{V1}{2})}$$

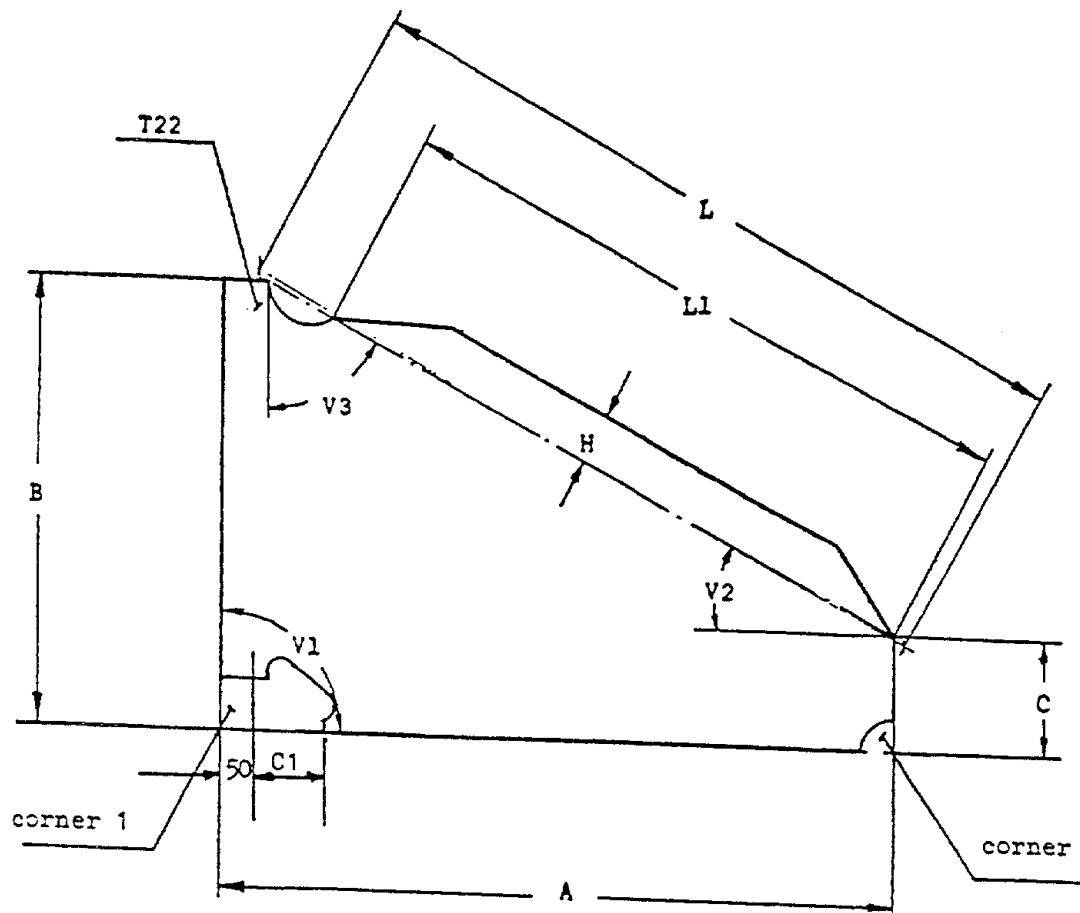
ALT 1	RO
ALT 2	R100
ALT 3	R150

Bracket Type BCA



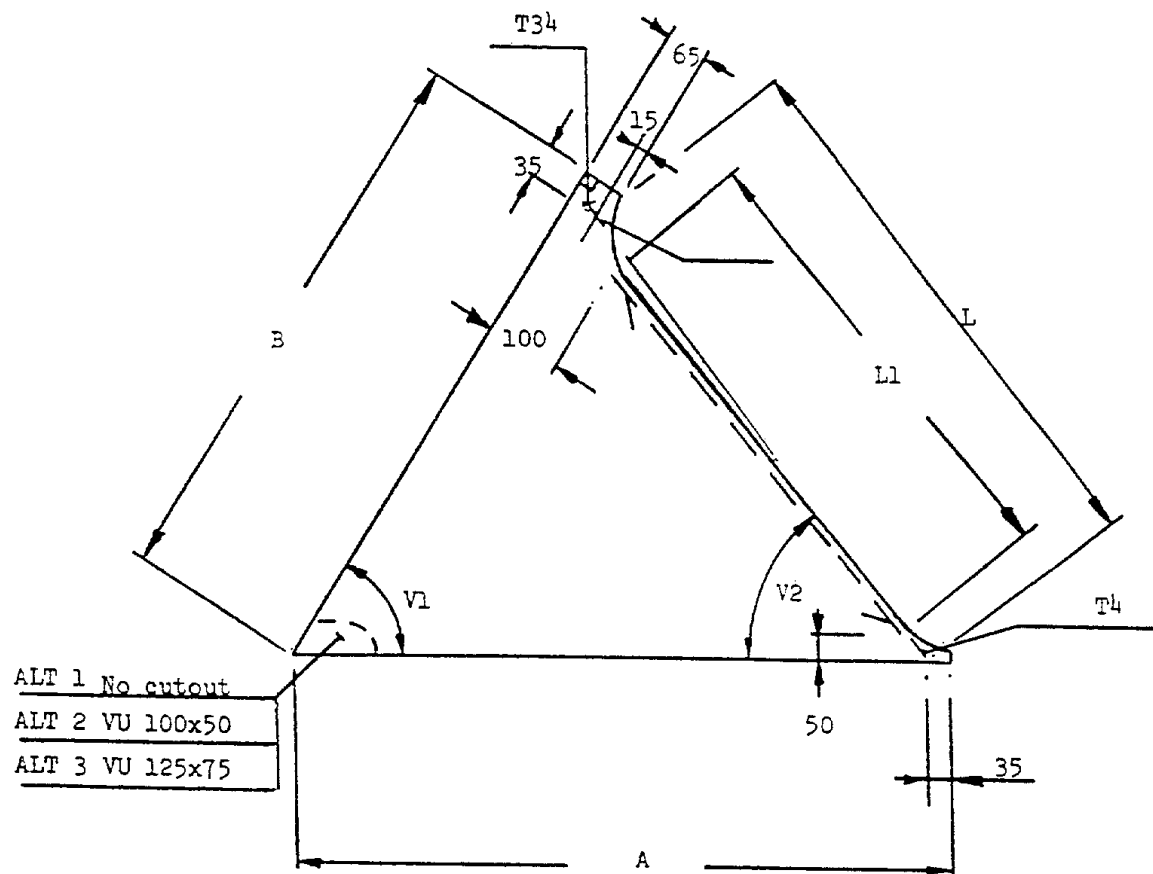
Alt	Cutouts	
	Corner 1	Corner 2
1	No cut-out	KS10
2	No cut-out	R50
3	No cut-out	R75
4	VU 100x50	KS10
5	VU 100x50	R50
6	VU 100x50	R75
7	VU 125x75	KS10
8	VU 125x75	R50
9	VU 125x75	R75

Bracket Type BHK

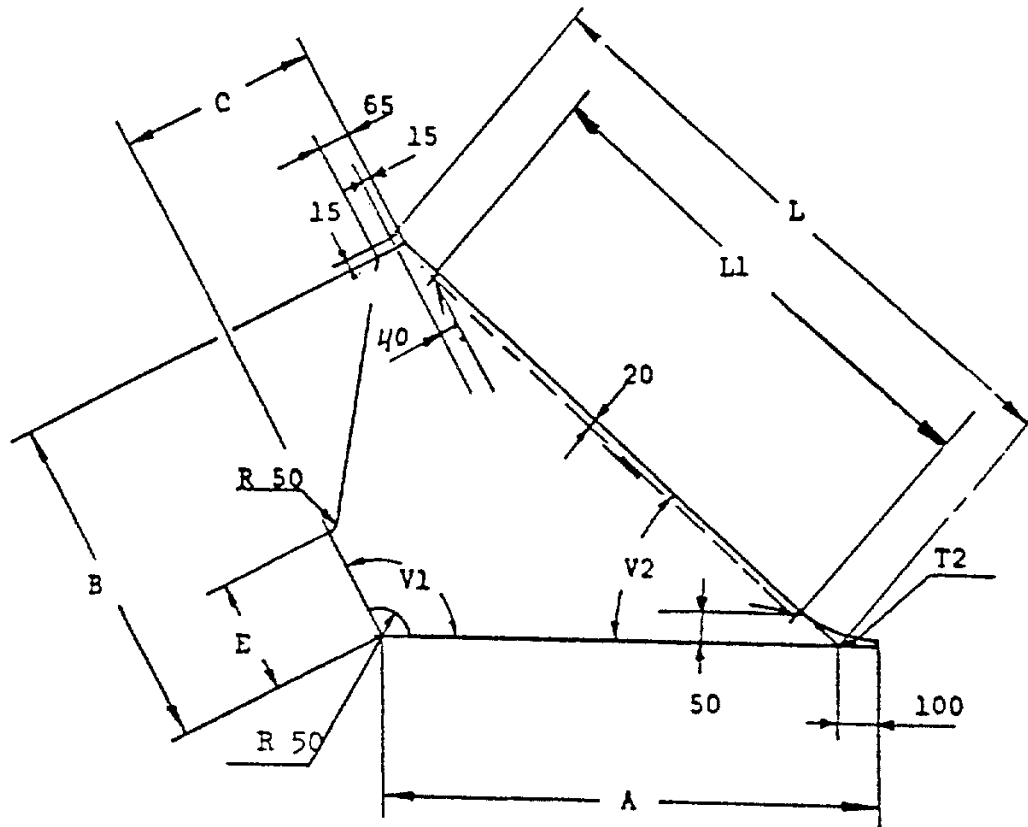


Alt.	Cutouts	
	Corner 1	Corner 2
4	VOB 150	KS10
5	VOB 150	R50
6	VOB 150	R75

Bracket Types BAB

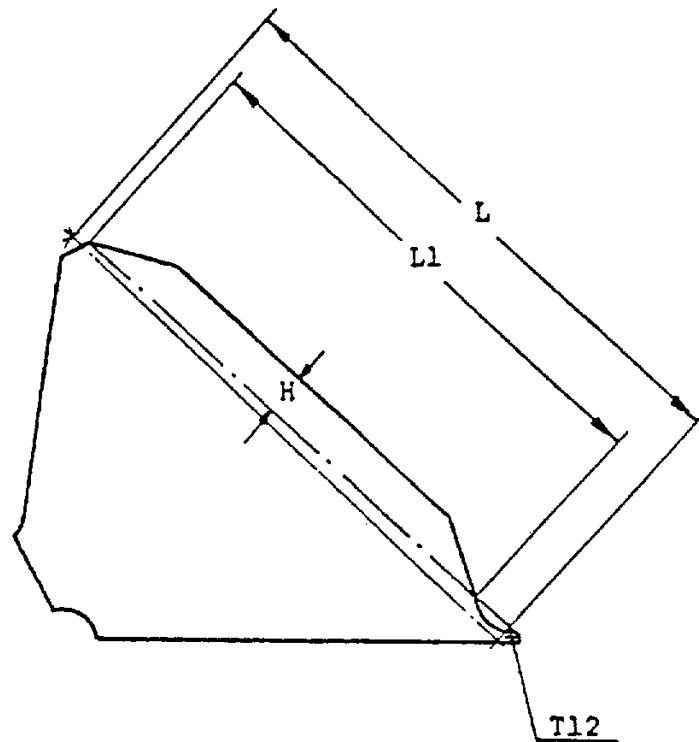


Brackets Types EA, EAK

Type EA

Brackets Types EA, EAK

Type EAK



Brackets Types EA, EAK

Description

1. The A-, B- and C-dimensions are determined by H1 and V1.
2. The length of the overlap of the bracket is measured in line with the profile and must be $1.5 \times H1$.
- 2a. At bevel cut profile the length of the overlap is given by the measure H2. See fig 2.
3. The min. width of the bracket where the profile ends is measured at right angles to the profile s and must be $0.5 \times H1$ and $0.5 \times H2$ respectively. V2 and V3 must not, however, exceed 75° . If $V2 \leq 45^\circ$ set $X = 15$, otherwise 0.
- 3a. M1 must be given at least 176 irrespective of the real measure to seam.
4. If $L \leq 800$ mm, one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
5. The flat bar to be of the same quality as that of the plate of the bracket.
6. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, to be 90×12 .
7. When flanging, corresponding flange areas, where $t = 14$ mm to be used rounded off to the nearest standard flange (199/500 20/- page 2).
8. All welds to be  where nothing else is given on the class drawing.
9. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Note 1-3 apply only if nothing else is given on the class drawing.

Example of Assembly

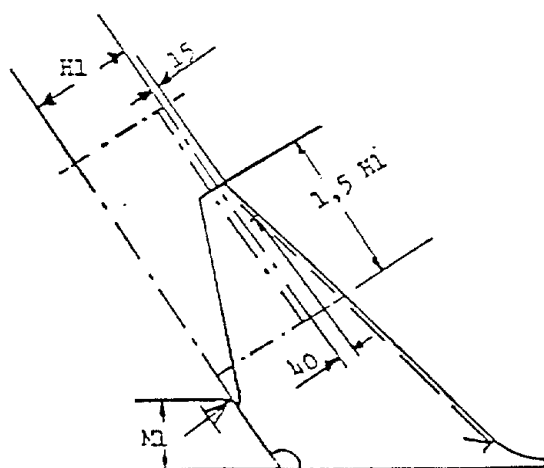


Fig 1

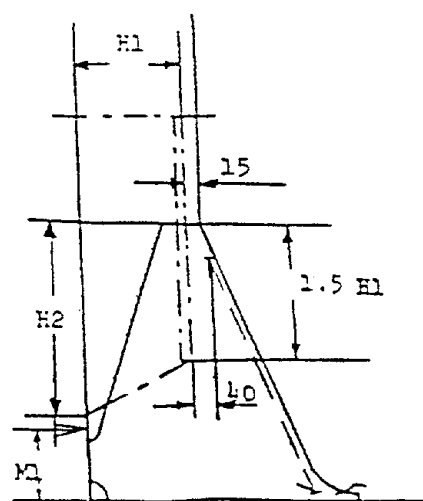
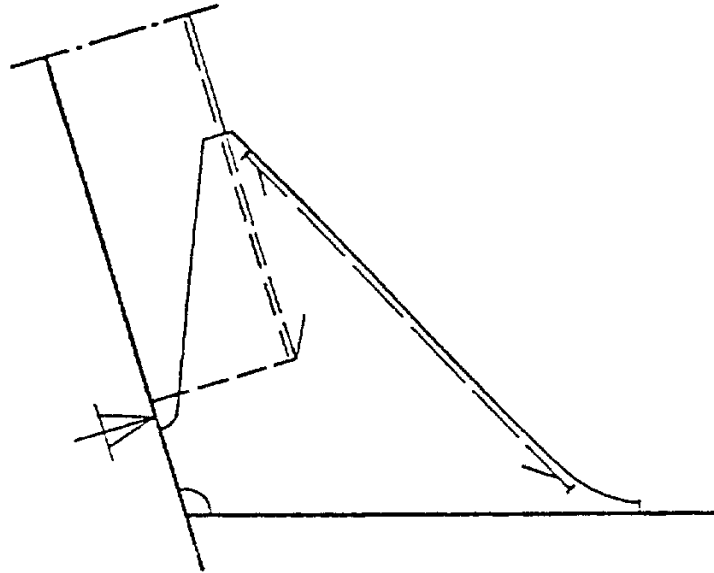


Fig 2

Brackets Types EA, EAK

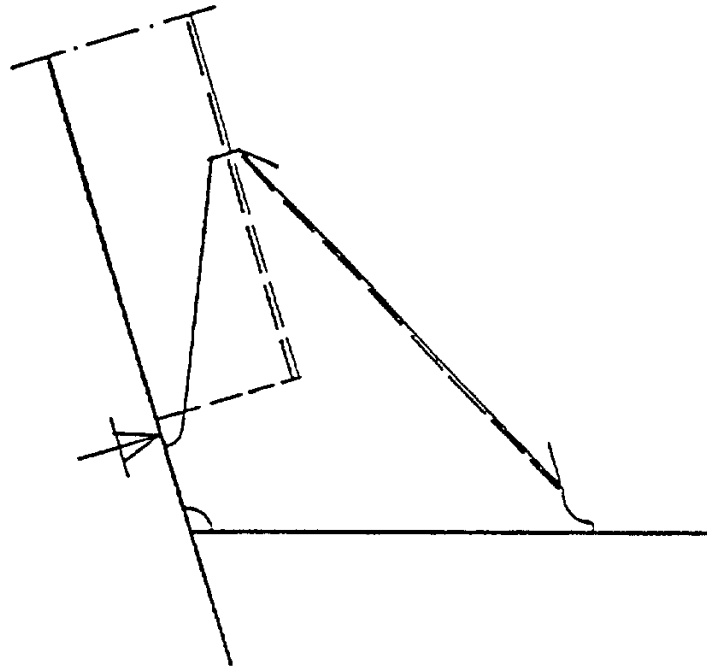
Connections for Type EA (40)



Alt.	Cut-out
2	R50

Brackets Types EA, EAK

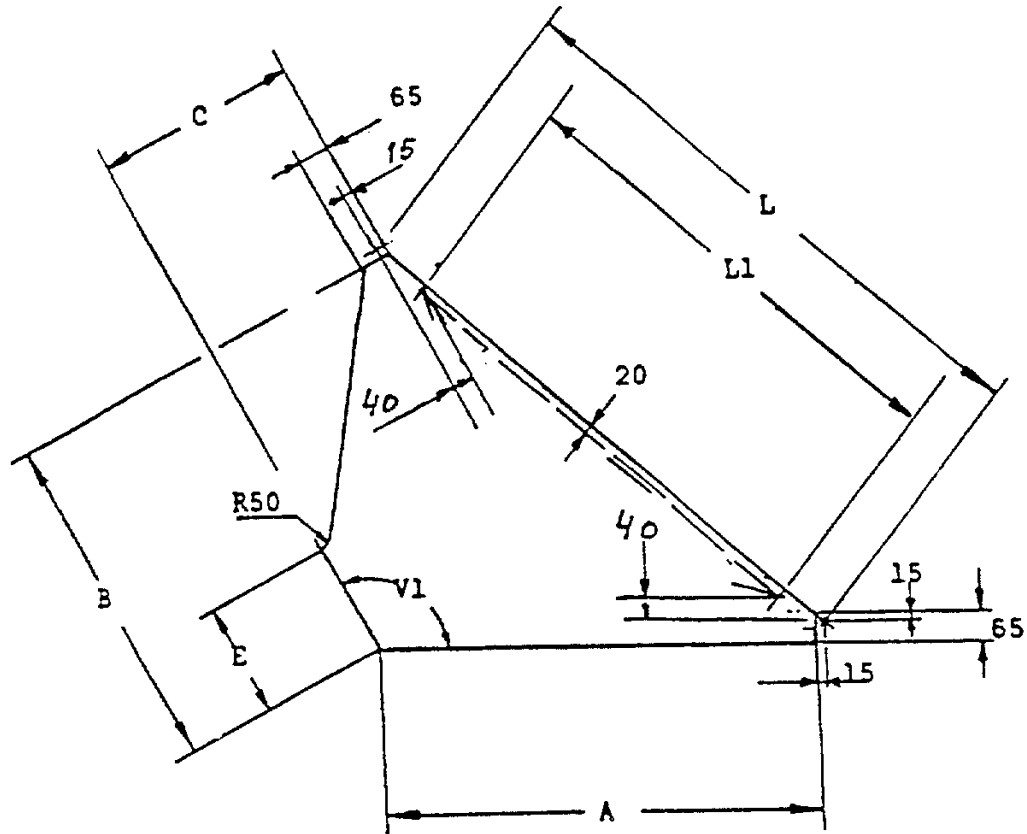
Connections for Type EAK



Alt.	Cut-out
2	R50

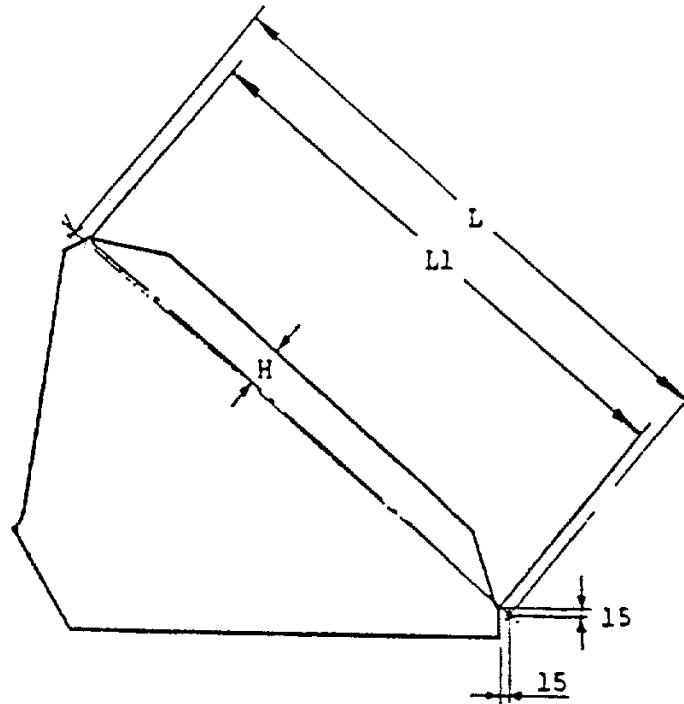
Brackets Types EB, EBK

Type EB



Brackets Types EB, EBK

Type EBK



For dimensions not shown see page 1.

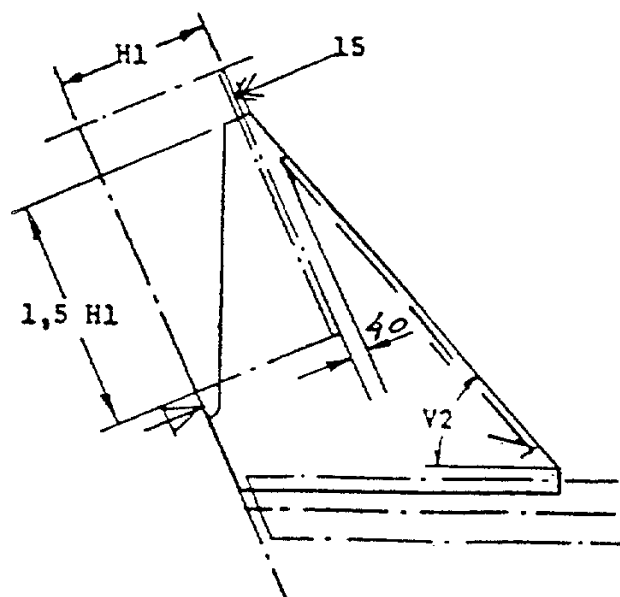
Brackets Types EB, EBK

Description

1. The A-, B- and C-dimensions are determined by H1 and V1.
2. The length of the overlap of the bracket is measured in line with the profile and must be $1.5 \times H1$.
3. The min width of the bracket where the profile ends, is measured at right angles to the profile and must be $0.5 \times H1$, V2 should, however, not exceed 75° .
4. If $L \leq 800$ mm, then one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
5. The flat bar to be of the same quality as that of the plate of the bracket.
6. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, is 90×12 mm.
7. When flanging, corresponding flange areas, where $t = 14$ mm to be used rounded off to the nearest standard flange (199/500 20/- page 2).
8. All welds to be  where nothing else is given on the class drawing.
9. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

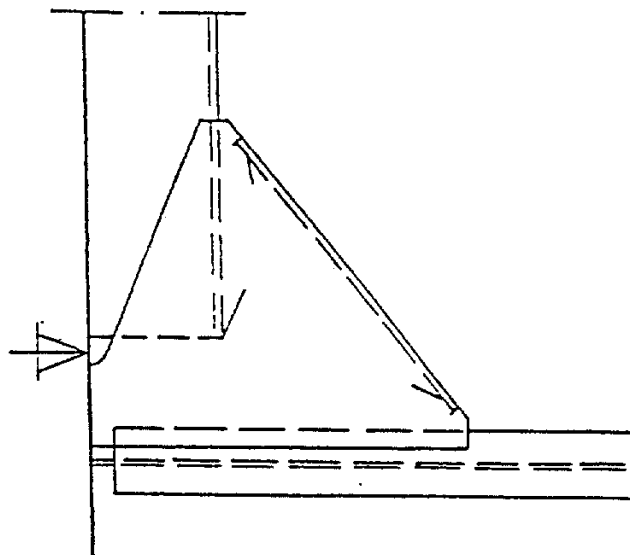
Note 1-3 apply only if nothing else is given on the class drawing.

Example of Assembly



Brackets Types EB, EBK

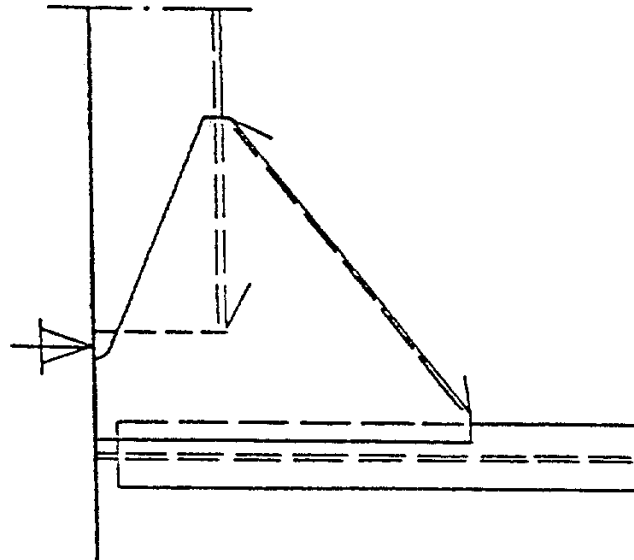
Connection for Type EB



Alt.	Cut-out
1	No cut-out

Brackets Types EB, EBK

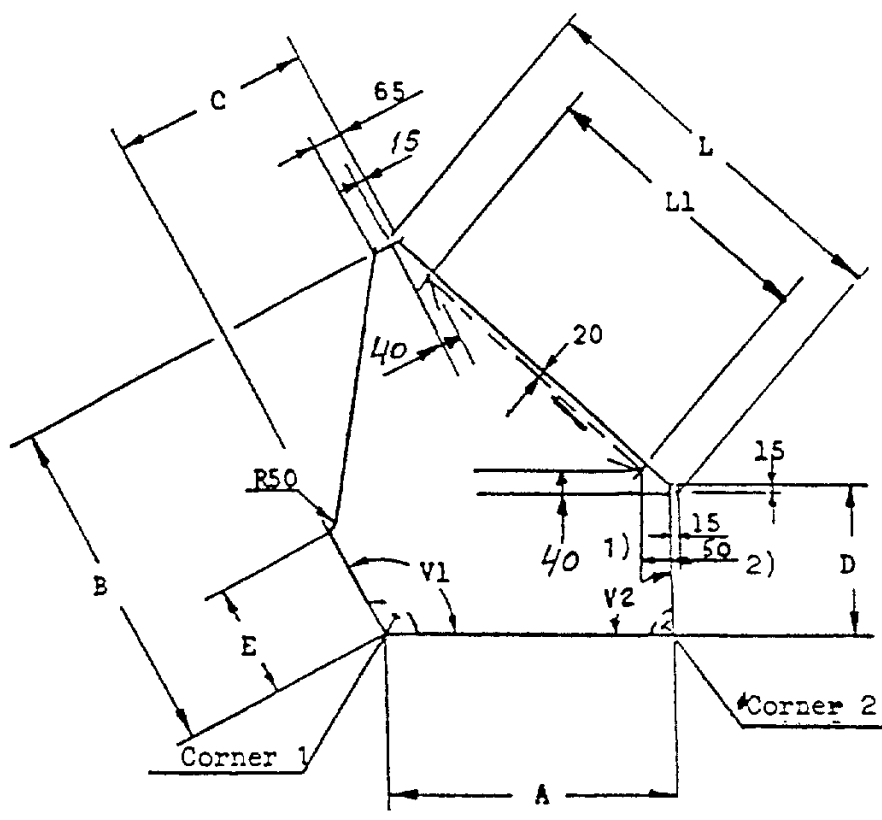
Connection for Type EE



Alt.	Cut-out
1	No cut-out

Brackets Types EC, ECK

Type EC

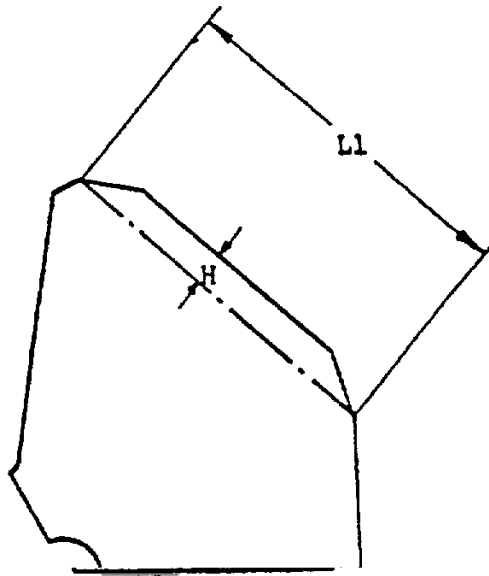


Alt.	Cut-out	
	Corner 1	Corner 2
1	No cut-out	No cut-out
2	R50	No cut-out
3	R50	KS10
4	R50	R50
5	R50	R75

- 1) Valid for alt 1-3.
2) Valid for alt 4 alt 5.

Brackets Types EC, ECK

Type ECK



For dimensions not shown see page 1.

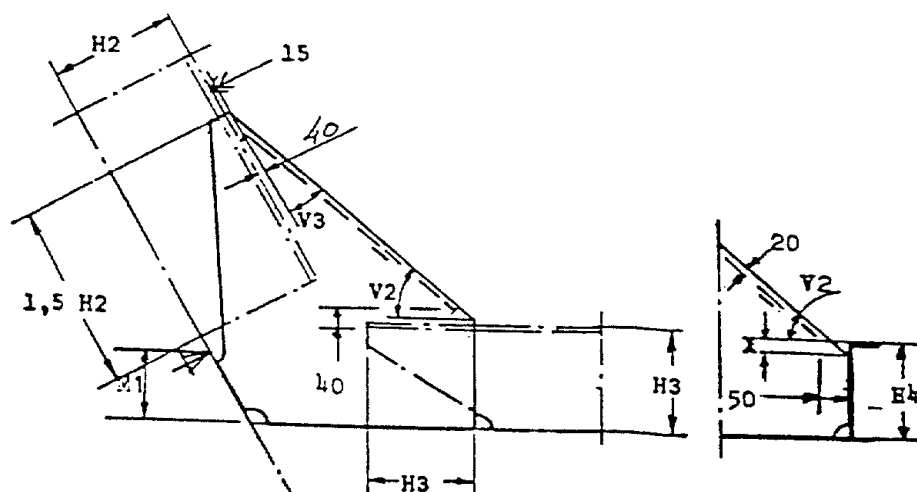
Brackets Types EC, ECK

Description

1. The A-, B- and C-dimensions are determined by H2 and V1.
- 1a The D-dimension is determined by H3 or H4.
2. The overlap length of the bracket is measured in line with the profile and must be $1.5 \times H2$ and $1 \times H3$, respectively.
3. The min width of the bracket where the profile ends, is measured at right angles to the profile and must be $0.5 \times H2$ and $0.5 \times H3$, respectively.
V2 and V3 must, however, not exceed 75° .
If $V2 \geq 45^\circ$, set $X = 15$, otherwise 0.
- 3a M1 must be given at least min 176, irrespective of the real distance to the seam.
4. If $L \leq 800$ mm, then one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
5. The flat bar to be of the same quality as that of the plate of the bracket.
6. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, is 90×12 .
7. When flanging, corresponding flange areas, where $t = 14$ mm to be used rounded off to the nearest standard flange (199/500 20/- page 2).
8. All welds to be  4 where nothing else is given on the class drawing.
9. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

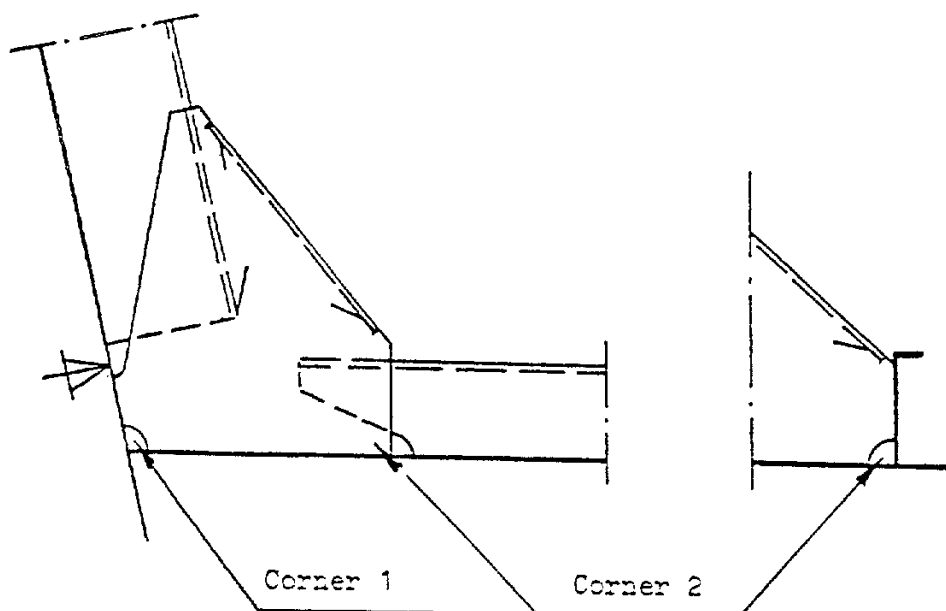
Note 1-3 apply only if nothing else is given on the class drawing.

Example of Assembly



Brackets Types EC, ECK

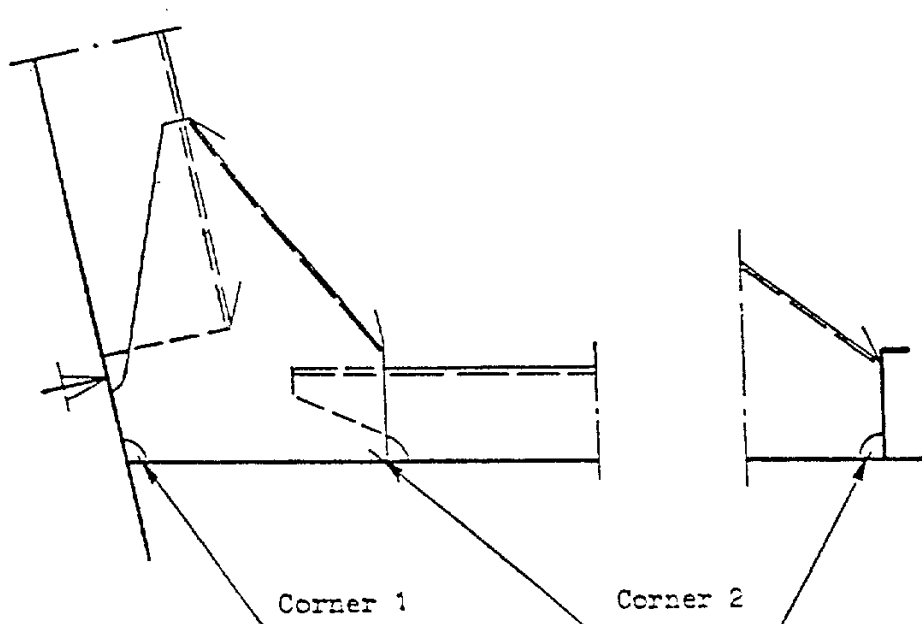
Connections for Type EC



Alt.	Cut-out	
	Corner 1	Corner 2
1	0	0
2	R50	0
3	R50	KS8
4	R50	R50
5	R50	R75

Brackets Types EC, ECK

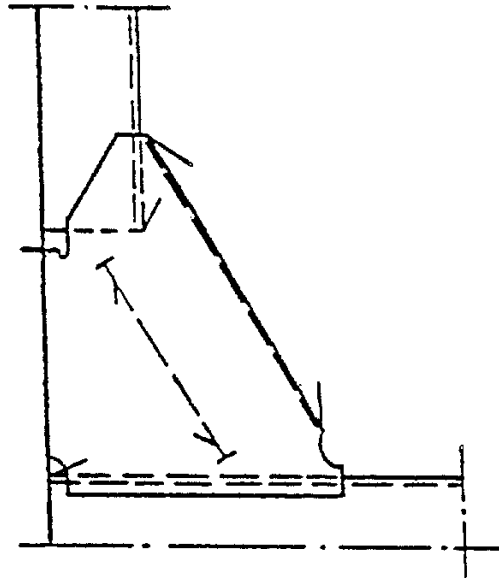
Connections for Type EC



Alt.	Cut-out	
	Corner 1	Corner 2
1	0	0
2	R50	0
3	R50	KS8
4	R50	R50
5	R50	R75

Brackets Types EGK

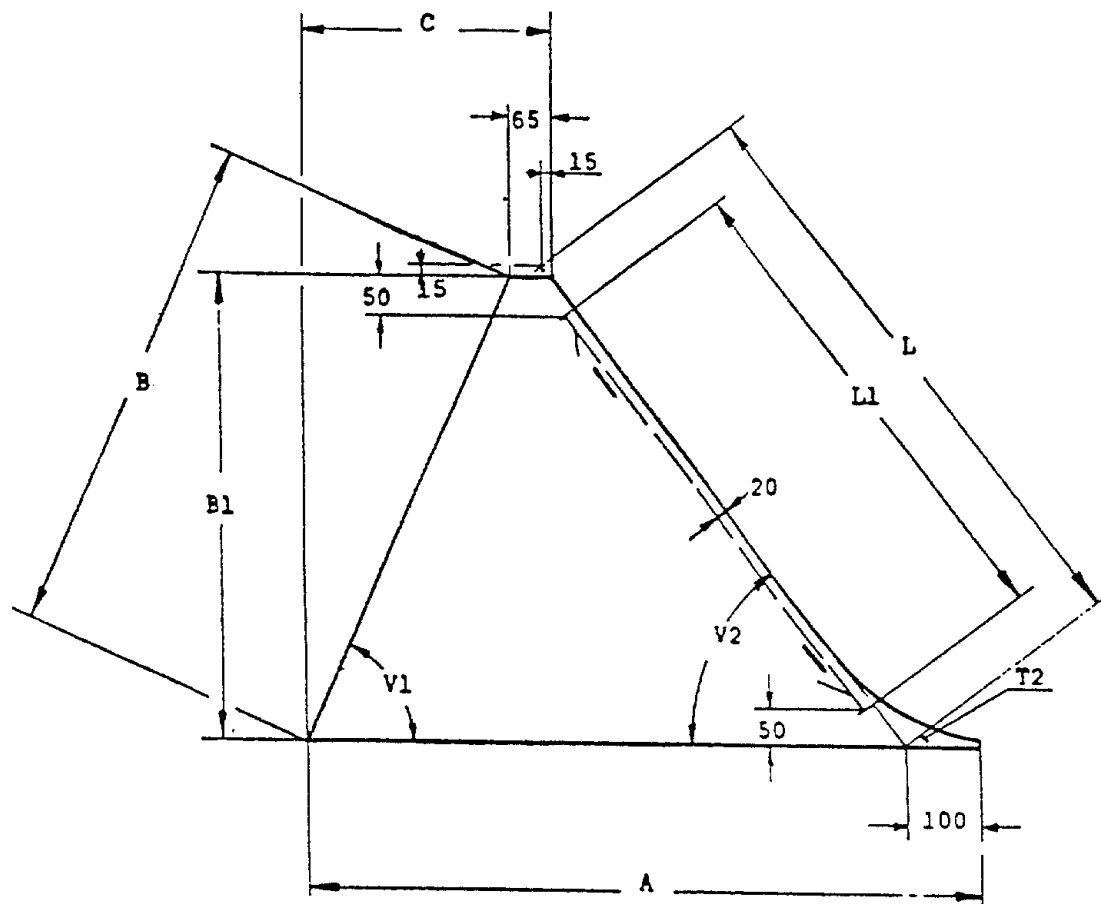
Connections for Type EGK



Alt.	Cut-out
1	No cut-out
2	VU 125 x 75

Brackets Types GA, GAK

Type GA

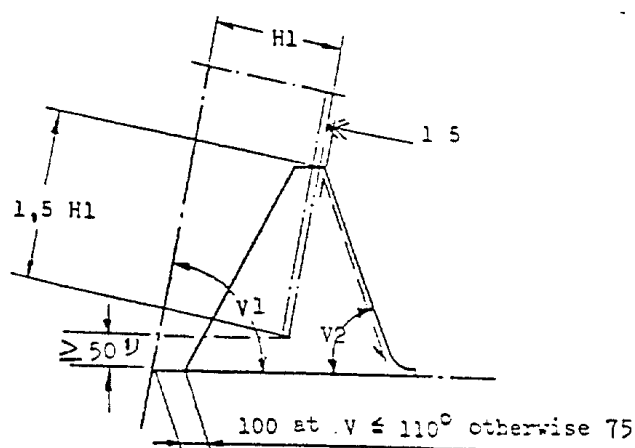


Brackets Types GA, GAK

Description

1. The A-, B- and C-dimensions are determined by H1 and V1.
- 1a The C-dimension is measured parallel with the A-dimension.
2. The overlap length of the bracket is measured in line with the profile and must be $1.5 \times H1$.
3. The min width of the bracket where the profile ends, is measured at right angles to the profile and must be $0.5 \times H21$.
V2 must not, however, exceed 75° .
4. If $L \leq 800$ mm, then one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
5. The flat bar to be of the same quality as that of the plate of the bracket.
6. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, is 90×12 mm.
7. When flanging, corresponding flange areas, where $t = 14$ mm to be used rounded off to the nearest standard flange (199/500 20/- page 2).
8. All welds to be  where nothing else is given on the class drawing.
9. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

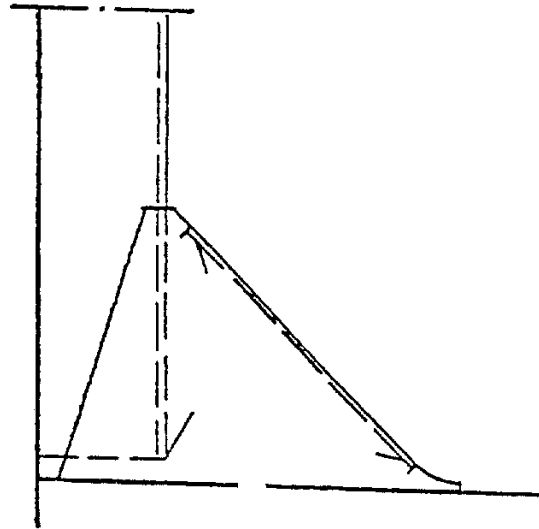
Note 1-3 apply only if nothing else is given on the class drawing.



1) If < 50 , use bracket type BA.

Brackets Types GA, GAK

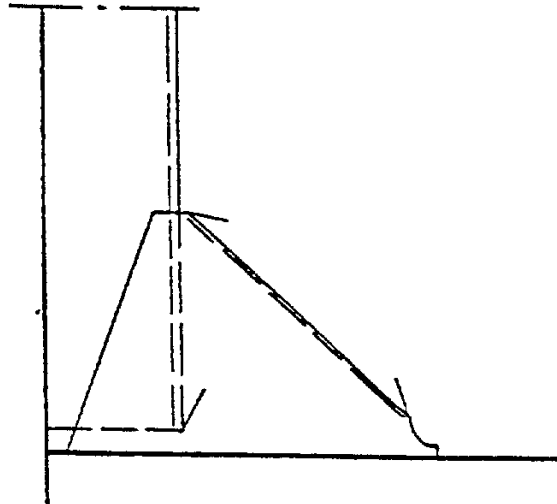
Connections for Type GA



Only one alternative (=1).

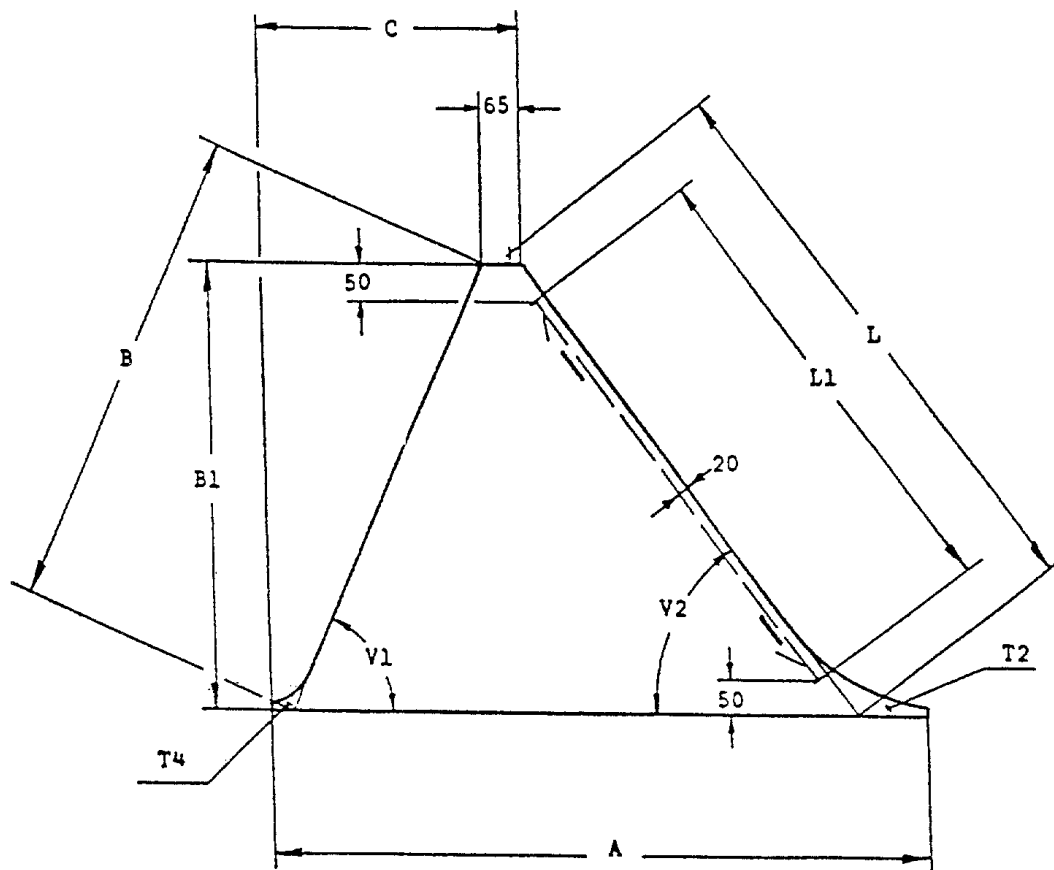
Brackets Types GA, GAK

Connection for Type GAK



Only one alternative (=1).

Brackets Type GB



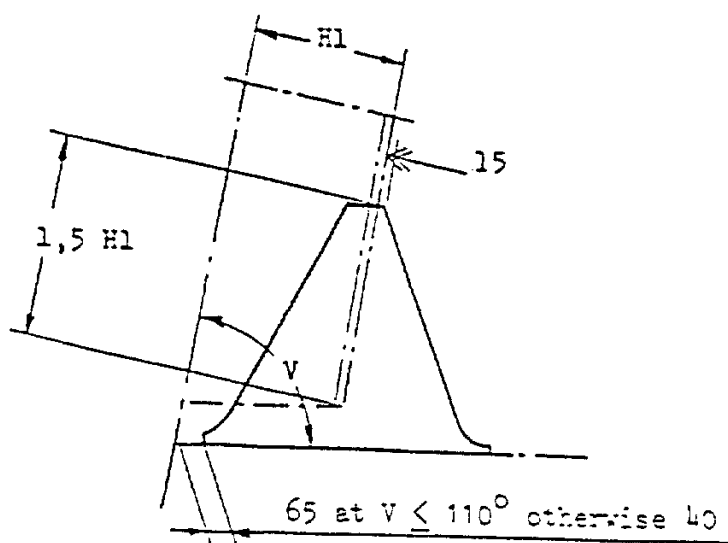
Brackets Type GB

Description

1. The A-, B- and C-dimensions are determined by H1 and V1.
- 1a The C-dimension is measured parallel with the A-dimension.
2. The overlap length of the bracket is measured in line with the profile and must be $1.5 \times H1$.
3. The min width of the bracket where the profile ends, is measured at right angles to the profile and must be $0.5 \times H1$.
V2 must not, however, exceed 75° .
4. If $L \geq 800$ mm, then one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
5. The flat bar to be of the same quality as that of the plate of the bracket.
6. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, is 90×12 mm.
7. All welds to be  4 when nothing else is given on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

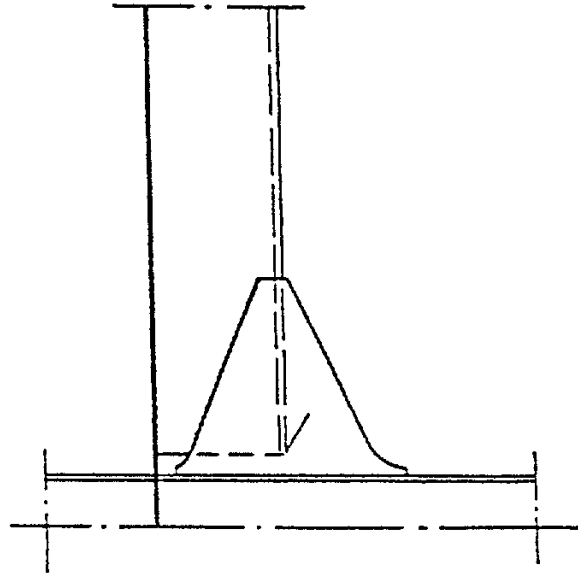
Note 1-3 apply only if nothing else is given on the class drawing.

Example of Assembly



Brackets Type GB

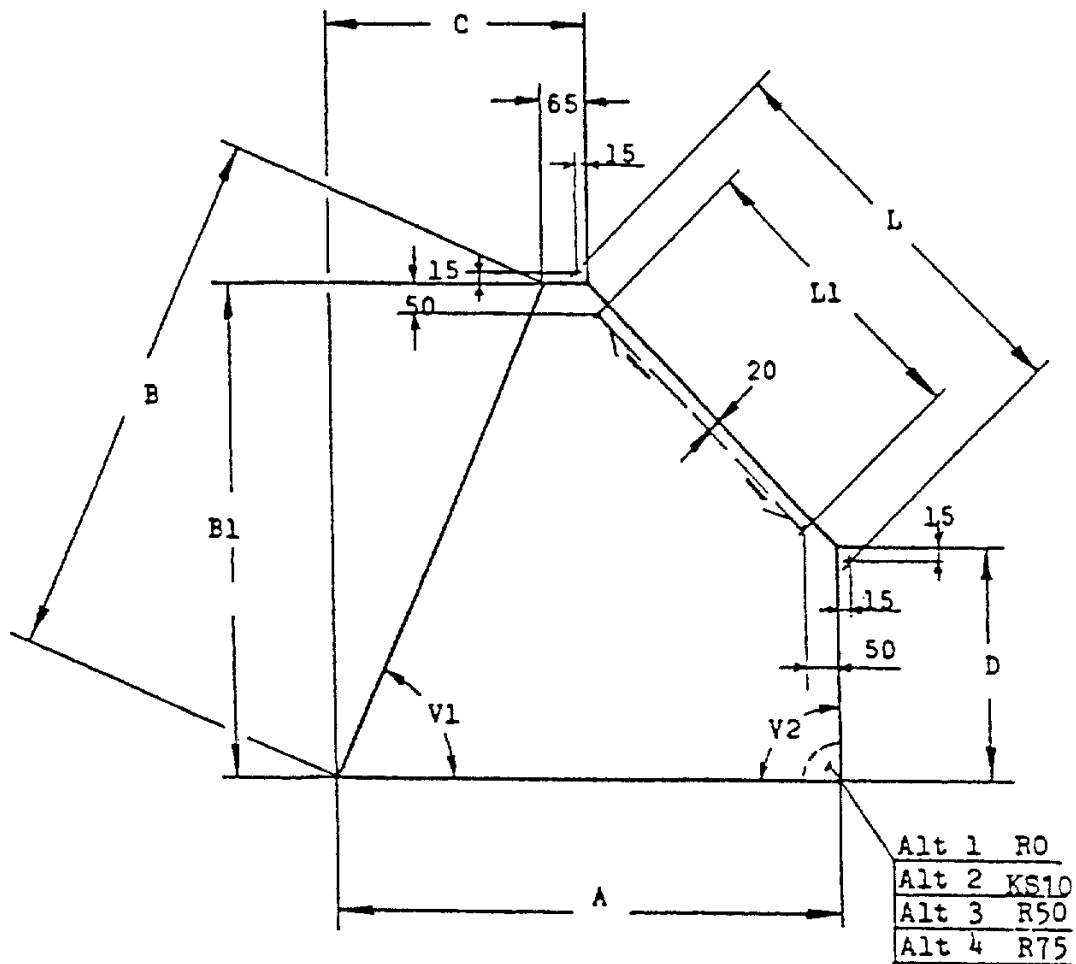
Connections for Type GE



Only one alternative (= 1).

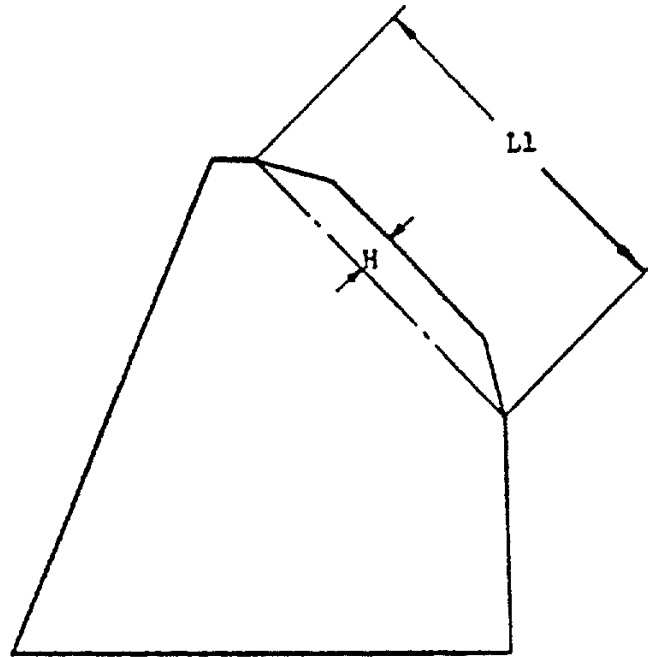
Brackets Types GC, GCK

Type GC



Brackets Types GC, GCK

Type GCK



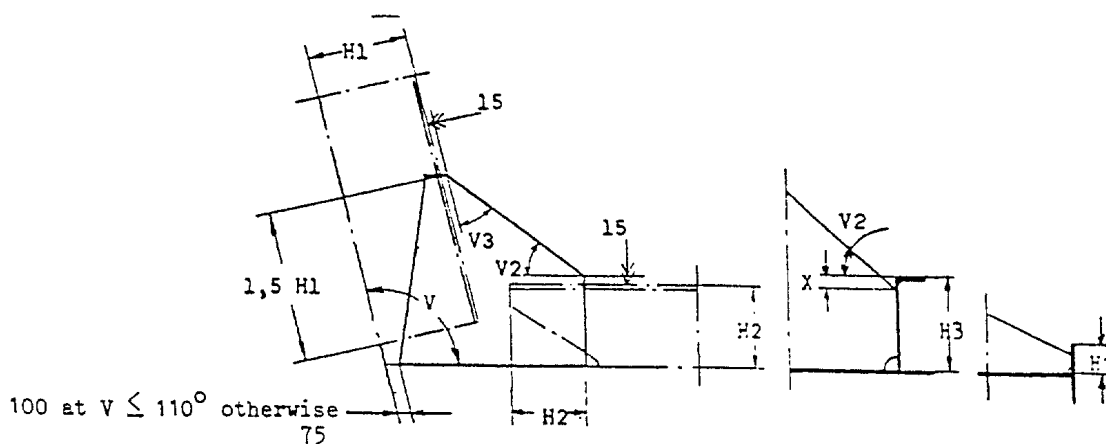
For dimensions not given see page 169.

Brackets Types GC, GCK

Description

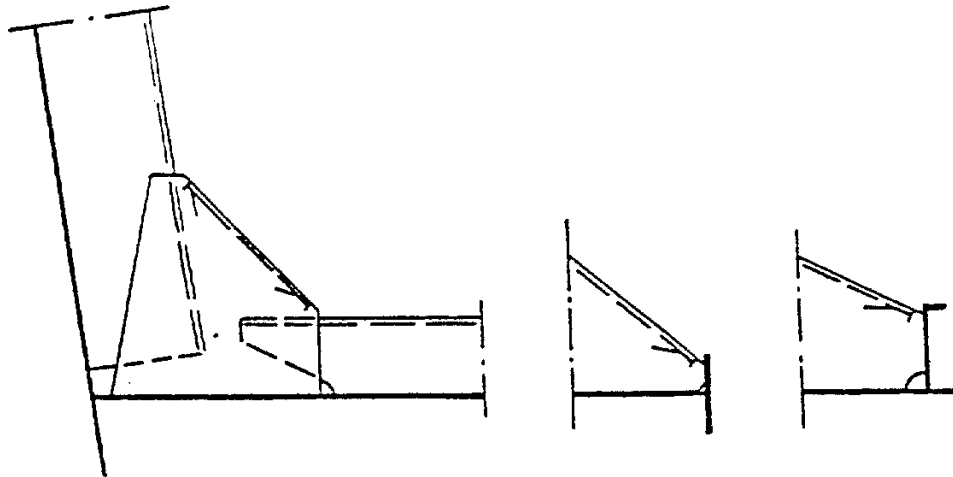
1. The A- and B-dimensions are determined by H1 and V1.
- 1a The B-dimension is measured parallel with the A-dimension.
- 1b The D-dimension are determined by H2 or H3.
2. The overlap length of the bracket is measured in line with the profile and must be $1.5 \times H1$ and $1.0 \times H2$ respectively.
3. The min width of the bracket where the profile ends, is measured at right angles to the profile and must be $0.5 \times H1$ and $0.5 \times H2$ respectively.
V2 and V3 must not, however, exceed 75° .
If $V2 \geq 45^\circ$ set $X = 15$, otherwise 0.
4. If $L \geq 800$ mm, then one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
5. The flat bar to be of the same quality as that of the plate of the bracket.
6. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, is 90×12 mm.
7. When flanging, the corresponding flange area is used.
8. All welds to be  4 when nothing else is given on the class drawing.
9. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Example of Assembly



Brackets Types GC, GCK

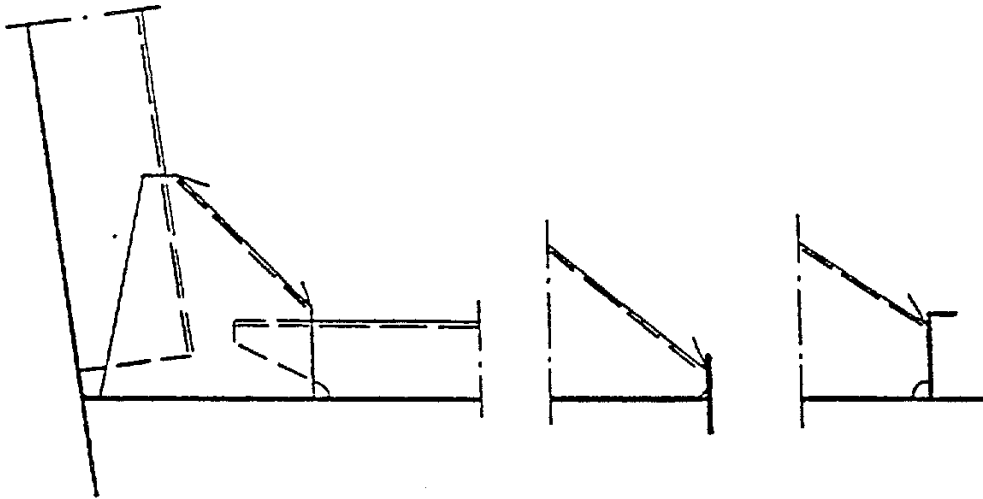
Connections for Type GC



Alt.	Cut-out
1	R0
2	KS8
3	R50
4	R75

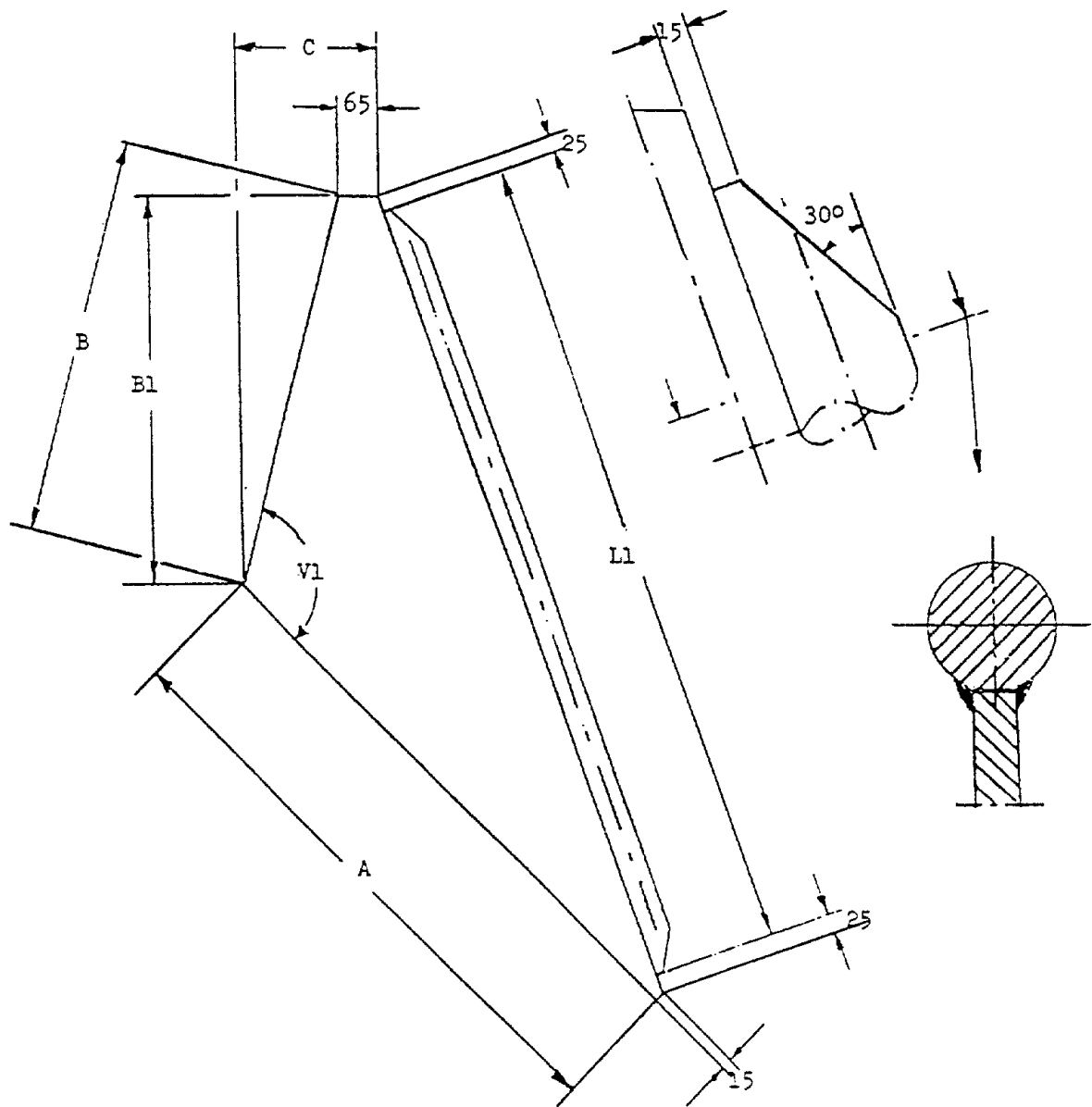
Brackets Types GC, GCK

Connections for Type GC



Alt.	Cut-out
1	R0
2	KS8
3	R50
4	R75

Brackets Type GD

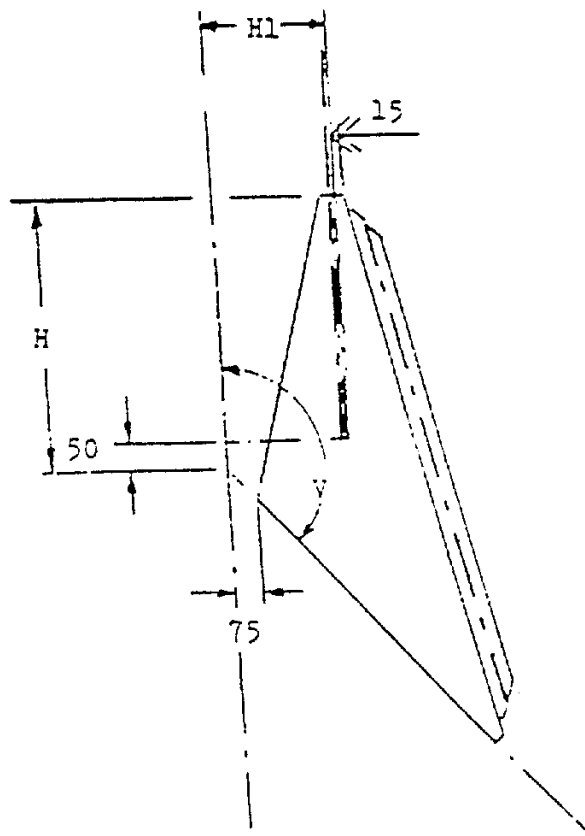


Brackets Type GD (55)

Description

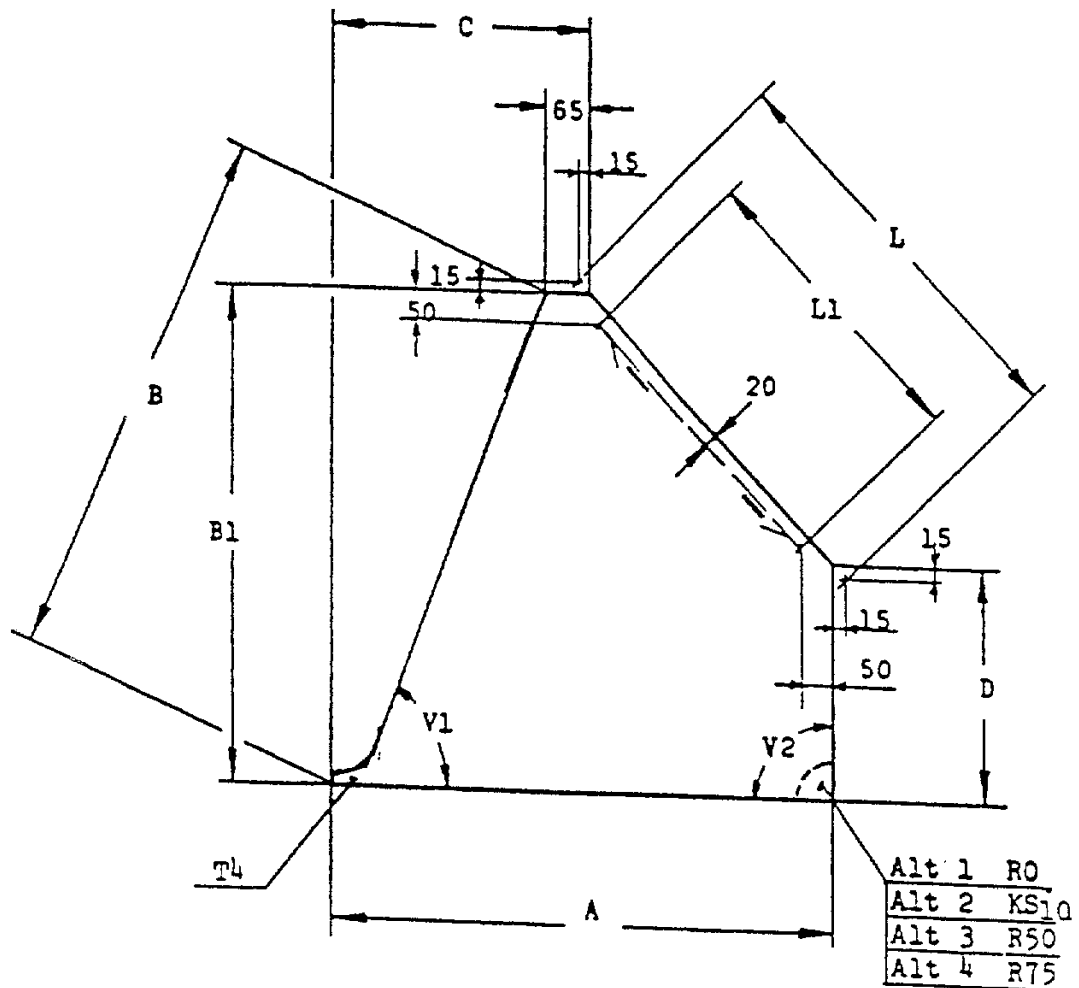
1. The A- and H-measures according to the class drawing.
2. The B1-measure = $75 \times \tan (V-9C) + H$.
3. The B-measure = $\frac{B1}{\cos(V-V1)}$
4. The C-measure = $H1-75+15$ and is measured perpendicular to the profile.
5. The length of the overlap of the bracket is measured in line with the profile and must be H-50.
6. From the class drawing is also obtained: plate thickness, dimensions of round bar, quality of material and erection weld.

Example of Assembly



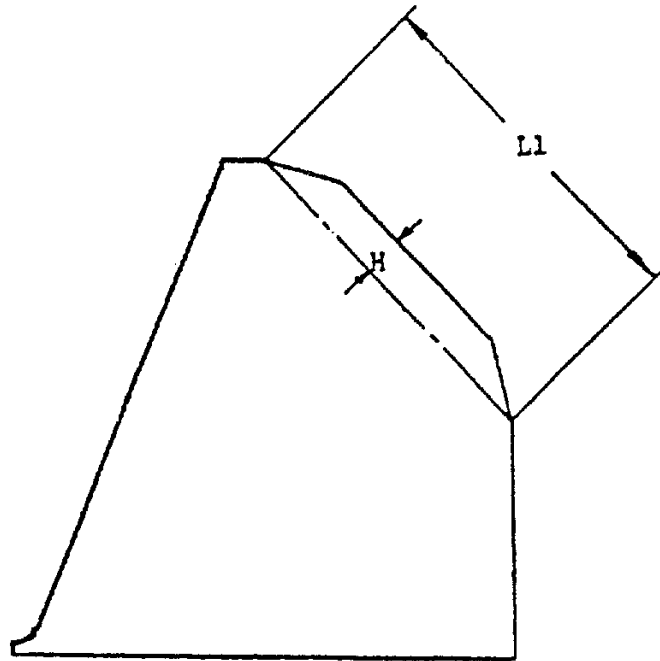
Brackets Type GE, GEK

Type GE



Bracket Types GE, GEK

Type GEK



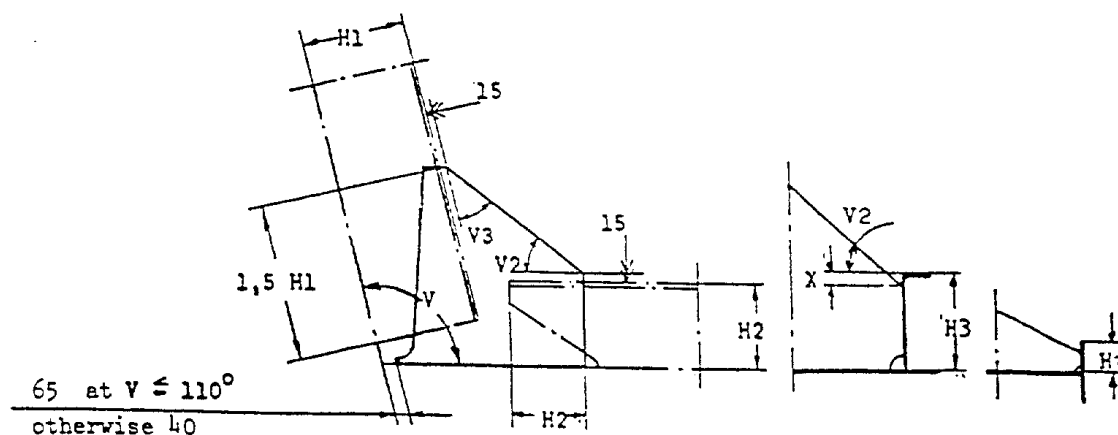
For dimensions not given see page 1.

Brackets Type GE, GEK

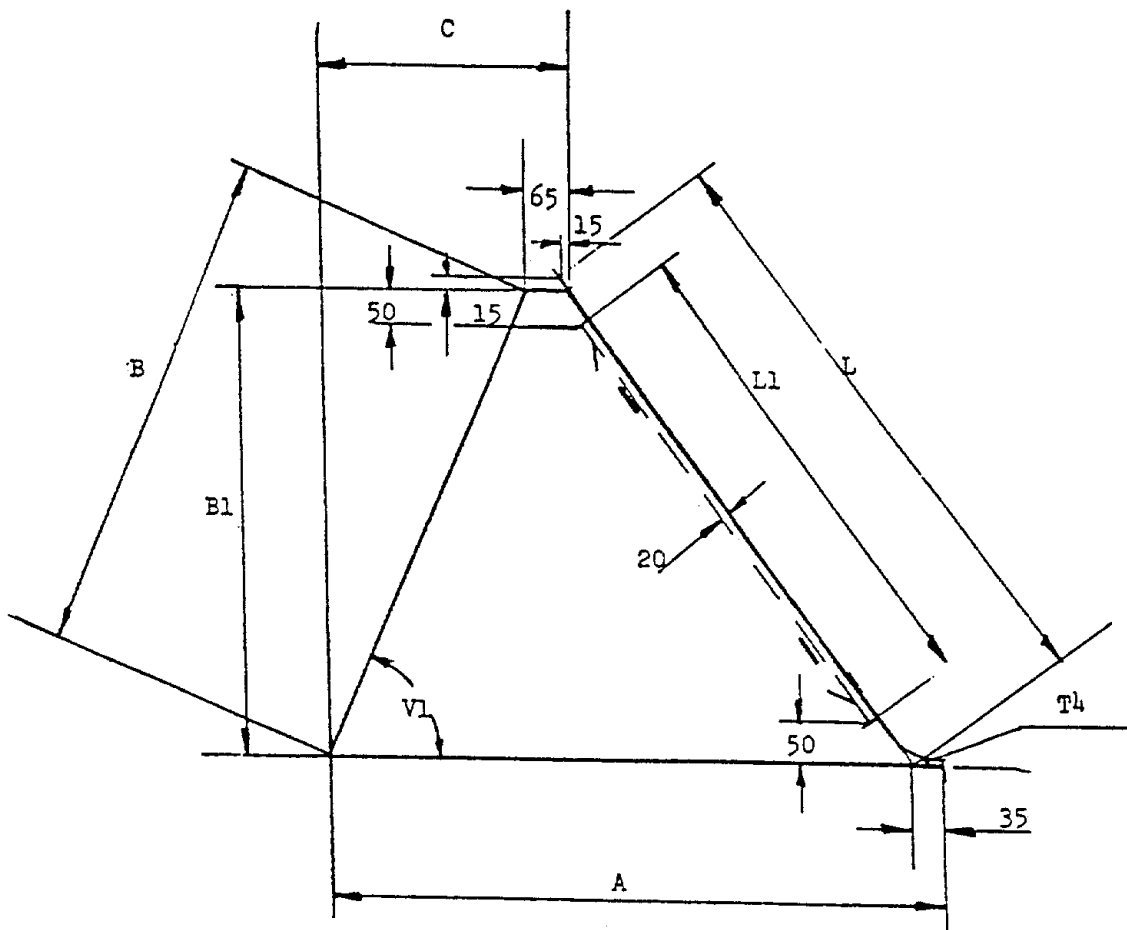
Description

1. The A- and B-dimensions are determined by H1 and V1.
- 1a The B-dimension is measured parallel with the A-dimension.
- 1b The D-dimension are determined by H2 or H3.
2. The overlap length of the bracket is measured in line with the profile and must be $1.5 \times H1$ and $1.0 \times H2$ respectively.
3. The min width of the bracket where the profile ends, is measured at right angles to the profile and must be $0.5 \times H1$ and $0.5 \times H2$ respectively.
V2 and V3 must not, however, exceed 75° .
If $V2 \geq 45^\circ$ set $X = 15$, otherwise 0.
4. If $L \leq 800$ mm, then one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
5. The flat bar to be of the same quality as that of the plate of the bracket.
6. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, is 90×12 mm.
7. When flanging, the corresponding flange area is used.
8. All welds to be  4 when nothing else is given on the class drawing.
9. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Example of Assembly



Brackets Type GAA



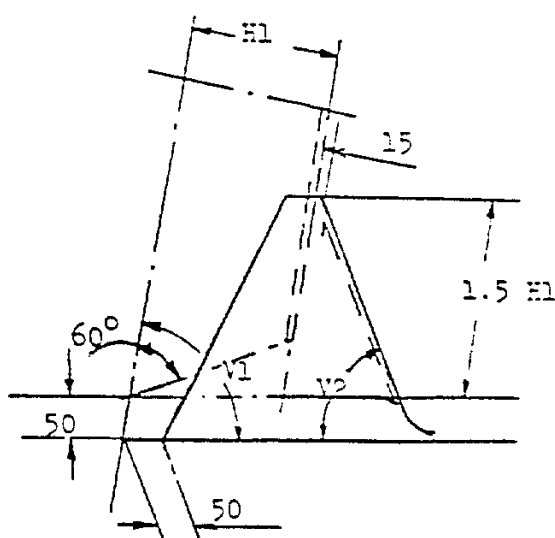
Brackets Type GAA,GAAS

Description

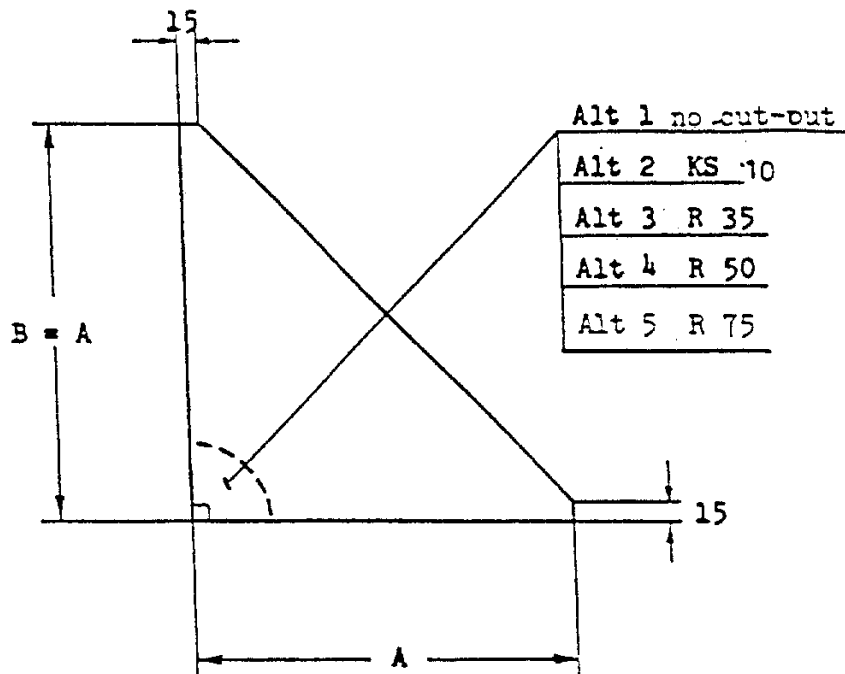
1. The A-, B- and C-dimensions are determined by H1 and V1.
- 1a The C-dimension is measured parallel with the A-dimension.
2. The overlap length of the bracket is measured in line with the profile and must be $1.5 \times H1$.
3. The min width of the bracket where the profile ends, is measured at right angles to the profile and must be $0.5 \times H1$ and $0.5 \times H2$ respectively.
V2 must not, however, exceed 75° .
4. If $L \geq 800$ mm, then one flat bar is required.
The area of this bar to be $0.14 \times L \text{ cm}^2$.
5. The flat bar to be of the same quality as that of the plate of the bracket.
6. The above area to be rounded off downwards to a suitable recommended standard dimension, the smallest bar, however, is 90×12 mm.
7. All welds to be  4 when nothing else is given on the class drawing.
8. From the class drawing is also obtained:
plate thickness, quality of material and erection weld.

Note 1-3 apply only if nothing else is given on the class drawing.

Example of Assembly

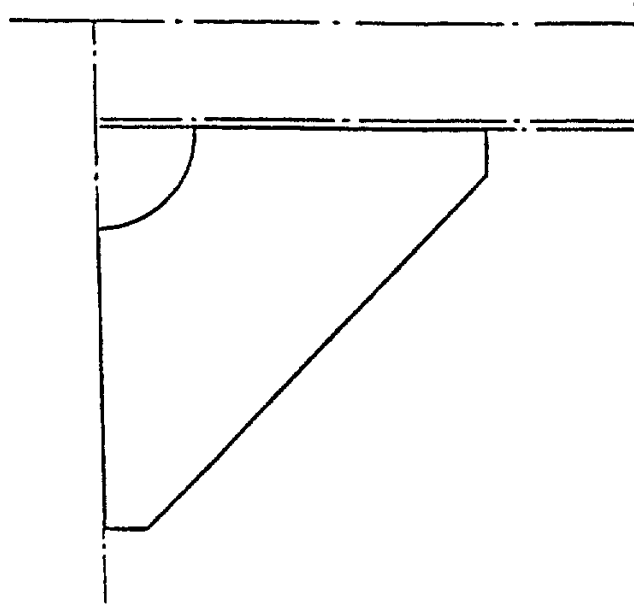


Brackets Type KL



Brackets Type KL

Example of Assembly

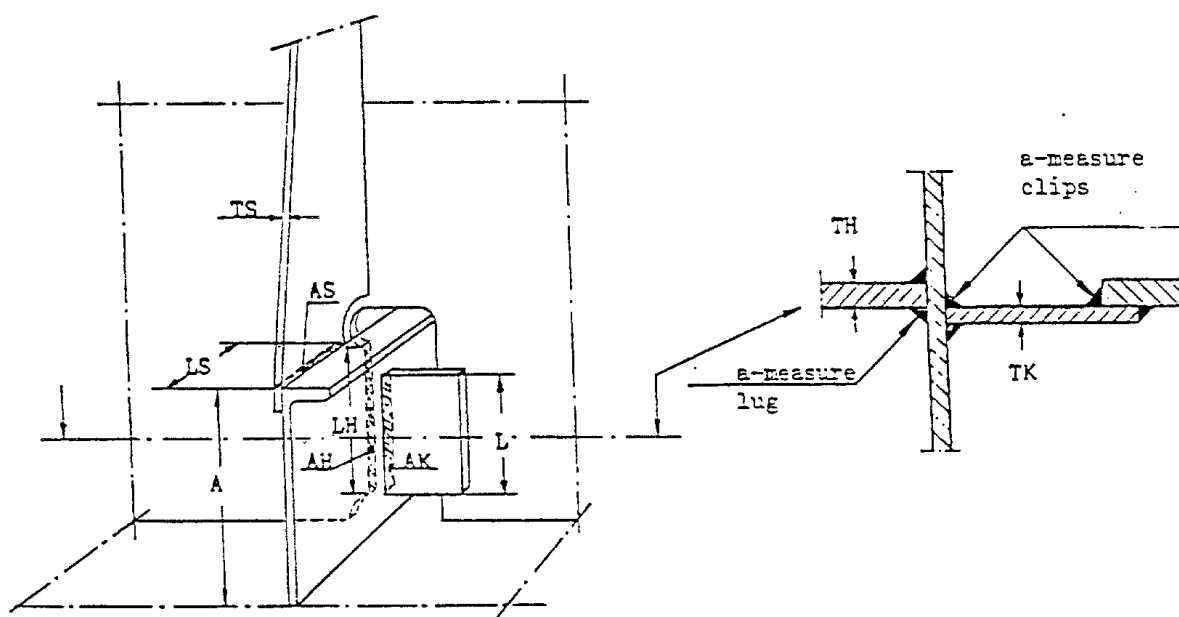


Assembly of Profiles *)

Description

1.

By the assembly area is meant the total cross-sectional area for the stiffener or bracket at the top of the profile and the shear area at the lug and the clips, which is required for securely to fix a profile to the main connection at dimensioning loading



Sketch showing the principle of attachment where:

Cross-sectional area for stiffener or bracket $= AS = LS \times TS$

Shear area at lug $= AS = LH \times TH$

Shear area at clips $= AS = L \times TK$

*) The rules in this document are valid at the Kockum Shipyard and are normally used in SBH under user control with the only exception for the case when automatic dimensioning of clips is performed by the SBH-system.

Assembly of Profiles

2.

The assembly will never be stronger than the weakest link in the connection and thus, if the requirement for the assembly area normally shall be met, we get the following standards:

- a) a-size stiffener or bracket = $0.5 \times TS$
- b) a-size lug = $0.5 \times TH$, max. however 8 mm
- c) a-size = $0.5 \times TK$, max. however 8 mm

For the clips we finish off with the following:

- d) Quality of clips at double assembly should be the same as for the main connection.
- e) Thickness of clips $TK \leq TH$, however a max. of 15 mm is normal.
- f) Length of clips $L \leq 0.5 \times A$, where A is the height of the profile in the plane of the main connection.
- g) Overlap of clips = 35 - 65 mm.

Points b, c and e do not apply to collars for tight connections.

3.

Other assemblies may occur, which automatically will increase the connection area, for instance,

- Edge brackets
- Partial walls
- Collars at watertight connections
- Collars at re-inforcements of main connections

These extra connection are of course dimensioned in accordance with the requirements for their function, while the possibility of reducing the length of the clips in the first two instances is normally not used.

4.

Normally the assembly area is given in cm² such as $AS + AH + AK$. AS is usually known, since we used for instance standard brackets with a known thickness.

AH is obtained from a known cut-out type and given profile dimensions.

We thus also obtain AK, which if we choose clip code 010 will give a clip with $L > 0.5 \times A$ adjusted to the nearest standard dimension, assuming that we know TH.

5.

Normally the rule $L > 0.5 \times A$ applies to clips when fastening lugs, provided that the profile is of "mild steel". If the profile is of "HT-36 steel", then as a rule of thumb, the clip should be given the length $L > 0.6 \times A$.

6.

If we on the back of the profile in place of a lug use a clip, then the corresponding clip length should be used, if nothing else is given.

Assembly of Profiles

7.

For the evaluation of the clip sizes we give an example with a lug-fastened angle profile:

Start with the code 000
 If not suitable try code 010 with $G1=60$ $F1=\min 75$
 010 $G1=40$ $F1=\min 75$
 020 $G1=60$ $F1=0$
 020 $G1=40$ $F1=0$
 060

If none of these are suitable, closing up is only left, i. e. altered cut-outs and clip code 048.

8.

In the program, the fastening area, a , is used, defined as lug + clip, thus without bracket on top.

If $a = 0$ follow the example in the point 7.

If $a = 0$ we get:

For cutouts types 52 and 53 full clip according to the clip code. For the other cut-out types, if there is HT-steel in the profile, we choose clip length = $0.6 \times A$ if possible, otherwise the length is reduced so that clearance is obtained, however, not less than $0.5 \times A$ if this does not manage to follow the example in point 7.

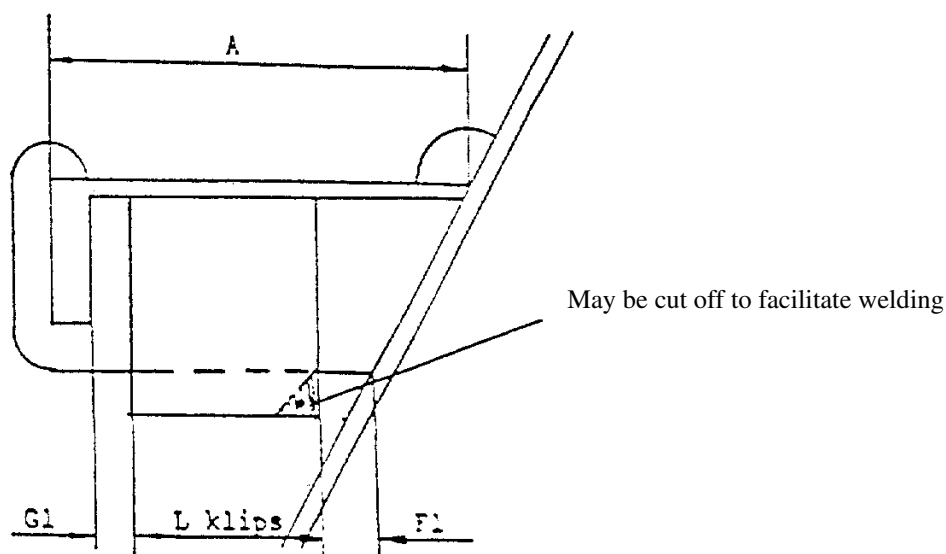
9.

If the profile is at an angle to the plate, it can be difficult to get a clip below it. In order to get several alternatives for assembly, we can in such instances tolerate less clearance between clip - flange or clip - plate.

The example in point 7 with a lug fastened angle selection can then be written (see also point 8):

Starting with code 000
 If not suitable try code 010 give e g $G1=60$ $F1=\min 75$
 010 with $G1=40$ $F1=40$
 050 $G1=0$ $F1=40$
 050 $G1=0$ $F1=40$

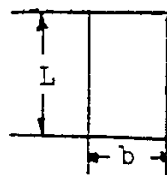
and the length reduced to a minimum of $0.5 \times A$.



If this is not suitable try code 060 with $G1=0$ $F1$ and the length $0.5 \times A$.

Assembly of Profiles

Clips



b	
mm	cm rounded off
60	6
80	8
100	10
130	13
150	15
180	18
200	20
225	22
275	27
300	30

L 1)		Ass area cm ² weld 5.5	Ass area cm ² weld 7.0
mm	cm rounded off		
75	7	8,2	10,5
100	10	11,0	14,0
125	12	13,7	17,5
150	15	16,5	21,0
175	17	19,2	24,5
200	20	22,0	28,0
225	22	24,7	31,5
250	25	27,5	35,0
275	27	30,2	38,5
300	30	33,0	42,0
325	32	35,7	45,5
350	35	38,5	49,0
375	37	41,2	52,5
400	40	44,0	56,0
425	42	46,7	59,5
450	45	49,5	63,0
475	47	52,2	66,5
500	50	55,0	70,0
525	52	57,7	73,5
550	55	60,5	77,0
575	57	63,2	80,5
600	60	66,0	84,0
625	62	68,7	87,5
650	65	71,5	91,0
675	67	74,2	94,5
700	70	77,0	98,0
725	72	79,7	101,5
750	75	82,5	105,0
775	77	85,2	108,5
800	80	88,0	112,0
825	82	90,7	115,5
850	85	93,5	119,0
875	87	96,2	122,5
900	90	99,0	126,0
925	92	101,7	129,5
950	95	104,5	133,0
975	97	107,2	136,5
1000	100	111,0	140,0

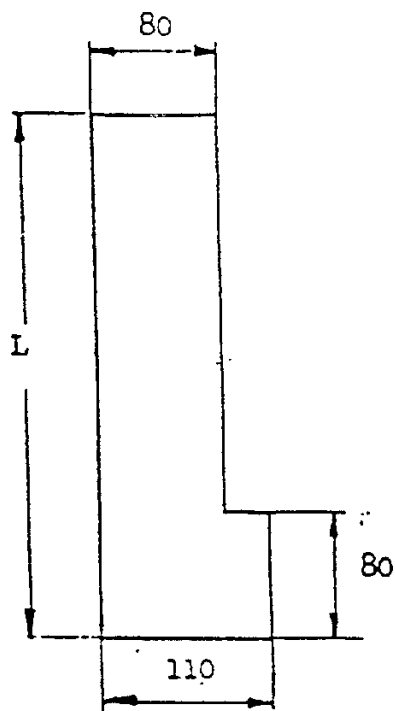
Designation: Clip b (in cm) / L (in cm) - thickness material quality (material quality not given for A-quality). The material quality is always the same as in the corresponding web or bulkhead.

Example: Clip 6/7 - 12.5 AH

1) Note: The above listing uses recommended lengths, if necessary, others may be used. In these cases choose even 10's of mm. Designation as above.

Assembly of Profiles

Clips 11



KMW Designation: Clips 11

L
210
230
250
270
320
400
500
600
650
700
800
900
1000

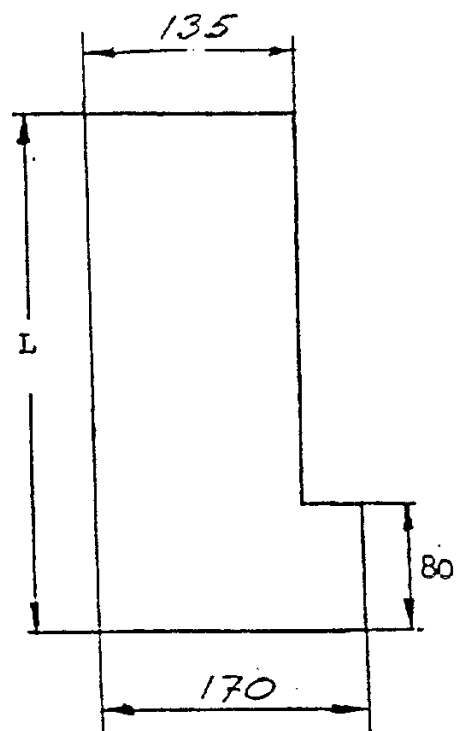
Designation:

Clips 11/L (in cm) - thickness, material quality (not given for quality A).
Material quality is always the same as for the corresponding web or bulkhead.

Example:

Clips 11/21 - 12.5 high tensile steel AH 36.

Clips 17

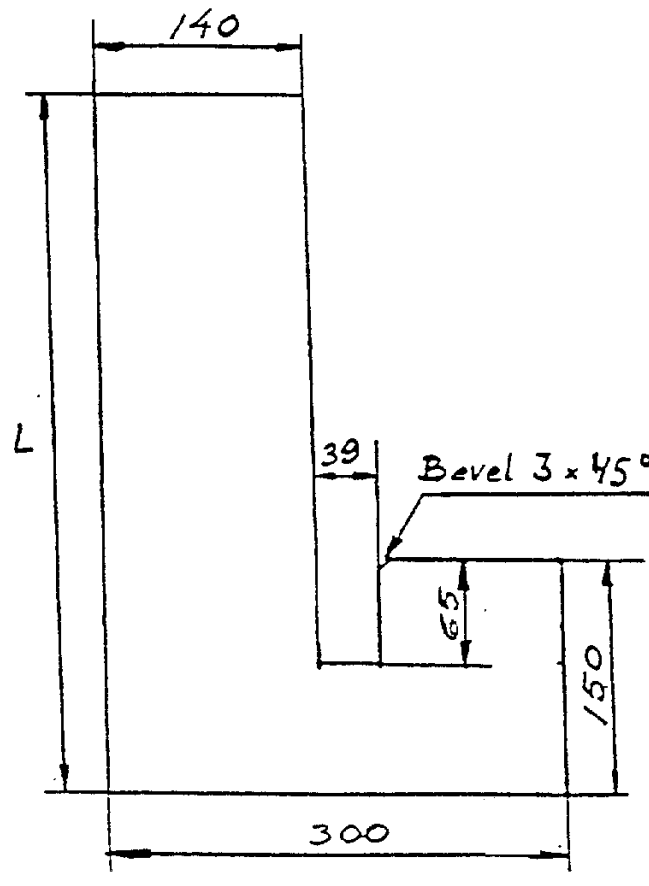


KMW Designation: Clips 17

L
210
230
250
270
320
400
500
600
650
700
800
900
1000

Assembly of Profiles

Clips



KMV Designation: Clips 30

L
380
405
430
455

Example: Clips 30/38

Assembly of Profiles

Clips

Main area of Usage for Various Clip Width

B-100 Used on flat bars and on the back of L-bars.

B-110 (Page 2) used at tight penetrations.

B-130 Used on T-bars, with flange 65–80 mm and at tight penetrations for flat bars, on the flange at T-bars and on the back and flange of L-bars.

B-150 Used on T-bars with flange 160 mm and at flange side of L-bars at 90 mm flange and on the back of L-bars of profile type 35.

B-170 (Page 2) used at tight penetrations with flat bars.

B-180 Used on T-bars with flange 200 mm and on flange side of L-bars at flange 100 mm.

B-200 Used on T-bars with flange 225 and 250 mm and on flange side of L-bars at flange 120 mm.

B-225 Used on flange side of L-bars at flange 150 mm.

B-275 Used on flange side of L-bars at flange 200 and 225 mm.

B-300 (Page 3) used on L-bars flange for sealing around the flange.

Table 1 - Sealing collars (Quality the same as bulkhead)

Plate thickness in bulkhead mm	Sealing collar thickness mm	Burnt from plate thickness mm	Marked	Weld
11 - 12	Min 11	11 - 12	11	The same as for bulkhead plate
12,5 - 14,5	Min 12,5	12,5 - 14,5	12,5	
15 - 17,5	Min 15,0	15 - 17,5	15,0	
18 - 21,5	Min 18,0	18 - 21,5	18,0	
22 - 26,5	Min 22,0	22 - 26	22,0	

Table 2 - Fastening clips (Quality the same as in web)

Plate thickness in web mm	Clip thickness mm	Burnt from plate thickness mm	Marked	Weld
11 - 12	Min 11,0	11 - 12,0	11	5,5
12,5 - 14,5	Min 12,5	12,5 - 14,5	12,5	6,5
15 - 17,5	Min 15,0	15,0 - 17,5	15,0	7,5
18 - 21,5 ¹⁾	Min 18,0	18,0 ²⁾	18,0	9,0

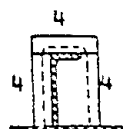
¹⁾ For greater plate thickness see respective class drawing.

²⁾ Thicker plates may be used in special cases.

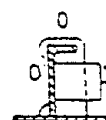
Positioning of Clips

	First digit	Second digit	Third digit
1			
2			
3	 	 	
4	 	 	
5			
6			
7	 	 	
8			
9			

Example:



Code 444



Code 010

Positioning of Clips

Description

The clip code is made up for a max of 3 clips at each connection, which gives a three-digit combination. See page 1.

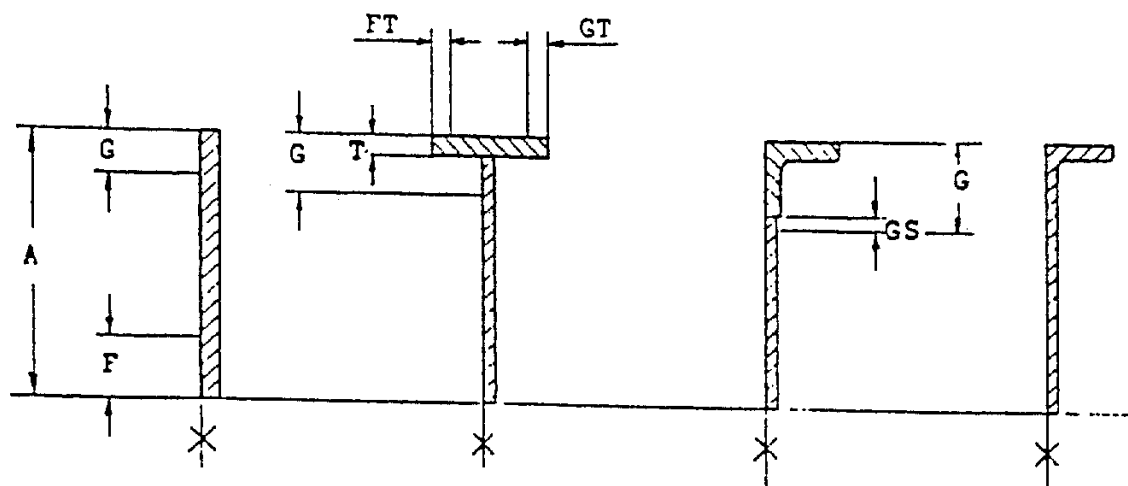
First digit gives clip position on the mould side.

Second digit gives clip position on the non-mould side.

Third digit gives clip position on the profile top.

If clips are omitted at any position, this is indicated by 0.

Positioning of Clips



For the exact positioning of clips, the following instances may occur:

$F = 0$

$F = 20$ at drainage cut-outs measured from the top edge of the drainage cut-out.

$F = 40$

$F = 50$ for $A \leq 225$

$F = 75$ for $A > 225$

$G = 0$

$G = 20$ at flat bars and at the mould side of L-profiles.

$GS = 20$ at the opposite side to the mould side of welded L-profiles.

$G = T$ at T-sections and L-sections sides opposite to the mould side.

$G = 40 + T$ at T-sections and L-sections sides opposite to the mould side

$G = 60 + T$ at T-sections and L-sections sides opposite to the mould side

Note: For welded angle profiles is first chosen $GS = 20$ for the positioning of clips on the side opposite to the mould side. If by this the clip area will become insufficient, choose $G = 60 + T$ in place.

FT respectively $GT = 0$

FT respectively $GT = 20$

